

KUS			PIPING MATERIALS CLASSIFICATION INDEX								Project No.	01000			
											Doc. No.	01000-MA-SPC-0002			
											Contr. Doc.No.				
											Rev.	R00			
Rev.	Piping Class	Service	Design Condition		C.A. (mm)	Pipe		Flange			Valve Body	Trim Mat.	Gasket	Bolting/Nuts	Remarks
			T (°C)	P (barg)		Size(in)	Material	Rating	Face	Material					
D01	A1A	LP Steam													
		Low pressure condensate													
		Methanol													
		Boiler feed water													
		Suspect Condensate	80	16.5											
		GT Washing Water	100	12											
		GT Closed Cooling Water Supply	143	8.6											
		GT Closed Cooling Water Return	-28/175	12											
		LP Boiler Blowdown	-28/340	6.0+FV											
		Oily Water	-28/340	14.0*+FV	1.5	1/2"~16"	CS	CL150	RF/B16.5	ASTM A105	CS	TR #8	SWG,304/FG,CR:CS	A193 GR. B7/ A194 GR. 2H	2,*
		Boiler Intermittent Blowdown	404	0.2											
		HRSG Burner Cooling Air	-28/140	15.0											
		Safety Valve vent	120	6											
		Inter.Drum Vent	340	6.0+FV											
		Process													
		Primary demineralised water													
PAM															
Phosphate															
Recycled water															
Ammonia solution															
	A1AV	Steam Turbine Outlet (Vacuum)	BY DEC	BY DEC	1.5	1/2"~66"	CS	CL150	RF/B16.5 RF/B16.47 A	ASTM A105	CS	TR #8	SWG,304/FG,CR:CS	A193 GR. B7/ A194 GR. 2H	2
	A1B	Isopropanol	120	6											
Safety Valve to Atmosphere		120	3	1.5	1/2"~ 3"	CS	CL150	RF/B16.5	ASTM A105	CS	TR #8	Flat Ring , Compressed Fiber Without Asbestos	A193 GR. B7/ A194 GR. 2H	5	
Tea		120	6.0+FV												
	A1D	Process													
Silone		-28/80	16.7												
Waste Water		-28/150	14.4												
Purge Gas		-28/275	10.5												
Liquid Fuel		-28/290	6.4												
Fuel Oil		250	6.0+FV	1.5	1/2"~84"	CS	CL150	RF/B16.5 RF/B16.47 A	ASTM A105	CS	TR #8	SWG,304/FG, CR:CS,IR:SS	A193 GR. B7/ A194 GR. 2H	2	
Wash Oil		80	6.0+FV												
Regeneration Air		120	6.0+FV												
	A1DV	Methanol													
2%~4%NaOH Caustic soda															
Process		-28/80	14.0+FV												
Waste Water		120	8.1+FV	1.5	1/2"~84"	CS	CL150	RF/B16.5 RF/B16.47 A	ASTM A105	CS	TR #8	SWG,304/FG, CR:CS,IR:SS	A193 GR. B7/ A194 GR. 2H	2	
Process (Vacuum)		160	8.1+FV												
			-28/320	8.1+FV											
			-28/170	3.5+FV											
			-28/340	6+FV											

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			T (°C)	P (barg)		Size(in)	Material	Rating	Face	Material					
	A1E	Hydrogen Hydrocarbon Hydrogen Rich Offgas Dryer Regeneration Effluent LP Fuel Gas	-28/80 -28/290 -28/315	14.4 10.5 6.4	1.5	1/2"~24"	CS	CL150	RF/B16.5	ASTM A105	CS	TR #1	SWG,304/FG, CR:CS,IR:SS	A193 GR. B7/ A 194 GR. 2H	2
	A1K	Dry Flare Wet Flare	250 -28/150 -28/250	3.5 9.0 3.5	1.5	1/2"~30"	CS	CL150	RF/B16.5 RF/B16.47 A	ASTM A105	CS	TR #1	SWG,304/FG,CR:CS	A193 GR. B7/ A194 GR. 2H	2
	A1R	50%NaOH Caustic soda liquid Caustic liquid Urea	50 60	6 8	3	1/2"~8"	CS	CL150	RF/B16.5	ASTM A105	CS	TR #2	SWG,304/FG,CR:CS	A193 GR. B7/ A194 GR. 2H	3
	A1S	98%H2SO4 Acid liquid	50	6	3	1/2"~8"	CS	CL150	RF/B16.5	ASTM A105	CS	TR #13	SWG,316/FG, CR:SS,IR:SS	A193 GR. B7/ A194 GR. 2H	
	A1W	Slurry drainage Regeneration effluent Neutralizing water	70	11	3	1/2"~24"	CS	CL150	RF/B16.5	ASTM A105	CS	TR #5	COMPOSITE GASKET, 304+GRAF	A193 GR. B7/ A194 GR. 2H	
	A2A	Atmosphere Vent Catalyst Cooling Water Return Cooling Water Supply Dry Flare Cold Liquid Drain Low Pressure Nitrogen—4.0 barg Process Plant Air Powder Propylene Refrigerant Safety Valve to Atmosphere Nitrogen Dry air Propane Plant Air Vent to Atmosphere Propylene flare Natural gas Oxygen-enriched air Flare Discharge Dryer Regeneration Injection Gas Dry Out Air in PDH Unit Washing Oil	-38/80 -38/80 -45/80 -48/80 -38/70 70 -38/120 120 -38/140 -48/160 -48/200 -38/305 -38/315 -38/427 -48/80 -38/145 -38/425	-0.05/0.1 10 16 16.7 10 10.5 6 3 14 6 6 10 6.4 3 7 10 3.5	1.5	1/2"~60"	LTCS	CL150	RF/B16.5	ASTM A350 GR.LF2 CL1	LTCS	TR #10	SWG,304/FG,CR:SS ,IR:SS	A320 GR.L7/ A194 GR. 4	1

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Rev.	Piping Class	Service	Design Condition		C.A.	Pipe		Flange			Valve Body	Trim Mat.	Gasket	Bolting/Nuts	Remarks
			T (°C)	P (barg)	(mm)	Size(in)	Material	Rating	Face	Material					
	A21A	Atmosphere Vent Demineralized Water Dryer Regeneration Heating Recycled water PAM Phosphate Filter outlet water Oxygen Scavenger Ammonia Pellet water Wash Water	60 70 80 80 -38/80 -38/275	16.8 11 -0.05/3 12 14.4 10.5	0.5	1/2"~24"	304/304L SS	CL150	RF LJ/B16.5	ASTM A182 GR. F304/304L	316SS	TR #10	Flat Ring,PTFE Envelope SPW GASKET FOR Dryer Regeneration Heating	A193 GR. B7/ A194 GR. 2H	
	A2D	Wash Oil Process Offgas	-48/80 -38/145 -38/150	14.4 10 14.4	1.5	1/2"~24"	LTCS	CL150	RF/B16.5	ASTM A350 GR. LF2 CL. 1	LTCS	TR #10	SWG,304/FG,CR:SS	A320 GR. L7 A 194 GR. 4	1
	A21D	Lube Oil (Vendor Package) Lube Oil	60 120	4.5 ATM	0.5	1/2"~12"	304/304LSS	CL150	RF/B16.5	ASTM A182 GR. F304/304L	316SS	TR #10	COMPRESSED NON- ASBESTOS, FLAT RING	A320 GR.B8 CL2 A194 GR..8	
	A2DV	Injection Gas/Nitrogen (Vacuum)	-48/80 -38/145 -38/170	14.4 10 14.4	1.5	1/2"~24"	LTCS	CL150	RF/B16.5	ASTM A350 GR. LF2 CL. 1	LTCS	TR #10	SWG,304/FG,CR:SS	A320 GR. L7 A 194 GR. 4	1
	A2E	Dryer regeneration effluent offgas, Hydrogen flare	-38/80 -38/120	14.4 14	1.5	1/2"~24"	LTCS	CL150	RF/B16.5	ASTM A350 GR. LF2 CL. 1	LTCS	TR #10	SWG,304/FG,CR:SS	A320 GR. L7 A 194 GR. 4	1
	A21K	Atmosphere Vent Catalyst Isopropanol Liquid/Melttable Additive Low Pressure Nitrogen—4.0 barg Peroxide Safety Valve to Atmosphere LT Process Liquid Drain Propylene Propylene flare Methanol flare	-38/80 -38/80 -38/80 -48/80 120 -38/120 -50/150	6 -0.05/0.1 6.0+FV 16.7 6.0+FV 6 12	0.5	1/2"~30"	304/304L SS (Impact Tested)	CL150	RF/B16.5 RF/B16.47 A	ASTM A182 GR. F304/304L	316SS	TR #10	SWG,304/FG,CR:SS ,IR:SS	A320 GR. B8 CL2/ A194 GR. 8	
	A21L	LT Process Cold Flare Cold Liquid Drain Ethylene Refrigerant Liquid Nitrogen Ethylene flare Ethylene	-105/80 -105/80 -103/80 -196	11.6+FV 14.4 7 10	0.5	1/2"~24"	304/304LSS (Impact Tested)	CL150	RF/B16.5	ASTM A182 GR. F304/304L	316SS	TR #10	SWG,304/FG,CR:SS ,IR:SS	A320 GR.B8 CL2 A194 GR..8	
	A23B	DMDS PAC	DEC 60	DEC 8	0.5	1/2"~16"	316/316LSS	CL150	RF/B16.5	ASTM A182 GR.F316/316L	316SS	TR #16	SPW GASKET 316LSS/PTFE FILLE CR- SS	A320 GR.B8 CL2 A194 GR..8	

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			T (°C)	P (barg)	(mm)	Size(in)	Material	Rating	Face	Material					
	A23D	Hydraulic Oil Seal Oil (after filter)	DEC	DEC	0.5	1/2"~16"	316/316LSS	CL150	RF/B16.5	ASTM A182 GR. F316/316L,	316SS	TR #10	FILLED OR REINFORCED PTFE, FLAT RING	A320 GR.B8 CL2 A194 GR..8	
	A24P	Low Pressure Nitrogen—4.0 barg Powder Solid/Powder Additive	-38/120 -38/120 -38/80	6 6 -0.05/0.1	0.5	1/2"~24"	304/304L SS	CL150	RF LJ	ASTM A182 GR. F304/304L	316SS	TR #10	Flat Gasket Fiber Without Asbestos	A193 GR. B7/ A194 GR. 2H	
	A24P1	Low Pressure Nitrogen—4.0 barg PP Pellets Conveying (with and w/o Solids)	-38/120	6	0.5	1/2"~24"	304/304L SS	CL150	RF LJ	ASTM A182 GR. F304/304L	316SS	TR#10	Flat Gasket Fiber Without Asbestos	A193 GR. B7/ A194 GR. 2H	
	A24P2	Low Pressure Nitrogen—4.0 barg PP Powder Conveying (with and w/o Solids)	-38/120	0.2	0.5	1/2"~4"	304/304L SS	CL150	RF LJ	ASTM A182 GR. F304/304L	316SS	TR #10	Flat Gasket Fiber Without Asbestos	A193 GR. B7/ A194 GR. 2H	
	A24P3	Low Pressure Nitrogen—4.0 barg	-38/120	6	0.5	1/2"~24"	304/304L SS	CL150	RF LJ	ASTM A182 GR. F304/304L	316SS	TR#10	Flat Gasket Fiber Without Asbestos	A193 GR. B7/ A194 GR. 2H	HOLD
	A01S	Sour water Ro reject water 1%~3%H2SO4 Acid liquid Acid liquid	50 60 70	6 10 10	0	1/2"~24"	CS LINED PTFE	CL150	RF/B16.5	CS LINED PTFE	CS LINED FEP		FLAT GASKET, PTFE ENVELOPE	A193 GR. B7/ A194 GR. 2H	
	A02T	Sodium hypochlorite pipeline Corrosion inhibitor pipeline Coagulant pipeline Flocculation pipeline Scale inhibitor pipeline Reducing agent pipeline Ammonia Phosphate(AP)	AMB	10	0	1/2"~8"	PVDF	—	—	PVDF	PVDF	EPDM	MFR	MFR	
	A03T	Sodium hypochlorite pipeline Corrosion inhibitor pipeline Coagulant pipeline Flocculation pipeline Scale inhibitor pipeline Ammonia Phosphate(AP) Reducing agent pipeline	AMB	10	0	1/2"~8"	PVC-U	—	—	PVC-U	PVC-U	EPDM	MFR	MFR	4
	B1A	MP Condensate MP Steam Natural Gas	-28/80 -28/360	23 + FV 17.2+FV	1.5	1/2"~24"	CS	CL300	RF/B16.5	ASTM A105	CS	TR #8	SWG,304/FG,CR:CS ,IR:SS	A193 GR. B7 A 194 GR. 2H	2
	B1B	Ethylene Refrigerant Process	-28/105 -28/145	34.2 18.5	1.5	1/2"~30"	CS	CL300	RF/B16.5 RF/B16.47 A	ASTM A105	CS	TR #1	SWG,304/FG,CR:CS ,IR:SS	A193 GR. B7 A 194 GR. 2H	2
	B1E	Hydrogen Hydrogen (from PSA)	80 -28/80 -28/125 -28/80	42 23 25 44	1.5	1/2"~24"	CS	CL300	RF/B16.5	ASTM A105	CS	TR #1	SWG,304/FG,CR:CS ,IR:SS	A193 GR. B7/ A194 GR. 2H	2

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			T (°C)	P (barg)	(mm)	Size(in)	Material	Rating	Face	Material					
	B2A	Cooling Water Return Cooling Water Supply Dry Flare Cold Liquid Drain Process Process (Propane) Process (Recycle Propylene) Propylene Refrigerant Ammonia Propane Liquid Propane Propane Gas Propane Steam Instrument air Propylene Ethylene Natural gas	-38/70 70 -45/80 -48/80 -48/110 -48/115 135 -48/140 -48/160 -45/80	31.5 40 45 34.2 34.2 22 45 17.5 40 25	1.5	1/2"~20"	LTCS	CL300	RF/B16.5	ASTM A350 GR.LF2 CL1	LTCS	TR #10	SWG,304/FG,CR:SS ,IR:SS	A320 GR.L7/ A194 GR. 4	1
	B2B	Ethylene Refrigerant Process	-38/105 -38/145	34.2 18.5	1.5	1/2"~24"	LTCS	CL300	RF/B16.5	ASTM A350 GR. LF2 CL. 1	LTCS	TR #10	SWG,304/FG,CR:SS ,IR:SS	A320 GR. L7 A 194 GR. 4	1
	B2E	Hydrogen	-38/80 -38/80	42 44	3	1/2"~12"	LTCS	CL300	RF/B16.5	ASTM A350 GR. LF2 CL. 1	LTCS	TR #10	SWG,304/FG,OR:SS ,IR:SS	A320 GR. L7 A 194 GR. 4	1
	B21A	Oxygen Scavenger Filter outlet water	-38/70 70 80	31.5 25 24	0.5	1/2"~24"	304SS	CL300	RF/B16.5	ASTM A182 GR. F304SS	316SS	TR #10	FLAT GASKET, TELFON ENVELOPE	A193 GR. B7 A 194 GR. 2H	
	B21D	Lube oil	80	20	0.5	1/2"~12"	304SS	CL300	RF/B16.5	ASTM A182 GR. F304SS	316SS	TR #10	SWG,304/FG,CR:SS ,IR:SS	A320 GR.B8 CL2 A194 GR..8	
	B21K	Propylene Ethylene	-48/80 -50/150	34.2 25	0.5	1/2"~24"	304/304LSS (Impact Tested)	CL300	RF/B16.5	ASTM A182 GR. F304/304L	316SS	TR #10	SWG,304/FG,CR:SS ,IR:SS	A320 GR.B8 CL2 A194 GR..8	
	B21L	Ethylene Refrigerant (Liquid) Cold Liquid Drain Process Ethylene Ethylene flare	-104/80 -105/110 -105/80	25 34.5 23	0.5	1/2"~24"	304/304LSS (Impact Tested)	CL300	RF/B16.5	ASTM A182 GR. F304/304L	316SS	TR #10	SWG,304/FG, CR:SS ,IR:SS	A320 GR.B8 CL2 A194 GR..8	
	B27A	High Temperature Process HC (Vacuum)	-38/649	2.76 + FV	0.0	1/2"~78"	321HSS	CL300	RTJ/B16.5 RTJ/B16.47A	ASTM A182 GR. F321H	316SS	347H TRIM FULL HF SEAT	SOLID OVAL RING, 347HSS	HEAVY HEX NUTS HASTELLOY-X, ASME SB572 UNS N06002	
	B27B	Regeneration Air	-28/649 -28/704	1 2.6	0.0	1/2"~96"	321HSS	CL300	RTJ/B16.5 RTJ/B16.47A	ASTM A182 GR. F321H	316SS	347H TRIM FULL HF SEAT	SOLID OVAL RING, 347HSS	HEAVY HEX NUTS HASTELLOY-X, ASME SB572 UNS N06002	
	B54H	LP Steam @ PDH Reactor Reduction Gas @ PDH Reactor	-28/340	2.76 + FV	3	1/2"~24"	1-1/4CR –1/2MO	CL300	RTJ/B16.5	ASTM A182-F11 CL2	1-1/4CR– 1/2MO	TR #8	OVAL RING ,5 CR-1/2 MO, MAX HARDNESS BHN 130	A193 GR. B16, A 194 GR.8	2

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			T (°C)	P (barg)	(mm)	Size(in)	Material	Rating	Face	Material					
	B54K	Gas Turbine Exhaust * DEC to confirm based on selected gas turbine.	-28/578	0.5	1.5	1/2"~90"	1-1/4CR –1/2MO	CL300	RTJ/B16.5 RTJ/B16.47A	ASTM A182-F11 CL2	1-1/4CR– 1/2MO	TR #8	SOLID OVAL RING, 5 Chromium	A193 GR. B16, A 194 GR.8	2
	B09A	Propane	HOLD	45	1.9	12"	X56N	CL300	RF/B16.5	ASTM A694 GR.F56	316SS	TR #10	SWG,304/FG, ,IR:SS	CR:SS A193 GR. B7 A 194 GR. 2H	
D01	D1A	Boiler-Feed-Water HP Steam HP Condensate HP Boiler Blowdown HP Boiler Continuous Blowdown	-28/145 143 275 -28/427 -28/80	70 + FV 78 58.3 50+FV 80	1.5	1/2"~24"	CS	CL600	RF/B16.5	ASTM A105	CS	TR #8	SWG,304/FG, ,IR:SS	CR:CS A193 GR. B7 A 194 GR. 2H	2
	D1B	HP Boiler Blowdown Process	DEC	DEC	3	1/2"~14"	CS	CL600	RF/B16.5	ASTM A105	CS	TR #8	SWG,304/FG, ,IR:SS	CR:CS A193 GR. B7 A 194 GR. 2H	2
	D2A	Process Natural gas	-48/80 -38/80	56.5 61	1.5	1/2"~12"	LTCS	CL600	RF/B16.5	A350 GR.LF2 CL1	LTCS	TR #10	SWG,304/FG,OR:SS ,IR:SS	A320 GR. L7/ A194 GR. 4	1
	D21A	Phosphate	80	60	0.5	1/2"~2"	304SS	CL600	RF/B16.5	ASTM A182 GR.F304	316SS	TR #10	SWG,304/FG,OR:SS ,IR:SS	A193 GR. B7 A 194 GR. 2H	1
	D21K	Catalyst High Pressure Nitrogen—40.0 barg Isopropanol Cold Liquid Drain Low Pressure Nitrogen—4.0 barg Process Powder Silane Tea Washing Oil Propylene Recycle Gas	-38/80 -38/80 -48/80 80 -50/80 -48/160	48 50+FV 56.5 48+FV 60 45	0.5	1/2"~12"	304/304L SS (Impact Tested)	CL600	RF/B16.5	ASTM A182 GR. F304/304L	316SS	TR #10	SWG,304/FG,OR:SS ,IR:SS	A320 GR. B8 CL2/ A194 GR. 8	5
	D23B	Chemical Injection (Phosphate)	-38/80	49.5	0.5	1/2"~8"	316/316LSS	CL600	RF/B16.5	ASTM A182 GR. F316/316L	316SS	TR #16	SPW GASKET 316LSS/PTFE FILLER CR-316LSS IR-316LSS	A320 GR. B8M A 194 GR.8	
	D24L	Catalyst Cold Liquid Drain	-48/80 -48/80	48 54	0.5	1/2"~3"	316L SS	CL600	RF/B16.5	ASTM A182 GR.F316L	316LSS	TR #10	SWG,316/FG,OR:CS ,IR:SS	A320 GR. B8 CL2/ A194 GR. 8	
	D24P	High Pressure Nitrogen—40.0 barg Powder Propylene	-48/160	50	0.5	1/2"~16"	304/304L SS	CL600	RF/B16.5	ASTM A182 GR. F304/304L	316SS	TR #12	SWG,304/FG,OR:SS ,IR:SS	A320 GR. B8 CL2/A194 GR. 8	
	E2A	Natural gas(FG) Propane	-38/80 -45/80	75 75	1.5	1/2"~12"	LTCS	CL900	RTJ/B16.5	ASTM A350 GR.LF2	LTCS	TR #16	OVAL RING, 316SS	A320 GR. L7/ A194 GR. 4	1
	E54K	High pressure steam HP Clean Condensate	490	54	1.5	1/2"~6"	1-1/4CR –1/2MO	CL900	RTJ/B16.5	ASTM A182-F11 CL2	1-1/4CR– 1/2MO	TR #15	OVAL RING, 316SS	A193 GR. B7/ A194 GR. 2H	
	F2A	Carbon Dioxide High Pressure Nitrogen—40.0 barg Powder	80 -48/160	165 165	1.5	1/2"~2 1/2"	LTCS	CL1500	RF/B16.5	ASTM A350 GR.LF2 CL1	LTCS	TR #10	SWG,304/FG,OR:SS ,IR:SS	A320 GR.L7/ A194 GR. 4	
	F23A	Hydraulic Oil Seal Oil	DEC	DEC	0.5	1/2"~4"	316/316LSS	CL1500	RF/B16.5	ASTM A182 GR. F316/316L,	316SS	TR #16	SPW GASKET 316LSS/PTFE FILLER CR-316L IR-316LSS	A320 GR. B8M A 194 GR.8	

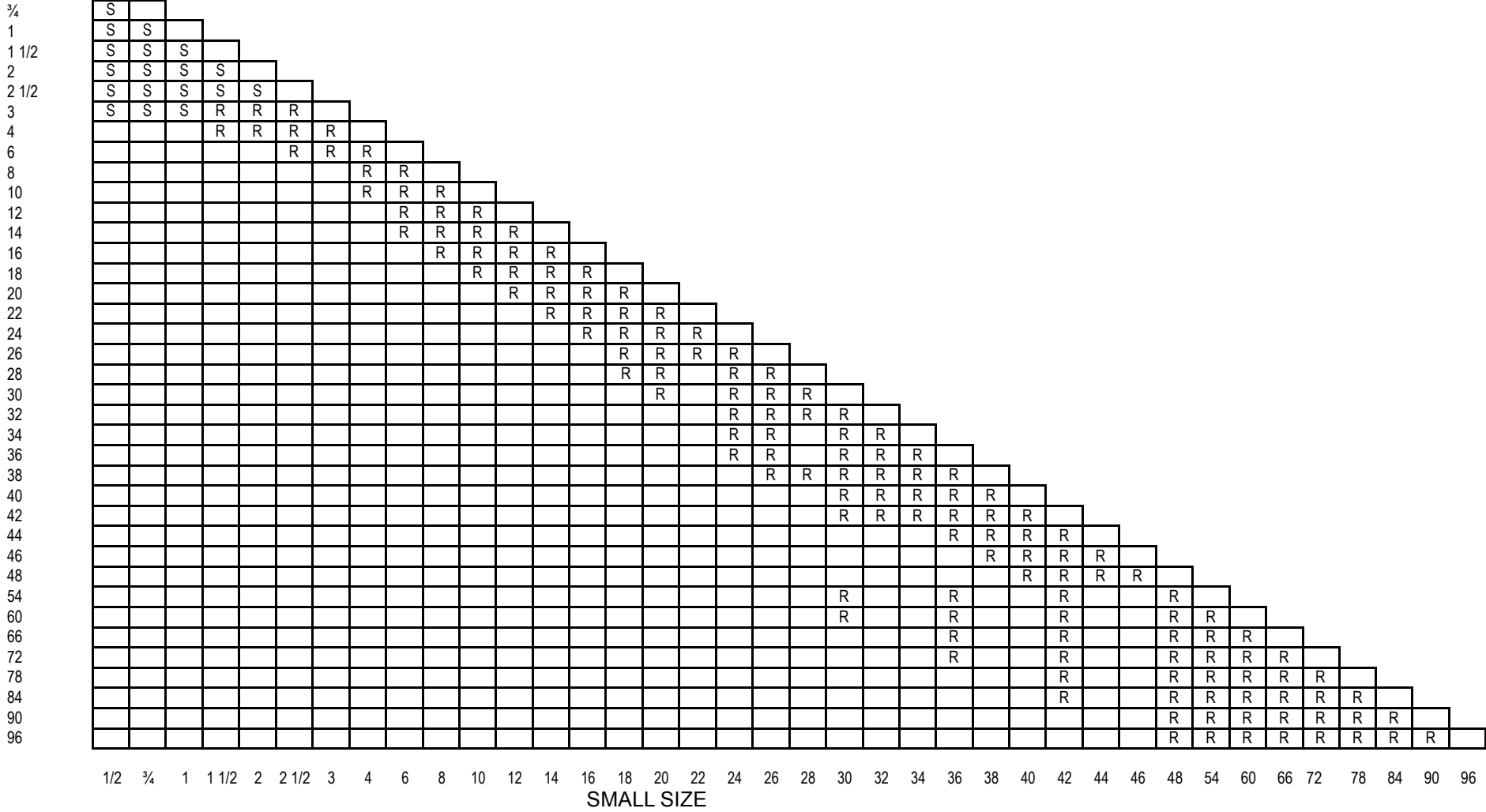
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	G82E	Hydrogen	-38/220 -28/85	210 220	0.5	1/2"~12"	316L SS	CL2500	RF/B16.5	ASTM A182 GR.F316L	316LSS	TR #10	OVAL RING, 316SS	A320 GR.L7/ A194 GR.4	
	H1A	Cooling Water Return Cooling Water Supply Safety Valve to Atmosphere Blow down Chemical Drain Service Water Waste Water Non-Contaminated Sewer Oily Contaminated Rainy Water Drain Fire Water (Aboveground) Hot water High pressure fire water (Aboveground) Raw water Pre-treatment outlet water Service water Recoveryed water Ro washing water Chemical Drain Service Water Clean surface water Wet Air Oily wastewater Sanitary wastewater Blower air Retentant Chemical Sewer Off gas Return sludge Excess activated sludge Liquid slop oil Oily sludge Scum Caustic liquid Urea(Hold) Cooling Water (UG)	60 70 80 -28/80 0/60 100 120 -28/125 -28/140 -28/210 -28/340	-0.035/2 10.5 10 10.5 10 8 8 6.5 14 10.0+FV 6+FV	3	1/2"~72"	CS	CL150	RF/B16.5 RF/B16.47 A	ASTM A105	CS	TR #1	COMPOSITE GASKET, 304+GRAF	A193 GR. B7/ A194 GR. 2H	2
D01	H2A	Service Water Waste Water Blow down Chemical Drain Service Water Non-Contaminated Sewer Oily Contaminated Rainy Water Drain Fire Water (Aboveground) Cooling Water Return Cooling Water Supply Saturation Air High pressure fire water (underground)	-38/80 -28/80 -48/80 120 -28/125 -28/210 -28/340	14 10.5 12 11 6.5 10+FV 6+FV	3	1/2"~16"	LTCS	CL150	RF/B16.5	ASTM A350 GR.LF2 CL1	LTCS	TR #10	SWG,304/FG,OR:SS ,IR:SS	A320 GR.L7/ A194 GR. 4	1

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											Rev.	R00			
Rev.	Piping Class	Service	Design Condition		C.A.	Pipe		Flange			Valve Body	Trim Mat.	Gasket	Bolting/Nuts	Remarks
			T (°C)	P (barg)	(mm)	Size(in)	Material	Rating	Face	Material					
	H3A	Instrument Air Potable Water Fire Water	-38/70 -38/60 70 80	10 12 10 12	1.5	1/2"~16"	LTCS Galv.	CL150	RF/B16.5	ASTM A350 GR.LF2 CL1 GALV.	LTCS	TR #10	Flat Gasket Fiber Without Asbestos	A320 GR.L7/ A194 GR. 4	1
	H3B	Fire Water	0/60	12	0	1/2"~12"	CS Galv.	CL150	RF/B16.5	ASTM A105	CS	TR #8	COMPOSITE GASKET, 304+GRAF	A193 GR. B7/ A194 GR. 2H	
D01	H61B	Service Water (UG) Oily Contaminated Rainy Water(UG) Clean Surface Water (UG) High Salt Water Clean Surface Water	0/40	10	0	2"~80"	FRP	CL150	FF	FRP	CS	TR #1	Flat Gasket Fiber Without Aabestos	A193 GR. B7/ A194 GR. 2H	
D01	H61C	RAW(UG)	0/40	18	0	16"-20"	FRP	CL300	RF	FRP	Ductile Iron	BRONZE	Flat Gasket Fiber Without Aabestos	A193 GR. B7/ A194 GR. 2H	
D01	H61D	Potabl Water (UG)	0/40	8	0	OD: 20mm- 160mm	HDPE	CL150	FF	HDPE	CS	TR #1	Flat Gasket Fiber Without Aabestos	A193 GR. B7/ A194 GR. 2H	
D01	H61E	Fire Water & Foam (UG)	0/40	16	0	4"-24"	FRP	CL150	FF	FRP	Ductile Iron	BRONZE	Flat Gasket Fiber Without Aabestos	A193 GR. B7/ A194 GR. 2H	
	S82A	Carbon Dioxide Catalyst Liquid/Melttable Additive Cold Liquid Drain Process	80 -48/80 120 -48/80 -48/80	6 54 175+FV 54 54	0	1/4"~1"	DUPLEX 2205	-	-	-	316SS	TR #10	SWG,316L/FG,OR:CS ,IR:SS	A193 GR. B7/ A194 GR. 2H	5 Tubing
	T62A	Nalco Oxidant Reductant Scale inhibitor HCl	38	6	0	1/2"~2"	PVC	CL150	FF	PVC	PVC	TFE	Flat Ring, VITON	A193 GR. B7/ A194 GR. 2H	

KUS			PIPING MATERIALS CLASSIFICATION INDEX									Project No.		01000		
												Doc. No.		01000-MA-SPC-0002		
												Contr. Doc.No.				
												Rev.		R00		
Rev.	Piping Class	Service	Design Condition		C.A.	Pipe		Flange			Valve Body	Trim Mat.	Gasket	Bolting/Nuts	Remarks	
			T (°C)	P (barg)	(mm)	Size(in)	Material	Rating	Face	Material						
Note: 1. These low alloy steel shall satisfy the impact (toughness) requirements of ASME B31.3 Para. 323.2.2, as a minimum. Impact test requirements shall be based on a temperature of -48 °C coincident with the maximum design pressure. 2. All carbon and low alloy steel shall satisfy the impact (toughness) requirements of ASME B31.3 Para. 323.2.2, as a minimum. Impact test requirements shall be based on a temperature of -28°C coincident with the maximum design pressure. Additional requirements, if any, for prevention of brittle fracture shall be adhered to if required by local codes. Detailed engineering shall evaluate piping (e.g., control valve bypasses, dead legs, sampling lines, PSV inlets, etc.) where temperature could go below -28°C due to no/low flow conditions (refer to Winterization). 3. PWHT required for all welds. 4. This CLASS is for indoor use only. * Just for temporary condition. 5.The vacuum condition is instantaneous and occasional except A1B(0.5"-1"),D21K(0.5"-1") and S82A(1"). Full vacuum valve will be considered to be used in A1B(0.5"-1"),D21K(0.5"-1") and S82A(1"). 6. Reducer selection shall be according with the reducer chart.																

KUS Karabatan Utility Solutions	REDUCER CHART	Project	01000
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		Rev.	R00

LAGER SIZE



KUS Karabatan Utility Solutions	REDUCER CHART	Project	01000
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		Rev.	R00

	LEGEND	NOTES	REV
S	Swage Nipple		
R	Reducer		

PROJECT NUMBER	01000
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CONTR. DOCUM. NO.	

PIPING CLASS	REV
A1A	R00

SERVICES	LP Steam	Boiler feed water	Oil Water	Suspect Condensate	Inter.Drum Vent	REV
	LP Condensate	GT Washing Water	HRSG Burner Cooling Air	GT Closed Cooling Water Supply/Return	Process	
	Methanol	LP Boiler Blowdown	Safety Valve vent	Boiler Intermittent Blowdown	PAM	
	Phosphate	Recycled waster	Ammonia solution	Primary demineralised waster		

PIPING MATERIAL	CARBON STEEL	VALVE MATERIAL	CARBON STEEL	WELD EXAMINATION		REV
RATING / FACING	ASME CL150 / RF	TRIM MATERIAL	13 CR/HF (API #8)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.50 mm (0.059")	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*

SIZE

TEMP P °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
-28 to 38	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65
93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93
149	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86
204	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79
260	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72
316	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65
343	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62
404	6.33	6.33	6.33	6.33	6.33	6.33	6.33	6.33	6.33	6.33	6.33	6.33	6.33	6.33	6.33	6.33

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES RTFE SEAT RATINGS (bar)

TEMP(C)	< NPS 2	>NPS2	REV
-28 to 38	19.65	19.65	
93	17.93	17.93	
149	15.86	15.86	
204	5.49	3.45	

BUTTERFLY RTFE SEAT RATINGS(bar)

TEMP(C)	ALL SIZES	REV
-28 to 38	19.65	
93	17.93	
149	15.86	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" – 1 1/2" 2" – 6" 8" – 16" 18" – 24"	80 STD STD STD	PE BE BE BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, API 5L GRADE B, DSAW, ASME B36.10M, E _j =0.95		D01
NIPPLES	1/2" – 1 1/2" 1/2" – 1 1/2"	160 80	TOE PBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
SWAGES CON&ECC	1/2" – 2"	80	BBE	CS, ASTM A234 GRADE WPB-S, MSS-SP-95	1	
OLET SOCKOLET WELDOLET	1/2" – 1 1/2" 2" – 8"	3000 LB AS PIPE	SOCW WELD	CS, ASTM A105, MSS-SP-97 CS, ASTM A105, MSS-SP-97		
PLUG	1/2" – 2"	THRD		CS, ASTM A105, ASME B16.11, ROUND HEAD	279	
CAP	1/2" – 1 1/2" 2" – 16" 18" – 24"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9		D01

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
90° ELBOW	1/2" – 1 1/2" 2" – 6" 8" – 16" 18" – 24"	3000 LB AS PIPE AS PIPE AS PIPE	SOCW WELD WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9			D01
45° ELBOW	1/2" – 1 1/2" 2" – 6" 8" – 16" 18" – 24"	3000 LB AS PIPE AS PIPE AS PIPE	SOCW WELD WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9			D01
COUPLING COUPLING COUPLING(RED)	1/2" – 1 1/2" 1/2" – 1 1/2"	3000LB 3000LB	SOCW SOCW	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11			
TEE	1/2" – 1 1/2"	3000LB	SOCW	CS, ASTM A105, ASME B16.11			
	2" – 6" 8" – 16" 18" – 24"	AS PIPE AS PIPE AS PIPE	WELD WELD WELD	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9			D01
TEE (RED)	1/2" – 1 1/2"	3000LB	SOCW	CS, ASTM A105, ASME B16.11			
REDUCER							
CON	2" – 6" 8" – 16" 18" – 24"	AS PIPE AS PIPE AS PIPE	WELD WELD WELD	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9			D01
ECC	2" – 6" 8" – 16" 18" – 24"	AS PIPE AS PIPE AS PIPE	WELD WELD WELD	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9			D01
FLANGE							
SOCKETWELD	1/2" – 1 1/2" 1/2" – 1 1/2"	CL150 CL300	RF RF	CS, ASTM A105, ASME B16.5 CS, ASTM A105, ASME B16.5		278	
WELDNECK	2" – 24" 2" – 24"	CL150 CL300	RF RF	CS, ASTM A105.ASME B16.5 THICKNESS AS PIPE CS, ASTM A105.ASME B16.5 THICKNESS AS PIPE		278	D01
BLIND FLG	1/2"- 24" 1/2"- 24"	CL150 CL300	RF RF	CS, ASTM A105.ASME B16.5 CS, ASTM A105.ASME B16.5		278	D01
MISCELLANEOUS							
DRIP RING	1" – 12"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, 38.100 mm (1-1/2") THICK WITH ¾"			
FIGURE 8 BLANK	1" – 12"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48			D01
PADDLE BLANK	14" – 24"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48			D01
PADDLE SPACER	14" – 24"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48			D01
VALVES					TAG	277	
GATE	1/2" – 1 1/2"	CL800	SOCW	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVGB68S0108 ACAD0ZZ		
	1/2" – 1 1/2"	CL800	SOCW	CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT	PVVGF68S0108 ACAD0ZZ	275	
	1/2" – 2"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVGB62R0108 ACAD0ZZ	24	
	2"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8),FULL PORT	PVVGF62R0308 ACAC0ZZ	275	
	3"– 14"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8),FULL PORT	PVVGF62R0308 BCAC0ZZ	275	
	16" – 24"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8),FULL PORT, GEAR OP.	PVVGF62R0308 BCAC5ZZ	275	D01
GLOBE	1/2" – 1 1/2"	CL800	SOCW	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVLB68S0108A CAD0ZZ		
	2" – 12"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVLB62R0308 BCAC0ZZ		
	14"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8), GEAR OP.	PVVLB62R0308 BCAC5ZZ		
LIFT CHECK	½" – 1 ½"	CL800	SOCW	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVCL68S0108 NCAD0ZZ		
WAFER CHECK	2" – 16"	CL150	WAF- RF	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVCD62W0308 XNZA0ZZ		
PLUG	1/2" – 4"	CL150	RF	CS BODY WITH 316 SS PLUG, PTFE SLEEVE	PVVPC62R037D ACAB1ZZ	27,43,326	
	6" – 14'	CL150	RF	CS BODY WITH 316 SS PLUG, PTFE SLEEVE, GEAR OP.	PVVPC62R037D BCAB5ZZ	27,43,326	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
BALL	1/2" – 1 1/2"	CL600	BW	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS	PVVBH66E0171 ACAF0ZZ	27,55,43, 326	
	1/2" – 1 1/2"	CL600	BW	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT	PVVBF66E0171 ACAF0ZZ	27,55,43 326	
	1/2" – 1 1/2"	CL150	RF	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT, SHORT PATTERN	PVVBK62R0171 ACAF0ZZ	27,43,326	
	2" – 3"	CL150	RF	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT, SHORT PATTERN	PVVBK62R0371 BCAF1ZZ	27,43,326	
	3" – 6"	CL150	WAF- RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT	PVVUB62W0399 BCAG1ZZ	27,43,326	
	8" – 16"	CL150	WAF- RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT, GEAR OP.	PVVUB62W0399 BCAG3ZZ	27,43,326	
	3" – 6"	CL150	LUG- RF	CS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUB62L0399 BCAG1ZZ	27,43,326	
	8" – 16"	CL150	LUG- RF	CS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUB62L0399 BCAG3ZZ	27,43,326	
BOLTING	ALL	CL150		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEXNUTS			
	ALL	CL300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEXNUTS			
GASKET SPIRAL WOUND	1/2" - 16"	CL150		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, CS CENTERING RING, ASME B16.20 / ASME B16.5			
	1/2" - 16"	CL300		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, CS CENTERING RING, ASME B16.20 / ASME B16.5			

	NOTES				REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD	SIZES		
			1/2" – 11/2"		
			2" – 16"		
			1/2" – 11/2"		
			2" – 16"		

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
24	Use when required at flanged equipment nozzle, or where noted on P&IDs or Modules in Piping Standard Drawings.	
27	TFE, PTFE seated valves are limited to maximum 204°C.	
43	Only to be used as indicated on the P&IDs .	
55	Supplier shall weld equal length of line class seamless plain end nipples on each butt weld end of the valve, overall length to be 250mm.	
275	Only to be used where full port gate valve is required as indicated on P&IDs .	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME CL300 flanges shall only be used to match higher rated flange connections at control valves and special equipment.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drain requiring plugs, connections to tube fittings and for connections to threaded proprietary items. See referenced Modules in Piping Standard Drawings.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	

1/2	T														
3/4	RT	T													
1	RT	RT	T												
1 1/2	RT	RT	RT	T											
2	A	RT	RT	RT	T										
3	A	A	A	A	S	T									
4	A	A	A	A	S	S	T								
6	A	A	A	A	S	S	S	T							
8	A	A	A	A	S	S	S	S	T						
10	A	A	A	A	S	S	S	S	S	T					
12	A	A	A	A	S	S	S	S	S	S	T				
14	A	A	A	A	S	S	S	S	S	S	S	T			
16	A	A	A	A	S	S	S	S	S	S	S	S	T		
	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16		

BRANCH LEGEND		NOTES	REV
T	Tee (B.W. for two inch and larger, S.W. for 1 11/2" and smaller)		
A	Sockolet		
RT	Reducing Tee (B.W. for two inch and larger, S.W. for 1 11/2" and smaller)		
S	Stub-in without Reinforcing Pad shall be set-in type as shown in detail "2" or "4" of Figure 328.5.4D of ASME B31.3		

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
c.	Weldolet may be used in lieu of stub-in if required and verified by flexibility analysis and shown on the piping design drawing.	
d.	A PAD has been selected to provide reinforcement for this branch connection subject to the most severe coincident condition of internal pressure and fluid temperature seen in this specification. However, the specific pressure-temperature condition for this connection may eliminate the need for reinforcement. In addition, when the diameter-to-thickness ratio is restricted by paragraph 304.3.1 (3) (b) of ASME B31.3, finite element analysis (FEA) shall be performed .	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
A1AV	R00

SERVICES	Steam Turbine Outlet (Full Vacuum)				REV

PIPING MATERIAL	CARBON STEEL	VALVE MATERIAL	CARBON STEEL	WELD EXAMINATION		REV
RATING / FACING	ASME CLASS 150 / RF	TRIM MATERIAL	13 CR/HF (API #8)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.5 mm	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)*

SIZE																
TEMP. °C	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20	24
-28 to 38	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65
93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93
149	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88
204	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79
260	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72
316	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65
343	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62
371	7.58	7.58	7.58	7.58	7.58	7.58	7.58	7.58	7.58	7.58	7.58	7.58	7.58	7.58	7.58	7.58

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE TATING

BALL VALVES RTFE SEAT RATINGS (Bar)

TEMP(C)	< NPS 2	REV
-28 to 38	19.65	
93	17.93	
149	15.86	
204	5.49	

BUTTERFLY RTFE SEAT RATINGS(Bar)

TEMP(C)	ALL SIZES	REV
-28 to 38	19.65	
93	17.93	
149	15.86	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	½ " - 1½"	80	PE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	2" - 16"	STD	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	18" - 48"	STD	BE	CS, API 5L GRADE B, EFW, ASME B36.10M (Ej=1)		
	36" - 48"	12.7MM	BE	CS, ASTM A671 GRADE CC60 CL 12, EFW, STR SM, ASME B36.10M(Ej=1)		D01
	54"	14.27MM	BE	CS, ASTM A671 GRADE CC60 CL 12, EFW, STR SM, ASME B36.10M(Ej=1)		D01
	60"	15.88MM	BE	CS, ASTM A671 GRADE CC60 CL 12, EFW, STR SM, ASME B36.10M(Ej=1)		D01
	66"	17.48MM	BE	CS, ASTM A671 GRADE CC60 CL 12, EFW, STR SM, ASME B36.10M(Ej=1)		D01
NIPPLE	½ " - 1½"	160	TOE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	½ " - 1½"	160	TBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	½ " - 1½"	80	PBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
SWAGE	½ " - 1½"	160	TSE	CS, ASTM A234 GRADE WPB, MSS-SP-95	1	
	½ " - 1½"	80	BBE	CS, ASTM A234 GRADE WPB, MSS-SP-95		
	½ " - 1½"	160	TSE	CS, ASTM A234 GRADE WPB, MSS-SP-95		
	½ " - 1½"	80	BBE	CS, ASTM A234 GRADE WPB, MSS-SP-95		
COUPLING	½ " - 1½"	CL3000	SOCW	CS, ASTM A105, ASME B16.11		
FULL COUPLING	½ " - 2"	CL3000	THRD	CS, ASTM A105, ASME B16.11	279	
OLET	½ " - 1 ½"	CL3000	WELD	CS, ASTM A105, MSS-SP-97		
SOCKOLET	½ " - 2"	CL3000	WELD	CS, ASTM A105, MSS-SP-97	279	
THREDOLET	2" - 10"	AS PIPE	WELD	CS, ASTM A105, MSS-SP-97		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
PLUG ROUND HEAD	½" - 2"		THRD	CS, ASTM A105, ASME B16.11, ROUND HEAD			
CAP	½" - 1½" ½" - 2" 2" - 48" 54" - 66"	CL3000 CL3000 AS PIPE AS PIPE	SOCW THRD BW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-S, ASME B16.9 (AS PER DETAIL DRAWING)		279	
90° ELBOW SW LR	½" - 1½" 2" - 16" 18" - 48" 54" - 66"	CL3000 AS PIPE AS PIPE AS PIPE	SOCW BW BW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)			
45° ELBOW	½" - 1½" 2" - 16" 18" - 48" 54" - 66"	CL3000 AS PIPE AS PIPE AS PIPE	SOCW BW BW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)			
TEE	½" - 1½" 2" - 16" 18" - 48" 54" - 66"	CL3000 AS PIPE AS PIPE AS PIPE	SOCW BW BW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)			
RED TEE	½" - 1½" 2" - 16" 18" - 48" 54" - 66"	CL3000 AS PIPE AS PIPE AS PIPE	SOCW BW BW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)			
REDUCER CON	2" - 16" 18" - 48" 54" - 66"	AS PIPE AS PIPE AS PIPE	BW BW BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)			
ECC	2" - 16" 18" - 48" 54" - 66"	AS PIPE AS PIPE AS PIPE	BW BW BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)			
FLANGE SOCKETWELD	½" - 1½" ½" - 1½"	CL150 CL300	RF RF	CS, ASTM A105, ASME B16.5 CS, ASTM A105, ASME B16.5		278	
WELDNECK	2" - 24" 30" - 60" 66" 2" - 24" 30" - 48" 66"	CL150 CL150 CL150 CL300 CL300 CL300	RF RF RF RF RF RF	CS, ASTM A105, ASME B16.5 THICKNESS AS PIPE CS, ASTM A105, ASME B16.47 SERIES A THICKNESS AS PIPE CS, ASTM A105 (AS PER DETAIL DRAWING) CS, ASTM A105, ASME B16.5 THICKNESS AS PIPE CS, ASTM A105, ASME B16.47 SERIES A THICKNESS AS PIPE CS, ASTM A105 (AS PER DETAIL DRAWING)		278 278 278	
BLIND FLG	½" - 24" 30" - 60" 66" ½" - 24" 30" - 48" 66"	CL150 CL150 CL150 CL300 CL300 CL300	RF RF RF RF RF RF	CS, ASTM A105, ASME B16.5 CS, ASTM A105, ASME B16.47 SERIES A THICKNESS AS PIPE CS, ASTM A105 (AS PER DETAIL DRAWING) CS, ASTM A105, ASME B16.5 CS, ASTM A105, ASME B16.47 SERIES A THICKNESS AS PIPE CS, ASTM A105 (AS PER DETAIL DRAWING)		278 278 278	
DRIP RING	1" - 12"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, 1-1/2" THICK WITH ¾" SOCW TAPS (STANDARD DRAWINGS)			
FIGURE 8 BLANK+SPACER	1" - 12" 14" - 24"	CL150 CL150	FF FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48 CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48			
VALVES GATE	½" - 1 ½" ½" - 1 ½" ½" - 1 ½"	CL800 CL800 CL800	SOCW SOCW T/SW	CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT CS BODY WITH 13 CR / HF TRIM (API #8)	TAG PVVGB68S0108 ACAD0VV PVVGF68S0108 ACAD0VV PVVGB68D0108 ACAD0VV	48,277 275 4,279	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
GATE	½" – 1 ½"	CL800	T/SW	CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT	PVVG68D0108 ACAD0VV	4,275,279	
	½" - 2"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVG62R0108 ACAD0VV	24	
	2"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT	PVVG62R0308 ACAD0VV	275	
	3" – 14"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT	PVVG62R0308 BCAC0VV		
	16" – 24"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT, GEAR OP.	PVVG62R0308 BCAC5VV		
GLOBE	½" – 1 ½"	CL800	SOCW	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVL68S0108A CAD0VV		
	2" – 12"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVL62R0308 BCAC0VV		
	14"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8) , GEAR OP.	PVVL62R0308 BCAC5VV		
LIFT CHECK	½" – 1 ½"	CL800	SOCW	CS BODY WITH 13 CR TRIM (API # 8)	PVWCL68S0108 NCAD0VV		
WAFER DUAL PLATE CHECK	2" – 24"	CL150	RF	CS BODY WITH 13 CR PLATES, METAL SEATS	PVWCD62W0308 XNZA0VV		
BALL	½" – 1 ½"	CL600	BW	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS	PVVBH66E0171 ACAF0VV	27,55,43, 326	
	½" – 1 ½"	CL600	BW	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT	PVVB66E0171 ACAF0VV	27,55,43, 326	
	½" – 1 ½"	CL600	T/SW	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS	PVVBH66D0171 ACAF0VV	27,56,43, 326	
	½" - 2"	CL150	RF	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT, SHORT PATTERN	PVVBK62R0371 BCAF1VV	27,43,326	
WAF BUTTERFLY	3" – 6"	CL150	RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT	PVVUB62W0399 BCAG1VV	27,43,326	
	8" – 24"	CL150	RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT, GEAR OP.	PVVUB62W0399 BCAG3VV	27,43,326	
LUG BUTTERFLY	3" – 6"	CL150	RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT	PVVUB62L0399 BCAG1VV	27,43,326	
	8" – 24"	CL150	RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT, GEAR OP.	PVVUB62L0399 BCAG3VV	27,43,326	
BOLT/NUTS	ALL	CL150		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEX NUTS			
	ALL	CL300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEX NUTS			
GASKETS SPIRAL WOUND	½" - 24"	CL150		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, CS CENTERING RING, ASME B16.20 / ASME B16.5			
	½" - 24"	CL300		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, CS CENTERING RING , ASME B16.20 / ASME B16.5			

	NOTES				REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD	SIZES		
			½" – 2"		
			3" – 66"		
			½" – 1 ½"		
			2" – 66"		

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
4	Socket-weld x Threaded gate valve shall be used for valved pressure, instrument, vent and drain connections only. Piping system end shall be socketweld and instrument end shall be threaded.	
23	Use line class weld-neck flangers(born to AS PIPE pipe) with radius taps in pipe.	
24	Use when required at flanged equipment nozzle,or where noted on P&IDs or Modules in Piping Standard Drawings.	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
43	Only to be used as indicated on the P&IDs .	
48	Valves shall be suitable for full vacuum	
55	Supplier shall weld equal length of line class seamless plain end nipples on each butt weld end of the valve, overall length to be 250 mm.	
56	Supplier shall weld one piece of line class seamless plain end nipples on the socketweld end of the socketweld x threaded valve, overall length to be 200mm.	
275	Only to be used where full port gate valve is required as indicated on P&IDs .	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME Class 300 flanges shall only be used to AS PIPE higher rated flange connections at control valves and special equipment.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections(when applicable), gage glass piping, vents and drain requiring plugs, ,connections to tube fittings and for connections to threaded proprietary items. See referenced Modules in Piping Standard Drawings.	
290	Use when required for maintenance, flanged equipment nozzles, or where noted on the P&ID.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
A1B	R00

					REV
SERVICES	TEA	Isopropanol	Safety Valve to Atmosphere		
					REV
PIPING MATERIAL	CARBON STEEL	VALVE MATERIAL	CARBON STEEL	WELD EXAMINATION	
RATING / FACING	ASME CL150 / RF	TRIM MATERIAL	13 CR/HF (API #8)	STRESS RELIEF	NOT REQUIRED
CORROSION ALLOWANCE	1.50 mm(0.059")	CODE	ASME B31.3	BENDING	NOT ALLOWED

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*

REV

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2
-29 to 38	20.09	20.09	20.09	20.09	20.09
93	18.25	18.25	18.25	18.25	18.25
149	16.21	16.21	16.21	16.21	16.21
204	14.07	14.07	14.07	14.07	14.07

BALL VALVES
 RTFE SEAT RATINGS (bar)

TEMP(C)	≤NPS2	>NPS2	REV
-29 to 38	19.65	19.65	
93	17.93	17.93	
149	15.86	15.86	
204	5.49	3.45	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" – 1-1/2" 2" – 3"	80 40	PE BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
NIPPLES	1/2" – 1-1/2"	160	TOE	CS, ASTM 106 GRADE B, SMLS, ASME B36.10M		
SWAGES	1/2" – 1-1/2" 1/2" – 1-1/2"	160 80	TSE BBE	CS, ASTM A234 GRADE WPB-S, SMLS, MSS-SP-95 CS, ASTM A234 GRADE WPB-S, SMLS, MSS-SP-95	1	
OLET WELDOLET	2" – 3"	AS PIPE	WELD	CS, ASTM A105, MSS-SP-97		
CAP	1/2" – 1-1/2" 2" – 3"	AS PIPE AS PIPE	WELD WELD	CS, ASTM A234 GRADE WPB-S , ASME B16.9 CS, ASTM A234 GRADE WPB-S , ASME B16.9		
90° ELBOW	1/2" – 1-1/2" 2" – 3"	AS PIPE AS PIPE	WELD WELD	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S , ASME B16.9		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
45° ELBOW	1/2" – 1-1/2" 2" – 3"	AS PIPE AS PIPE	WELD WELD	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
TEE	1/2" – 1 1/2" 2" – 3"	AS PIPE AS PIPE	WELD WELD	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
RED TEE	1/2" – 1 1/2" 2" – 3"	AS PIPE AS PIPE	WELD WELD	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
REDUCER CON ECC	2" – 3" 2" – 3"	AS PIPE AS PIPE	WELD WELD	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
FLANGE WELDNECK BLIND	1/2" – 3" 1/2" – 3"	CL150 CL150	RF RF	CS, ASTM A105, ASME B16.5 CS, ASTM A105, ASME B16.5			
MISCELLANEOUS FIGURE 8 BLANK	1" – 3"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48			
VALVES	BALL	1/2" – 1"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8), Full Port, Floating Ball, VACUUM	TAG PVVBA62R0108 AVAF1VV	342,277 326
	BALL	1-1/2" – 4"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8), Full Port, Floating Ball	PVVBA62R0108 AVAF1ZZ	326
	GLOBE	1/2" – 1-1/2"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8), Parabolic Disc, VACUUM	PVVLB62R0108 AVAD0VP	329 329
		2" – 2"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8), Parabolic Disc	PVVLB62R0308 BVAC0PD	
	LIFT CHECK	1/2" – 1-1/2"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8), SS Spring, Horizontal & Vert. Upward	PVVCL62R0108 NVAD0ZZ	
	WAFFER CHECK	2" – 2"	CL150	WAF- RF	CS BODY WITH 13 CR / HF TRIM (API #8), Inconel-X-Spring	PVVCD62W0308 XNZAOZZ	
BOLTING	STUD	ALL	CL150		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A194 GRADE 2H HEAVY HEX NUTS, ASME B18.2.1/ASME B18.2.2		
	STUD	ALL	CL300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A194 GRADE 2H HEAVY HEX NUTS, ASME B18.2.1/ASME B18.2.2		
GASKETS	1/2" – 3"	CL150	RF	FLAT RING, COMPRESSED NON-ASBESTOS (GARLOCK BLUE GARD STYLE 3000 OR EQUAL), 1.6mm (1/16") THK, ASME B16.21 / ASME B16.5			
	1/2" – 3"	CL300	RF	FLAT RING, COMPRESSED NON-ASBESTOS (GARLOCK BLUE GARD STYLE 3000 OR EQUAL), 1.6mm (1/16") THK, ASME B16.21 / ASME B16.5			

NOTES		REV	
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES	SIZES
		FLANGES	1/2" – 1-1/2"
		BUTT WELD	2" – 2"
		BUTT WELD	1/2" – 1-1/2"

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
6	The flanges, gasket and bolting materials on all vents, drains, pressure, instrument, or flow connections shall be furnished by Piping/BOM.	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
326	This resilient seated valve does not in all cases meet full flange rating,See valve MAWP in this specification.	
329	Use only in bypass for regulating valves.	
342	Valves shall be suitable for TEA services as stated in Valve Purchasing Specification.	

REV
BRANCH CONNECTION CHART

HEADER SIZE

0.5	T					
0.75	RT	T				
1	RT	RT	T			
1.5	RT	RT	RT	T		
2	W	W	W	W	T	
3	W	W	W	W	RT	T

BRANCH SIZE

Branch Connections:

BRANCH

BRANCH LEGEND		NOTES	REV
T	Tee (B.W)		
W	Weldolet (PWHT required at branch attachment welds when the thickness through the weld in any plane through the branch is greater than 38.100 mm (1.5") for P1 material per ASME B31.3.		
RT	Reducing Tee (B.W.)		

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	108184-0000-MA-SPC-0002

PIPING CLASS	REV
A1D	R00

						REV
SERVICES	Process	Wash Oil	Methanol	2%~4%NaOH Caustic soda		
	Purge gas	Silane	Liquid Fuel	Fuel Oil		
	Regeneration Air	Waste Water				

						REV
PIPING MATERIAL	CARBON STEEL	VALVE MATERIAL	CARBON STEEL	WELD EXAMINATION		
RATING / FACING	ASME CL150 / RF	TRIM MATERIAL	13 CR/HF (API #8)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.50 mm (0.059")	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*

REV

TEMP P. C	SIZE																		
	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20	24	30	36	48
-28 to 38	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.42	19.65	19.65
93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93
149	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86
204	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79
260	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72
316	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65

TEMP.C	54	66	72	84
-28 to 38	18.29	17.1	17.72	16.91
93	17.93	17.1	17.72	16.91
149	15.86	15.86	15.86	15.86
204	13.79	13.79	13.79	13.79
260	11.72	11.72	11.72	11.72
316	9.65	9.65	9.65	9.65

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES RTFE SEAT RATINGS (bar)

TEMP(C)	< NPS 2	REV
-28 to 38	19.65	
93	17.93	
149	15.86	
204	5.49	

BUTTERFLY RTFE SEAT RATINGS(bar)

TEMP(C)	ALL SIZES	REV
-28 to 38	19.65	
93	17.93	
149	15.86	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" - 2"	XS	PE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	3" - 24"	STD	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	30" - 36"	STD	BE	CS, ASTM A671 GRADE CC60 CL12, EFW, STR SM, ASME B36.10M, Ej=1.0		
	42" - 48"	XS	BE	CS, ASTM A671 GRADE CC60 CL12, EFW, STR SM, ASME B36.10M, Ej=1.0		
	54"	12.7 MM	BE	CS, ASTM A671 GRADE CC60 CL12, EFW, STR SM, ASME B36.10M, Ej=1.0		
	66"	14.27 MM	BE	CS, ASTM A671 GRADE CC60 CL12, EFW, STR SM, ASME B36.10M, Ej=1.0	548	
	72"	15.88 MM	BE	CS, ASTM A671 GRADE CC60 CL12, EFW, STR SM, ASME B36.10M, Ej=1.0	548	
	84"	17.48 MM	BE	CS, ASTM A671 GRADE CC60 CL12, EFW, STR SM, ASME B36.10M, Ej=1.0	548	
	84"	23.83 MM	BE	CS, ASTM A671 GRADE CC60 CL12, EFW, STR SM, ASME B36.10M, Ej=1.0	548,706	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
NIPPLES	1/2" - 1 1/2"	XS	PBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M, 3" (80mm) & 6" (150mm) LONG	503	
	1/2" - 1 1/2"	160	PBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M, 3" (80mm) & 6" (150mm) LONG	03	
	1/2" - 1 1/2"	160	TOE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M, 3" (80mm) & 6" (150mm) LONG		
SWAGES CON&ECC	1/2" - 1 1/2" 1/2" - 2" 3" - 4" (1/2" - 2")	160 XS STD x XS	TSE PBE BLE x PSE	CS, ASTM A234 GRADE WPB, SMLS, MSS-SP-95 CS, ASTM A234 GRADE WPB, SMLS, MSS-SP-95 CS, ASTM A234 GRADE WPB, SMLS, MSS-SP-95		
OLET SOCKOLET WELDOLET THREDOLET SWELBOWLET 45 LATROLET 45 BW LATROLET WELDOLET	1/2" - 1 1/2" 2" - 8" 1/2" - 1 1/2" 1/2" - 1 1/2" 1/2" - 1 1/2" 2" - 12" 2" - 36"	3000 LB AS PIPE 3000 LB 3000 LB 3000 LB	SOCW WELD WELD WELD SOCW WELD WELD	CS, ASTM A105, MSS-SP-97 CS, ASTM A105, MSS-SP-97 CS, ASTM A105, MSS-SP-97 CS, ASTM A105, MSS-SP-97 CS, ASTM A105, MANUFACTURER STANDARD, MSS-SP-97 CS, ASTM A105, MANUFACTURER STANDARD, MSS-SP-97, TO MATCH PIPE CS, ASTM A105, MSS-SP-97		
PLUG	1/2" - 1 1/2" 1/2" - 1 1/2"	THRD PE		CS, ASTM A105, ASME B16.11, ROUND HEAD CS, ASTM A105, ASME B16.11, BARSTOCK, ROUND HEAD		
CAP	1/2" - 1 1/2" 1/2" - 1 1/2" 2" - 48" 54" - 84"	3000 LB 3000 LB	SOCW THRD WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)	548, 701	
90° ELBOW	1/2" - 1 1/2" 2" - 24" 30" - 48" 54" - 84"	3000 LB AS PIPE AS PIPE AS PIPE	SOCW WELD WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)	548, 701	
45° ELBOW	1/2" - 1 1/2" 2" - 24" 30" - 48" 54" - 84"	3000 LB AS PIPE AS PIPE AS PIPE	SOCW WELD WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)	548, 701	
COUPLING	1/2" - 1 1/2"	3000LB	SOCW	CS, ASTM A105, ASME B16.11		
TEE	1/2" - 1 1/2" 2" - 24" 30" - 48" 54" - 84"	3000LB AS PIPE AS PIPE AS PIPE	SOCW WELD WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)	548, 701	
RED TEE	1/2" - 1 1/2" 2" - 24" 30" - 48" 54" - 84"	3000LB AS PIPE AS PIPE AS PIPE	SOCW WELD WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)	548, 701	
REDUCER CON ECC	2" - 24" 30" - 48" 54" - 84" 2" - 24" 30" - 48" 54" - 84"	AS PIPE AS PIPE AS PIPE AS PIPE AS PIPE AS PIPE	WELD WELD WELD WELD WELD WELD	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING) CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
FLANGE							
SOCKETWELD	½" – 1 1/2"	CL150	RF	CS, ASTM A105, ASME B16.5			
WELDNECK	2" – 24"	CL150	RF	CS, ASTM A105, ASME B16.5			
BLIND	½" – 24"	CL150	RF	CS, ASTM A105, ASME B16.5			
WELDNECK	30" – 60"	CL150	RF	CS, ASTM A105, ASME B16.47 SERIES A		702	
WELDNECK	62" – 84"	CL150	RF	CS, ASTM A105 (AS PER DETAIL DRAWING)			
BLIND	30" – 60"	CL150	RF	CS, ASTM A105, ASME B16.47 SERIES A		702	
BLIND	62" – 84"	CL150	RF	CS, ASTM A105, (AS PER DETAIL DRAWING)			
SOCKETWELD	½" – 1 1/2"	CL300	RF	CS, ASTM A105, ASME B16.5		12	
WELDNECK	2" – 24"	CL300	RF	CS, ASTM A105, ASME B16.5		12	
BLIND	½" – 24"	CL300	RF	CS, ASTM A105, ASME B16.5		12	
PR WN ORIFICE	1" – 24"	CL300	RF	CS, ASTM A105 W/ ½" SOCKETWELD TAPS, ASME B16.36			
MISCELLANEOUS							
FIGURE 8 BLANK	½" – 12"	CL150	RF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48			
PADDLE BLANK	14" – 24"	CL150	RF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48			
PADDLE SPACER	14" – 24"	CL150	RF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48			
VALVES					TAG	277	
GATE	½" – 1-1/2"	CL800	SOCW	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVGB68S0108 ACAD0ZZ	06, 03	
GATE	½" – 1-1/2"	CL800	T/SW	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVGB68D0108 ACAD0ZZ	15	
GATE	½" – 1-1/2"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8), Conventional Port	PVVGB62R0108 ACAD0ZZ	06, 15	
GATE	½" – 1-1/2"	CL300	RF	CS BODY WITH 13 CR / HF TRIM (API #8), Conventional Port	PVVGB63R0108 ACAD0ZZ		
GATE	2" – 14"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT	PVVGf62R0308 BCAC0ZZ		
GATE	16" – 24"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT, GEAR OP.	PVVGf62R0308 BCAC5ZZ		
GATE	30" – 48"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT, GEAR OP.	PVVGf62K0308 BCAC5ZZ		
GLOBE	½" – 1-1/2"	CL800	SOCW	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVLB68S0108A CAD0ZZ		
GLOBE	2" – 12"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVLB62R0308 BCAC0ZZ		
GLOBE	14" – 16"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8) , GEAR OP.	PVVLB62R0308 BCAC5ZZ	61	
LIFT CHECK	½" – 1-1/2"	CL800	SOCW	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVCL68S0108 NCAD0ZZ	62	
SWING CHECK	2" – 10"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVCS62R0308 NCAC0ZZ	63	
DUAL PLT CHECK	12" – 24"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8), DBL FLGD	PVVCD62R032C XCAA0ZZ	62	
SWING CHECK	12" – 24"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVCS62R0308 NCAC0ZZ	63	
DUAL PLT CHECK	30" – 48"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8), DBL FLGD, Series-A	PVVCD62K032C XCAA0ZZ		
BUTTERFLY	2" – 6"	CL150	RF	CS Body WITH Resilient Metal SEAT, HF SEAT, DBL FLGD	PVVUB62R0368 BCAG0ZZ		
BUTTERFLY	8" – 24"	CL150	RF	CS Body WITH Resilient Metal SEAT, HF SEAT, DBL FLGD, GEAR OP.	PVVUB62R0368 BCAG3ZZ		
BUTTERFLY	30" – 48"	CL150	RF	CS Body WITH Resilient Metal SEAT, HF SEAT, DBL FLGD, Series-A, GEAR OP.	PVVUB62K0368 BCAG3ZZ	09	
BALL	¾" – 1-1/2"	CL150	RF	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, RP, SHORT PATTERN, Lever	PVVB62R0171 ECAF1ZZ	09,326	
BALL	2" – 6"	CL150	RF	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, RP, SHORT PATTERN, Lever	PVVB62R0371 ECAF1ZZ	09,326	
BALL	8" – 8"	CL150	RF	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, RP, SHORT PATTERN, GEAR OP.	PVVB62R0371 ECAF3ZZ	09,326	
BALL	10" – 24"	CL150	RF	CS Body WITH RPTFE SEAT, Nickel Plated Trunnion Ball, RP, LONG PATTERN, GEAR OP.	PVVB62R037C ECAF3ZZ	06, 03, 326	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
BOLTING	ALL	CL150		ALLOY STEEL, ASTM A193 GRADE B7, ASTM A194 GRADE 2H HEAVY HEX NUTS		
	ALL	CL300		ALLOY STEEL, ASTM A193 GRADE B7, ASTM A194 GRADE 2H HEAVY HEX NUTS		
GASKET	½" – 24"	CL150	RF	LOW STRESS SPIRAL WOUND, ASME B16.20, ASME B16.5, TP 304 SS WDG, W/FLEXIBLE GRAPHITE FILLER, TP 304 SS INNER RING, CS OUTER RING		
	30" – 60"	CL150	RF	LOW STRESS SPIRAL WOUND, ASME B16.20, ASME B16.47 SERIES A, TP 304 SS WDG, W/FLEXIBLE GRAPHITE FILLER, TP 304 SS INNER RING, CS OUTER RING		
	62" – 84"	CL150	RF	LOW STRESS SPIRAL WOUND, ASME B16.20, (AS PER DETAIL DRAWING), TP 304 SS WDG, W/FLEXIBLE GRAPHITE FILLER, TP 304 SS INNER RING, CS OUTER RING		
	½" – 24"	CL300	RF	SPIRAL WOUND, ASME B16.20, ASME B16.5, TP 304 SS WDG, W/FLEXIBLE GRAPHITE FILLER, TP 304 SS INNER RING, CS OUTER RING		

NOTE	SPECIFIC PIPING NOTES	REV
03.	Threaded joints are permitted only at the outlet of vent and drain valves, hydrostatic connections, outlet instrument take-off valves, and to match equipment.	
06.	These valves shall be used only for vent, drain, and instrument connections.	
09.	Pressure and temperature rating can be limited by certain components permitted by this Practice. Manufacturer's recommended pressure-temperature restrictions shall be consulted.	
12.	These flanges shall be used only to match Class 300 flange connections at control valves and special equipment.	
15.	These valves shall be used if mating to flanged nozzles.	
53.	This item shall be used for orifice flange tap connections only.	
61.	These check valves shall be installed in a horizontal position with cover up.	
62.	These check valves shall be installed in a horizontal position with cover up or in a vertical position with upward flow.	
63.	These check valves shall be installed in a horizontal position with hinge pin vertical or in a vertical position with upward flow.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	
502.	The pad thickness equals run pipe thickness, the pad width equals ½ of branch O.D. These items can be replaced with a Weldolet or Reducing Tee depending on engineering requirements, availability and economics.	
503.	Use this item for Branch Nipple which terminate at the valve or continue as tubing. (eg. For instrument connections, valved vents & drains, etc.).	
510.	The pad thickness equals run pipe thickness, the pad width equals ½ of branch O.D. These items can be replaced with a Latrolet or Lateral depending on engineering requirements, availability and economics.	
531.	Examination shall be in accordance with requirements in "Standard for Piping Welding", refer to document number: 108184-0000-PI-SPC-0006.	
548.	Pipe and pipe components thicker than 0.500" may require impact-tested materials. See ASME B31.3, Paragraph 323.2.2.	
701.	Manufacturer of fittings above 48" shall provide proof that fittings supplied meet requirements of para 2.2 of ASME B16.9 for review by Buyer.	
702.	These flanges have been calculated based on process design conditions on an individual basis.	
706.	For 45° branch connection 84" header with 66" branch a thicker wall is required as per ASME B31.3, paragraph 304.3.1.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCMU. NO.	

PIPING CLASS	REV
A1DV	R00

						REV
SERVICES	Process (with vacuum)	Process				
	Waste Water					

						REV
PIPING MATERIAL	CARBON STEEL	VALVE MATERIAL	CARBON STEEL	WELDEXAMINATION		
RATING / FACING	ASME CL150 / RF	TRIM MATERIAL	13 CR/HF (API #8)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.50 mm (0.059")	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*

REV

TE MP. C	SIZE																		
	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20	24	30	36	48
-28 to 38	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65
93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93
149	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86
204	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79
260	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72
316	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65
343	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62

TEMP.C	54	60	66	72	78	84
-28 to 38	17.63	17.92	18.15	18.3	18.44	18.59
93	17.63	17.92	17.93	17.93	17.93	17.93
149	15.86	15.86	15.86	15.86	15.86	15.86
204	13.79	13.79	13.79	13.79	13.79	13.79
260	11.72	11.72	11.72	11.72	11.72	11.72
316	9.65	9.65	9.65	9.65	9.65	9.65
343	8.62	8.62	8.62	8.62	8.62	8.62

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES RTFE SEAT RATINGS (bar)

TEMP(C)	< NPS 2	REV
-28 to 38	19.65	
93	17.93	
149	15.86	
204	5.49	

BUTTERFLY RTFE SEAT RATINGS(bar)

TEMP(C)	ALL SIZES	REV
-28 to 38	19.65	
93	17.93	
149	15.86	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" - 2"	XS	PE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	3" - 24"	STD	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	30"	STD	BE	CS, ASTM A671 GRADE CC60 CL12, EFW, STR SM, ASME B36.10M, Ej=1.0		
	36" - 48"	XS	BE	CS, ASTM A671 GRADE CC60 CL12, EFW, STR SM, ASME B36.10M, Ej=1.0		
	54"	14.27 MM	BE	CS, ASTM A671 GRADE CC60 CL12, EFW, STR SM, ASME B36.10M, Ej=1.0	548	
	60"	15.88 MM	BE	CS, ASTM A671 GRADE CC60 CL12, EFW, STR SM, ASME B36.10M, Ej=1.0	548	
	66"	17.48 MM	BE	CS, ASTM A671 GRADE CC60 CL32, EFW, STR SM, ASME B36.10M, Ej=1.0	548	
	72"	19.05 MM	BE	CS, ASTM A671 GRADE CC60 CL32, EFW, STR SM, ASME B36.10M, Ej=1.0	548,700	
	78"	20.62 MM	BE	CS, ASTM A671 GRADE CC60 CL32, EFW, STR SM, ASME B36.10M, Ej=1.0	548,700	
	84	22.23 MM	BE	CS, ASTM A671 GRADE CC60 CL 32, EFW, STR SM, ASME B36.10M, Ej=1.0	700	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
NIPPLES	1/2" - 1 1/2"	XS	PBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M, 3" (80mm) & 6" (150mm) LONG	503	
	1/2" - 1 1/2"	160	PBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M, 3" (80mm) & 6" (150mm) LONG	03	
	1/2" - 1 1/2"	160	TOE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M, 3" (80mm) & 6" (150mm) LONG		
SWAGES CON&ECC	1/2" - 1 1/2"	160	TSE	CS, ASTM A234 GRADE WPB, SMLS, MSS-SP-95		
	1/2" - 2"	XS	PBE	CS, ASTM A234 GRADE WPB, SMLS, MSS-SP-95		
	3" - 4" (1/2" - 2")	STD x XS	BLE x PSE	CS, ASTM A234 GRADE WPB, SMLS, MSS-SP-95		
OLET SOCKOLET WELDOLET THREDOLET SWELBOWLET	1/2" - 1 1/2" 2" - 8" 1/2" - 1 1/2" 1/2" - 1 1/2"	3000 LB AS PIPE 3000 LB 3000 LB	SOCW WELD WELD WELD	CS, ASTM A105, MSS-SP-97 CS, ASTM A105, MSS-SP-97 CS, ASTM A105, MSS-SP-97 CS, ASTM A105, MSS-SP-97		
45 LATROLET 45 BW LATROLET WELDOLET	1/2" - 1 1/2" 2" - 12" 2" - 36"	3000 LB	SOCW WELD WELD	CS, ASTM A105, MANUFACTURER STANDARD, MSS-SP-97 CS, ASTM A105, MANUFACTURER STANDARD, MSS-SP-97, TO MATCH PIPE CS, ASTM A105, MSS-SP-97		
PLUG	1/2" - 1 1/2" 1/2" - 1/2"	THRD PE		CS, ASTM A105, ASME B16.11, ROUND HEAD CS, ASTM A105, ASME B16.11, BARSTOCK, ROUND HEAD		
CAP	1/2" - 1 1/2" 1/2" - 1 1/2" 2" - 48" 54" - 84"	3000 LB 3000 LB	SOCW THRD WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)	548, 701	
90° ELBOW	1/2" - 1 1/2" 2" - 24" 30" - 48" 54" - 84"	3000 LB AS PIPE AS PIPE AS PIPE	SOCW WELD WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)	548, 701	
45° ELBOW	1/2" - 1 1/2" 2" - 24" 30" - 48" 54" - 84"	3000 LB AS PIPE AS PIPE AS PIPE	SOCW WELD WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)	548, 701	
COUPLING	1/2" - 1 1/2"	3000LB	SOCW	CS, ASTM A105, ASME B16.11		
TEE	1/2" - 1 1/2" 2" - 24" 30" - 48" 54" - 84"	3000LB AS PIPE AS PIPE AS PIPE	SOCW WELD WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)	548, 701	
RED TEE	1/2" - 1 1/2" 2" - 24" 30" - 48" 54" - 84"	3000LB AS PIPE AS PIPE AS PIPE	SOCW WELD WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)	548, 701	
REDUCER CON	2" - 24" 30" - 48" 54" - 84"	AS PIPE AS PIPE AS PIPE	WELD WELD WELD	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)	548, 701	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
ECC	2" – 24" 30" – 48" 54" – 84"	AS PIPE AS PIPE AS PIPE	WELD WELD WELD	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 LTCS, ASTM 420 GRADE WPL6-W, ASME B16.9 (AS PER DETAIL DRAWING)	548, 701	
FLANG						
SOCKETWELD	½" – 1 1/2"	CL150	RF	CS, ASTM A105, ASME B16.5		
WELDNECK	2" – 24"	CL150	RF	CS, ASTM A105, ASME B16.5		
BLIND	½" – 24"	CL150	RF	CS, ASTM A105, ASME B16.5		
WELDNECK	30" – 60"	CL150	RF	CS, ASTM A105, ASME B16.47 SERIES A	702	
WELDNECK	62" – 84"	CL150	RF	CS, ASTM A105 (AS PER DETAIL DRAWING)		
BLIND	30" – 60"	CL150	RF	CS, ASTM A105, ASME B16.47 SERIES A	702	
BLIND	62" – 84"	CL150	RF	CS, ASTM A105 (AS PER DETAIL DRAWING)		
SOCKETWELD	½" – 1.5"	CL300	RF	CS, ASTM A105, ASME B16.5	12	
WELDNECK	2" – 24"	CL300	RF	CS, ASTM A105, ASME B16.5	12	
BLIND	½" – 24"	CL300	RF	CS, ASTM A105, ASME B16.5	12	
PR WN ORIFICE	1" – 24"	CL300	RF	CS, ASTM A105 W/ ½" SOCKETWELD TAPS, ASME B16.36		
MISCELLANEOUS						
FIGURE 8 BLANK	½" – 12"	CL150	RF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48		
PADDLE BLANK	14" – 24"	CL150	RF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48		
PADDLE SPACER	14" – 24"	CL150	RF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48		
VALVES						
					TAG	277
GATE	½" – 1-1/2"	CL800	SOCW	CS Body WITH 13 CR TRIM, HF SEAT (API #8) , Vacuum Service	PVVB68S0108 ACAD0VV	
GATE	½" – 1-1/2"	CL800	T/SW	CS Body WITH 13 CR TRIM, HF SEAT (API #8) , Vacuum Service	PVVB68D0108 ACAD0VV	06, 03
GATE	½" – 1-1/2"	CL150	RF	CS Body WITH 13 CR TRIM, HF SEAT (API #8) , Conventional Port, Vacuum Service	PVVB62R0108 ACAD0VV	15
GATE	½" – 1-1/2"	CL300	RF	CS Body WITH 13 CR TRIM, HF SEAT (API #8) , Conventional Port, Vacuum Service	PVVB63R0108 ACAD0VV	06, 15
GATE	2" – 14"	CL150	RF	CS Body WITH 13 CR TRIM, HF SEAT (API #8), FULL PORT, Vacuum Service	PVVG62R0308 BCAC0VV	
GATE	16" – 24"	CL150	RF	CS Body WITH 13 CR TRIM, HF SEAT (API #8), FULL PORT, GEAR OP. , Vacuum Service	PVVG62R0308 BCAC5VV	
GATE	30" – 48"	CL150	RF	CS Body WITH 13 CR TRIM, HF SEAT (API #8), FULL PORT, GEAR OP. , Vacuum Service	PVVG62K0308 BCAC5VV	
GLOBE	½" – 1-1/2"	CL800	SOCW	CS Body WITH 13 CR TRIM, HF SEAT (API #8) , Vacuum Service	PVVL68S0108A CAD0VV	
GLOBE	2" – 12"	CL150	RF	CS Body WITH 13 CR TRIM, HF SEAT (API #8) , Vacuum Service	PVVL62R0308 BCAC0VV	
GLOBE	14" – 16"	CL150	RF	CS Body WITH 13 CR TRIM, HF SEAT (API #8) , GEAR OP. , Vacuum Service	PVVL62R0308 BCAC5VV	
LIFT CHECK	½" – 1-1/2"	CL800	SOCW	CS Body WITH 13 CR TRIM, HF SEAT (API #8) , Vacuum Service	PVVC68S0108 NCAD0VV	61
SWING CHECK	2" – 10"	CL150	RF	CS Body WITH 13 CR TRIM, HF SEAT (API #8) , Vacuum Service	PVVC62R0308 NCAC0VV	62
DUAL PLT CHECK	12" – 24"	CL150	RF	CS Body WITH 410 SS DISC, HF SEAT (API #8), DBL FLGD, Vacuum Service	PVVC62R032C XCAA0VV	63
SWING CHECK	12" – 24"	CL150	RF	CS Body WITH 13 CR TRIM, HF SEAT (API #8) , Vacuum Service	PVVC62R0308 NCAC0VV	62
DUAL PLT CHECK	30" – 48"	CL150	RF	CS Body WITH 410 SS DISC, HF SEAT (API #8), DBL FLGD, Series-A, Vacuum Service	PVVC62K032C XCAA0VV	63
BUTTERFLY	2" – 6"	CL150	RF	CS Body WITH Resilient Metal SEAT, HF SEAT, DBL FLGD, Vacuum Service	PVVUB62R0368 BCAG0VV	
BUTTERFLY	8" – 24"	CL150	RF	CS Body WITH Resilient Metal SEAT, HF SEAT, DBL FLGD, GEAR OP. , Vacuum Service	PVVUB62R0368 BCAG3VV	
BUTTERFLY	30" – 48"	CL150	RF	CS Body WITH Resilient Metal SEAT, HF SEAT, DBL FLGD, Series-A, GEAR OP. , Vacuum Service	PVVUB62K0368 BCAG3VV	
BUTTERFLY	72"	CL150	RF	CS Body WITH Resilient Metal SEAT, HF SEAT, DBL FLGD, Man STD, GEAR OP. , Vacuum Service	PVVUB62K0368 BCAG3VV	
BALL	¾" – 1-1/2"	CL150	RF	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, RP, SHORT PATTERN, Lever, Vacuum Service	PVVB62R0171 ECAAF1VV	09,326
BALL	2" – 6"	CL150	RF	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, RP, SHORT PATTERN, Lever, Vacuum Service	PVVB62R0371 ECAAF1VV	09,326
BALL	8" – 8"	CL150	RF	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, RP, SHORT PATTERN, GEAR OP. , Vacuum Service	PVVB62R0371 ECAAF3VV	09,326

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
BALL	10" – 24"	CL150	RF	CS Body WITH RPTFE SEAT, Nickel Plated Trunnion Ball, RP, LONG PATTERN, GEAR OP. , Vacuum Service	PVVBE62R037C ECAF3VV	09,326	
NEEDLE	½" – 1-1/2"	CL800	SOCW	CS Body WITH 13 CR TRIM, HF SEAT (API #8) , Vacuum Service	PVVNB68S0108 ACAN0VV		
BOLTING	ALL	CL150		ALLOY STEEL, ASTM A193 GRADE B7, ASTM A194 GRADE 2H HEAVY HEX NUTS			
	ALL	CL300		ALLOY STEEL, ASTM A193 GRADE B7, ASTM A194 GRADE 2H HEAVY HEX NUTS			
GASKET	½" – 24"	CL150	RF	LOW STRESS SPIRAL WOUND, ASME B16.20, ASME B16.5, TP 304 SS WDG, W/FLEXIBLE GRAPHITE FILLER, TP 304 SS INNER RING, CS OUTER RING			
	30" – 60"	CL150	RF	LOW STRESS SPIRAL WOUND, ASME B16.20, ASME B16.47 SERIES A, TP 304 SS WDG, W/FLEXIBLE GRAPHITE FILLER, TP 304 SS INNER RING, CS OUTER RING			
	62" – 84"	CL150	RF	LOW STRESS SPIRAL WOUND, ASME B16.20, (AS PER DETAIL DRAWING), TP 304 SS WDG, W/FLEXIBLE GRAPHITE FILLER, TP 304 SS INNER RING, CS OUTER RING			
	½" – 24"	CL300	RF	SPIRAL WOUND, ASME B16.20, ASME B16.5, TP 304 SS WDG, W/FLEXIBLE GRAPHITE FILLER, TP 304 SS INNER RING, CS OUTER RING			

NOTE	SPECIFIC PIPING NOTES	REV
03.	Threaded joints are permitted only at the outlet of vent and drain valves, hydrostatic connections, outlet instrument take-off valves, and to match equipment.	
06.	These valves shall be used only for vent, drain, and instrument connections.	
09.	Pressure and temperature rating can be limited by certain components permitted by this Practice. Manufacturer's recommended pressure-temperature restrictions shall be consulted.	
12.	These flanges shall be used only to match Class 300 flange connections at control valves and special equipment.	
15.	These valves shall be used if mating to flanged nozzles.	
53.	This item shall be used for orifice flange tap connections only.	
61.	These check valves shall be installed in a horizontal position with cover up.	
62.	These check valves shall be installed in a horizontal position with cover up or in a vertical position with upward flow.	
63.	These check valves shall be installed in a horizontal position with hinge pin vertical or in a vertical position with upward flow.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	
502.	The pad thickness equals run pipe thickness, the pad width equals ½ of branch O.D. These items can be replaced with a Weldolet or Reducing Tee depending on engineering requirements, availability and economics.	
503.	Use this item for Branch Nipple which terminate at the valve or continue as tubing. (eg. For instrument connections, valved vents & drains, etc.).	
510.	The pad thickness equals run pipe thickness, the pad width equals ½ of branch O.D. These items can be replaced with a Latrolet or Lateral depending on engineering requirements, availability and economics.	
531.	Examination shall be in accordance with requirements in "Standard for Piping Welding", refer to document number: 108184-0000-PI-SPC-0006.	
548.	Pipe and pipe components thicker than 0.500" may require impact-tested materials. See ASME B31.3, Paragraph 323.2.2.	
700.	As per ASME B31.3, figure 323.2.2 A, curve C material shall be impact tested above approximately 17 mm Wall thickness (for a minimum design temperature of -29 degrees C). Above 17 mm Wall Thickness normalized material (see note 3 at figure 323.2.2 A) or Carbon Steel for Low temperature Service shall be used.	
702.	These flanges have been calculated based on process design conditions on an individual basis.	
705.	Valves to be suitable for FV at 704°C.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
A1E	R00

SERVICES	Hydrogen	Hydrocarbon	Hydrogen Rich Offgas,		REV
	Dryer Regeneration Effluent		LP Fuel Gas		

PIPING MATERIAL	CARBON STEEL	VALVE MATERIAL	CARBON STEEL	WELDEXAMINATION		REV
RATING / FACING	ASME CLASS 150 / RF	TRIM MATERIAL	13 CR (API #1)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.5 mm	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)*

SIZE

TEMP. °C	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20	24
-28 to 38	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65
93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93
149	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88	15.88
204	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79
260	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72
316	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65
343	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62	8.62

BALL VALVES RTFE SEAT RATINGS (Bar)

TEMP(C)	< NPS 2	REV
-28 to 38	19.65	
93	17.93	
149	15.86	
204	5.52	

BUTTERFLY RTFE SEAT RATINGS(Bar)

TEMP(C)	ALL SIZES	REV
-28 to 38	19.65	
93	17.93	
149	15.86	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2 " 3/4 " - 1 1/2 " 2" - 16" 18" - 24"	160 80 STD STD	PE PE BE BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, API 5L GRADE B, EFW, ASME B36.10M, Ej=0.95		
NIPPLE	TOE 1/2 " - 3/4 " 1" - 1 1/2 " TBE 1/2 " - 3/4 " 1" - 1 1/2 " PBE 1/2 " 3/4 " - 1 1/2 "	XXS 160 XXS 160 160 80	TOE TOE TBE TBE PBE PBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10 CS, ASTM A106 GRADE B, SMLS, ASME B36.10 CS, ASTM A106 GRADE B, SMLS, ASME B36.10 CS, ASTM A106 GRADE B, SMLS, ASME B36.10 CS, ASTM A106 GRADE B, SMLS, ASME B36.10 CS, ASTM A106 GRADE B, SMLS, ASME B36.10		
SWAGE	CON 1/2 " - 3/4 " 1" - 1 1/2 " 1/2 " 3/4 " - 1 1/2 " ECC 1/2 " - 3/4 " 1" - 1 1/2 " 1/2 " 3/4 " - 1 1/2 "	XXS 160 160 80 XXS 160 160 80	TSE TSE BBE BBE TSE TSE BBE BBE	CS, ASTM A234 GRADE WPB, MSS-SP-95 CS, ASTM A234 GRADE WPB, MSS-SP-95 CS, ASTM A234 GRADE WPB, MSS-SP-95 CS, ASTM A234 GRADE WPB, MSS-SP-95 CS, ASTM A234 GRADE WPB, MSS-SP-95 CS, ASTM A234 GRADE WPB, MSS-SP-95 CS, ASTM A234 GRADE WPB, MSS-SP-95 CS, ASTM A234 GRADE WPB, MSS-SP-95	1	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
COUPLING FULL COUPLING	½"	CL6000	SOCW	CS, ASTM A105, ASME B16.11		
	¾" - 1 ½"	CL3000	SOCW	CS, ASTM A105, ASME B16.11	279	
FULL COUPLING	½" - ¾"	CL6000	THRD	CS, ASTM A105, ASME B16.11	279	
	1" - 2"	CL3000	THRD	CS, ASTM A105, ASME B16.11		
OLET						
SOCKOLET	½" - ¾"	CL6000	WELD	CS, ASTM A105, MSS-SP-97		
	1" - 1 ½"	CL3000	WELD	CS, ASTM A105, MSS-SP-97		
THREDOLET	½" - ¾"	CL6000	WELD	CS, ASTM A105, MSS-SP-97	279	
	1" - 2"	CL3000	WELD	CS, ASTM A105, MSS-SP-97	279	
WELDOLET	2" - 8"	AS PIPE	WELD	CS, ASTM A105, MSS-SP-97		
PLUG ROUND HEAD	½" - 2"		THRD	CS, ASTM A105, ASME B16.11, ROUND HEAD		
CAP	½"	CL6000	SOCW	CS, ASTM A105, ASME B16.11		
	¾" - 1 ½"	CL3000	SOCW	CS, ASTM A105, ASME B16.11	279	
	½" - ¾"	CL6000	THRD	CS, ASTM A105, ASME B16.11	279	
	1" - 2"	CL3000	THRD	CS, ASTM A105, ASME B16.11		
	2" - 24"	AS PIPE	BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9		
90° ELBOW						
SW	½"	CL6000	SOCW	CS, ASTM A105, ASME B16.11		
	¾" - 1 ½"	CL3000	SOCW	CS, ASTM A105, ASME B16.11		
LR. BW	2" - 16"	AS PIPE	BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9		
	18" - 24"	AS PIPE	BW	CS, ASTM A234 GRADE WPB-W, ASME B16.9		
45° ELBOW						
	½"	CL6000	SOCW	CS, ASTM A105, ASME B16.11		
	¾" - 1 ½"	CL3000	SOCW	CS, ASTM A105, ASME B16.11		
	2" - 16"	AS PIPE	BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9		
	18" - 24"	AS PIPE	BW	CS, ASTM A234 GRADE WPB-W, ASME B16.9		
TEE						
	½"	CL6000	SOCW	CS, ASTM A105, ASME B16.11		
	¾" - 1 ½"	CL3000	SOCW	CS, ASTM A105, ASME B16.11		
	2" - 16"	AS PIPE	BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9		
	18" - 24"	AS PIPE	BW	CS, ASTM A234 GRADE WPB-W, ASME B16.9		
RED TEE	¾" - 1 ½"	CL3000	SOCW	CS, ASTM A105, ASME B16.11		
	2" - 16"	AS PIPE	BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9		
	18" - 24"	AS PIPE	BW	CS, ASTM A234 GRADE WPB-W, ASME B16.9		
REDUCER						
CON	2" - 16"	AS PIPE	BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9		
	18" - 24"	AS PIPE	BW	CS, ASTM A234 GRADE WPB-W, ASME B16.9		
ECC	2" - 16"	AS PIPE	BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9		
	18" - 24"	AS PIPE	BW	CS, ASTM A234 GRADE WPB-W, ASME B16.9		
FLANGE						
SOCKETWELD	½" - 1 ½"	CL150	RF	CS, ASTM A105, ASME B16.5	278	
	½" - 1 ½"	CL300	RF	CS, ASTM A105, ASME B16.5		
WELDNECK	2" - 24"	CL150	RF	CS, ASTM A105, ASME B16.5 THICKNESS AS PIPE	278	
	2" - 24"	CL300	RF	CS, ASTM A105, ASME B16.5 THICKNESS AS PIPE		
BLIND FLG	½" - 24"	CL150	RF	CS, ASTM A105, ASME B16.5	278	
	½" - 24"	CL300	RF	CS, ASTM A105, ASME B16.5		
DRIP RING	1" - 12"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, 1-1/2" THICK WITH ¾" SOCW TAPS (STANDARD DRAWINGS)		
FIGURE 8	1" - 12"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48		
BLANK+SPACER	14" - 24"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
VALVES						41, 277	
GATE	½" - 1 ½"	CL800	SOCW	CS BODY WITH 13 CR TRIM (API #1)	TAG PVVGB68S0101 ACAD0ZZ		
	½" - 1 ½"	CL800	T/SW	CS BODY WITH 13 CR TRIM (API #1)	PVVGB68D0101 ACAD0ZZ	4	
	½" - 2"	CL150	RF	CS BODY WITH 13 CR TRIM (API #1)	PVVGB62R0101 ACAD0VZZ	24	
	2"	CL150	RF	CS BODY WITH 13 CR TRIM (API #1), FULL PORT	PVVG62R0301 ACAD0ZZ	275	
	3" - 14"	CL150	RF	CS BODY WITH 13 CR TRIM (API #1), FULL PORT	PVVG62R0301 BCAC0ZZ		
	16" - 24"	CL150	RF	CS BODY WITH 13 CR TRIM (API #1), FULL PORT, GEAR OP.	PVVG62R0301 BCAC5ZZ		
GLOBE	½" - 1 ½"	CL800	SOCW	CS BODY WITH 13 CR TRIM (API #1)	PVVL68S0101A CAD0ZZ		
	2" - 12"	CL150	RF	CS BODY WITH 13 CR TRIM (API #1)	PVVL62R0301 BCAC0ZZ		
	14"	CL150	RF	CS BODY WITH 13 CR TRIM (API #1), GEAR OP.	PVVL62R0301 BCAC5ZZ		
PISTON CHECK	½" - 1 ½"	CL800	SOCW	CS BODY WITH 13 CR TRIM (API #1)	PVVCY68S0101 NCAD0ZZ		
WAFER DUAL PLATE CHECK	2" - 24"	CL150	RF	CS BODY WITH 13 CR PLATES, METAL SEATS	PVVC62W0301 XNZA0ZZ		
BALL	½" - 1 ½"	CL600	BW	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS	PVVBH66E0171 ACAF0ZZ	27,55,43, 326	
	½" - 1 ½"	CL600	BW	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT	PVVB66E0171 ACAF0ZZ	27,55,43, 326	
	½" - 1 ½"	CL600	T/SW	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS	PVVBK62R0171 ACAF0ZZ	27,56,43, 326	
	½" - 2"	CL150	RF	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT, SHORT PATTERN	PVVBK62R0171 BCAF1ZZ	27,43,326	
WAF BUTTERFLY	3" - 6"	CL150	RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT	PVVUB62W0364 ANAG0ZZ	27,43,326	
	8" - 24"	CL150	RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT, GEAR OP.	PVVUB62W0364 ANAG3ZZ	27,43,326	
LUG BUTTERFLY	3" - 6"	CL150	RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT	PVVUB62L0364 ANAG0ZZ	27,43,326	
	8" - 24"	CL150	RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT, GEAR OP.	PVVUB62L0364 ANAG3ZZ	27,43,326	
BOLT/NUTS	ALL	CL150		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEXNUTS			
	ALL	CL300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEXNUTS			
GASKETS SPIRAL WOUND	½" - 24"	CL150		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, CS CENTERING RING AND 304SS INTER RING, ASME B16.20 / ASME B16.5			
	½" - 24"	CL300		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, CS CENTERING RING AND 304SS INTER RING, ASME B16.20 / ASME B16.5			

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD	
			SIZES
			½" - 2"
			3" - 24"
			½" - 1 ½"
			2" - 24"

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
4	Socket-weld x Threaded gate valve shall be used for valved pressure, instrument, vent and drain connections only. Piping system end shall be socketweld and instrument end shall be threaded.	
24	Use when required at flanged equipment nozzles, or where noted on P&IDs or Modules.	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
41	Valve shall be suitable for hydrogen service as stated in HOLD.	
43	Only to be used as indicated on the P&IDs.	
55	Supplier shall weld equal length of line class seamless plain end nipples on each buttweld end of the valve, overall length to be 250 mm.	
56	Supplier shall weld only a piece of line class seamless plain end nipple on the socketweld d end of the socketweld x threadedvalve, overall length to be 200 mm.	
275	Only to be used where full port gate valve is required as indicated on P&ID's.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME Class 300 flanges shall only be used to AS PIPE higher rated flange connections at control valves and special equipment.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drains requiring plugs, connections to tube fittings and for connections to threaded proprietary items.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
A1K	R00

						REV
SERVICES	WET FLARE	Dry Flare				

PIPING MATERIAL	CARBON STEEL	VALVE MATERIAL	CARBON STEEL	WELD EXAMINATION		REV
RATING / FACING	ASME CL150 / RF	TRIM MATERIAL	13 CR (API #1)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.50 mm (0.059")	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*

REV

SIZE

TEMP P °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24	30	36
-28-38	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	17.84	14.85
93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.84	14.85
149	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	14.85
204	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79
260	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BUTTERFLY
RTFE SEAT RATINGS(bar)

TEMP(C)	ALL SIZES	REV
-28 to 38	19.65	
93	17.93	
149	15.86	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	½" – 1 ½" 2" – 16" 18"– 36"	80 STD STD	PE BE BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, API 5L GRADE B, DSAW, ASME B36.10M, Ej=0.95		
NIPPLES	½" – 2" ½" – 1 ½"	160 80	TOE PBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
SWAGES CON & ECC	½" – 2"	80	BBE	CS, ASTM A234 GRADE WPB-S, SMLS, MSS-SP-95	1	
OLET 45° LATROLET	½" – 1 ½" 2" – 12"	3000 LB AS PIPE	SOCW WELD	CS, ASTM A105, MSS-SP-97 CS, ASTM A105, MSS-SP-97, TO MATCH PIPE		
PLUG	½" – 2"		THRD	CS, ASTM A105, ASME B16.11, ROUND HEAD	279	
CAP	½" – 1 ½" 2" – 16" 18"– 36"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
90° ELBOW	1/2" – 1 1/2" 2" – 16" 18" – 36"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9		
45° ELBOW	1/2" – 1 1/2" 2" – 16" 18" – 36"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9		
COUPLING COUPLING COUPLING(RED)	1/2" – 1 1/2" 1/2" – 1 1/2"	3000 LB 3000 LB	SOCW SOCW	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11		
45° S.W. LATERAL	1/2" – 1 1/2"	3000 LB	SOCW	CS, ASTM A105, MSS-SP-97, MFR STD		
45° LATERAL	2" – 16" 18" – 36"	AS PIPE AS PIPE	WELD WELD	CS, ASTM A234 GRADE WPB-S, ASME B16.9, ASME B31.3, MFR STD CS, ASTM A234 GRADE WPB-W, ASME B16.9, ASME B31.3, MFR STD		
45° SW LATERAL(RED)	1/2" – 1 1/2"	3000 LB	SOCW	CS, ASTM A105, MSS-SP-97, MFR STD		
45° LATERAL (RED)	2" – 16" 18" – 36"	AS PIPE AS PIPE	WELD WELD	CS, ASTM A234 GRADE WPB-S, ASME B16.9, ASME B31.3, MFR STD CS, ASTM A234 GRADE WPB-W, ASME B16.9, ASME B31.3, MFR STD		
REDUCER CON	2" – 16" 18" – 36"	AS PIPE AS PIPE	WELD WELD	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9		
ECC	2" – 16" 18" – 36"	AS PIPE AS PIPE	WELD WELD	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9		
FLANGE SOCKETWELD	1/2" – 1 1/2" 1/2" – 1 1/2"	CL150 CL300	RF RF	CS, ASTM A105, ASME B16.5 CS, ASTM A105, ASME B16.5	278	
WELDNECK	2" – 24" 26" – 36" 2" – 24"	CL150 CL150 CL300	RF RF RF	CS, ASTM A105, ASME B16.5 CS, ASTM A105, ASME B16.47 SERIES A CS, ASTM A105, ASME B16.5	278	
BLIND FLG	1/2" – 24" 26" – 36" 1/2" – 24"	CL150 CL150 CL300	RF RF RF	CS, ASTM A105, ASME B16.5 CS, ASTM A105, ASME B16.47 SERIES A CS, ASTM A105, ASME B16.5	278	
MISCELLANEOUS DRIP RING	1" – 12"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, 38.100 mm (1-1/2") WITH 3/4" SOCKETWELD TAPS		
FIGURE 8 BLANK	1" – 12"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48		
PADDLE BLANK	14" – 24"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48		
PADDLE SPACER	14" – 24"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48		
VALVES GATE	1/2" – 1 1/2" 1/2" – 1 1/2" 1/2" – 1 1/2" 2"	CL800 CL800 CL150 CL150	SOCW SOCW RF RF	CS BODY WITH 13 CR TRIM (API #1) CS BODY WITH 13 CR TRIM (API #1), FULL PORT CS BODY WITH 13 CR TRIM (API #1) CS BODY WITH 13 CR TRIM (API #1), FULL PORT	277 275 24 43, 275	
				TAG PVVGB68S0101 ACAD0ZZ PVVGF68S0101 ACAD0ZZ PVVGB62R0101 ACAD0ZZ PVVGF62R0301 ACAC0ZZ		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
GATE	3" – 14"	CL150	RF	CS BODY WITH 13 CR TRIM (API #1), FULL PORT	PVVGF62R0301 BCAC0ZZ	33	
	16" – 24"	CL150	RF	CS BODY WITH 13 CR TRIM (API #1), FULL PORT, GEAR OP.	PVVGF62R0301 BCAC5ZZ		
	30" – 36"	CL150	RF	CS BODY WITH 13 CR TRIM (API #1), FULL PORT, GEAR OP.	PVVGF62K0301 BCAC5ZZ		
GLOBE	½" – 1 ½"	CL800	SOCW	CS BODY WITH 13 CR TRIM (API #1)	PVVLB68S0101A CAD0ZZ		
	2" – 12"	CL150	RF	CS BODY WITH 13 CR TRIM (API #1)	PVVLB62R0301 BCAC0ZZ		
	14"	CL150	RF	CS BODY WITH 13 CR TRIM (API #1), GEAR OP.	PVVLB62R0301 BCAC5ZZ		
LIFT CHECK	½" – 1 ½"	CL800	SOCW	CS BODY WITH 13 CR TRIM (API #1)	PVVCL68S0101 NCAD0ZZ	33	
WAFER CHECK	2" – 24"	CL150	WAF-RF	CS BODY WITH 13 CR PLATES, METAL SEATS	PVVCD62W0301 XNZA0ZZ		
	30" – 36"	CL150	WAF-RF	CS BODY WITH 13 CR PLATES, METAL SEATS	PVVCD62I0301X NZA0ZZ		
WAF BUTTERFLY	2" – 6"	CL150	WAF-RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT	PVVUB62W0399 BCAG1ZZ	27, 43,326	
	8" – 24"	CL150	WAF-RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT, GEAR OP.	PVVUB62W0399 BCAG3ZZ	27, 43,326	
	30" – 36"	CL150	WAF-RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT, GEAR OP.	PVVUB62I0399B CAG3ZZ	27, 33, 43,326	
LUG BUTTERFLY	2" – 6"	CL150	LUG-RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT	PVVUB62L0399 BCAG1ZZ	27, 43,326	
	8" – 24"	CL150	LUG-RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT, GEAR OP.	PVVUB62L0399 BCAG3ZZ	27, 43,326	
	30" – 36"	CL150	LUG-RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT, GEAR OP.	PVVUB62O0399 BCAG3ZZ	27, 33, 43,326	
BOLTING	ALL	CL150		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEX NUTS			
	ALL	CL300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEX NUTS			
GASKET	½" – 24"	CL150		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, CS CENTERING RING, ASME B16.20 / ASME B16.5			
	½" – 24"	CL300		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, CS CENTERING RING, ASME B16.20 / ASME B16.5			
	30" – 36"	CL150		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, CS CENTERING RING AND 304SS INNER RING, ASME B16.20/ ASME B16.47 SERIES A			
	30" – 36"	CL300		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, CS CENTERING RING AND 304SS INNER RING, ASME B16.20 / ASME B16.47 SERIES A			

	NOTES			REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD	SIZES	
			½" – 1 ½"	
			2" – 36"	
			½" – 1 ½"	
			2" – 36"	

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
24	Use when required at flanged equipment nozzle, or where noted on P&IDs or Modules in Piping Standard Drawings.	
27	Teflon seated valves are limited to maximum 204 °C. For pressure and temperatures outside table consideration should be given to alternative seat materials in consultation with manufacturers.	
33	Flanged valves and equipment, larger than NPS 24, shall have flange drilling and bolting patterns conforming to series A of ASME B16.47.	
43	Only to be used as indicated on the P&IDs .	
275	Only to be used where full port gate valve is required as indicated on P&IDs .	
277	All valves shown below have detailed descriptions provided in the Valve Specification..	
278	ASME CL300 flanges shall only be used to match higher rated flange connections at control valves and special equipment.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drain requiring plugs, connections to tube fittings and for connections to threaded proprietary items. See referenced Modules in Piping Standard Drawings.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR DOCUM. NO.	

PIPING CLASS	REV
A1R	R00

SERVICES	50%NaOH Caustic soda liquid(CL)	Caustic liquid	Urea			REV

PIPING MATERIAL	CARBON STEEL	VALVE MATERIAL	CARBON STEEL	WELD EXAMINATION	HOLD	REV
RATING / FACING	ASME CLASS 150 / RF	TRIM MATERIAL	304 (API 2#)	STRESS RELIEF	PWHT	
CORROSION ALLOWANCE	3.0 mm (0.118")	CODE	ASME B31.3	BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (MPa)*

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16
-28 to 38	1.965	1.965	1.965	1.965	1.965	1.965	1.965	1.965	1.965				
93	1.793	1.793	1.793	1.793	1.793	1.793	1.793	1.793	1.793				

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES
 RTFE SEAT RATINGS (Mpa)

TEMP(C)	< NPS 3	REV
-28 to 38	1.965	
93	1.793	
149		
204		

BUTTERFLY
 RTFE SEAT RATINGS(Mpa)

TEMP(C)	ALL SIZES	REV
-28 to 38		
93		
149		
204		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" – 1 1/2" 2" – 6" 8"	80 STD STD	PE BE BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
NIPPLES	1/2" – 1 1/2"	80	PBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
SWAGES CON&ECC CON&ECC	1/2" – 1 1/2" 1/2" – 4"	80 AS PIPE	PBE BLE/P SE	CS, ASTM A234 GRADE WPB, MSS-SP-95 CS, ASTM A234 GRADE WPB, MSS-SP-95	702	
OLET SOCKOLET	1/2" – 1 1/2"	CL3000	SOCW	CS, ASTM A105, MSS-SP-97		
CAP	1/2" – 1 1/2" 2" – 8"	CL3000 AS PIPE	SOCW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9		
90° ELBOW	1/2" – 1 1/2" 2" – 6" 8"	CL3000 AS PIPE AS PIPE	SOCW BW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9		
45° ELBOW	1/2" – 1 1/2" 2" – 6" 8"	CL3000 AS PIPE AS PIPE	SOCW BW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
COUPLING COUPLING COUPLING(RD)	1/2" – 1 1/2" 1/2" – 1 1/2"	CL3000 CL3000	SOCW SOCW	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11			
TEE RED TEE	1/2" – 1 1/2" 2" – 6" 8" 1/2" – 1 1/2" 2" – 8"	CL3000 AS PIPE AS PIPE CL3000 AS PIPE	SOCW BW BW SOCW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
REDUCER CON ECC	2" – 6" 8" 2" – 6" 8"	AS PIPE AS PIPE AS PIPE AS PIPE	BW BW BW BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
FLANGE SOCKETWELD WELDNECK BLIND FLG	1/2" – 1 1/2" 1/2" – 1 1/2" 2" – 8" 2" – 8" 1/2"- 8" 1/2"- 8"	CL150 CL300 CL150 CL300 CL150 CL300	RF RF RF RF RF RF	CS, ASTM A105, ASME B16.5 CS, ASTM A105, ASME B16.5 CS, ASTM A105, ASME B16.5 THICKNESS AS PIPE CS, ASTM A105, ASME B16.5 THICKNESS AS PIPE CS, ASTM A105, ASME B16.5 CS, ASTM A105, ASME B16.5		278 278 278	
MISCELLANEOUS FIGURE 8 BLANK	1" – 8"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48			
VALVES GATE GLOBE LIFT CHECK SWING CHECK BALL	1/2" – 1 1/2" 2"-8" 1/2" – 1 1/2" 2" – 8" 1/2" – 1 1/2" 2" – 8" 1/2" – 1 1/2" 2" – 3"	CL150 CL150 CL150 CL150 CL150 CL150 CL150 CL150	RF RF RF RF RF RF RF RF	CS BODY WITH 304 TRIM (API #2) CS BODY WITH 304 TRIM (API #2) ,FULL PORT CS BODY WITH 304 TRIM (API #2) CS BODY WITH 304 TRIM (API #2) CS BODY WITH 304 TRIM (API # 2) (HORIZ. ONLY) CS BODY WITH 304 TRIM (API # 2) CS BODY WITH 304 SS BALL & STEM, RTFE SEATS , FULL PORT CS BODY WITH 304 SS BALL & STEM, RTFE SEATS, FULL PORT		TAG PVVGB62R0102 FCAD0ZZ PVVGB62R0302 FCAC0ZZ PVVLB62R0102F CAD0ZZ PVVLB62R0302F CAK0ZZ PVVCL62R0102 NCAD0ZZ PVVCS62R0302 NCAL0ZZ PVVBA62R0169 EJAX0ZZ PVVBA62R0369 EJAX0ZZ	277 27,326 27,326
BOLTING	ALL ALL	CL150 CL300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEXNUTS ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEXNUTS		278	
GASKET SPIRAL WOUND	1/2"- 8" 1/2"- 8"	CL150 CL300		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, CS CENTERING RING, ASME B16.20 / ASME B16.5 SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, CS CENTERING RING, ASME B16.20 / ASME B16.5		278	

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD	SIZES 1/2" – 1 1/2" 2" – 8" 1/2" – 1 1/2" 2" – 8"

NOTE	SPECIFIC PIPING NOTES	REV
27	TFE,PTFE seated valves are limited to maximum 204°C.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME Class 300 flanges shall only be used to match higher rated flange connections at control valves and special equipment.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	
702	Applied in the case where the small end is 1 1/2 inch and smaller, and the larger end is two inch and larger.	

REV
BRANCH CONNECTION CHART

HEADER SIZE

1/2	T																			
3/4	RT	T																		
1	RT	RT	T																	
1 1/2	RT	RT	RT	T																
2	A	RT	RT	RT	T															
3	A	A	A	A	RT	T														
4	A	A	A	A	RT	RT	T													
6	A	A	A	A	S	RT	RT	T												
8	A	A	A	A	S	S	RT	RT	T											
10																				
12																				
14																				
16																				
	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16							

BRANCH SIZE

BRANCH LEGEND		NOTES	REV
T	Tee (B.W. for two inch and larger, S.W. for 1 11/2" and smaller)		
A	Sockolet (Note e.)		
RT	Reducing Tee (B.W. for two inch and larger, S.W. for 1 11/2" and smaller)		
S	Stub-in without Reinforcing Pad shall be set-in type as shown in detail "2" or "4" of Figure 328.5.4D of ASME B31.3		D01

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
c.	Weldolet may be used in lieu of stub-in if required and verified by flexibility analysis and shown on the piping design drawing.	
e.	PWHT required at all branch attachment welds when the thickness through the weld in any plane through the branch is greater than 38.1 mm (1.5") for P1 material per ASME B31.3 and greater than 19 mm (0.75") for P1 materials per ASME B31.1.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
A1S	R00

SERVICES	98% H ₂ SO ₄ Acid liquid (AL)					REV

PIPING MATERIAL	CARBON STEEL	VALVE MATERIAL	CARBON STEEL	WELDE EXAMINATION	HOLD	REV
RATING / FACING	ASME CLASS 150 / RF	TRIM MATERIAL	Alloy 20(API 13#)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	3.0 mm (0.118")	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (MPa)*

REV

SIZE

TEMP P °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16
-28 to 38	1.965	1.965	1.965	1.965	1.965	1.965	1.965	1.965	1.965				
93	1.793	1.793	1.793	1.793	1.793	1.793	1.793	1.793	1.793				

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES RTFE SEAT RATINGS (Mpa)

TEMP(C)	< NPS 3	REV
-28 to 38	1.965	
93	1.793	
149		
204		

BUTTERFLY RTFE SEAT RATINGS(Mpa)

TEMP(C)	ALL SIZES	REV
-28 to 38		
93		
149		
204		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" – 1 1/2" 2" – 6" 8"	80 STD STD	PE BE BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
NIPPLES	1/2" – 1 1/2"	80	PBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
SWAGES CON&ECC CON&ECC	1/2" – 1 1/2" 1/2" – 4"	80 AS PIPE	PBE BLE/P SE	CS, ASTM A234 GRADE WPB, MSS-SP-95 CS, ASTM A234 GRADE WPB, MSS-SP-95	702	
OLET SOCKOLET	1/2" – 1 1/2"	CL3000	SOCW	CS, ASTM A105, MSS-SP-97		
CAP	1/2" – 1 1/2" 2" – 8"	CL3000 AS PIPE	SOCW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9		
90° ELBOW	1/2" – 1 1/2" 2" – 6" 8"	CL3000 AS PIPE AS PIPE	SOCW BW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9		
45° ELBOW	1/2" – 1 1/2" 2" – 6" 8"	CL3000 AS PIPE AS PIPE	SOCW BW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
COUPLING COUPLING COUPLING (RD)	1/2" – 1 1/2" 1/2" – 1 1/2"	CL3000 CL3000	SOCW SOCW	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11			
TEE	1/2" – 1 1/2" 2" – 6" 8"	CL3000 AS PIPE AS PIPE	SOCW BW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
RED TEE	1/2" – 1 1/2" 2" – 8"	CL3000 AS PIPE	SOCW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
REDUCER							
CON	2" – 6" 8"	AS PIPE AS PIPE	BW BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
ECC	2" – 6" 8"	AS PIPE AS PIPE	BW BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
FLANGE							
SOCKETWELD	1/2" – 1 1/2" 1/2" – 1 1/2"	CL150 CL300	RF RF	CS, ASTM A105, ASME B16.5 CS, ASTM A105, ASME B16.5		278	
WELDNECK	2" – 8" 2" – 8"	CL150 CL300	RF RF	CS, ASTM A105, ASME B16.5 THICKNESS AS PIPE CS, ASTM A105, ASME B16.5 THICKNESS AS PIPE		278	
BLIND FLG	1/2"- 8" 1/2"- 8"	CL150 CL300	RF RF	CS, ASTM A105, ASME B16.5 CS, ASTM A105, ASME B16.5		278	
MISCELLANEOUS FIGURE 8 BLANK PADDLE BLANK PADDLE SPACER	1" – 8"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48			
VALVES							
GATE	1/2" – 1 1/2" 2" – 8"	CL150 CL150	RF RF	CS BODY WITH Alloy 20 TRIM (API #13) CS BODY WITH Alloy 20 TRIM (API #13), FULL PORT	TAG PVVGB62R0113 FCAD0ZZ PVVGB62R0313 FCAC0ZZ	277	
GLOBE	1/2" – 1 1/2" 2" – 8"	CL150 CL150	RF RF	CS BODY WITH Alloy 20 TRIM (API #13) CS BODY WITH Alloy 20 TRIM (API #13)	PVVLB62R0113F CAD0ZZ PVVLB62R0313F CAK0ZZ		
LIFT CHECK	1/2" – 1 1/2"	CL150	RF	CS BODY WITH Alloy 20 TRIM (API #13) (HORIZ. ONLY)	PVVCL62R0113 NCAD0ZZ		
SWING CHECK	2" – 8"	CL150	RF	CS BODY WITH Alloy 20 TRIM (API #13)	PVVC62R0313 NCAL0ZZ		
BALL	1/2" – 1 1/2" 2" – 3"	CL150 CL150	RF RF	CS BODY WITH Alloy 20 BALL & STEM, RTFE SEATS, FULL PORT CS BODY WITH Alloy 20 BALL & STEM, RTFE SEATS, FULL PORT	PVVBA62R0358 FCAX0ZZ PVVBA62R0358 FCAX0ZZ	27,326 27,326	
BOLTING							
	ALL ALL	CL150 CL300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEXNUTS ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEXNUTS			
GASKET							
SPIRAL WOUND	1/2"- 8" 1/2"- 8"	CL150 CL300		SPIRAL WOUND, 316 SS W/GRAPHITE FILLER, SS CENTERING RING & SS INNER RING, ASME B16.20 / ASME B16.5 SPIRAL WOUND, 316 SS W/GRAPHITE FILLER, SS CENTERING RING & SS INNER RING, ASME B16.20 / ASME B16.5			

NOTES			REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD	SIZES
			1/2" – 1 1/2"
			2" – 8"
			1/2" – 1 1/2" 2" – 8"

NOTE	SPECIFIC PIPING NOTES	REV
27	TFE,PTFE seated valves are limited to maximum 204 °C.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME Class 300 flanges shall only be used to match higher rated flange connections at control valves and special equipment.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	
702	Applied in the case where the small end is 1 1/2 inch and smaller, and the larger end is two inch and larger.	

REV
BRANCH CONNECTION CHART

HEADER SIZE

1/2	T														
3/4	RT	T													
1	RT	RT	T												
1 1/2	RT	RT	RT	T											
2	A	RT	RT	RT	T										
3	A	A	A	A	RT	T									
4	A	A	A	A	RT	RT	T								
6	A	A	A	A	S	RT	RT	T							
8	A	A	A	A	S	S	RT	RT	T						
10															
12															
14															
16															
	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16		

BRANCH SIZE

BRANCH LEGEND		NOTES	REV
T	Tee (B.W. for two inch and larger, S.W. for 1 1/2" and smaller)		
A	Sockolet		
RT	Reducing Tee (B.W. for two inch and larger, S.W. for 1 1/2" and smaller)		
S	Stub-in without Reinforcing Pad shall be sit-in type as shown in detail "2" or "4" of Figure 328.5.4D of ASME B31.3		
P	Stub-in with Reinforcing Pad shall be set -in type as shown in detail "2" or "4" of Figure 328.5.4D of ASME B31.3 (Note d.)		D01

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
c.	Weldolet may be used in lieu of stub-in if required and verified by flexibility analysis and shown on the piping design drawing.	
d.	A PAD has been selected to provide reinforcement for this branch connection subject to the most severe coincident condition of internal pressure and fluid temperature seen in this specification. However, the specific pressure-temperature condition for this connection may eliminate the need for reinforcement. In addition, when the diameter-to-thickness ratio is restricted by paragraph 304.3.1 (3) (b) of ASME B31.3, finite element analysis (FEA) shall be performed .	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
A1W	R00

SERVICES						REV
	Slurry drainage					
	Regeneration effluent					

PIPING MATERIAL	CARBON STEEL	VALVE MATERIAL	CARBON STEEL	WELD EXAMINATION	HOLD	REV
RATING / FACING	ASME CLASS 150 / RF	TRIM MATERIAL	HF (API 5#)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	3.0 mm (0.118")	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (MPa)*

SIZE

TEMP P °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
-28 to 38	1.965	1.965	1.965	1.965	1.965	1.965	1.965	1.965	1.965	1.965	1.965	1.965	1.965	1.965	1.965	1.965
93	1.793	1.793	1.793	1.793	1.793	1.793	1.793	1.793	1.793	1.793	1.793	1.793	1.793	1.793	1.793	1.793

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES RTFE SEAT RATINGS (Mpa)

TEMP(C)	ALL SIZES	REV
-28 to 38	1.965	
93	1.793	
149		
204		

BUTTERFLY RTFE SEAT RATINGS (Mpa)

TEMP(C)	ALL SIZES	REV
-28 to 38		
93		
149		
204		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" – 1 1/2" 2" – 6" 8" – 16" 18" – 24"	80 STD STD 20	BE BE BE BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, API 5L GRADE B, EFW, ASME B36.10M, Ej=0.85		
NIPPLES	1/2" – 1 1/2"	80	BBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
SWAGES CON&ECC CON&ECC	1/2" – 1 1/2" 1/2" – 4"	80 AS PIPE	BBE BBE	CS, ASTM A234 GRADE WPB, MSS-SP-95 CS, ASTM A234 GRADE WPB, MSS-SP-95	702	
OLET WELDOLET	1/2" – 1 1/2"		BW	CS, ASTM A105, MSS-SP-97	1001	
CAP	1/2" – 1 1/2" 2" – 24"	AS PIPE AS PIPE	BW BW	CS, ASTM A105, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9		
90° ELBOW	1/2" – 1 1/2" 2" – 16" 18" – 24"	AS PIPE AS PIPE AS PIPE	BW BW BW	CS, ASTM A105, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9		
45° ELBOW	1/2" – 1 1/2" 2" – 16" 18" – 24"	AS PIPE AS PIPE AS PIPE	BW BW BW	CS, ASTM A105, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
TEE	1/2" – 1 1/2" 2" – 16" 18" – 24"	AS PIPE AS PIPE AS PIPE	BW BW BW	CS, ASTM A105, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9		
RED TEE	1/2" – 1 1/2" 2" – 16"	AS PIPE AS PIPE	BW BW	CS, ASTM A105, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9		
REDUCER						
CON	2" – 16" 18" – 24"	AS PIPE AS PIPE	BW BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9		
ECC	2" – 16" 18" – 24"	AS PIPE AS PIPE	BW BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9		
FLANGE						
WELDNECK	1/2" – 24"	CL150	RF	CS, ASTM A105, ASME B16.5 THICKNESS AS PIPE	278	
BLIND FLG	1/2" – 24"	CL300	RF	CS, ASTM A105, ASME B16.5 THICKNESS AS PIPE		
	1/2" – 24"	CL150	RF	CS, ASTM A105, ASME B16.5	278	
	1/2" – 24"	CL300	RF	CS, ASTM A105, ASME B16.5		
MISCELLANEOUS						
FIGURE 8 BLANK	1" – 12"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48		
PADDLE BLANK	14" – 24"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48		
PADDLE SPACER	14" – 24"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48		
VALVES						
GATE	1/2" – 1 1/2"	CL800	BW	CS BODY WITH HF TRIM (API #5), FULL PORT	277	
	1/2" – 1 1/2"	CL150	RF	CS BODY WITH HF TRIM (API #5), FULL PORT	24	
	2" – 12"	CL150	RF	CS BODY WITH HF TRIM (API #5), FULL PORT		
	14" – 24"	CL150	RF	CS BODY WITH HF TRIM (API #5), FULL PORT, GEAR OP.		
GLOBE	1/2" – 1 1/2"	CL800	BW	CS BODY WITH HF TRIM (API #5)		
	3" – 12"	CL150	RF	CS BODY WITH HF TRIM (API #5)		
	14"	CL150	RF	CS BODY WITH HF TRIM (API #5), GEAR OP.		
LIFT CHECK	1/2" – 1 1/2"	CL150	BW	CS BODY WITH HF TRIM (API #5) (HORIZ. ONLY)		
SWING CHECK	2" – 24"	CL150	RF	CS BODY WITH HF TRIM (API #5)		
BALL						
	1/2" – 3"	CL150	RF	CS BODY WITH 316 SS BALL & STEM, HF TRIM (API #5), FULL PORT		
BOLTING						
	ALL	CL150		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEXNUTS		
	ALL	CL300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEXNUTS		
GASKET						
	1/2" – 14"	CL150		1.5 mm COMPOSITE GASKET, 304+GRAF, ASME B16.21/ ASME B16.5		
	16" – 24"	CL150		3.0 mm COMPOSITE GASKET, 304+GRAF, ASME B16.21/ ASME B16.5		
	1/2" – 14"	CL300		1.5 mm COMPOSITE GASKET, 304+GRAF, ASME B16.21/ ASME B16.5		
	16" – 24"	CL300		3.0 mm COMPOSITE GASKET, 304+GRAF, ASME B16.21/ ASME B16.5		

NOTE	SPECIFIC PIPING NOTES	REV
24	Use when required at flanged equipment nozzle, or where noted on P&IDs or Modules in Piping Standard Drawings.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME Class 300 flanges shall only be used to match higher rated flange connections at control valves and special equipment.	
702	Applied in the case where the small end is 1 1/2 inch and smaller, and the larger end is two inch and larger.	
1001	The nominal size, schedule of main pipes and branches must be indicated when order is placed.	

HEADER SIZE																
1/2	T															
3/4	RT	T														
1	RT	RT	T													
1 1/2	RT	RT	RT	T												
2	W	RT	RT	RT	T											
3	W	W	W	W	RT	T										
4	W	W	W	W	RT	RT	T									
6	W	W	W	W	S	RT	RT	T								
8	W	W	W	W	S	S	RT	RT	T							
10	W	W	W	W	S	S	S	RT	RT	T						
12	W	W	W	W	S	S	S	RT	RT	RT	T					
14	W	W	W	W	S	S	S	S	RT	RT	RT	T				
16	W	W	W	W	S	S	S	S	S	RT	RT	RT	T			
18	W	W	W	W	S	S	S	S	S	S	RT	RT	RT	T		
20	W	W	W	W	S	S	S	S	S	S	S	RT	RT	RT	T	
24	W	W	W	W	S	S	S	S	S	S	S	S	RT	RT	RT	T
	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20	24

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PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
b.	For 45° branch connections, see 45° Branch Table. The exception is the branches from heater wash cone lines on 'TLE' type heaters where 45° lateral stub-in Branch Weld is acceptable for atmospheric drain conditions.	
c.	Weldolet may be used in lieu of stub-in if required and verified by flexibility analysis and shown on the piping design drawing.	
d.	A PAD has been selected to provide reinforcement for this branch connection subject to the most severe coincident condition of internal pressure and fluid temperature seen in this specification. However, the specific pressure-temperature condition for this connection may eliminate the need for reinforcement. In addition, when the diameter-to-thickness ratio is restricted by paragraph 304.3.1 (3) (b) of ASME B31.3, finite element analysis (FEA) shall be performed .	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
A2A	R00

SERVICES						REV
Dry Flare	Propylene Refrigerant	LP Nitrogen	Dry air	Atmosphere Vent		
Cold Liquid Drain	Catalyst	Plant Air	Cooling Water Return	Cooling Water Supply		
Process	Powder	Safety Valve to Atmosphere	Nitrogen	Washing Oil		
Propane	Propylene flare	Plant Air Vent to Atmosphere	Natural gas	Oxygen-enriched air		
Flare Discharge	Dry Out Air in PDH Unit	Dryer Regeneration Injection Gas				

PIPING MATERIAL	LT CARBON STEEL (IMPACT TESTED)	VALVE MATERIAL	LT CARBON STEEL	WELD EXAMINATION		REV
RATING / FACING	ASME CL150 / RF	TRIM MATERIAL	316 SS (API #10)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.50 mm (0.059")	CODE	ASME B31.3	BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*

SIZE

TEMP. °C	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20	24	30	36	48	54	60
-48 to 38	18.2 7	18.2 7	18.2 7	18.27	18.2 7	18.2 7	18.2 7	18.2 7	18.2 7	18.2 7	18.2 7	18.2 7	18.2 7	18.2 7	18.2 7	18.2 7	18.2 7	18.2 7	18.2 7	18.2 7	18.2 7
93	17.2 4	17.2 4	17.2 4	17.24	17.2 4	17.2 4	17.2 4	17.2 4	17.2 4	17.2 4	17.2 4	17.2 4	17.2 4	17.2 4	17.2 4	17.2 4	17.2 4	17.2 4	17.2 4	17.2 4	17.2 4
149	15.8	15.8	15.8	15.8	15.8	15.8	15.8	15.8	15.8	15.8	15.8	15.8	15.8	15.8	15.8	15.8	15.8	15.8	15.8	15.8	15.8
204	13.6 6	13.6 6	13.6 6	13.66	13.6 6	13.6 6	13.6 6	13.6 6	13.6 6	13.6 6	13.6 6	13.6 6	13.6 6	13.6 6	13.6 6	13.6 6	13.6 6	13.6 6	13.6 6	13.6 6	13.6 6
260	11.7 2	11.7 2	11.7 2	11.72	11.7 2	11.7 2	11.7 2	11.7 2	11.7 2	11.7 2	11.7 2	11.7 2	11.7 2	11.7 2	11.7 2	11.7 2	11.7 2	11.7 2	11.7 2	11.7 2	11.7 2
316	9.84	9.84	9.84	9.84	9.84	9.84	9.84	9.84	9.84	9.84	9.84	9.84	9.84	9.84	9.84	9.84	9.84	9.84	9.84	9.84	9.84
343	8.65	8.65	8.65	8.65	8.65	8.65	8.65	8.65	8.65	8.65	8.65	8.65	8.65	8.65	8.65	8.65	8.65	8.65	8.65	8.65	8.65
371	7.56	7.56	7.56	7.56	7.56	7.56	7.56	7.56	7.56	7.56	7.56	7.56	7.56	7.56	7.56	7.56	7.56	7.56	7.56	7.56	7.56
399	6.53	6.53	6.53	6.53	6.53	6.53	6.53	6.53	6.53	6.53	6.53	6.53	6.53	6.53	6.53	6.53	6.53	6.53	6.53	6.53	6.53
427	5.42	5.42	5.42	5.42	5.42	5.42	5.42	5.42	5.42	5.42	5.42	5.42	5.42	5.42	5.42	5.42	5.42	5.42	5.42	5.42	5.42

BALL VALVES RTFE SEAT RATINGS (bar)

TEMP(C)	< NPS 2	>NPS2	REV
-28 to 38	18.27	18.27	
93	17.24	17.24	
149	15.86	15.86	
204	5.49	3.45	

BUTTERFLY RTFE SEAT RATINGS(bar)

TEMP(C)	ALL SIZES	REV
-28 to 38	18.27	
93	17.24	
149	15.86	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	½" - 1 ½" 2" - 24" 30" - 36" 42"-54" 60"	80 STD STD 12.7 MM 14.27 MM	PE BE BE BE BE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A671 GRADE CC60 CL22 W/S2 AT -48 °C, ASME B36.10M, Ej=1.0 LTCS, ASTM A671 GRADE CC60 CL22 W/S2 AT -48 °C, ASME B36.10M, Ej=1.0 LTCS, ASTM A671 GRADE CC60 CL22 W/S2 AT -48 °C, ASME B36.10M, Ej=1.0	143	
NIPPLES	½" - 1 ½" ½" - 1 ½"	160 80	TOE PBE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M	143	
SWAGES CON & ECC	½" - 2"	80	BLE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95	1,143	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
OLET					143	
SOCKOLET	½" – 1 ½"	3000 LB	SOCW	LTCS, ASTM A350 GRADE LF2 CL1, MSS-SP-97		
WELDOLET	2" – 8"	AS PIPE	WELD	LTCS, ASTM A350 GRADE LF2 CL1, MSS-SP-97		
45° LATROLET	½" – 1 ½"	3000 LB	SOCW	LTCS, ASTM A350 GRADE LF2 CL1, MSS-SP-97		
	2" – 8"	AS PIPE	WELD	LTCS, ASTM A350 GRADE LF2 CL1, MSS-SP-97, TO MATCH PIPE		
PLUG					143	
	½" – 2"	THRD		LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11, ROUND HEAD	279	
CAP					143	
	½" – 1 ½"	3000 LB	SOCW	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11		
	½" – 1 ½"	3000 LB	THRD	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11	279	
	2" – 48"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		
	54"–60"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-W , (ASME B16.9)	61	
90° ELBOW					143	
	½" – 1 ½"	3000 LB	SOCW	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11		
	2" – 24"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		
	30" – 48"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-W, ASME B16.9		
	54"–60"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-W , (ASME B16.9)	61	
45° ELBOW					143	
	½" – 1 ½"	3000 LB	SOCW	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11		
	2" – 24"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		
	30" – 48"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-W, ASME B16.9		
	54" – 60"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-W , (ASME B16.9)	61	
COUPLING					143	
COUPLING	½" – 1 ½"	3000LB	SOCW	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11		
COUPLING(RED)	½" – 1 ½"	3000LB	SOCW	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11		
COUPLING	½" – 1 ½"	3000LB	THRD	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11	279	
COUPLING(RED)	½" – 1 ½"	3000LB	THRD	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11	279	
45° S.W. LATERAL TEE	½" – 1 ½"	3000 LB	SOCW	LTCS, ASTM A350 GRADE LF2 CL1, MSS-SP-97, MFR STD	143	
45° LATERAL TEE	2" – 24"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9, ASME B31.3, MFR STD		
	30"–48"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-W, ASME B16.9, ASME B31.3, MFR STD		
	54"–60"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-W , (ASME B16.9) , MFR STD	61	
45° SW LATERAL TEE(RED)	½" – 1 ½"	3000 LB	SOCW	LTCS, ASTM A350 GRADE LF2 CL1, MSS-SP-97, MFR STD		
45° LATERAL TEE (RED)	2" – 16"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9, ASME B31.3 , MFR STD		
	30"–48"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-W, ASME B16.9, ASME B31.3, MFR STD		
	54"–60"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-W , (ASME B16.9) , ASME B31.3, MFR STD	61	
TEE					143	
	½" – 1 ½"	3000LB	SOCW	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11		
	2" – 24"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		
	30" – 48"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-W, ASME B16.9		
	54"–60"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-W , (ASME B16.9)	61	
RED	½" – 1 ½"	3000LB	SOCW	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11		
	2" – 24"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		
	30" – 48"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-W, ASME B16.9		
	54"–60"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-W , (ASME B16.9)	61	
REDUCER					143	
CON	2" – 24"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		
	30" – 48"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-W, ASME B16.9		
	54"–60"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-W , (ASME B16.9)	61	
ECC	2" – 24"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		
	30" – 48"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-W, ASME B16.9		
	54"–60"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-W , (ASME B16.9)	61	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
FLANGE						143	
SOCKETWELD	½" – 1 ½"	CL150	RF	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5			
	½" – 1 ½"	CL300	RF	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5		278	
WELDNECK	2" – 24"	CL150	RF	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5			
	30" – 60"	CL150	RF	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.47 SERIES A			
WELDNECK	2" – 24"	CL300	RF	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5		278	
BLIND FLG	½" – 24"	CL150	RF	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5			
	30" – 60"	CL150	RF	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.47 SERIES A			
	½" – 24"	CL300	RF	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5		278	
MISCELLANEOUS							
DRIP RING	1" – 12"	CL150	FF	LTCS, ASTM A516 GRADE 70 S-5, NORMALIZED, 38.100 mm (1-1/2") THICK WITH ¾" SOCW TAPS			
FIGURE 8 BLANK	1" – 8"	CL150	FF	LTCS, ASTM A516 GRADE 70 S-5, NORMALIZED, ASME B16.48			
FIGURE 8 BLANK	10" – 12"	CL150	FF	LTCS, ASTM A516 GRADE 70 S-5, NORMALIZED, ASME B16.48			
PADDLE BLANK	14" – 24"	CL150	FF	LTCS, ASTM A516 GRADE 70 S-5, NORMALIZED, ASME B16.48			
PADDLE SPACER	14" – 24"	CL150	FF	LTCS, ASTM A516 GRADE 70 S-5, NORMALIZED, ASME B16.48			
VALVES					TAG	143,277	
GATE	½" – 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API #10)	PVVB68S0710 ACLD0ZZ		
	½" – 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVG68S0710 ACLD0ZZ	275	
	½" – 2"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10)	PVVB62R0710 ACLD0ZZ	24	
	2"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVG62R1110 BCLC0ZZ		
	3" – 14"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVG62R1110 BCLC0ZZ	801	
	16" – 24"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT, GEAR OP.	PVVG62R1110 BCLC5ZZ	801	
	30" – 48"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT, GEAR OP.	PVVG62K1110 BCLC5ZZ	33,801	
GLOBE	½" – 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API #10)	PVVL68S0710A CLD0ZZ		
	2" – 12"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10)	PVVL62R1110 BCLC0ZZ	801	
	14"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10), GEAR OP.	PVVL62R1110 BCLC5ZZ	801	
LIFT CHECK	½" – 1 ½"	CL800	RF	LTCS BODY WITH 316 SS TRIM (API #10) (HORIZ. ONLY)	PVVCL68S0710 NCLD0ZZ		
WAFFER CHECK	2" – 24"	CL150	WAF- RF	LTCS BODY WITH 316 SS PLATES, METAL SEATS	PVVCD62W1110 XNZA0ZZ	801	
	30" – 60"	CL150	WAF- RF	LTCS BODY WITH 316 SS PLATES, METAL SEATS	PVVCD62I1110X NZA0ZZ	33	
WAF BUTTERFLY	3" – 6"	CL150	WAF- RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUB62W1199 BCLG1ZZ	27,43,801, 326	
	8" – 24"	CL150	WAF- RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUB62W1199 BCLG3ZZ	27,43,801, 326	
	30" – 48"	CL150	WAF- RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUB62I1199B CLG3ZZ	27,33,43, 801,326	
LUG BUTTERFLY	3" – 6"	CL150	LUG- RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUB62L1199 BCLG1ZZ	27,43,328, 801,326	
	8" – 24"	CL150	LUG- RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUB62L1199 BCLG3ZZ	27,43,328, 801,326	
	30" – 60"	CL150	LUG- RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUB62O1199 BCLG3ZZ	27,33,43, 328, 801,326	
BALL	½" – 1 ½"	CL600	T/SW	LTCS BODY WITH 316 SS BALL&STEM, FULL PORT	PVVB66D0710 ACLF0S2	27,43,56, 279,326	
BALL	½" – 1 ½"	CL600	SOCW	LTCS BODY WITH 316 SS BALL&STEM, FULL PORT	PVVB66S0710 ACLF0ZZ	27,43,55, 326	
BALL	2" – 4"	CL150	RF	LTCS BODY WITH 316 SS BALL&STEM, RTFE SEATS, SHORT PATTERN	PVVB62R1110 BCLF1ZZ	27,43,326, 801	
BALL	6" – 14"	CL150	RF	LTCS BODY WITH 316 SS BALL&STEM, RTFE SEATS, SHORT PATTERN, WORM GEAR	PVVB62R1110 BCLF3ZZ	27,43,326, 801	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
NEEDLE	1/2" – 1 1/2"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API #10)	PVVNB68S0710 ACLN0ZZ		
BOLTING	ALL	CL150		ALLOY STEEL, ASTM A320 GR.L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS			
	ALL	CL300		ALLOY STEEL, ASTM A320 GR.L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS			
GASKET SPIRAL WOUND	1/2" – 24"	CL150		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, SS INNER & SS OUTER RING, ASME B16.20 / ASME B16.5			
	1/2" – 24"	CL300		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, SS INNER & SS OUTER RING, ASME B16.20 / ASME B16.5			
	30" – 60"	CL150		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, SS INNER & SS OUTER RING, ASME B16.20 / ASME B16.47 SERIES A			
	30" – 60"	CL300		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, SS INNER & SS OUTER RING, ASME B16.20 / ASME B16.47 SERIES A			

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD	SIZES
			1/2" – 1 1/2" 2" – 60" 1/2" – 1 1/2" 2" – 60"

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
24	Use when required at flanged equipment nozzle, or where noted on P&IDs or Modules in Piping Standard Drawings.	
27	TFE, PTFE seated valves are limited to maximum 204°C.	
33	Flanged valves and equipment, larger than NPS 24, shall have flange drilling and bolting patterns conforming to series A of ASME B16.47.	
43	Only to be used as indicated on the P&IDs .	
55	Supplier shall weld equal length of line class seamless plain end nipples on each socketweld end of the valve, overall length to be 250mm.	
56	Supplier shall weld one piece of line class seamless plain end nipple on the socketweld end of the socketweld x threaded valve, overall length to be 200mm.	
61	Fittings and branch connections above NPS 48 diameter, outside the jurisdiction of the specified industry standard, shall be designed or approved by the Buyer's piping engineer. See piping design guidelines or specialty item data sheets.	
143	In shop and/or field fabrication welds (girth, branch and fillet), all weld metal deposits and heat affected zone (HAZ) shall meet impact test requirements for low temperature service (at -48°C for L.T.C.S., at -105°C for semi-cryogenic S.S. "K" specs and for cryogenic S.S. "L" specs) per Paragraph 323 of ASME B31.3.	
275	Only to be used where full port gate valve is required as indicated on P&IDs .	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME CL300 flanges shall only be used to match higher rated flange connections at control valves and special equipment.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drain requiring plugs, connections to tube fittings and for connections to threaded proprietary items. See referenced Modules in Piping Standard Drawings.	
326	This resilient seated valve does not in all cases meet full flange rating. See valve MAWP in this specification	
328	Lug and wafer valves shall only be used at 315°C (600°F) and lower for hydrocarbon and dangerous service and as noted on P&ID. LCB	
801	body valves are limited to maximum 340°C. (If the design temperature is above 340°C, the valve shall be considered as special item)	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	108

PIPING CLASS	REV
A21A	R00

SERVICES	REV
Demineralized Water	Wash water
Phosphate	Recycled water
Pellet water	Filter outlet water
	Dryer regeneration heating
	Oxygen Scavenger
	PAM
	Ammonia
	Atmosphere Vent

PIPING MATERIAL	304/304L SS DUAL GRADE	VALVE MATERIAL	316 SS STAINLESS STEEL	WELD EXAMINATION	PER 108184-0000-PI-SPC-0006	REV
RATING / FACING	ASME CL150 / RFLJ	TRIM MATERIAL	316 SS (API# 10)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	0.50 mm(0.02")	CODE		BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*

REV

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
-38 to 38	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96
93	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86
149	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13
204	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10
260	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72
315	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES RTFE SEAT RATINGS (bar)

TEMP(C)	NPS ≤ 2	REV
-38 ~ 38	18.96	
93	16.2	
149	14.82	
204	N/A	

BUTTERFLY RTFE SEAT RATINGS(bar)

TEMP(C)	ALL SIZES	REV
-38 ~ 38	18.96	
93	16.2	
149	14.82	
204	3.44	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2"-1-1/2" 2" 3"-12" 14"-24"	40S 40S 10S 10S	PE BE BE BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, EFW, ASME B36.19M Ej=1.0		
NIPPLES	1/2" - 1-1/2"	40S	PBE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M		
SWAGES	1/2" - 2"	40S	BBE	SS, ASTM A403 GRADE WP304/304L-S, SMLS, MSS-SP-95		
OLET SOCKOLET SOCELBOLET WELDOLET	1/2" - 1-1/2" 1/2" - 1-1/2" 2" - 8"	3000 LB 3000 LB As pipe	WELD WELD WELD	SS, ASTM A182 GRADE F304/304L, MSS-SP-97 SS, ASTM A182 GRADE F304/304L, MSS-SP-97 SS, ASTM A182 GRADE F304/304L, MSS-SP-97		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PLUG	½" – 2"		THRD	SS, ASTM A182 GRADE F304/304L, ASME B16.11, ROUND HEAD	279	
COUPLING COUPLING (RD)	½" – 1.1/2" ½" – 1.1/2"	3000 LB 3000 LB	SOCW SOCW	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A182 GRADE F304/304L, ASME B16.11		
CAP	½" – 1.1/2" 2" – 12" 14" – 24"	3000 LB As pipe As pipe	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		
90° ELBOW	½" – 1.1/2" 2" – 12" 14" – 24"	3000 LB As pipe As pipe	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		
45° ELBOW	1/2"-1-1/2" 2" – 12" 14" – 24"	3000 LB As pipe As pipe	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		
UNION (GJ)	½" – 1.1/2"	3000 LB	SOCW	SS, ASTM A182 GRADE F304/304L, ASME B16.11		
TEE	1/2"-1-1/2" 2" – 12" 14" – 24"	3000 LB As pipe As pipe	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		
RED	½" – 1.1/2" 2" – 12" 14" – 24"	3000 LB As pipe As pipe	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		
REDUCER CON ECC	2" – 12" 14" – 24" 2" – 12" 14" – 24"	As pipe As pipe As pipe As pipe	WELD WELD WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		
LAP JOINT STUB END	2" – 24"	As pipe	WELD	SS, ASTM A403 GRADE CR304/304L-SMLS, MSS-SP-43 (TYPE A—A)		
FLANGE SOCKETWELD LAP JOINT WELDNECK BLIND	½' – 1.1/2" ½" – 1.1/2" 2" – 24" 2" – 24" 2" – 24" 2" – 24" ½" – 24" ½" – 24"	CL150 CL300 CL150 CL300 CL150 CL300 CL150 CL300	RF RF FF FF RF RF RF RF	SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 CS, ASTM A105, ASME B16.5 CS, ASTM A105, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5	52 278 PREFERRED 501 PREFERRE 501 278 278	
MISCELLANEOUS DRIP RING FIGURE 8 BLANK PADDLE BLANK PADDLE SPACER	1" – 12" 1" – 12" 14" – 24" 14" – 24"	CL150 CL150 CL150 CL150	FF FF FF FF	SS, ASTM A240 GRADE 304/304L, 38.100 mm (1-1/2") THICK W/ SOCW TAPS SS, ASTM A240 GRADE 304/304L, ASME B16.48 SS, ASTM A240 GRADE 304/304L, ASME B16.48 SS, ASTM A240 GRADE 304/304L, ASME B16.48		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
VALVES						277	
GATE	½ – 1.1/2"	CL800	SOCW	SS BODY WITH 316 SS TRIM (API #10)	TAG PVVGB68S3810 ACAD0ZZ		
GATE	½ – 1.1/2"	CL800	SOCW	SS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVGF68S3810 ACAD0ZZ	275	
GATE	½ – 2"	CL150	RF	SS BODY WITH 316 SS TRIM (API #10)	PVVB62R3810 ACAD0ZZ	24	
GATE	½ – 1.1/2"	CL150	RF	SS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVGF62R3810 ACAD0ZZ	275	
GATE	2" – 14"	CL150	RF	SS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVGF62R5410 BCAC0ZZ		
GATE	16" – 24"	CL150	RF	SS BODY WITH 316 SS TRIM (API #10), FULL PORT, GEAR OP.	PVVGF62R5410 BCAC5ZZ		
GLOBE	½ – 1.1/2"	CL800	SOCW	SS BODY WITH 316 SS TRIM (API #10)	PVVL68S3810A CAD0ZZ		
GLOBE	2" – 12"	CL150	RF	SS BODY WITH 316 SS TRIM (API #10)	PVVL62R5410 BCAC0ZZ		
GLOBE	14"	CL150	RF	SS BODY WITH 316 SS TRIM (API #10), GEAR OP.	PVVL62R5410 BCAC5ZZ		
SWINGCHECK	½ – 1.1/2"	CL800	SOCW	SS BODY WITH 316 SS TRIM (API #10)	PVVCS68S3810 NCAD0ZZ		
WAFERCHECK	2" – 24"	CL150	WAF- RF	SS BODY WITH 316 SS PLATES, METAL SEATS	PVVCD62W5410 XNZA0ZZ		
BALL	½ – 1.1/2"	CL600	BW	SS BODY WITH 316 SS BALL & STEM, PTFE SEATS	PVVBH66E3864 ACAF0B2	27,43, 55,326	
BALL	½ – 1.1/2"	CL600	BW	SS BODY WITH 316 SS BALL & STEM, PTFE SEATS, FULL PORT	PVVBF66E3864 ACAF0B2	27,55,275, 326	
BALL	½ – 1.1/2"	CL600	T/SW	SS BODY WITH 316 SS BALL & STEM, PTFE SEATS	PVVBH66D3864 ACAF0S2	27,43, 56,326	
BALL	½ – 2"	CL150	RF	SS BODY WITH 316 SS BALL & STEM, PTFE SEATS, FULL PORT, SHORT PATTERN	PVVBK62R3864 ACAF0ZZ	27,43,326	
BALL	3" – 4"	CL150	RF	SS BODY WITH 316 SS BALL & STEM, PTFE SEATS, FULL PORT, SHORT PATTERN	PVVBK62R5464 BCAF1ZZ	27,43,326	
WAF BUTTERFLY	3" – 6"	CL150	WAF- RF	SS BODY WITH 316 SS DISC. PTFE/316 SS SEAT	PVVUB62W5499 BCAG1ZZ	27,43,326	
WAF BUTTERFLY	8" – 24"	CL150	WAF- RF	SS BODY WITH 316 SS DISC. PTFE/316 SS SEAT, GEAR OP.	PVVUB62W5499 BCAG3ZZ	27,43,326	
LUG BUTTERFLY	3" – 6"	CL150	LUG- RF	SS BODY WITH 316 SS DISC. PTFE/316 SS SEAT	PVVUB62L5499 BCAG1ZZ	27,43,326	
LUG BUTTERFLY	8" – 24"	CL150	LUG- RF	SS BODY WITH 316 SS DISC. PTFE/316 SS SEAT, GEAR OP.	PVVUB62L5499 BCAG3ZZ	27,43,326	
BOLTING							
STUD	ALL	CL150		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A194 GRADE 2H HEAVY HEX NUTS			
STUD	ALL	CL300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A194 GRADE 2H HEAVY HEX NUTS			
GASKETS						54	
FLAT RING	½ – 24"	CL150		3 mm THICK TELFON ENVELOPE GASKET WITH NON-ASBESTOS FILLER, FLAT RING, ASME B16.21 / ASME B16.5			
	½ – 24"	CL300		3 mm THICK TELFON ENVELOPE GASKET WITH NON-ASBESTOS FILLER, FLAT RING, ASME B16.21 / ASME B16.5			
SPIRAL WOUND	½ – 24"	CL150		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, CS CENTERING RING, ASME B16.20 / ASME B16.5		802	

	NOTES		REV
JOINT CONNECTS			
MAINTENANCE		FLANGES	
MAINTENANCE		FLANGES	
NON-MAINTENANCE		SOCKETWELD	
NON-MAINTENANCE		BUTTWELD	
		SIZES	
		1/2" – 1.1/2"	
		2" – 24"	
		1/2" – 1.1/2"	
		2" – 24"	

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
24	Use when required at flanged equipment nozzles, or where noted on P&IDs or Modules.	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
43	Only to be used as indicated on the P&IDs.	
52	Class 150 (B16.5) flat face flanges may be used to match equipment connections as indicated on P&IDs only (S.W. for two inches and smaller, S.O. for three inches and larger), and full face line class gaskets shall be used.	
54	Flat face line class flanges may be used to match equipment connections as indicated on P&IDs only, and full face line class gaskets shall be used.	
55	Supplier shall weld equal length of line class seamless plain end nipples on each buttetweld end of the valve, overall length to be 250 mm.	
56	Supplier shall weld one piece of line class samless plain end nipple on the socketweld end of the socketweld x threaded valve,overall length to be 200mm.	
275	Only to be used where full port gate valve is required as indicated on P&IDs .	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME Class 300 flanges shall only be used to match higher rated flange connections at control valves and special equipment.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections(when applicable),gage glass piping, vents and drain requiring plugs, ,connections to tube fittings and for connections to threaded proprietary items. See referenced Modules in Piping Standard Drawings.	
326	This resilient seated valve does not in all cases meet full flange rating,See valve MAWP in this specification.	
501	Lap joint flange is the preferred joint. Weld necks shall be used only for stress purposes, cyclic conditions or at equipment nozzles.	
802	SPW Gasket W/Graphite Filler just use for Dryer Regeneration Heating	

BRANCH CONNECTION CHART	REV
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HEADER SIZE

[illegible]

BRANCH SIZE

BRANCH

BRANCH LEGEND		NOTES	REV
T	Tee (B.W. for two inch and larger, S.W. for 1 1½" and smaller)		
W	Weldolet (PWHT required at branch attachment welds when the thickness through the weld in any plane through the branch is greater than 38.100 mm (1.5") for P1 material per ASME B31.3.		
A	Sockolet		
RT	Reducing Tee (B.W. for two inch and larger, S.W. for 1 1½" and smaller)		
S	Stub-in without Reinforcing Pad shall be set-in type as shown in detail "2" or "4" of Figure 328.5.4D of ASME B31.3		
P	Stub-in with Reinforcing Pad shall be set-in type as shown in detail "2" or "4" of Figure 328.5.4D of ASME B31.3		

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
A2D	R00

					REV
SERVICES	Wash Oil	Process	Offgas		

						REV
PIPING MATERIAL	L.T. CARBON STEEL (IMPACT TESTED)	VALVE MATERIAL	L.T. CARBON STEEL	WELDEXAMINATION		
RATING / FACING	ASME CLASS 150 / RF	TRIM MATERIAL	316SS (API #10)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.5mm	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)*

SIZE																
TEMP. °C	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20	24
-48 to 38	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27
93	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24
149	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86
204	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BUTTERFLY RTFE SEAT RATINGS(Bar)

TEMP(C)	ALL SIZES	REV
-48 to 38	18.27	
93	17.24	
149	15.86	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" - 1 1/2" 2" - 16" 18" - 24"	80 STD STD	PE BE BE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A671 GRADE CC60 CLASS 22 W/S2 AT -48 °C, ASME B36.10M (Ej=1)	143	
NIPPLE	TOE 1/2" - 1 1/2" TBE 1/2" - 1 1/2" PBE 1/2" - 1 1/2"	160 160 80	TOE TBE PBE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10 LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10 LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10	143	
SWAGE	CON 1/2" - 1 1/2" 1/2" - 1 1/2" ECC 1/2" - 1 1/2" 1/2" - 1 1/2"	160 80 160 80	TSE BBE TSE BBE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95 LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95 LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95 LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95	1, 143	
COUPLING	FULL COUPLING 1/2" - 1 1/2" FULL COUPLING 1/2" - 2"	CL3000 CL3000	SOCW THRD	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11	143 279	
OLET	SOCKOLET 1/2" - 1 1/2" THREDOLET 1/2" - 2" WELDOLET 2" - 8"	CL3000 CL3000 AS PIPE	WELD WELD WELD	LTCS, ASTM A350 GRADE LF2 CLASS 1, MSS-SP-97 LTCS, ASTM A350 GRADE LF2 CLASS 1, MSS-SP-97 LTCS, ASTM A350 GRADE LF2 CLASS 1, MSS-SP-97	143 279	
PLUG	ROUND HEAD 1/2" - 2"		THRD	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11, ROUND HEAD	143	
CAP	1/2" - 1 1/2" 1/2" - 2" 2" - 24"	CL3000 CL3000 AS PIPE	SOCW THRD BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9	143 279	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
90° ELBOW						143	
SW	½" - 1 ½"	CL3000	SOCW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11			
LR	2" - 16"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
	18" - 24"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-W, ASME B16.9			
45° ELBOW						143	
	½" - 1 ½"	CL3000	SOCW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11			
	2" - 16"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
	18" - 24"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-W, ASME B16.9			
TEE						143	
	½" - 1 ½"	CL3000	SOCW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11			
	2" - 16"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
	18" - 24"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-W, ASME B16.9			
RED TEE							
	½" - 1 ½"	CL3000	SOCW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11			
	2" - 16"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
	18" - 24"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-W, ASME B16.9			
REDUCER						143	
CON	2" - 16"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
	18" - 24"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-W, ASME B16.9			
ECC	2" - 16"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
	18" - 24"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-W, ASME B16.9			
FLANGE						143,144	
SOCKETWELD	½" - 1 ½"	CL150	RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5			
	½" - 1 ½"	CL300	RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5		278	
WELDNECK	2" - 24"	CL150	RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5 THICKNESS AS PIPE			
	2" - 24"	CL300	RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5 THICKNESS AS PIPE		278	
BLIND FLG	½" - 24"	CL150	RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5			
	½" - 24"	CL300	RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5		278	
DRIP RING	1" - 12"	CL150	FF	CS, ASTM A516 GRADE 70 S-5, NORMALIZED, 1-1/2" THICK WITH ¾" SOCW TAPS (STANDARD DRAWINGS)			
FIGURE 8	1" - 12"	CL150	FF	CS, ASTM A516 GRADE 70 S-5, NORMALIZED, ASME B16.48			
BLANK+SPACER	14" - 24"	CL150	FF	CS, ASTM A516 GRADE 70 S-5, NORMALIZED, ASME B16.48			
VALVES						143, 277	
GATE	½" - 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API#10)	TAG PVVGB68S0710 ACLD0ZZ		
	½" - 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API#10), FULL PORT	PVVG68S0710 ACLD0ZZ	275	
	½" - 1 ½"	CL800	T/SW	LTCS BODY WITH 316 SS TRIM (API#10)	PVVGB68D0710 ACLD0ZZ	4	
	½" - 1 ½"	CL800	T/SW	LTCS BODY WITH 316 SS TRIM (API#10), FULL PORT	PVVG68D0710 ACLD0ZZ	4, 275	
	½" - 2"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API#10)	PVVGB62R0710 ACLD0ZZ	24	
	2"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API#10), FULL PORT	PVVG62R1110 BCLC0ZZ	43	
	3" - 14"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API#10), FULL PORT	PVVG62R1110 BCLC0ZZ		
	16" - 24"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API#10), FULL PORT, GEAR OP.	PVVG62R1110 BCLC5ZZ		
GLOBE	½" - 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API#10)	PVVLB68S0710A CLD0ZZ		
	2" - 12"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API#10)	PVVLB62R1110 BCLC0ZZ		
	14"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API#10), GEAR OP.	PVVLB62R1110 BCLC5ZZ		
LIFT CHECK	½" - 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API# 10) (HORIZ. ONLY)	PVVCL68S0710 NCLD0ZZ		
WAFER DUAL PLATE CHECK	2" - 24"	CL150	RF	LTCS BODY WITH 316 SS PLATES, METAL SEATS	PVVCD62W1110 NCLA0ZZ		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
WAF BUTTERFLY	3"-6"	CL150	RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUT62W1171 BCLG0ZZ	27,43,326	
	8"-24"	CL150	RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUT62W1171 BCLG3ZZ	27,43,326	
LUG BUTTERFLY	3"-6"	CL150	RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUT62L1171B CLG0ZZ	27,43,326	
	8"-24"	CL150	RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUT62L1171B CLG3ZZ	27,43,326	
BOLT/NUTS	ALL	CL150		ALLOY STEEL, ASTM A320 GRADE L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS			
	ALL	CL300		ALLOY STEEL, ASTM A320 GRADE L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS			
GASKETS SPIRAL WOUND	½" - 24"	CL150		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, 304SS CENTERING RING, ASME B16.20 / ASME B16.5		144	
	½" - 24"	CL300		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, 304SS CENTERING RING, ASME B16.20 / ASME B16.5			

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD	SIZES ½" - 2" 3" - 24" ½" - 1 ½" 2" - 24"

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
4	Socket-weld x Threaded gate valve shall be used for valved pressure, instrument, vent and drain connections only. Piping system end shall be socketweld and instrument end shall be threaded.	
24	Use when required at flanged equipment nozzles, or where noted on P&IDs or Modules.	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
43	Only to be used as indicated on the P&IDs.	
143	In shop and/or field fabrication welds (girth, branch and fillet), all weld metal deposits and heat affected zone (HAZ) shall meet impact test requirements for low temperature service (at -48°C for L.T.C.S., at -48°C for semi-cryogenic S.S. "K" specs and -105°C for cryogenic S.S. "L" specs) per Paragraph 323 of ASME B31.3.	
144	Flanges mated to equipment with aluminum flanges shall use 1/8 in. soft Durabla (Durlon 9000) gaskets and one piece of 1/8 in. thick NEMA G-10 Gr. H22033 glass cloth epoxy resin isolation washer with 1/32 in. thick full length NEMA G-10 isolation sleeve and 1/8 in. thick steel washer between the nut and isolation washer on the aluminum flanges. Flange bolt torque per Manufacturer's recommendations (the G-10 material is limited to 121°C maximum).	
275	Only to be used where full port gate valve is required as indicated on P&ID's.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME Class 300 flanges shall only be used to AS PIPE higher rated flange connections at control valves and special equipment.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drains requiring plugs, connections to tube fittings and for connections to threaded proprietary items. See referenced Piping Modules (Section VI) and Standard Drawings.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	

BRANCH CONNECTION CHART

HEADER SIZE

[illegible]

BRANCH

BRANCH LEGEND		NOTES	REV
T	Tee(B.W. for sizes three inch and larger, S.W. for two inch and smaller)		
RT	Reducing Tee (B.W. for three inch and larger, S.W. for two inch and smaller)		
W	Weldolet (PWHT required at branch attachment welds when the thickness through the weld in any plane through the branch is greater than 1.5" for P1 material per ASME B31.3.)		
A	Sockolet		
S	Stub-in with or without Reinforcing Pad shall be set-on type as shown in detail "1" or "3" of Figure 328.5.4D of ASME B31.3		
P	Reinforcing Pad (Note d.)		

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
c.	Weldolet may be used in lieu of a stub-in Branch Weld if verified and approved by pressure design and flexibility analysis and shown on the piping design drawing	
e	PWHT required at all branch attachment welds when the thickness through the weld in any plane through the branch is greater than 38.1 mm (1.5") for P1 material per ASME B31.3 and greater than 19 mm (0.75") for P1 materials per ASME B31.1.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
A21D	R00

SERVICES	LUBE OIL	Lube Oil (Vendor Package)			REV

PIPING MATERIAL	304/304L Dual Marked	VALVE MATERIAL	316SS	WELDEXAMINATION		REV
RATING / FACING	ASME CLASS 150 / RF	TRIM MATERIAL	316SS (API #10)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	0.5 mm	CODE	ASME B31.3	BENDING	PER PROJECT	

**FLANGE RATING &
MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)*
SIZE**

REV

TEMP (C)	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12
-28 TO 38	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96
93	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86
149	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13
204	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

**API 608 BALL VALVE
PTFE SEAT RATINGS PTFE (Bar)**

TEMP (C)	NPS < 2	2 < NPS < 6	REV
-28 TO 38	18.96	18.96	
93	15.86	15.86	
149	14.13	14.13	
204	N/A	N/A	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2 " - 2"	40S	BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M		
	3" - 12"	10S	BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M		
NIPPLE	PBE 1/2 " - 2"	40S	PBE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M		
SWAGE	CON 1/2 " - 2"	40S	BBE	SS, ASTM A403 GRADE WP 304/304L -S, MSS-SP-95	1,	
	ECC 1/2 " - 2"	40S	BBE	SS, ASTM A403 GRADE WP 304/304L -S, MSS-SP-95		
COUPLING	1/2 " - 2"	CL3000	THRD	SS, ASTM A182 GRADE F 304/304L, ASME B16.11	279	
FULL COUPLING	1/2 " - 2"	CL3000	THRD	SS, ASTM A182 GRADE F 304/304L, ASME B16.11	279	
OLET	1/2 " - 2"	CL3000	WELD	SS, ASTM A182 GRADE F 304/304L, MSS-SP-97	279	
THRD ELBOLET	1/2 " - 2"	CL3000	WELD	SS, ASTM A182 GRADE F 304/304L, MSS-SP-97	279	
WELDOLET	1/2 " - 8"	AS PIPE	WELD	SS, ASTM A182 GRADE F 304/304L, MSS-SP-97		
PLUG	1/2 " - 2"		THRD	SS, ASTM A182 GRADE F 304/304L, ASME B16.11, ROUND HEAD	279	
ROUND HEAD						
CAP	1/2 " - 2"	CL3000	THRD	SS, ASTM A182 GRADE F 304/304L, ASME B16.11	279	
	1/2 " - 12"	AS PIPE	BW	SS, ASTM A403 GRADE WP 304/304L -S, ASME B16.9		
90° ELBOW	1/2 " - 8"	AS PIPE	BW	SS, ASTM A403 GRADE WP 304/304L -S, ASME B16.9		
LR	10" - 12"	AS PIPE	BW	SS, ASTM A403 GRADE WP 304/304L -S, ASME B16.9		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
45° ELBOW	½" – 12"	AS PIPE	BW	SS,ASTM A403 GRADE WP 304/304L-S,ASME B16.9			
TEE	½" – 12"	AS PIPE	BW	SS,ASTM A403 GRADE WP 304/304L-S,ASME B16.9			
RED TEE	½" – 12"	AS PIPE	BW	SS,ASTM A403 GRADE WP 304/304L-S,ASME B16.9			
REDUCER	½" – 12"	AS PIPE	BW	SS,ASTM A403 GRADE WP 304/304L-S,ASME B16.9			
CON	½" – 12"	AS PIPE	BW	SS,ASTM A403 GRADE WP 304/304L-S,ASME B16.9			
ECC	½" – 12"	AS PIPE	BW	SS,ASTM A403 GRADE WP 304/304L-S,ASME B16.9			
FLANGE	½" – 12"	CL150	RF	SS,ASTM A182 GRADE F 304/304L,ASME B16.5 THICKNESS AS PIPE		54	
WELDNECK	½" – 12"	CL300	RF	SS,ASTM A182 GRADE F 304/304L,ASME B16.5 THICKNESS AS PIPE		278	
BLIND FLG	½" – 12"	CL150	RF	SS,ASTM A182 GRADE F 304/304L,ASME B16.5		278	
	½" – 12"	CL300	RF	SS,ASTM A182 GRADE F 304/304L,ASME B16.5			
DRIP RING	1" – 12"	CL150	FF	SS,ASTM A240 GRADE 304/304L(SOLUTION HEAT TREATED),1-1/2"THICK W/3/4"SOCW TAs(STANDARD DRWGS)			
FIGURE 8	1" – 12"	CL150	FF	SS,ASTM A240 GRADE 304/304L,ASME B16.48			
VALVES						277	
GATE	½" – 2"	CL150	RF	SS BODY WITH 316 SS TRIM(API#10)	TAG PVVGB62R3810 AKND0ZZ	275	
	2"	CL150	RF	SS BODY WITH 316 SS TRIM(API#10),FULL PORT	PVVGf62R5410 BKNC0ZZ		
	3" – 12"	CL150	RF	SS BODY WITH 316 SS TRIM(API#10),FULL PORT	PVVGf62R5410 BKNC0ZZ		
GLOBE	½" – 2"	CL150	RF	SS BODY WITH 316 SS TRIM(API#10)	PVVLb62R3810 AKNC0ZZ		
	3" – 12"	CL150	RF	SS BODY WITH 316 SS TRIM(API#10)	PVVLb62R5410 BKNC0ZZ		
SWING CHECK	½" – 2"	CL150	RF	SS BODY WITH 316 SS TRIM(API#10)	PVVCs62R3810 NKNA0ZZ		
	3" – 12"	CL150	RF	SS BODY WITH 316 SS TRIM(API#10)	PVVCs62R5410 NKNA0ZZ		
BALL	½" – 2"	CL150	RF	SS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT, SHORT PATTERN	PVVBQ62R3871 ACAF0ZZ	27,43,326	
	3" – 4"	CL150	RF	SS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT, SHORT PATTERN	PVVBQ62R5471 BCAF0ZZ	27,43,326	
BOLT/NUTS	ALL	CL150		ALLOY STEEL, ASTM A320 GRADE B8 CL2 WITH ASTM A 194 GRADE 8 HEAVY HEX NUTS			
	ALL	CL300		ALLOY STEEL, ASTM A320 GRADE B8 CL2 WITH ASTM A 194 GRADE 8 HEAVY HEX NUTS			
GASKETS						54	
SPIRAL WOUND	½" – 12"	CL150		1.6 mm THK COMPRESSED NON-ASBESTOS, FLAT RING, ASME B16.21 / ASME B16.5			
	½" – 12"	CL300		1.6 mm THK COMPRESSED NON-ASBESTOS, FLAT RING, ASME B16.21 / ASME B16.5			

	NOTES				REV
JOINT CONNECTS		FLANGES		SIZES	
MAINTENANCE		FLANGES		1/2" – 2"	
MAINTENANCE		FLANGES		3" – 24"	
NON-MAINTENANCE		BUTTWELD		1/2" – 1 1/2"	
NON-MAINTENANCE		BUTTWELD		2" – 24"	

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
43	Only to be used as indicated on the P&IDs.	
54	Flat face line class flanges may be used to AS PIPE equipment connections as indicated on P&IDs only, and full face line class gaskets shall be used.	
275	Only to be used where full port gate valve is required as indicated on P&ID's.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME Class 300 flanges shall only be used to AS PIPE higher rated flange connections at control valves and special equipment.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drains requiring plugs, connections to tube fittings and for connections to threaded proprietary items.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	

HEADER SIZE

BRANCH

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
A2DV	R00

					REV
SERVICES	Injection Gas(Vacuum)	Nitrogen (Vacuum)			

PIPING MATERIAL	L.T. CARBON STEEL (IMPACT TESTED)	VALVE MATERIAL	L.T. CARBON STEEL	WELDEXAMINATION		REV
RATING / FACING	ASME CLASS 150 / RF	TRIM MATERIAL	316SS (API#10)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.5 mm	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)*

REV

SIZE

TEMP. °C	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20	24
-48 to 38	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27
93	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24
149	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86
204	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BUTTERFLY RTFE SEAT RATINGS(Bar)

TEMP(C)	ALL SIZES	REV
-48 to 38	18.27	
93	17.24	
149	15.86	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2 " - 1 1/2 " 2" - 24"	80 STD	PE BE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M	143	
NIPPLE	1/2 " - 1 1/2 "	160	TOE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M	143	
	1/2 " - 1 1/2 "	160	TBE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M		
	1/2 " - 1 1/2 "	80	PBE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M		
SWAGE	1/2 " - 1 1/2 "	160	TSE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95	1, 143	
	1/2 " - 1 1/2 "	80	BBE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95		
	1/2 " - 1 1/2 "	160	TSE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95		
	1/2 " - 1 1/2 "	80	BBE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95		
COUPLING	1/2 " - 1 1/2 "	CL3000	SOCW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11	143	
FULL COUPLING	1/2 " - 2"	CL3000	THRD	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11	279	
OLET	1/2 " - 1 1/2 "	CL3000	WELD	LTCS, ASTM A350 GRADE LF2 CLASS 1, MSS-SP-97	143	
SOCKOLET	1/2 " - 2"	CL3000	WELD	LTCS, ASTM A350 GRADE LF2 CLASS 1, MSS-SP-97	279	
THREDOLET	2" - 10"	AS PIPE	WELD	LTCS, ASTM A350 GRADE LF2 CLASS 1, MSS-SP-97		
WELDOLET						
PLUG	1/2 " - 2"		THRD	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11, ROUND HEAD	143	
ROUND HEAD						
CAP	1/2 " - 1 1/2 "	CL3000	SOCW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11	143	
	1/2 " - 2"	CL3000	THRD	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11	279	
	2" - 24"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
90° ELBOW						143	
SW	½" - 1 ½"	CL3000	SOCW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11			
LR	2" - 24"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
45° ELBOW						143	
	½" - 1 ½"	CL3000	SOCW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11			
	2" - 24"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
TEE						143	
	½" - 1 ½"	CL3000	SOCW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11			
	2" - 24"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
RED TEE							
	½" - 1 ½"	CL3000	SOCW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11			
	2" - 24"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
REDUCER						143	
CON	2" - 24"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
ECC	2" - 24"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
FLANGE						143,144	
SOCKETWELD	½" - 1 ½"	CL150	RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5			
	½" - 1 ½"	CL300	RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5		278	
WELDNECK	2" - 24"	CL150	RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5 THICKNESS AS PIPE			
	2" - 24"	CL300	RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5 THICKNESS AS PIPE		278	
BLIND FLG	½" - 24"	CL150	RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5			
	½" - 24"	CL300	RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5		278	
DRIP RING	1" - 12"	CL150	FF	CS, ASTM A516 GRADE 70 S-5, NORMALIZED, 1-1/2" THICK WITH ¾" SOCW TAPS (STANDARD DRAWINGS)			
FIGURE 8	1" - 12"	CL150	FF	CS, ASTM A516 GRADE 70 S-5, NORMALIZED, ASME B16.48			
BLANK+SPACER	14" - 24"	CL150	FF	CS, ASTM A516 GRADE 70 S-5, NORMALIZED, ASME B16.48			
VALVES						143, 277	
GATE	½" - 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API #10)	TAG PVVGB68S0710 ACLD0VV		
	½" - 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVG68S0710 ACLD0VV	275	
	½" - 1 ½"	CL800	T/SW	LTCS BODY WITH 316 SS TRIM (API #10)	PVVGB68D0710 ACLD0VV	4	
	½" - 1 ½"	CL800	T/SW	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVG68D0710 ACLD0VV	4, 275	
	½" - 2"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10)	PVVGB62R0710 ACLD0VV	24	
	2"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVG62R1110 BCLC0VV	43	
	3" - 14"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVG62R1110 BCLC0VV		
	16" - 24"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT, GEAR OP.	PVVG62R1110 BCLC5VV		
GLOBE	½" - 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API #10)	PVVL68S0710A CLD0VV		
	2" - 12"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10)	PVVL62R1110 BCLC0VV		
	14"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10), GEAR OP.	PVVL62R1110 BCLC5VV		
LIFT CHECK	½" - 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API # 10) (HORIZ. ONLY)	PVVCL68S0710 NCLD0VV		
WAFER DUAL PLATE CHECK	2" - 24"	CL150	RF	LTCS BODY WITH 316 SS PLATES, METAL SEATS	PVVCD62W1110 NCLA0VV		
WAF BUTTERFLY	3" - 6"	CL150	RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUT62W1171 BCLG0VV	27,43,326	
	8" - 24"	CL150	RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUT62W1171 BCLG3VV	27,43,326	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
LUG BUTTERFLY	3"–6"	CL150	RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUT62L1171B CLG0VV	27,43,326	
	8"–24"	CL150	RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUT62L1171B CLG3VV	27,43,326	
BOLT/NUTS	ALL	CL150		ALLOY STEEL, ASTM A320 GRADE L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS			
	ALL	CL300		ALLOY STEEL, ASTM A320 GRADE L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS			
GASKETS SPIRAL WOUND	1/2"–24"	CL150		SPIRAL WOUND GASKET, 304 SS W/GRAPHITE FILLER, 304SS CENTERING RING, ASME B16.20 / ASME B16.5		144	
	1/2"–24"	CL300		SPIRAL WOUND GASKET, 304 SS W/GRAPHITE FILLER, 304SS CENTERING RING, ASME B16.20 / ASME B16.5			

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD	

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
4	Socket-weld x Threaded gate valve shall be used for valved pressure, instrument, vent and drain connections only. Piping system end shall be socketweld and instrument end shall be threaded.	
24	Use when required at flanged equipment nozzles, or where noted on P&IDs or Modules.	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
43	Only to be used as indicated on the P&IDs.	
143	In shop and/or field fabrication welds (girth, branch and fillet), all weld metal deposits and heat affected zone (HAZ) shall meet impact test requirements for low temperature service (at -48°C for L.T.C.S., at -48°C for semi-cryogenic S.S. "K" specs and -105°C for cryogenic S.S. "L" specs) per Paragraph 323 of ASME B31.3.	
144	Flanges mated to equipment with aluminum flanges shall use 1/8 in. soft Durabla (Durlon 9000) gaskets and one piece of 1/8 in. thick NEMA G-10 Gr. H22033 glass cloth epoxy resin isolation washer with 1/32 in. thick full length NEMA G-10 isolation sleeve and 1/8 in. thick steel washer between the nut and isolation washer on the aluminum flanges. Flange bolt torque per Manufacturer's recommendations (the G-10 material is limited to 121°C maximum).	
275	Only to be used where full port gate valve is required as indicated on P&ID's.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME Class 300 flanges shall only be used to AS PIPE higher rated flange connections at control valves and special equipment.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drains requiring plugs, connections to tube fittings and for connections to threaded proprietary items. See referenced Piping Modules (Section VI) and Standard Drawings.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
A2E	R00

			REV
SERVICES	Dryer regeneration effluent offgas	Hydrogen flare	

PIPING MATERIAL	L.T. CARBON STEEL (IMPACT TESTED)	VALVE MATERIAL	L.T. CARBON STEEL	WELDEXAMINATION	Per108184-0000-PI-SPC-0006	REV
RATING / FACING	ASME CLASS 150 / RF	TRIM MATERIAL	316SS (API#10)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.5mm	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)*

											REV
DESIGN TEMP	NPS 1/2 - 4	6	8	10	12	14	16	18	20	24	
-48 to 38	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	18.27	
93	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	17.24	
149	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	
204	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BUTTERFLY RTFE SEAT RATINGS(Bar)

TEMP(C)	ALL SIZES	REV
-48 to 38	18.27	
93	17.24	
149	15.86	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2"	160	PE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M	143	
	3/4" - 1 1/2"	80	PE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M		
	2" - 24"	STD	BE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M		
NIPPLE					143	
TOE	1/2" - 3/4"	XXS	TOE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M		
	1" - 2"	160	TOE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M		
TBE	1/2" - 3/4"	XXS	TBE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M		
	1" - 1 1/2"	160	TBE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M		
PBE	1/2"	160	PBE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M		
	3/4" - 1 1/2"	80	PBE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M		
SWAGE					1, 143	
CON	1/2" - 3/4"	XXS	TSE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95		
	1" - 1 1/2"	160	TSE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95		
	1/2"	160	BBE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95		
	3/4" - 1 1/2"	80	BBE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95		
ECC	1/2" - 3/4"	XXS	TSE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95		
	1" - 1 1/2"	160	TSE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95		
	1/2"	160	BBE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95		
	3/4" - 1 1/2"	80	BBE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95		
COUPLING					143	
FULL COUPLING	1/2"	CL6000	SOCW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11		
	3/4" - 1 1/2"	CL3000	SOCW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11		
FULL COUPLING	1/2" - 3/4"	CL6000	THRD	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11	279	
	1" - 2"	CL3000	THRD	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11	279	
OLET					143	
SOCKOLET	1/2" - 3/4"	CL6000	WELD	LTCS, ASTM A350 GRADE LF2 CLASS 1, MSS-SP-97		
	1" - 1 1/2"	CL3000	WELD	LTCS, ASTM A350 GRADE LF2 CLASS 1, MSS-SP-97		
THREDOLET	1/2" - 3/4"	CL6000	WELD	LTCS, ASTM A350 GRADE LF2 CLASS 1, MSS-SP-97	279	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
WELDOLET	1" - 2" 2" - 8"	CL3000 AS PIPE	WELD WELD	LTCS, ASTM A350 GRADE LF2 CLASS 1, MSS-SP-97 LTCS, ASTM A350 GRADE LF2 CLASS 1, MSS-SP-97		279	
PLUG ROUND HEAD	½" - 2"		THRD	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11, ROUND HEAD		143	
CAP	½"	CL6000	SOCW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11		143	
	¾" - 1 ½"	CL3000	SOCW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11			
	½" - ¾"	CL6000	THRD	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11		279	
	1" - 2"	CL3000	THRD	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11		279	
	2" - 24"	AS PIPE	BW	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
90° ELBOW	½"	CL6000	SOCW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11		143	
	¾" - 1 ½" 2" - 24"	CL3000 AS PIPE	SOCW BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
45° ELBOW	½" ¾" - 1 ½" 2" - 24"	CL6000 CL3000 AS PIPE	SOCW SOCW BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		143	
TEE	½" ¾" - 1 ½" 2" - 24"	CL6000 CL3000 AS PIPE	SOCW SOCW BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		143	
RED TEE	½" ¾" - 1 ½" 2" - 24"	CL6000 CL3000 AS PIPE	SOCW SOCW BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
REDUCER	CON ECC	2" - 24" 2" - 24"	AS PIPE AS PIPE	BW BW	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9	143	
FLANGE						143,144	
SOCKETWELD	½" - 1 ½"	CL150	RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5		278	
	½" - 1 ½"	CL300	RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5			
WELDNECK	2" - 24" 2" - 24"	CL150 CL300	RF RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5 THICKNESS AS PIPE LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5 THICKNESS AS PIPE		278	
BLIND FLG	½" - 24"	CL150	RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5			
	½" - 24"	CL300	RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5		278	
DRIP RING	1" - 12"	CL150	FF	CS, ASTM A516 GRADE 70 S-5, NORMALIZED, 1-1/2" THICK WITH ¾" SOCW TAPS (STANDARD DRAWINGS)			
FIGURE 8 BLANK+SPACER	1" - 12" 14" - 24"	CL150 CL150	FF FF	CS, ASTM A516 GRADE 70 S-5, NORMALIZED, ASME B16.48 CS, ASTM A516 GRADE 70 S-5, NORMALIZED, ASME B16.48			
VALVES						143,277	
GATE	½" - 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API #10)	TAG PVVGB68S0710 ACLD0ZZ	275	
	½" - 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVGF68S0710 ACLD0ZZ	4	
	½" - 1 ½"	CL800	T/SW	LTCS BODY WITH 316 SS TRIM (API #10)	PVVGB68D0710 ACLD0ZZ	4, 275	
	½" - 1 ½"	CL800	T/SW	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVGF68D0710 ACLD0ZZ	24	
	½" - 2"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10)	PVVGB62R0710 ACLD0ZZ	43	
	2"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVGF62R1110 BCLC0ZZ		
	3" - 14"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVGF62R1110 BCLC0ZZ		
	16" - 24"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT, GEAR OP.	PVVGF62R1110 BCLC5ZZ		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
GLOBE	½" - 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API #10)	PVVLB68S0710A CLD0ZZ	27,43,326	
	2" - 12"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10)	PVVLB62R1110 BCLC0ZZ		
	14"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10), GEAR OP.	PVVLB62R1110 BCLC5ZZ		
CHECK	½" - 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API # 10) (HORIZ. ONLY)	PVVCL68S0710 NCLD0ZZ		
WAFER CHECK	2" - 24"	CL150	RF	LTCS BODY WITH 316 SS PLATES, METAL SEATS	PVVCD62W1110 NCLA0ZZ		
WAF BUTTERFLY	3" - 6"	CL150	RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUT62W1171 BCLG0ZZ		
	8" - 24"	CL150	RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUT62W1171 BCLG3ZZ		
LUG BUTTERFLY	3" - 6"	CL150	RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUT62L1171B CLG0ZZ		
	8" - 24"	CL150	RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUT62L1171B CLG3ZZ		
BOLT/NUTS	ALL	CL150		ALLOY STEEL, ASTM A320 GRADE L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS			
	ALL	CL300		ALLOY STEEL, ASTM A320 GRADE L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS			
GASKETS SPIRAL WOUND	½" - 24"	CL150		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, 304SS CENTERING RING, ASME B16.20 / ASME B16.5		144	
	½" - 24"	CL300		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, 304SS CENTERING RING, ASME B16.20 / ASME B16.5			

	NOTES				REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD			
			SIZES		
			½" – 2"		
			3" – 24"		
			½" – 1 ½"		
			2" – 24"		

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
4	Socket-weld x Threaded gate valve shall be used for valved pressure, instrument, vent and drain connections only. Piping system end shall be socketweld and instrument end shall be threaded.	
24	Use when required at flanged equipment nozzles, or where noted on P&IDs or Modules.	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
43	Only to be used as indicated on the P&IDs.	
143	In shop and/or field fabrication welds (girth, branch and fillet), all weld metal deposits and heat affected zone (HAZ) shall meet impact test requirements for low temperature service (at -48°C for L.T.C.S., at -48°C for semi-cryogenic S.S. "K" specs and -105°C for cryogenic S.S. "L" specs) per Paragraph 323 of ASME B31.3.	
144	Flanges mated to equipment with aluminum flanges shall use 1/8 in. soft Durabla (Durlon 9000) gaskets and one piece of 1/8 in. thick NEMA G-10 Gr. H22033 glass cloth epoxy resin isolation washer with 1/32 in. thick full length NEMA G-10 isolation sleeve and 1/8 in. thick steel washer between the nut and isolation washer on the aluminum flanges. Flange bolt torque per Manufacturer's recommendations (the G-10 material is limited to 121°C maximum).	
275	Only to be used where full port gate valve is required as indicated on P&ID's.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME Class 300 flanges shall only be used to AS PIPE higher rated flange connections at control valves and special equipment.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drains requiring plugs, connections to tube fittings and for connections to threaded proprietary items. See referenced Piping Modules (Section VI) and Standard Drawings.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	

BRANCH CONNECTION CHART		REV
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HEADER SIZE

[illegible]

BRANCH

BRANCH LEGEND		NOTES	REV
T	Tee(B.W. for sizes three inch and larger, S.W. for two inch and smaller)		
RT	Reducing Tee (B.W. for three inch and larger, S.W. for two inch and smaller)		
W	Weldolet (PWHT required at branch attachment welds when the thickness through the weld in any plane through the branch is greater than 1.5" for P1 material per ASME B31.3.)		
A	Sockolet		
S	Stub-in with or without Reinforcing Pad shall be set-in type as shown in detail "1" or "3" of Figure 328.5.4D of ASME B31.3		
P	Reinforcing Pad (Note d.)		

PIPING BRANCH NOTES

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
c.	Weldolet may be used in lieu of a stub-in Branch Weld if verified and approved by pressure design and flexibility analysis and shown on the piping design drawing	
e	PWHT required at all branch attachment welds when the thickness through the weld in any plane through the branch is greater than 38.1 mm (1.5") for P1 material per ASME B31.3 and greater than 19 mm (0.75") for P1 materials per ASME B31.1.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
A21K	R00

SERVICES	Atmosphere Vent	Catalyst	Isopropanol	Liquid/Melttable Additive	Peroxide	REV
	Low Pressure Nitrogen—4.0 barg	LT Process	Liquid Drain	Propylene	Propylene flare	
	Safety Valve to Atmosphere	Methanol flare				

PIPING MATERIAL	304/304L SS DUAL GRADE	VALVE MATERIAL	316 SS STAINLESS STEEL	WELDEXAMINATION	PER 108184-0000-PI-SPC-0006	REV
RATING / FACING	ASME CL150 / RF	TRIM MATERIAL	316 SS (API #10)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	0.50 mm(0.02")	CODE		BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24	30
-48 to 38	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96
93	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86
149	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13
204	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES RTFE SEAT RATINGS (bar)

TEMP(C)	< NPS 2	>NPS2; < NPS 6	> NPS 6	REV
-48 to 38	18.96	18.96	18.96	
93	16.2	16.2	13.79	
149	14.82	14.82	7.58	
204	5.51	3.34	1.38	

BUTTERFLY RTFE SEAT RATINGS(bar)

TEMP(C)	ALL SIZES	REV
-48 to 38	18.96	
93	16.2	
149	14.82	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" – 1.1/2" 2" 3" – 12" 14" – 24" 30"	40S 40S 10S 10S 10S	PE BE BE BE BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, EFW, ASME B36.19M, Ej=1.0 SS, ASTM A358 GRADE TP 304/304L, CLASS 1 (DOUBLE BUTT SEAM), EFW, SOLUTION HEAT TREATED, ASME B36.19M Ej=1.0	143	
NIPPLES	1/2" – 1.1/2"	40S	PBE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M	143	
SWAGES	1/2" – 2"	40S	PBE	SS, ASTM A403 GRADE WP304/304L-S, MSS-SP-95	1,143	
OLET SOCKOLET SOCWELBOLET WELDOLET	1/2" – 1.1/2" 1/2" – 1.1/2" 2" – 8"	3000 LB 3000 LB AS PIPE	WELD WELD WELD	SS, ASTM A182 GRADE F304/304L, MSS-SP-97 SS, ASTM A182 GRADE F304/304L, MSS-SP-97 SS, ASTM A182 GRADE F304/304L, MSS-SP-97		
PLUG	1/2" – 2"		THRD	SS, ASTM A182 GRADE F304/304L, ASME B16.11, ROUND HEAD	143 279	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
CAP	½" – 1.1/2" 2" – 12" 14" – 30"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		143	
90° ELBOW	½" – 1.1/2" 2" – 12" 14" – 30"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		143	
45° ELBOW	½" – 1.1/2" 2" – 12" 14" – 30"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		143	
COUPLING COUPLING COUPLING (RED)	½" – 1.1/2" ½" – 1.1/2"	3000 LB 3000 LB	SOCW SOCW	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A182 GRADE F304/304L, ASME B16.11		143	
TEE	½" – 1.1/2" 2" – 12" 14" – 30"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		143	
RED	½" – 1.1/2" 2" – 12" 14" – 30"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9			
REDUCER CON	2" – 12" 14" – 30"	AS PIPE AS PIPE	WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9			
ECC	2" – 12" 14" – 30"	AS PIPE AS PIPE	WELD WELD	SS, ASTM A403 GRADE WP304/304LS, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.			
FLANGE SOCKETWELD WELDNECK WELDNECK BLIND	½" – 1.1/2" 2" – 24" 30" ½" – 24"	CL150 CL150 CL150 CL150	RF RF RF RF	SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.47 SERIES A SS, ASTM A182 GRADE F304/304L, ASME B16.5		143,144	
SOCKETWELD WELDNECK WELDNECK BLIND	½" – 1.1/2" 2" – 24" 30" ½" – 24"	CL300 CL300 CL300 CL300	RF RF RF RF	SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.47 SERIES A SS, ASTM A182 GRADE F304/304L, ASME B16.5		278 278 278 278	
MISCELLANEOUS DRIP RING	1" – 12"	CL150	FF	SS, ASTM A240 GRADE 304/304L (SOLUTION HEAT TREATED), 38.100 mm (1-1/2") THICK W/ ¾" SOCW TAPS			
PADDLE BLANK PADDLE &SPACER	1" – 24" 1" – 24"	CL150 CL150	FF FF	SS, ASTM A240 GRADE 304/304L (SOLUTION HEAT TREATED), ASME B16.48 SS, ASTM A240 GRADE 304/304L(SOLUTION HEAT TREATED), ASME B16.48		266 266	
VALVES					TAG	143, 277	
GATE	½" – 1½"	CL800	SOCW	SS BODY WITH 316 SS TRIM (API#10)	PVVB68S3810 ACND0ZZ	146,148	
	½" – 1½"	CL800	SOCW	SS BODY WITH 316 SS TRIM (API#10),FULL PORT	PVVG68S3810 ACND0ZZ	275,146, 148	
	½" – 1½"	CL800	T/SW	SS BODY WITH 316 SS TRIM (API#10)	PVVB68D3810 ACNM0ZZ	4, 146,148	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
GATE	½" – 1½"	CL800	T/SW	SS BODY WITH 316 SS TRIM (API#10),FULL PORT	PVVGF68D3810 ACND0ZZ	4,275, 146,148	
	½" – 2"	CL150	RF	SS BODY WITH 316 SS TRIM (API#10)	PVVGB62R3810 ACND0ZZ	24,146.148	
	2"	CL150	RF	SS BODY WITH 316 SS TRIM (API#10),FULL PORT	PVVG62R5410 BCNC0ZZ	43,146.148	
	3" – 14"	CL150	RF	SS BODY WITH 316 SS TRIM (API#10),FULL PORT	PVVG62R5410 BCNC0ZZ	146.148	
	16" – 24"	CL150	RF	SS BODY WITH 316 SS TRIM (API#10),FULL PORT , GEAR OP.	PVVG62R5410 BCNC5ZZ	146.148	
GLOBE	½" – 1½"	CL800	SOCW	SS BODY WITH 316 SS TRIM (API#10)	PVVLB68S3810A CND0ZZ	148	
	2" – 12"	CL150	RF	SS BODY WITH 316 SS TRIM (API#10)	PVVLB62R5410 BCNC0ZZ	148	
	14"	CL150	RF	SS BODY WITH 316 SS TRIM (API#10) , GEAR OP.	PVVLB62R5410 BCNC5ZZ	148	
SWING CHECK	½" – 1.1/2"	CL800	SOCW	SS BODY WITH 316 SS TRIM (API#10)	PVVCS68S3810 NCND0ZZ		
WAFER CHECK	2" – 24"	CL150	WAF- RF	SS BODY WITH 316 SS PLATES, METAL SEATS	PVVCD62W5410 XNZAOZZ		
WAF BUTTERFLY	3" – 6"	CL150	WAF- RF	SS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUB62W5499 BCNG1ZZ	27, 43, 148,326	
	8" – 24"	CL150	WAF- RF	SS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUB62W5499 BCNG3ZZ	27, 43, 148,326	
LUG BUTTERFLY	3" – 6"	CL150	LUG- RF	SS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUB62L5499 BCNG1ZZ	27, 43, 148,326	
	8" – 24"	CL150	LUG- RF	SS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUB62L5499 BCNG3ZZ	27, 43, 148,326	
BALL	½" – 1.1/2	CL600	BW	SS BODY WITH 316SS BALL&STEM,RTFE SEATS	PVVBH66E3871 ACNF0B2	27,55,43, 326	
BALL	½" – 1.1/2	CL600	BW	SS BODY WITH 316SS BALL&STEM,RTFE SEATS,FULL PORT	PVVBH66E3871 ACNF0B2	27,55,43, 275, 326	
BALL	½" – 1.1/2	CL600	T/SW	SS BODY WITH 316SS BALL&STEM,RTFE SEATS	PVVBH66D3871 ACNF0S2	27,56,43, 326	
BALL	½" – 1.1/2	CL150	RF	SS BODY WITH 316SS BALL&STEM,RTFE SEATS,FULL PORT,SHORTPATTERN	PVVBK62R3871 BCNF0ZZ	27,43,275, 326	
BALL	2" – 4"	CL150	RF	SS BODY WITH 316SS BALL&STEM,RTFE SEATS,SHORT PATTERN	PVVBK62R5471 BCNF1ZZ	27,43,326	
BALL	6" – 8"	CL150	RF	SS BODY WITH 316SS BALL&STEM,RTFE SEATS,SHORT PATTERN,GEAR OP.	PVVBK62R5471 BCNF3ZZ	27,43,326	
BALL	10" – 24"	CL150	RF	SS BODY WITH 316SS BALL&STEM,RTFE SEATS,SHORT PATTERN,GEAR OP.	PVVBK62R5471 BCNF3ZZ	27,43,326	
NEEDLE	½" – 1½"	CL800	SOCW	SS BODY WITH 316 SS TRIM (API#10)	PVVNB68S3810 ACNN0ZZ	148	
BOLTING	ALL	CL150		ALLOY STEEL, ASTM A320 GRADE B8 CL2 WITH ASTM A194 GRADE 8 HEAVY HEX NUTS			
	ALL	CL300		ALLOY STEEL, ASTM A320 GRADE B8 CL2 WITH ASTM A194 GRADE 8 HEAVY HEX NUTS			
GASKET	½" – 24"	CL150		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, 304 SS INNER RING & SSCENTERING RING, ASME B16.20 / ASME B16.5			

BRANCH CONNECTION CHART		REV
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HEADER SIZE

1/2	T
3/4	RT T
1	RT RT T
1½	RT RT RT T
2	A A A A T
3	A A A A RT T
4	A A A A RT RT T
6	A A A A S RT RT T
8	A A A A W W RT RT T
10	A A A A W W W RT RT T
12	A A A A W W W W RT RT T
14	A A A A W W W W W RT RT T
16	A A A A W W W W W P RT RT T
18	A A A A W W W W W W P RT RT T
20	A A A A W W W W W W P P P RT RT T
24	A A A A W W W W W W P P P P RT RT T
30	A A A A W W W W W W P P P P P RT RT T
	½ ¾ 1 1½ 2 3 4 6 8 10 12 14 16 18 20 24 30

BRANCH SIZE

BRANCH LEGEND		NOTES	REV
T	Tee (B.W. for two inch and larger, S.W. for 1 1½" and smaller)		
W	Weldolet (PWHT required at branch attachment welds when the thickness through the weld in any plane through the branch is greater than 38.100 mm (1.5") for P1 material per ASME B31.3.		
A	Sockolet		
RT	Reducing Tee (B.W. for two inch and larger, S.W. for 1 1½" and smaller)		
S	Stub-in without Reinforcing Pad shall be set-in type as shown in detail "2" or "4" of Figure 328.5.4D of ASME B31.3		
P	Stub-in with Reinforcing Pad shall be set-in type as shown in detail "2" or "4" of Figure 328.5.4D of ASME B31.3		

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
c.	Weldolet may be used in lieu of stub-in if required and verified by flexibility analysis and shown on the piping design drawing.	
d.	A PAD has been selected to provide reinforcement for this branch connection subject to the most severe coincident condition of internal pressure and fluid temperature seen in this specification. However, the specific pressure-temperature condition for this connection may eliminate the need for reinforcement. In addition, when the diameter-to-thickness ratio is restricted by paragraph 304.3.1 (3) (b) of ASME B31.3, finite element analysis (FEA) shall be performed .	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
A21L	R00

SERVICES	LT Process	Cold Flare	Ethylene Refrigerant	Cold Liquid Drain	Liquid Nitrogen	REV
	Ethylene flare	Ethylene				

PIPING MATERIAL	304/304L SS (IMPACT TESTED)	VALVE MATERIAL	316 SS	WELD EXAMINATION		REV
RATING / FACING	ASME CLASS 150 / RF	TRIM MATERIAL	316 SS (API #10)	STRESS RELIEF	As per code	
CORROSION ALLOWANCE	0.500 mm (0.02")	CODE		BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)*

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
-196 to 38	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96	18.96
93	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86
149	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13	14.13
204	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10	13.10

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES RTFE SEAT RATINGS (Bar)

TEMP(C)	< NPS 2	>NPS2; < NPS 6	> NPS 6	REV
-105 to 38	18.96	18.96	18.96	
93	15.86	15.86	13.79	
149	14.13	14.13	7.58	
204	5.51	3.34	1.38	

BUTTERFLY RTFE SEAT RATINGS(Bar)

TEMP(C)	ALL SIZES	REV
-105 to 38	18.96	
93	15.86	
149	14.13	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" – 1.1/2" 2" 3" – 12" 14" – 24"	40S 40S 10S 10S	PE BE BE BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, EFW, ASME B36.19M, Ej=1	143	
NIPPLES	1/2" – 1.1/2"	40S	PBE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M	143	
SWAGES	1/2" – 2"	40S	PBE	SS, ASTM A403 GRADE WP304/304L-S, MSS-SP-95	1,143	
OLET SOCKOLET SOCWELBOLET WELDOLET	1/2" – 1.1/2" 1/2" – 1.1/2" 2" – 24"	3000 LB 3000 LB AS PIPE	WELD WELD WELD	SS, ASTM A182 GRADE F304/304L, MSS-SP-97 SS, ASTM A182 GRADE F304/304L, MSS-SP-97 SS, ASTM A182 GRADE F304/304L, MSS-SP-97		
PLUG	1/2" – 2"		THRD	SS, ASTM A182 GRADE F304/304L, ASME B16.11, ROUND HEAD	143	
CAP	1/2" – 1.1/2" 2" – 12" 14" – 24"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9	143	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
90° ELBOW	½" – 1.1/2" 2" – 12" 14" – 24"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		143	
45° ELBOW	½" – 1.1/2" 2" – 12" 14" – 24"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		143	
COUPLING COUPLING	½" – 1.1/2"	3000 LB	SOCW	SS, ASTM A182 GRADE F304/304L, ASME B16.11		143	
TEE	½" – 1.1/2" 2" – 12" 14" – 24"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		143	
RED TEE	½" – 1.1/2" 2" – 12" 14" – 24"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9			
REDUCER CON	2" – 12" 14" – 24"	AS PIPE AS PIPE	WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9			
ECC	2" – 12" 14" – 24"	AS PIPE AS PIPE	WELD WELD	SS, ASTM A403 GRADE WP304/304LS, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.			
FLANGE SOCKETWELD WELDNECK BLIND	½" – 1.1/2" 2" – 24" ½" – 24"	CLASS 150 CLASS 150 CLASS 150	RF RF RF	SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5		143, 144	
SOCKETWELD WELDNECK BLIND	½" – 1.1/2" 2" – 24" ½" – 24"	CLASS 300 CLASS 300 CLASS 300	RF RF RF	SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5		278 278 278	
MISCELLANEOUS DRIP RING	1" – 12"	CLASS 150	FF	SS, ASTM A240 GRADE 304/304L (SOLUTION HEAT TREATED), 38.100 mm (1-1/2") THICK W/ ¾" SOCW TAPS			
PADDLE BLANK PADDLE SPACER	1" – 24" 1" – 24"	CLASS 150 CLASS 150	FF FF	SS, ASTM A240 GRADE 304/304L (SOLUTION HEAT TREATED), ASME B16.48 SS, ASTM A240 GRADE 304/304L (SOLUTION HEAT TREATED), ASME B16.48		266 266	
VALVES						143, 277	
GATE	½" – 1.1/2"	CLASS 800	SOCW	SS BODY WITH 316 SS TRIM (API #10), EXTENDED BONNET	TAG PVVGB68S3810 ACND0EV	146, 147, 148	
GATE	½" – ¾"	CLASS 800	T/SW	SS BODY WITH 316 SS TRIM (API #10)	PVVGF68S3810 ACND0ZZ	4, 146, 147, 148	
GATE	½" – 2"	CLASS 150	RF	SS BODY WITH 316 SS TRIM (API #10), EXTENDED BONNET	PVVG62R3810 ACND0EV	24, 146, 147, 148	
GATE	2" – 14"	CLASS 150	RF	SS BODY WITH 316 SS TRIM (API #10), FULL PORT, EXTENDED BONNET	PVVGF62R5410 BCNC0EV	146, 147, 148	
GATE	16" – 24"	CLASS 150	RF	SS BODY WITH 316 SS TRIM (API #10), FULL PORT, GO, EXTENDED BONNET	PVVGF62R5410 BCNC5EV	146, 147, 148	
GLOBE	½" – 1.1/2"	CLASS 800	SOCW	SS BODY WITH 316 SS TRIM (API #10), EXTENDED BONNET	PVVL68S3810A CND0EV	147, 148	
GLOBE	2" – 12"	CLASS 150	RF	SS BODY WITH 316 SS TRIM (API #10), EXTENDED BONNET	PVVL62R5410 BCNC0EV	147, 148	
GLOBE	14"	CLASS 150	RF	SS BODY WITH 316 SS TRIM (API #10), GO, EXTENDED BONNET	PVVL62R5410 BCNC5EV	147, 148	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
SWING CHECK	½" – 1.1/2"	CLASS 800	SOCW	SS BODY WITH 316 SS TRIM (API #10)	PVVC68S3810 NCND0ZZ		
SWING CHECK	2" – 24"	CLASS 150	RF	SS BODY WITH 316 SS TRIM (API #10)	PVVC62R5410 NCNC0ZZ		
WAF BUTTERFLY	3" – 6"	CLASS 150	RF	SS BODY WITH 316 SS DISC, PTFE/316 SS SEAT, EXTENDED BONNET	PVVUT62W5464 BCNG0EV	27, 43, 147, 148, 326	
WAF BUTTERFLY	8" – 24"	CLASS 150	RF	SS BODY WITH 316 SS DISC, PTFE/316 SS SEAT, EXTENDED BONNET, GEAR OP.	PVVUT62W5464 BCNG3EV	27, 43, 147, 148, 326	
LUG BUTTERFLY	3" – 6"	CLASS 150	RF	SS BODY WITH 316 SS DISC, PTFE/316 SS SEAT, EXTENDED BONNET	PVVUT62L5464B CNG0EV	27, 43, 147, 148, 326 328	
LUG BUTTERFLY	8" – 24"	CLASS 150	RF	SS BODY WITH 316 SS DISC, PTFE/316 SS SEAT, EXTENDED BONNET, GEAR OP.	PVVUT62L5464B CNG3EV	27, 43, 147, 148, 326, 328	
BOLTING	ALL	CLASS 150		ALLOY STEEL, ASTM A320 GRADE B8 CL2 WITH ASTM A194 GRADE 8 HEAVY HEX NUTS			
	ALL	CLASS 300		ALLOY STEEL, ASTM A320 GRADE B8 CL2 WITH ASTM A194 GRADE 8 HEAVY HEX NUTS			
GASKET	½" – 24"	CLASS 150		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, 304 SS INNER RING & SS CENTERING RING, ASME B16.20 / ASME B16.5		144	
	½" – 24"	CLASS 300		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, 304 SS INNER RING & CENTERING RING, ASME B16.20 / ASME B16.5		144	

	NOTES				REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD			
			SIZES		
			½" – 1.1/2"		
			2" – 24"		
			½" – 1 ½"		
			2" – 24"		

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
24	Use when required at flanged equipment nozzle, or where noted on P&IDs or Modules in Piping Standard Drawings.	
27	TFE, PTFE seated valves are limited to maximum 204°C.	
43	Only to be used as indicated on the P&IDs .	
143	In shop and/or field fabrication welds (girth, branch and fillet), all weld metal deposits and heat affected zone (HAZ) shall meet impact test requirements for low temperature service (at -48°C for L.T.C.S., at -48°C for semi-cryogenic S.S. "K" specs and -105°C for cryogenic S.S. "L" specs) per Paragraph 323 of ASME B31.3.	
144	Flanges mated to equipment with aluminum flanges shall use 3.175 mm (1/8") soft Durabla (Durlon 9000) gaskets and one piece of 3.175 mm (1/8") thick NEMA G-10 Gr. H22033 glass cloth epoxy resin isolation washer with 0.794 mm (1/32") thick full length NEMA G-10 isolation sleeve and 3.175 mm (1/8") thick steel washer between the nut and isolation washer on the aluminum flanges. Flange bolt torque per Manufacturer's recommendations (the G-10 material is limited to 121°C maximum).	
146	A 1/8 in. diameter bleed hole shall be provided in the disc from the cavity of the valve to the "vent" side of the valve only. An arrow with "vent" shall be stamped on the outside of the valve body indicating the direction of venting. (See P&ID's for direction of venting for various valve installations of cryogenic gate valves)	
147	Extended bonnet valves shall have the minimum extended bonnet lengths from the centerline of the valve bore to the bottom of the packing (stuffing box) as stated in 3.3 of VALVE SPECIFICATION Notes.	
148	Unless otherwise noted on the modules, valves shall not be installed in vertical lines. Valves in horizontal lines shall be installed with stem in vertical position or at least 45 degrees vertically from the horizontal centerline of the pipe.	
149	No extension bonnet required for valved vents, drains and not used in normal service.	
266	Blind and spacer shall be used instead of Figure 8 in cryogenic services.	
275	Only to be used where full port gate valve is required as indicated on P&IDs .	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME Class 300 flanges shall only be used to match higher rated flange connections at control valves and special equipment.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drain requiring plugs, connections to tube fittings and for connections to threaded proprietary items. See referenced Modules in Piping Standard Drawings.	
326	This resilient seated valve does not in all cases meet full flange rating.	
328	Lug valves shall only be used at 315°C (600°F) and lower for hydrocarbon and dangerous service and as noted on P&ID.	

BRANCH CONNECTION CHART

1/2	T																			
3/4	RT	T																		
1	RT	RT	T																	
1½	RT	RT	RT	T																
2	A	A	A	A	T															
3	A	A	A	A	RT	T														
4	A	A	A	A	RT	RT	T													
6	A	A	A	A	S	RT	RT	T												
8	A	A	A	A	W	W	RT	RT	T											
10	A	A	A	A	W	W	W	RT	RT	T										
12	A	A	A	A	W	W	W	W	RT	RT	T									
14	A	A	A	A	W	W	W	W	W	P	RT	T								
16	A	A	A	A	W	W	W	W	W	P	RT	RT	T							
18	A	A	A	A	W	W	W	W	W	P	P	RT	RT	T						
20	A	A	A	A	W	W	W	W	W	P	P	P	RT	RT	T					
24	A	A	A	A	W	W	W	W	W	P	P	P	P	RT	RT	T				
	½	¾	1	1½	2	3	4	6	8	10	12	14	16	18	20	24				

BRANCH SIZE

BRANCH LEGEND

PIPING BRANCH NOTES

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
A23B	R00

SERVICES	DMD5	PAC		REV

PIPING MATERIAL	316/316L SS DUAL MARKED STAINLESS STEEL	VALVE MATERIAL	316 SS	WELD EXAMINATION		REV
RATING / FACING	ASME CLASS150 / RF	TRIM MATERIAL	316SS (API #16)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	0.5mm	CODE	ASME B31.3	BENDING	PER PROJECT	

**FLANGE RATING &
MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)*
SIZE**

REV

TEMP (C)	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16
-48 TO 38	19.00	19.00	19.00	19.00	19.00	19.00	19.00	19.00	19.00	19.00	19.00	16.12	14.11
93	16.50	16.50	16.50	16.50	16.50	16.50	16.50	16.50	16.50	16.50	16.50	16.12	14.11
149	14.82	14.82	14.82	14.82	14.82	14.82	14.82	14.82	14.82	14.82	14.82	14.82	14.11
204	13.57	13.57	13.57	13.57	13.57	13.57	13.57	13.57	13.57	13.57	13.57	13.57	13.57

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2 " - 1 1/2 "	40S	PE	SS, ASTM A312 GRADE TP 316/316L, SMLS, ASME B36.19M		
	2"	40S	BE	SS, ASTM A312 GRADE TP 316/316L, SMLS, ASME B36.19M		
	3" - 12"	10S	BE	SS, ASTM A312 GRADE TP 316/316L, SMLS, ASME B36.19M		
	14" - 16"	10S	BE	SS, ASTM A312 GRADE TP 316/316L, EFW, ASME B36.19M, E _j =1		
NIPPLE	1/2 " - 1 1/2 "	80S	PBE	SS, ASTM A312 GRADE TP 316/316L, SMLS, ASME B36.19M		
SWAGE	1/2 " - 1 1/2 "	80S	PBE	SS, ASTM A403 GRADE WP 316/316L -S, MSS-SP-95	1,	
	1/2 " - 1 1/2 "	80S	PBE	SS, ASTM A403 GRADE WP 316/316L -S, MSS-SP-95		
COUPLING	1/2 " - 1 1/2 "	CL3000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
FULL COUPLING	1/2 " - 2"	CL3000	THRD	SS, ASTM A182 GRADE F 316/316L, ASME B16.11	279	
OLET	1/2 " - 1 1/2 "	CL3000	WELD	SS, ASTM A182 GRADE F 316/316L, MSS-SP-97		
SOCKOLET	1/2 " - 2"	CL3000	WELD	SS, ASTM A182 GRADE F 316/316L, MSS-SP-97	279	
THREDOLET	2" - 8"	AS PIPE	WELD	SS, ASTM A182 GRADE F 316/316L, MSS-SP-97		
WELDOLET						
PLUG	1/2 " - 2"		THRD	SS, ASTM A182 GRADE F 316/316L, ASME B16.11, ROUND HEAD		
ROUND HEAD						
CAP	1/2 " - 1 1/2 "	CL3000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
	1/2 " - 2"	CL3000	THRD	SS, ASTM A182 GRADE F 316/316L, ASME B16.11	279	
	2" - 16"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-S, ASME B16.9		
90° ELBOW	1/2 " - 1 1/2 "	CL3000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
SW	2" - 12"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-S, ASME B16.9		
LR	14" - 16"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-W, ASME B16.9		
45° ELBOW	1/2 " - 1 1/2 "	CL3000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
SW	2" - 12"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-S, ASME B16.9		
	14" - 16"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-W, ASME B16.9		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
TEE	½" - 1½"	CL3000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11			
	2" - 12"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-S, ASME B16.9			
	14" - 16"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-W, ASME B16.9			
RED TEE	½" - 1½"	CL3000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11			
	2" - 12"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-S, ASME B16.9			
	14" - 16"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-W, ASME B16.9			
REDUCER							
CON	2" - 12"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-S, ASME B16.9			
	14" - 16"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-W, ASME B16.9			
ECC	2" - 12"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-S, ASME B16.9			
	14" - 16"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-W, ASME B16.9			
FLANGE						54	
SOCKETWELD	½" - 1½"	CL150	RF	SS, ASTM A182 GRADE F 316/316L, ASME B16.5			
WELDNECK	2" - 16"	CL150	RF	SS, ASTM A182 GRADE F 316/316L, ASME B16.5 THICKNESS AS PIPE			
BLIND FLG	½" - 16"	CL150	RF	SS, ASTM A182 GRADE F 316/316L, ASME B16.5			
DRIP RING	1" - 16"	CL150	FF	SS, ASTM A240 GRADE 316/316L (SOLUTION HEAT TREATED), 1-1/2" THICK W/3/4" SOCW TAPS (STANDARD DRWGS)			
FIGURE 8	1" - 12"	CL150	FF	SS, ASTM A240 GRADE 316/316L, ASME B16.48			
BLANK & SPACER	14" - 16"	CL150	FF	SS, ASTM A240 GRADE 316/316L, ASME B16.48			
VALVES						277	
GATE	½" - 1½"	CL800	SOCW	SS BODY WITH 316SSTRIM(API#16)	TAG PVVGB68S3816 AJND0ZZ		
	½" - 1½"	CL800	SOCW	SS BODY WITH 316SSTRIM(API#16), FULL PORT	PVVGF68S3816 AJND0ZZ	275	
	½" - 2"	CL150	RF	SS BODY WITH 316SSTRIM(API#16)	PVVG62R3816 AJND0ZZ	24	
	2"	CL150	RF	SS BODY WITH 316SSTRIM(API#16), FULL PORT	PVVGF62R5416 BJNC0ZZ	43, 275,	
	3" - 14"	CL150	RF	SS BODY WITH 316SSTRIM(API#16), FULL PORT	PVVGF62R5416 BJNC0ZZ		
GLOBE	½" - 1½"	CL800	SOCW	SS BODY WITH 316SSTRIM(API#16)	PVVL68S3816A JND0ZZ		
	2" - 8"	CL150	RF	SS BODY WITH 316SSTRIM(API#16)	PVVL62R5416 BJNC0ZZ		
SWING CHECK	½" - 1½"	CL150	RF	SS BODY WITH 316SSTRIM(API#16)	PVVC62R5416 NJND0ZZ		
	2" - 16"	CL150	RF	SS BODY WITH 316SSTRIM(API#16)	PVVC62R5416 NJNC0ZZ		
WAFFER DUAL PLATE CHECK	2" - 16"	CL150	RF	SS BODY WITH 316 SS PLATES, METAL SEATS	PVVC62R5410 NJNA0ZZ	43	
BOLT/NUTS	ALL	CL150		ALLOY STEEL, ASTM A320 GRADE B8 CL2 WITH ASTM A 194 GRADE 8 HEAVY HEX NUTS			
GASKETS SPIRAL WOUND	½" - 12"	CL150		SPIRAL WOUND GASKET, 316L SS W/ PTFE FILLER, 316L CENTERING RING, ASME B16.20 / ASME B16.5		54	

	NOTES				REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD	SIZES		
			1/2" – 2"		
			3" – 16"		
			1/2" – 1 1/2"		
			2" – 16"		

BRANCH LEGEND		NOTES	REV
T	Tee(B.W. for sizes three inch and larger, S.W. for two inch and smaller)		
RT	Reducing Tee (B.W. for three inch and larger, S.W. for two inch and smaller)		
W	Weldolet (PWHT required at branch attachment welds when the thickness through the weld in any plane through the branch is greater than 1.5" for P1 material per ASME B31.3.)		
A	Sockolet		
S	Stub-in with or without Reinforcing Pad shall be set-in type as shown in detail "1" or "3" of Figure 328.5.4D of ASME B31.3		
P	Reinforcing Pad (Note d.)		

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
c.	Weldolet may be used in lieu of a stub-in Branch Weld if verified and approved by pressure design and flexibility analysis and shown on the piping design drawing	
e	PWHT required at all branch attachment welds when the thickness through the weld in any plane through the branch is greater than 38.1 mm (1.5") for P1 material per ASME B31.3 and greater than 19 mm (0.75") for P1 materials per ASME B31.1.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
A23D	R00

SERVICES	Hydraulic Oil(after filter)	Seal Oil(after filter)	REV

PIPING MATERIAL	316/316L SS DUAL MARKED STAINLESS STEEL	VALVE MATERIAL	316 SS	WELDEXAMINATION	REV
RATING / FACING	ASME CLASS150 / RF	TRIM MATERIAL	316SS (API #10)	STRESS RELIEF	PER CODE
CORROSION ALLOWANCE	0.5mm	CODE	ASME B31.3	BENDING	PER PROJECT

**FLANGE RATING &
MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)*
SIZE**

TEMP (C)	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16
-48 TO 38	19.00	19.00	19.00	19.00	19.00	19.00	19.00	19.00	19.00	19.00	19.00	16.12	14.11
93	16.50	16.50	16.50	16.50	16.50	16.50	16.50	16.50	16.50	16.50	16.50	16.12	14.11
149	14.82	14.82	14.82	14.82	14.82	14.82	14.82	14.82	14.82	14.82	14.82	14.82	14.11
204	13.57	13.57	13.57	13.57	13.57	13.57	13.57	13.57	13.57	13.57	13.57	13.57	13.57

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2 " - 1 1/2 "	40S	PE	SS, ASTM A312 GRADE TP 316/316L, SMLS, ASME B36.19M		
	2"	40S	BE	SS, ASTM A312 GRADE TP 316/316L, SMLS, ASME B36.19M		
	3" - 12"	10S	BE	SS, ASTM A312 GRADE TP 316/316L, SMLS, ASME B36.19M		
	14" - 16"	10S	BE	SS, ASTM A312 GRADE TP 316/316L, EFW, ASME B36.19M, E _j =1		
NIPPLE	1/2 " - 1 1/2 "	80S	PBE	SS, ASTM A312 GRADE TP 316/316L, SMLS, ASME B36.19M		
SWAGE	1/2 " - 1 1/2 "	80S	PBE	SS, ASTM A403 GRADE WP 316/316L -S, MSS-SP-95	1,	
	1/2 " - 1 1/2 "	80S	PBE	SS, ASTM A403 GRADE WP 316/316L -S, MSS-SP-95		
COUPLING	1/2 " - 1 1/2 "	CL3000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
FULL COUPLING	1/2 " - 2"	CL3000	THRD	SS, ASTM A182 GRADE F 316/316L, ASME B16.11	279	
OLET	1/2 " - 1 1/2 "	CL3000	WELD	SS, ASTM A182 GRADE F 316/316L, MSS-SP-97		
SOCKOLET	1/2 " - 2"	CL3000	WELD	SS, ASTM A182 GRADE F 316/316L, MSS-SP-97	279	
THREDOLET	2" - 8"	AS PIPE	WELD	SS, ASTM A182 GRADE F 316/316L, MSS-SP-97		
WELDOLET						
PLUG	1/2 " - 2"		THRD	SS, ASTM A182 GRADE F 316/316L, ASME B16.11, ROUND HEAD		
ROUND HEAD						
CAP	1/2 " - 1 1/2 "	CL3000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
	1/2 " - 2"	CL3000	THRD	SS, ASTM A182 GRADE F 316/316L, ASME B16.11	279	
	2" - 16"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-S, ASME B16.9		
90° ELBOW	1/2 " - 1 1/2 "	CL3000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
SW	2" - 12"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-S, ASME B16.9		
LR	14" - 16"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-W, ASME B16.9		
45° ELBOW	1/2 " - 1 1/2 "	CL3000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
SW	2" - 12"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-S, ASME B16.9		
	14" - 16"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-W, ASME B16.9		

BRANCH CONNECTION CHART	REV
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[illegible]

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BRANCH LEGEND		NOTES	REV
T	Tee(B.W. for sizes three inch and larger, S.W. for two inch and smaller)		
RT	Reducing Tee (B.W. for three inch and larger, S.W. for two inch and smaller)		
W	Weldolet (PWHT required at branch attachment welds when the thickness through the weld in any plane through the branch is greater than 1.5" for P1 material per ASME B31.3.)		
A	Socket		
S	Stub-in with or without Reinforcing Pad shall be set-on type as shown in detail "1" or "3" of Figure 328.5.4D of ASME B31.3		
P	Reinforcing Pad (Note d.)		

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
c.	Weldolet may be used in lieu of a stub-in Branch Weld if verified and approved by pressure design and flexibility analysis and shown on the piping design drawing	
e	PWHT required at all branch attachment welds when the thickness through the weld in any plane through the branch is greater than 38.1 mm (1.5") for P1 material per ASME B31.3 and greater than 19 mm (0.75") for P1 materials per ASME B31.1.	

PROJECT NUMBER	01000
CLIENT	01000-MA-SPC-0002
LOCATION	

PIPING CLASS	REV
A24P	R00

			REV
SERVICES	Low Pressure Nitrogen—4.0 barg	Powder Additive	
	PP Powder	Solid Additives	

						REV
PIPING MATERIAL	304/304L SS DUAL GRADE	VALVE MATERIAL	316SS STAINLESS STEEL	WELD EXAMINATION		
RATING / FACING	ASME CL150 / RF	TRIM MATERIAL	316SS (API # 10)	STRESS RELIEF	NOT REQUIRED	
CORROSION ALLOWANCE	0.50mm (0.02")	CODE	ASME B31.3	BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (MPa)*

REV

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
-38	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11
38	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91
93	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66
149	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24
204	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22

BALL VALVES

RTFE SEAT RATINGS (Mpa)

TEMP(C)	≤NPS 2	>NPS 2; ≤NPS 6	REV
-38 to 38	18.96	18.96	
93	16.20	16.20	
149	14.82	14.82	
204	5.51	3.44	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2"– 1.1/2"	40S	BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M	343,344,345	548
	2" –12"	10S	BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M		
	14" –24"	10S	BE	SS, ASTM A312 GRADE TP 304/304L, EFW, ASME B36.19M Ej=1.0		
NIPPLES	1/2"– 1.1/2"	40S	BBE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M	344	
SWAGES	1/2"–1.1/2"	80S	BBE	SS, ASTM A403 GRADE WP304/304L-S, SMLS, MSS-SP-95	1, 344	
	1/2"–1.1/2"	40S	BBE	SS, ASTM A403 GRADE WP304/304L-S, SMLS, MSS-SP-95		
CAP	1/2"– 12" 14" – 24"	AS PIPE AS PIPE	WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9	330344	
STUB END	1/2"– 12"	AS PIPE	WELD	(SPECIAL DESIGN)SS, ASTM A403 GRADE WP304/304L-S, MSS-SP-43(SEE PIPING STANDARD DRWG Figure 1)	330, 344	
	14" – 24"	AS PIPE	WELD	(SPECIAL DESIGN)SS, ASTM A403 GRADE WP304/304L-W, MSS-SP-43(SEE PIPING STANDARD DRWG Figure 1)		
90° ELBOW	1/2"– 12" 14" – 24"	AS PIPE AS PIPE	WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9	330,344	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
90° BENDS	2" – 12" 14" – 24" 2" – 12" 14" – 24"	AS PIPE AS PIPE AS PIPE AS PIPE	WELD WELD WELD WELD	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M, 10X DIAMETER SS, ASTM A312 GRADE TP 304/304L, EFW, ASME B36.19M, 10X DIAMETER SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M, 5X DIAMETER SS, ASTM A312 GRADE TP 304/304L, EFW, ASME B36.19M, 5X DIAMETER		330,344	
45° ELBOW	1/2" – 12" 14" – 24"	AS PIPE AS PIPE	WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		330,344	
TEE TEE(RED) 45° LATERAL 45° LATERAL(RED)	1/2" – 12" 14" – 24" 1/2" – 12" 14" – 24" 1/2" – 12" 14" – 24" 1/2" – 12" 14" – 24"	AS PIPE AS PIPE AS PIPE AS PIPE AS PIPE AS PIPE AS PIPE AS PIPE	WELD WELD WELD WELD WELD WELD WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9, MFR STD SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9, MFR STD SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9, MFR STD SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9, MFR STD		330,344	
REDUCER CON ECC	3/4" – 12" 14" – 24" 3/4" – 12" 14" – 24"	AS PIPE AS PIPE AS PIPE AS PIPE	WELD WELD WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		330,344	
FLANGE LAP JOINT SLIP-ON BLIND PAD FLANGE	1/2" – 1.1/2" 2" – 24" 1/2" – 1.1/2" 2" – 24" 1/2" – 1.1/2" 2" – 16" 1/2" – 24" 3"	CL150 CL150 CL300 CL300 CL150 CL150 CL300 CL300 CL150	RF RF RF RF RF RF RF RF RF	(SPECIAL DESIGN)CS,ASTM A105, ASME B16.5(SEE PIPING STANDARD DRWG Figure 1) (SPECIAL DESIGN)CS,ASTM A105, ASME B16.5(SEE PIPING STANDARD DRWG Figure 1) (SPECIAL DESIGN)CS,ASTM A105, ASME B16.5(SEE PIPING STANDARD DRWG Figure 1) (SPECIAL DESIGN)CS,ASTM A105, ASME B16.5(SEE PIPING STANDARD DRWG Figure 1) SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5		334 374 374 374 374	
MISCELLANEOUS FIGURE 8 BLANK PADDLE BLANK PADDLE SPACER	1" – 12" 14" – 24" 14" – 24"	CL150 CL150 CL150	FF FF FF	SS, ASTM A182 GRADE F304/304L, ASME B16.48 SS, ASTM A182 GRADE F304/304L, ASME B16.48 SS, ASTM A182 GRADE F304/304L, ASME B16.48			
VALVES SLIDE SLIDE BALL BALL GLOBE	3" – 14" 16" – 24" 1/2" – 4" 6" 2" – 10"	CL150 CL150 CL150 CL150 CL150	RF RF RF RF RF	G-ALZN 10 SI 8 MG BODY,PLATE MATERIAL A240 GR 316, TRIM:316 G-ALZN 10 SI 8 MG BODY,PLATE MATERIAL A240 GR 316, TRIM:316, GEAR OP. SS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT,SHORT PATTERN. SS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT,SHORT PATTERN,GEAR OP. SS BODY WITH 316 SS TRIM(API #10)		332,277 27,326 27,326	
					TAG PVVKB62R9B2D BCAU0ZZ PVVKB62R9B2D BCAU5ZZ PVVKB62R3871 ECAFOZZ PVVKB62R5471 ECAF3ZZ PVVLB62R5410 BCAC0ZZ		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
T-NEEDLE VALVE	½ – 1.1/2"	CL800	THRD	SS BODY WITH 316 SS TRIM(API #10)	PVVNT68N3810 ACAN0ZZ		
SWINGCHECK	½ – 1.1/2"	CL150	RF	SS BODY WITH 316 SS TRIM(API #10)	PVVCS62R3810 NCAD0ZZ		
LUG CHECK	2"-12"	CL150	LUG- RF	SS BODY WITH 316 SS PLATES,METAL SEATS	PVVCD62L5410 XNZA0ZZ		
BUTTERFLY	2" -6"	CL150	RF	SS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUB62R5471 BCAG0ZZ	326	
BUTTERFLY	8" -12"	CL150	RF	SS BODY WITH 316 SS DISC, RTFE/316 SS SEAT,GEAR OP	PVVUB62R5471 BCAG3ZZ	326	
BOLTING	ALL	CL150		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A194 GRADE 2H HEAVY HEX NUTS ,ASME B18.2.1/ASME B18.2.2			
	ALL	CL300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A194 GRADE 2H HEAVY HEX NUTS, ASME B18.2.1/ASME B18.2.2			
GASKET	1/2"– 24"	CL150	RF	COMPRESSED NON ASBESTOS(ARAMIND FIBER/NBR RUBBER,1.6mm (1/16") THK, ASME B16.21 / ASME B16.5			
	1/2"– 24"	CL150	RF	(SPECIAL DESIGN)SILICON-RUBBER 2.0mm THK, ASME B16.21 / ASME B16.5 (SEE PIPING STANDARD DRWG Figure 1)			
	1/2"– 24"	CL300	RF	COMPRESSED NON ASBESTOS(ARAMIND FIBER/NBR RUBBER,1.6mm (1/16") THK, ASME B16.21 / ASME B16.5			

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES BUTTWELD BUTTWELD	SIZES 1/2"–1.1/2" 2"–24" 1/2"–1.1/2" 2"–24"

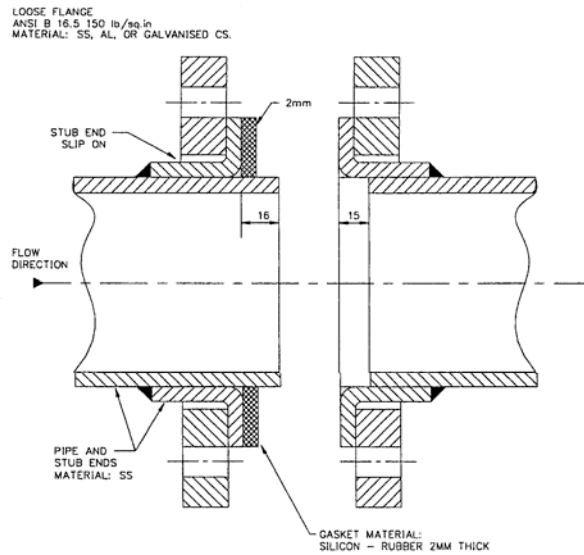
NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size..	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
326	This resilient seated valve does not in all cases meet full flange rating. See valve MAWP in this specification.	
330	The installation length of elbows must be 1500mm including the cylindrical part for PP Powder/Pellets and any Conveyings Service.	
332	Valves shall be selected as indicated in the P&ID(For PP Powder and any Conveyings Service,ball valves are used up to DN150(NPS 6) and the use of check valves is not forseen).	
334	Flange connection may be replaced by not disconnectable sleeve construction via sketch.	
344	VENDOR TO CONFIRM THE FOLLOWING :For Powder Conveying—Required roughness inside pipe and elbows shall be Ra 40-70um for SS.Minimum radius of bends shall be 5Xdiameter,where required.	
345	Grouding is required for PP pellets and Powder conveying piping.	
374	Flanges shall only be used at valve and equipment connections.	
548	ASME A312 EFW pipe shall be examined by spot radiography.The requirement of spot radiograph shall be specified in the requisition" for all 150# and 300# A312 EFW piping classes ending with suffix "L" and piping classes A24P,A24P2 which are used only for purity purposes.	

Branch Connections: (TO BE PROVIDED DURING DETIAL DESIGN)

Branch like O-lets, Stub-ins and other 90 degree branches not allowed because of sensitive service. In case of unavoidable need ,Branches have to be carry out with respect to requirement of special service.

Flange Connection with Slip-On Flanges for PP Pellets Conveying Sevice

Figure 1: Example of flange connection for conveying pipe - Slip on flange (special design)



PROJECT NUMBER	01000
CLIENT	01000-MA-SPC-0002
LOCATION	

PIPING CLASS	REV
A24P1	R00

						REV
SERVICES	PP Pellets Conveying (with and w/o Solids)					
	Low Pressure Nitrogen—4.0 barg					
						REV
PIPING MATERIAL	304/304L SS DUAL GRADE	VALVE MATERIAL	316 SS STAINLESS STEEL	WELD EXAMINATION		
RATING / FACING	ASME CL150 / RF	TRIM MATERIAL	316SS (API # 10)	STRESS RELIEF	NOT REQUIRED	
CORROSION ALLOWANCE	0.500mm(0.02")	CODE	ASME B31.3	BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
-29	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11
38	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91
93	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66
149	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24
204	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES
 RTFE SEAT RATINGS (bar)

TEMP(C)	≤NPS 2	>NPS2 ; ≤NPS 6	REV
-29 to 38	18.96	18.96	
93	16.20	16.20	
149	14.82	14.82	
204	5.51	3.44	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE					333	
	1/2" –1.1/2"	40S	BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M		
	2" –12"	10S	BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M		
	14" –24"	10S	BE	SS, ASTM A312 GRADE TP 304/304L, EFW, ASME B36.19M Ej=1.0	548	
NIPPLES	1/2" – 1.1/2"	80S	TOE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M	333	
SWAGES CON & ECC	1/2" –1.1/2"	80S	TSE	SS, ASTM A403 GRADE WP304/304L-S, SMLS, MSS-SP-95	1,333	
	1/2" –1.1/2"	40S	BBE	SS, ASTM A403 GRADE WP304/304L-S, SMLS, MSS-SP-95		
CAP	1/2" –1.1/2" 2" –12" 14" –24"	AS PIPE AS PIPE AS PIPE	WELD WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9	330,333	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
90° BENDS	1/2"–1.1/2" 2"–12" 14"–24"	AS PIPE AS PIPE AS PIPE	WELD WELD WELD	SS, ASTM A312 GRADE TP304/304L-S,SMLS,ASME B36.19M,10X DIAMETER SS, ASTM A312 GRADE TP304/304L-S,SMLS,ASME B36.19M, 10X DIAMETER SS, ASTM A312 GRADE TP304/304L, EFW,ASME B36.19M, 10X DIAMETER		330,333	
45° BENDS	1/2"–1.1/2" 2"–12" 14"–24"	AS PIPE AS PIPE AS PIPE	WELD WELD WELD	SS, ASTM A312 GRADE TP304/304L-S,SMLS,ASME B36.19M,10X DIAMETER SS, ASTM A312 GRADE TP304/304L-S,SMLS,ASME B36.19M, 10X DIAMETER SS, ASTM A312 GRADE TP304/304L, EFW, ASME B36.19M, 10X DIAMETER		330,333	
SLEEVE	1/2"–1.1/2" 2"–12" 14"–24"		WELD WELD WELD	(SPECIAL DESIGN)SS, ASTM A312 GRADE TP304/304L-S,SMLS, ASME B16.19M, (SEE PIPING STANDARD DRWG Figure 2) (SPECIAL DESIGN)SS, ASTM A312 GRADE TP304/304L-S,SMLS, ASME B16.19M, (SEE PIPING STANDARD DRWG Figure 2) (SPECIAL DESIGN)SS, ASTM A312 GRADE TP304/304L, EFW, ASME B16.19M, (SEE PIPING STANDARD DRWG Figure 2)		330,333	
STUB END	1/2"–1.1/2" 2"–24"	AS PIPE AS PIPE	WELD WELD	(SPECIAL DESIGN)SS, ASTM A403 GRADE WP304/304L-S, TYPE A, MSS-SP-43(SEE PIPING STANDARD DRWG Figure 1) (SPECIAL DESIGN)SS, ASTM A403 GRADE WP304/304L-W,TYPE A, MSS-SP-43(SEE PIPING STANDARD DRWG Figure 1)		330,333	
TEE 45° LATERAL LATERAL(RED)	1/2"–12" 14"–24" 1/2"-12" 14"-24"	AS PIPE AS PIPE AS PIPE AS PIPE	WELD WELD WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9, MFR STD SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9, MFR STD SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9, MFR STD SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9, MFR STD		330,333	
REDUCER CON ECC	3/4"-12" 14"-24" 3/4"-12" 14"-24"	AS PIPE AS PIPE AS PIPE AS PIPE	WELD WELD WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		330,333	
FLANGE LAP JOINT SLIP-ON BLIND	1/2"–1.1/2" 2"–24" 1/2"–1.1/2" 2"–24" 1/2"–1.1/2" 2–16" 1/2"–1.1/2" 2"–16" 1/2"–24" 1/2"–24"	CLASS 150 CLASS 150 CLASS 300 CLASS 300 CLASS 150 CLASS 150 CLASS 300 CLASS 300 CLASS 150 CLASS 300	RF RF RF RF RF RF RF RF	(SPECIAL DESIGN)CS,ASTM A105, ASME B16.5(SEE PIPING STANDARD DRWG Figure 1) (SPECIAL DESIGN)CS,ASTM A105, ASME B16.5(SEE PIPING STANDARD DRWG Figure 1) (SPECIAL DESIGN)CS,ASTM A105, ASME B16.5(SEE PIPING STANDARD DRWG Figure 1) (SPECIAL DESIGN)CS,ASTM A105, ASME B16.5(SEE PIPING STANDARD DRWG Figure 1) SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5		334,279 374 374	
VALVES SLIDE	3"–14"	CL150	RF	G-ALZN 10 SI 8 MG BODY,PLATE MATERIAL A240 GR 316	TAG PVVKB62R9B2D BCAU0ZZ	332,277	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
SLIDE	16" – 16"	CL150	RF	G-ALZN 10 SI 8 MG BODY,PLATE MATERIAL A240 GR 316, GEAR OP.	PVVKB62R9B2D BCAU5ZZ	
SWING CHECK	½ – 1.1/2"	CL150	RF	SS BODY WITH 316 SS TRIM ,RENEWABLE DISC	PVVCS62R3810 NCAD0ZZ	
BALL	1/2" – 4"	CL150	RF	SS BODY WITH 316 SS BALL & STEM, RPFE SEATS, FULL PORT,SHORT PATTERN.	PVVVK62R3871 ECAF0ZZ	27,326
BALL	6"	CL150	RF	SS BODY WITH 316 SS BALL & STEM, RPFE SEATS, FULL PORT,SHORT PATTERN,GEAR OP.	PVVVK62R5471 ECAF3ZZ	27,326
BOLTING						
STUD	ALL	CLASS 150		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A194 GRADE 2H HEAVY HEX NUTS,ASME B18.2.1/ASME B18.2.2		
STUD	ALL	CLASS 300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A194 GRADE 2H HEAVY HEX NUTS,ASME B18.2.1/ASME B18.2.2		
GASKETS						
	1/2" –24"	CLASS 150	RF	COMPRESSED NON ASBESTOS(Garlock blue gard style 3000 or equal),1.6mm (1/16") THK, ASME B16.21 / ASME B16.5		
	1/2" – 24"	CLASS 150	RF	(SPECIAL DESIGN)SILICON-RUBBER 2.0mm THK, ASME B16.21 / ASME B16.5 (SEE PIPING STANDARD DRWG Figure 1)		
	1/2" – 24"	CLASS 300	RF	COMPRESSED NON ASBESTOS(ARAMIND FIBER/NBR RUBBER),1.6mm (1/16") THK, ASME B16.21 / ASME B16.5		

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES BUTTWELD BUTTWELD	SIZES 1/2" –1.1/2" 2" –24" 1/2" –1.1/2" 2" –24"

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed DN40(1.1/2" inch size)and large end does not exceedDN100(4" inch size).	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections(when applicable),gage glass piping, vents and drain requiring plugs, ,connections to tube fittings and for connections to threaded proprietary items. See referenced Modules in Piping Standard Drawings.	
326	This resilient seated valve does not in all cases meet full flange rating. See valve MAWP in this specification.	
330	The installation length of elbows must be 1500mm including the cylindrical part for PP Powder/Pellets and any Conveyings Service.	
331	Instead of orifice flanges ,use line class flanges with radius taps in pipe.	
332	Valves shall be selected as indicated in the P&ID(For PP Powder and any Conveyings Service,ball valves are used up to DN150(NPS 6) and the use of check valves is not forseen).	
333	VENDOR TO CONFIRM THE FOLLOWING :For Powder Conveying—Required roughness inside pipe and elbows shall be Ra 0.5-0.8um.	
334	radius of bends shall be 10X diameter.	
374	Flange connection may be replaced by not disconnectable sleeve construction via sketch.	
548	Flanges shall not be used at valve and equipment connections.	
	ASME A312 EFW pipe shall be examined by spot radiography.The requirement of spot radiograph shall be specified in the requisition" for all 150# and 300# A312 EFW piping classes ending with suffix "L" and piping classes A24P,A24P and A24P1 which are used only for purity purposes.	

BRANCH CONNECTION: (TO BE PROVIDED DURING DETIAL DESIGN)

Branch like O-lets, Stub-ins and other 90 degree branches not allowed because of sensitive service. In case of unavoidable need ,Branches have to be carry out with respect to requirement of special service.

Flange Connection with Slip-On Flanges for PP Pellets Conveying Service

Figure 1: Example of flange connection for conveying pipe - Slip on flange (special design)

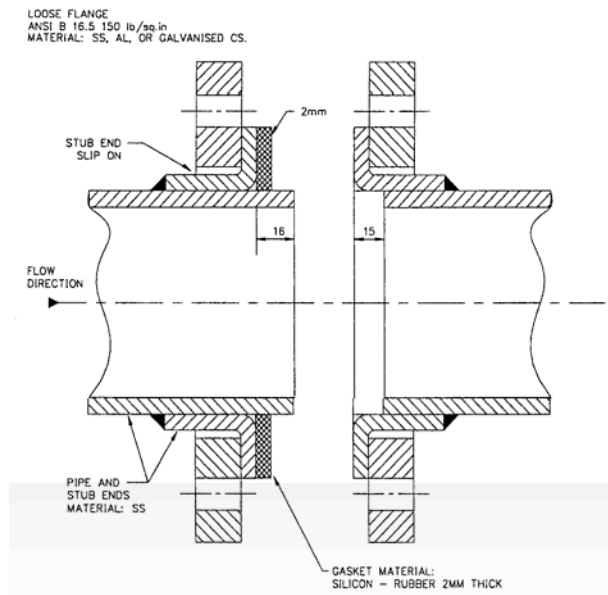
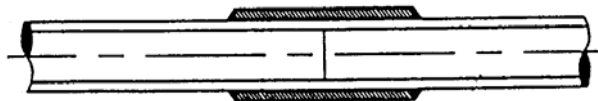


Figure 2: Example of flange connection for conveying pipe – sleeve construction



PROJECT NUMBER	01000
CLIENT	01000-MA-SPC-0002
LOCATION	

PIPING CLASS	REV
A24P2	R00

		REV
SERVICES	Low Pressure Nitrogen—4.0 barg	
	PP Powder Conveying (with and w/o Solids)	

						REV
PIPING MATERIAL	304/304L SS DUAL GRADE	VALVE MATERIAL	316 SS STAINLESS STEEL	WELD EXAMINATION		
RATING / FACING	ASME CL150 / RF	TRIM MATERIAL	316SS (API # 10)	STRESS RELIEF	NOT REQUIRED	
CORROSION ALLOWANCE	0.50 mm(0.02")	CODE	ASME B31.3	BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*

REV

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
-38	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11
38	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91
93	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66
149	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24
204	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22

BALL VALVES
 RTFE SEAT RATINGS (Mpa)

TEMP(C)	≤NPS 2	>NPS2 ; ≤NPS 6	REV
-38 to 38	18.96	18.96	
93	16.20	16.20	
149	14.82	14.82	
204	5.51	3.44	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" –1.1/2"	40S	BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M	333	
	2" –12"	10S	BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M		
	14" –24"	10S	BE	SS, ASTM A312 GRADE TP 304/304L, EFW, ASME B36.19M Ej=1.0	548	
NIPPLES	1/2" – 1.1/2"	80S	TOE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M	333	
SWAGES CON & ECC	1/2" –1.1/2"	80S	TSE	SS, ASTM A403 GRADE WP304/304L-S, SMLS, MSS-SP-95	1,333	
	1/2" –1.1/2"	40S	BBE	SS, ASTM A403 GRADE WP304/304L-S, SMLS, MSS-SP-95		
CAP	1/2" –12" 14" –24"	AS PIPE AS PIPE	WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		
90° BENDS	1/2" –1.1/2" 2" –12" 14" –24"	AS PIPE AS PIPE AS PIPE	WELD WELD WELD	SS, ASTM A312 GRADE TP304/304L-S,SMLS,ASME B36.19M,10X DIAMETER SS, ASTM A312 GRADE TP304/304L-S,SMLS,ASME B36.19M, 10X DIAMETER SS, ASTM A312 GRADE TP304/304L,EFW,ASME B36.19M, 10X DIAMETER	330,333	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
45° BENDS	1/2"–1.1/2" 2"–12" 14"–24"	AS PIPE AS PIPE AS PIPE	WELD WELD WELD	SS, ASTM A312 GRADE TP304/304L-S,SMLS,ASME B36.19M,10X DIAMETER SS, ASTM A312 GRADE TP304/304L-S,SMLS,ASME B36.19M, 10X DIAMETER SS, ASTM A312 GRADE TP304/304L,EFW,ASME B36.19M, 10X DIAMETER		330,333	
SLEEVE	1/2"–1.1/2" 2"–12" 14"–24"		WELD WELD WELD	(SPECIAL DESIGN)SS, ASTM A312 GRADE TP304/304L-S,SMLS, ASME B16.19M, (SEE PIPING STANDARD DRWG Figure 2) (SPECIAL DESIGN)SS, ASTM A312 GRADE TP304/304L-S,SMLS, ASME B16.19M, (SEE PIPING STANDARD DRWG Figure 2) (SPECIAL DESIGN)SS, ASTM A312 GRADE TP304/304L ,EFW, ASME B16.19M, (SEE PIPING STANDARD DRWG Figure 2)		330,333	
STUB END	1/2"– 12" 14" – 24"	AS PIPE AS PIPE	WELD WELD	(SPECIAL DESIGN)SS, ASTM A403 GRADE WP304/304L-S, TYPE A, MSS-SP-43(SEE PIPING STANDARD DRWG Figure 1) (SPECIAL DESIGN)SS, ASTM A403 GRADE WP304/304L-W, TYPE A,MSS-SP-43(SEE PIPING STANDARD DRWG Figure 1)		330,333	
TEE 45° LATERAL LATERAL(RED)	1/2"–12" 14" – 24" 1/2"-12" 14"- 24"	AS PIPE AS PIPE AS PIPE AS PIPE	WELD WELD WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9, MFR STD SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9, MFR STD SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9, MFR STD SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9, MFR STD		330,333	
REDUCER CON	3/4"-12" 14"- 24"	AS PIPE AS PIPE	WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		330,333	
ECC	3/4"-12" 14"- 24"	AS PIPE AS PIPE	WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9			
FLANGE LAP JOINT SLIP-ON BLIND ORIFICE	1/2"–1.1/2" 2" – 24" 1/2"–1.1/2" 2" – 24" 1/2" – 1.1/2" 2 – 16" 1/2" – 1.1/2" 2"–16" 1/2" –24" 1/2" –24" 1" – 24"	CL150 CL150 CL300 CL300 CL150 CL150 CL300 CL300 CL150 CL300 CL300	RF RF RF RF RF RF RF RF RF RF RF	(SPECIAL DESIGN)CS,ASTM A105, ASME B16.5(SEE PIPING STANDARD DRWG Figure 1) (SPECIAL DESIGN)CS,ASTM A105, ASME B16.5(SEE PIPING STANDARD DRWG Figure 1) (SPECIAL DESIGN)CS,ASTM A105, ASME B16.5(SEE PIPING STANDARD DRWG Figure 1) (SPECIAL DESIGN)CS,ASTM A105, ASME B16.5(SEE PIPING STANDARD DRWG Figure 1) SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5		334,279 374 374 5,331	
VALVES SLIDE SLIDE SWINGCHECK BALL	3" – 14" 16" – 16" 1/2" – 1.1/2" 1/2" – 4"	CL150 CL150 CL150 CL150	RF RF RF RF	G-ALZN 10 SI 8 MG BODY,PLATE MATERIAL A240 GR 316 G-ALZN 10 SI 8 MG BODY,PLATE MATERIAL A240 GR 316, GEAR OP. SS BODY WITH 316 SS TRIM ,RENEWABLE DISC SS BODY WITH 316 SS BALL & STEM, RPFE SEATS, FULL PORT,SHORT PATTERN.		332,277 27,326	
				TAG	PVVKB62R9B2D BCAU0ZZ PVVKB62R9B2D BCAU5ZZ PVVCS62R3810 NCAD0ZZ PVVBK62R3871 ECAFOZZ		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
BALL	6"	CL150	RF	SS BODY WITH 316 SS BALL & STEM, RPFE SEATS, FULL PORT, SHORT PATTERN, GEAR OP.	PVV BK62R5471 ECAF3ZZ	27,326	
BOLTING							
STUD	ALL	CL150		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A194 GRADE 2H HEAVY HEX NUTS, ASME B18.2.1/ASME B18.2.2			
STUD	ALL	CL300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A194 GRADE 2H HEAVY HEX NUTS, ASME B18.2.1/ASME B18.2.2			
GASKETS							
	1/2" – 24"	CL150	RF	COMPRESSED NON ASBESTOS (Garlock blue gard style 3000 or equal), 1.6mm (1/16") THK, ASME B16.21 / ASME B16.5			
	1/2" – 24"	CL150	RF	(SPECIAL DESIGN) SILICON-RUBBER 2.0mm THK, ASME B16.21 / ASME B16.5 (SEE PIPING STANDARD DRWG)			
	1/2" – 24"	CL300	RF	COMPRESSED NON ASBESTOS (ARAMIND FIBER/NBR RUBBER), 1.6mm (1/16") THK, ASME B16.21 / ASME B16.5			

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES BUTTWELD BUTTWELD	SIZES 1/2" – 1.1/2" 2" – 24" 1/2" – 1.1/2" 2" – 24"

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
5	See Piping Standard Drawings/Module sheets for instrument applications and dimensional data.	
27	TFE, PTFE seated valves are limited to maximum 204°C.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drain requiring plugs, connections to tube fittings and for connections to threaded proprietary items. See referenced Modules in Piping Standard Drawings.	
326	This resilient seated valve does not in all cases meet full flange rating. See valve MAWP in this specification.	
330	The installation length of elbows must be 1500mm including the cylindrical part for PP Powder/Pellets and any Conveyings Service.	
331	Instead of orifice flanges, use line class flanges with radius taps in pipe.	
332	Valves shall be selected as indicated in the P&ID (For PP Powder and any Conveyings Service, ball valves are used up to DN150 (NPS 6) and the use of check valves is not forseen).	
333	VENDOR TO CONFIRM THE FOLLOWING: For Powder Conveying—Required roughness inside pipe and elbows shall be Ra 40 – 70 µm (for SS). radius of bends shall be 10X diameter.	
334	Flange connection may be replaced by not disconnectable sleeve construction via sketch.	
374	Flanges shall not be used at valve and equipment connections.	
548	ASME A312 EFW pipe shall be examined by spot radiography. The requirement of spot radiograph shall be specified in the requisition for all 150# and 300# A312 EFW piping classes ending with suffix "L" and piping classes A24P, A24P and A24P1 which are used only for purity purposes.	

BRANCH CONNECTION: (TO BE PROVIDED DURING DETIAL DESIGN)

Branch like O-lets, Stub-ins and other 90 degree branches not allowed because of sensitive service. In case of unavoidable need ,Branches have to be carry out with respect to requirement of special service.

Flange Connection with Slip-On Flanges for PP Pellets Conveying Sevice

Figure 1: Example of flange connection for conveying pipe - Slip on flange (special design)

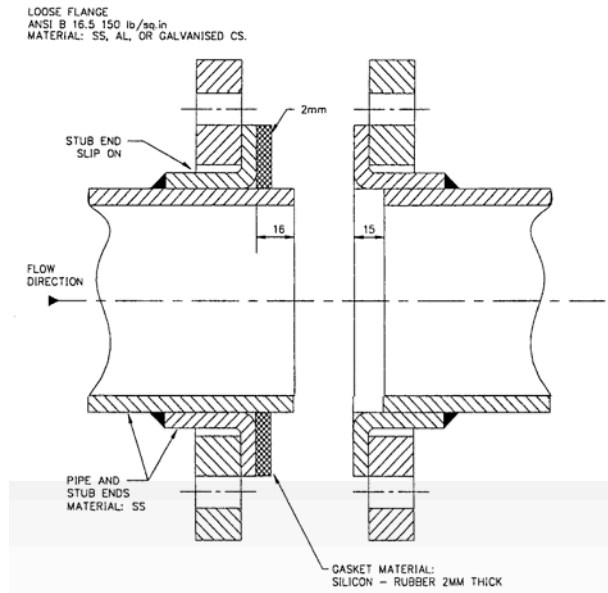
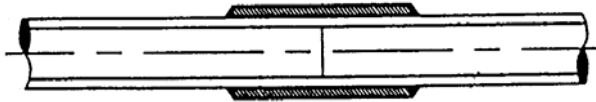


Figure 2: Example of flange connection for conveying pipe – sleeve construction



PROJECT NUMBER	01000
CLIENT	01000-MA-SPC-0002
LOCATION	

PIPING CLASS	REV
A24P3	R00

						REV
SERVICES	Low Pressure Nitrogen—4.0 barg					

						REV
PIPING MATERIAL	304/304L SS DUAL GRADE	VALVE MATERIAL	316 SS STAINLESS STEEL	WELD EXAMINATION		
RATING / FACING	ASME CL150 / RF	TRIM MATERIAL	316SS (API # 10)	STRESS RELIEF	NOT REQUIRED	
CORROSION ALLOWANCE	0.500mm(0.02")	CODE	ASME B31.3	BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*

REV

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
-29	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11	16.11
38	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91	15.91
93	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66	13.66
149	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24	12.24
204	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22	11.22

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES
RTFE SEAT RATINGS (bar)

TEMP(C)	≤NPS 2	>NPS2 ; ≤NPS 6	REV
-29 to 38	18.96	18.96	
93	16.20	16.20	
149	14.82	14.82	
204	5.51	3.44	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" –1.1/2"	40S	BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M	333	
	2" –12"	10S	BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M	548	
	14" –24"	10S	BE	SS, ASTM A312 GRADE TP 304/304L, EFW, ASME B36.19M Ej=1.0		
NIPPLES	1/2" – 1.1/2"	80S	TOE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M	333	
SWAGES CON & ECC	1/2" –1.1/2"	80S	TSE	SS, ASTM A403 GRADE WP304/304L-S, SMLS, MSS-SP-95	1,333	
	1/2" –1.1/2"	40S	BBE	SS, ASTM A403 GRADE WP304/304L-S, SMLS, MSS-SP-95		
CAP	1/2" –1.1/2" 2" –12" 14" –24"	AS PIPE AS PIPE AS PIPE	WELD WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9	330,333	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
90° BENDS	1/2"–1.1/2" 2"–12" 14"–24"	AS PIPE AS PIPE AS PIPE	WELD WELD WELD	SS, ASTM A312 GRADE TP304/304L-S,SMLS,ASME B36.19M,10X DIAMETER SS, ASTM A312 GRADE TP304/304L-S, SMLS,ASME B36.19M, 10X DIAMETER SS, ASTM A312 GRADE TP304/304L, EFW,ASME B36.19M, 10X DIAMETER		330,333	
45° BENDS	1/2"–1.1/2" 2"–12" 14"–24"	AS PIPE AS PIPE AS PIPE	WELD WELD WELD	SS, ASTM A312 GRADE TP304/304L-S,SMLS,ASME B36.19M,10X DIAMETER SS, ASTM A312 GRADE TP304/304L-S,SMLS,ASME B36.19M, 10X DIAMETER SS, ASTM A312 GRADE TP304/304L, EFW, ASME B36.19M, 10X DIAMETER		330,333	
SLEEVE	1/2"–1.1/2" 2"–12" 14"–24"		WELD WELD WELD	(SPECIAL DESIGN)SS, ASTM A312 GRADE TP304/304L-S,SMLS, ASME B16.19M, (SEE PIPING STANDARD DRWG Figure 2) (SPECIAL DESIGN)SS, ASTM A312 GRADE TP304/304L-S,SMLS, ASME B16.19M, (SEE PIPING STANDARD DRWG Figure 2) (SPECIAL DESIGN)SS, ASTM A312 GRADE TP304/304L, EFW, ASME B16.19M, (SEE PIPING STANDARD DRWG Figure 2)		330,333	
STUB END	1/2"–1.1/2" 2"–24"	AS PIPE AS PIPE	WELD WELD	(SPECIAL DESIGN)SS, ASTM A403 GRADE WP304/304L-S, TYPE A, MSS-SP-43(SEE PIPING STANDARD DRWG Figure 1) (SPECIAL DESIGN)SS, ASTM A403 GRADE WP304/304L-W,TYPE A, MSS-SP-43(SEE PIPING STANDARD DRWG Figure 1)		330,333	
TEE 45° LATERAL LATERAL(RED)	1/2"–12" 14"–24" 1/2"-12" 14"-24"	AS PIPE AS PIPE AS PIPE AS PIPE	WELD WELD WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9, MFR STD SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9, MFR STD SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9, MFR STD SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9, MFR STD		330,333	
REDUCER CON ECC	3/4"-12" 14"-24" 3/4"-12" 14"-24"	AS PIPE AS PIPE AS PIPE AS PIPE	WELD WELD WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-W, ASME B16.9		330,333	
FLANGE LAP JOINT SLIP-ON BLIND	1/2"–1.1/2" 2"–24" 1/2"–1.1/2" 2"–24" 1/2"–1.1/2" 2–16" 1/2"–1.1/2" 2"–16" 1/2"–24" 1/2"–24"	CLASS 150 CLASS 150 CLASS 300 CLASS 300 CLASS 150 CLASS 150 CLASS 300 CLASS 300 CLASS 150 CLASS 300	RF RF RF RF RF RF RF RF	(SPECIAL DESIGN)CS,ASTM A105, ASME B16.5(SEE PIPING STANDARD DRWG Figure 1) (SPECIAL DESIGN)CS,ASTM A105, ASME B16.5(SEE PIPING STANDARD DRWG Figure 1) (SPECIAL DESIGN)CS,ASTM A105, ASME B16.5(SEE PIPING STANDARD DRWG Figure 1) (SPECIAL DESIGN)CS,ASTM A105, ASME B16.5(SEE PIPING STANDARD DRWG Figure 1) SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5		334,279 374 374	
VALVES SLIDE	3"–14"	CL150	RF	G-ALZN 10 SI 8 MG BODY,PLATE MATERIAL A240 GR 316	TAG PVVKB62R9B2D BCAU0ZZ	332,277	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
SLIDE	16" – 16"	CL150	RF	G-ALZN 10 SI 8 MG BODY,PLATE MATERIAL A240 GR 316, GEAR OP.	PVVKB62R9B2D BCAU5ZZ	
SWING CHECK	½ – 1.1/2"	CL150	RF	SS BODY WITH 316 SS TRIM ,RENEWABLE DISC	PVVCS62R3810 NCAD0ZZ	
BALL	1/2" – 4"	CL150	RF	SS BODY WITH 316 SS BALL & STEM, RPFE SEATS, FULL PORT,SHORT PATTERN.	PVVVK62R3871 ECAF0ZZ	27,326
BALL	6"	CL150	RF	SS BODY WITH 316 SS BALL & STEM, RPFE SEATS, FULL PORT,SHORT PATTERN,GEAR OP.	PVVVK62R5471 ECAF3ZZ	27,326
BOLTING						
STUD	ALL	CLASS 150		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A194 GRADE 2H HEAVY HEX NUTS,ASME B18.2.1/ASME B18.2.2		
STUD	ALL	CLASS 300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A194 GRADE 2H HEAVY HEX NUTS,ASME B18.2.1/ASME B18.2.2		
GASKETS						
	1/2" –24"	CLASS 150	RF	COMPRESSED NON ASBESTOS(Garlock blue gard style 3000 or equal),1.6mm (1/16") THK, ASME B16.21 / ASME B16.5		
	1/2" – 24"	CLASS 150	RF	(SPECIAL DESIGN)SILICON-RUBBER 2.0mm THK, ASME B16.21 / ASME B16.5 (SEE PIPING STANDARD DRWG Figure 1)		
	1/2" – 24"	CLASS 300	RF	COMPRESSED NON ASBESTOS(ARAMIND FIBER/NBR RUBBER),1.6mm (1/16") THK, ASME B16.21 / ASME B16.5		

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES BUTTWELD BUTTWELD	SIZES 1/2" –1.1/2" 2" –24" 1/2" –1.1/2" 2" –24"

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed DN40(1.1/2" inch size)and large end does not exceedDN100(4" inch size).	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections(when applicable),gage glass piping, vents and drain requiring plugs, ,connections to tube fittings and for connections to threaded proprietary items. See referenced Modules in Piping Standard Drawings.	
326	This resilient seated valve does not in all cases meet full flange rating. See valve MAWP in this specification.	
330	The installation length of elbows must be 1500mm including the cylindrical part for PP Powder/Pellets and any Conveyings Service.	
331	Instead of orifice flanges ,use line class flanges with radius taps in pipe.	
332	Valves shall be selected as indicated in the P&ID(For PP Powder and any Conveyings Service,ball valves are used up to DN150(NPS 6) and the use of check valves is not forseen).	
333	VENDOR TO CONFIRM THE FOLLOWING :For Powder Conveying—Required roughness inside pipe and elbows shall be Ra 0.5-0.8um.	
334	radius of bends shall be 10X diameter.	
374	Flange connection may be replaced by not disconnectable sleeve construction via sketch.	
548	Flanges shall not be used at valve and equipment connections.	
	ASME A312 EFW pipe shall be examined by spot radiography.The requirement of spot radiograph shall be specified in the requisition" for all 150# and 300# A312 EFW piping classes ending with suffix "L" and piping classes A24P,A24P and A24P1 which are used only for purity purposes.	

BRANCH CONNECTION: (TO BE PROVIDED DURING DETIAL DESIGN)

Branch like O-lets, Stub-ins and other 90 degree branches not allowed because of sensitive service. In case of unavoidable need ,Branches have to be carry out with respect to requirement of special service.

Flange Connection with Slip-On Flanges for PP Pellets Conveying Sevice

Figure 1: Example of flange connection for conveying pipe - Slip on flange (special design)

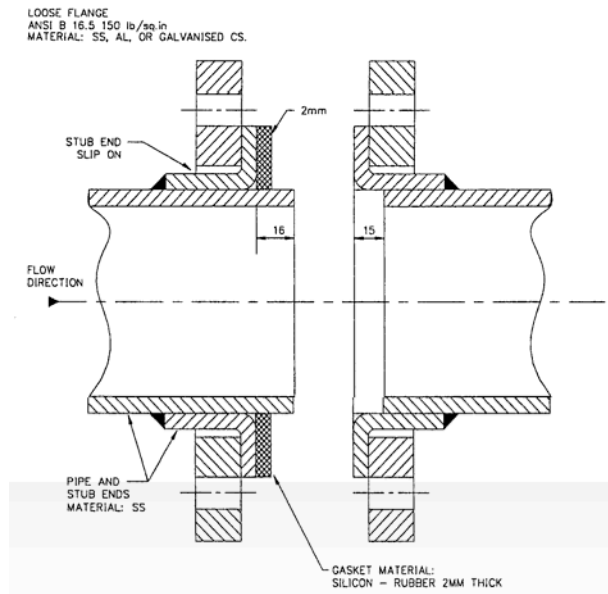
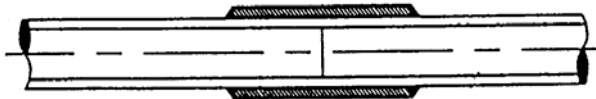


Figure 2: Example of flange connection for conveying pipe – sleeve construction



PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
A01S	R00

SERVICES	Sour water	Ro reject water	Acid liquid	1%~3%H ₂ SO ₄ Acid liquid	REV

PIPING MATERIAL	CS LINED (PTFE)	VALVE MATERIAL	CS LINED (FEP)	WELD EXAMINATION	HOLD	REV
RATING / FACING	ASME CLASS 150 / RF	TRIM MATERIAL	CS LINED (FEP)	STRESS RELIEF	NOT REQUIRED	
CORROSION ALLOWANCE	0.0 mm (0.0")	CODE	ASME B31.3	BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (MPa)*

REV

SIZE

TEMP P °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
-28 to 38																
93																

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES
RTFE SEAT RATINGS (Mpa)

TEMP(C)	< NPS 2	REV
-28 to 38		
93		
149		
204		

BUTTERFLY
RTFE SEAT RATINGS(Mpa)

TEMP(C)	ALL SIZES	REV
-28 to 38		
93		
149		
204		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1"	80	RF/CL 150	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 3.50mm) , ASME B16.5	175,176	
	1 1/2" – 4"	STD	RF/CL 150	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 4.00mm) , ASME B16.5		
	6" – 24"	STD	RF/CL 150	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 6.50mm) , ASME B16.5		
90° ELBOW	1"	AS PIPE	RF/CL 150	CS, ASTM A234 GRADE WPB, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 3.50mm), ASME B16.5	175,176	
	1 1/2" – 4"	AS PIPE	RF/CL 150	CS, ASTM A234 GRADE WPB, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 4.00mm), ASME B16.5		
	6" – 24"	AS PIPE	RF/CL 150	CS, ASTM A234 GRADE WPB, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 6.50mm), ASME B16.5		
45° ELBOW	1"	AS PIPE	RF/CL 150	CS, ASTM A234 GRADE WPB, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 3.50mm), ASME B16.5	175,176	
	1 1/2" – 4"	AS PIPE	RF/CL 150	CS, ASTM A234 GRADE WPB, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 4.00mm), ASME B16.5		
	6" – 24"	AS PIPE	RF/CL 150	CS, ASTM A234 GRADE WPB, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 6.50mm), ASME B16.5		
TEE	1"	AS PIPE	RF/CL 150	CS, ASTM A234 GRADE WPB, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 3.50mm), ASME B16.5	175,176	
	1 1/2" – 4"	AS PIPE	RF/CL 150	CS, ASTM A234 GRADE WPB, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 4.00mm), ASME B16.5		
	6" – 24"	AS PIPE	RF/CL 150	CS, ASTM A234 GRADE WPB, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 6.50mm), ASME B16.5		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
RED TEE	1" 1 1/2" – 4" 6" – 24"	AS PIPE AS PIPE AS PIPE	RF/CL 150 RF/CL 150 RF/CL 150	CS, ASTM A234 GRADE WPB, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 3.50mm), ASME B16.5 CS, ASTM A234 GRADE WPB, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 4.00mm), ASME B16.5 CS, ASTM A234 GRADE WPB, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 6.50mm), ASME B16.5			
REDUCER CON ECC	1" 1 1/2" – 4" 6" – 24" 1" 1 1/2" – 4" 6" – 24"	AS PIPE AS PIPE AS PIPE AS PIPE AS PIPE AS PIPE	RF/CL 150 RF/CL 150 RF/CL 150 RF/CL 150 RF/CL 150 RF/CL 150	CS, ASTM A234 GRADE WPB, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 3.50mm), ASME B16.5 CS, ASTM A234 GRADE WPB, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 4.00mm), ASME B16.5 CS, ASTM A234 GRADE WPB, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 6.50mm), ASME B16.5 CS, ASTM A234 GRADE WPB, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 3.50mm), ASME B16.5 CS, ASTM A234 GRADE WPB, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 4.00mm), ASME B16.5 CS, ASTM A234 GRADE WPB, PTFE LINED PER ASTM F1545 (LINING THICKNESS MIN. 6.50mm), ASME B16.5		175,176	
FLANGE BLIND FLG	1" – 24"	CL150	RF	CS, ASTM A105, BONDED PTFE, ASME B16.5		35,174,175	
MISCELLANEOUS FIGURE 8 BLANK							
VALVES GATE	1" – 1 1/2" 2" – 12" 14" – 24"	CL150 CL150 CL150	RF RF RF	A216 Gr.WCB BODY LINED WITH FEP (MOLDED RF). A216 Gr.WCB BODY LINED WITH FEP (MOLDED RF). A216 Gr.WCB BODY LINED WITH FEP (MOLDED RF), GEAR OP.	TAG PVVGB62R9097 EKAD0ZZ PVVGB62R9097 EKAC0ZZ PVVGB62R9097 EKAC5ZZ	277 35 35 35	
HORIZ. CHECK	1" – 24"	CL150	RF	A216 Gr.WCB BODY LINED WITH FEP (MOLDED RF).	PVVCL62R9097 NKAL0ZZ	35	
VERT. CHECK	1" – 24"	CL150	RF	A216 Gr.WCB BODY LINED WITH FEP (MOLDED RF).	PVVCM62R9097 NKAL0ZZ	35	
PLUG	1" – 4" 1" – 4"	CL150 CL150	RF RF	A216 Gr.WCB BODY LINED WITH FEP (MOLDED RF). A216 Gr.WCB BODY LINED WITH FEP (MOLDED RF) FURNISHED WITH STEM EXTENSION (VALVE BOX) AND T-HANDLE WRENCH.	PVVPC62R9097 EKAB0ZZ PVVPC62R9097 EKAB0EV	35,38 35,76,81, 181	
BALL	1" – 3"	CL150	RF	A216 Gr.WCB BODY LINED WITH FEP BALL & STEM, (MOLDED RF).	PVVBA62R9097 EKAX0ZZ	35	
WAF BUTTERFLY	2" – 4" 5" – 24"	CL150 CL150	RF RF	A216 Gr.WCB BODY LINED WITH FEP (MOLDED RF). A216 Gr.WCB BODY LINED WITH FEP (MOLDED RF), GEAR OP.	PVVUC62W9097 EKAG0ZZ PVVUC62W9097 EKAG3ZZ	35 35	
LUG BUTTERFLY	2" – 4" 5" – 24"	CL150 CL150	RF RF	A216 Gr.WCB BODY LINED WITH FEP (MOLDED RF). A216 Gr.WCB BODY LINED WITH FEP (MOLDED RF), GEAR OP.	PVVUC62L9097 EKAG0ZZ PVVUC62L9097 EKAG3ZZ	35 35	
DIAPHRAGM	1" – 4"	CL150	RF	A216 Gr.WCB BODY LINED WITH PTFE-EPDM DIAPHRAGM (MOLDED RF).	PVVDW62R9097 EKAM0ZZ	35	
BOLTING	ALL	CL150		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEX NUTS		174	
GASKET	1" – 24"	CL150		3 mm THICK PTFE ENVELOPE GASKET WITH NON-ASBESTOS FILLER, FLAT RING, ASME B16.21 / ASME B16.5 (USED ONLY BETWEEN PTFE RING SPACER AND STEEL FLANGE)			

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES	SIZES 1/2" – 1 1/2" 2" – 24"

NOTE	SPECIFIC PIPING NOTES	REV
35	All flanged joints and valve bonnets require safety shields after erection of the piping system.	
38	Use for first block valve from piping system to pressure instrument, flow, vent and drain connections.	
76	Depth of trench or bury (distance from ground line to bottom of pipe) or cover (distance from ground line to top of pipe) to be as indicated on P&ID or piping drawings.	
81	All underground valves shall be installed in a valve box.	
174	Bolted joints must be assembled and torqued in accordance with the manufacturer's recommendations for the plastic lined piping system.	
175	The manufacturer's recommendations and specifications must be followed during shipping, storing, handling, installation, erection, testing, final painting/insulating and for removal / maintenance requirements.	
176	Field Installation Note for plastic lined piping systems. Shop spools made to the required length with flanged ends are preferred and shall be used whenever possible. Field cutting and molding may be performed only when approved by the project and shall be done in accordance with the manufacturer's techniques, equipment and recommendations. Prior to installation, pipe liners and liner forming the flange gasket sealing surface of pipe and fittings shall be thoroughly examined for pinholes, cracks, gouges, nicks, or inclusions of foreign particles. If any of these items measured greater than 10% of liner thickness or reduce dielectric capacity below 12,500 volts, liner is subject to rejection.	
181	Buried block valves shall be supported with concrete pad designed by Piping Stress Engineering and Civil Engineering Groups.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
A02T	R00

SERVICES					REV
	Sodium hypochlorite pipeline	Flocculation pipeline	Ammonia Phosphate(AP)		
	Corrosion inhibitor pipeline	Reducing agent pipeline			
	Scale inhibitor pipeline				
	Coagulant pipeline				

PIPING MATERIAL	PVDF	VALVE MATERIAL	PVDF	WELD EXAMINATION	HOLD	REV
RATING / FACING	- / -	TRIM MATERIAL	EPDM	STRESS RELIEF	NOT REQUIRED	
CORROSION ALLOWANCE	0.0 mm (0.0")	CODE	PER PROJECT	BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (MPa)*

SIZE

TEMP P °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
38																
93																

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES RTFE SEAT RATINGS (Mpa)

TEMP(C)	< NPS 2	REV
-28 to 38		
93		
149		
204		

BUTTERFLY RTFE SEAT RATINGS(Mpa)

TEMP(C)	ALL SIZES	REV
-28 to 38		
93		
149		
204		

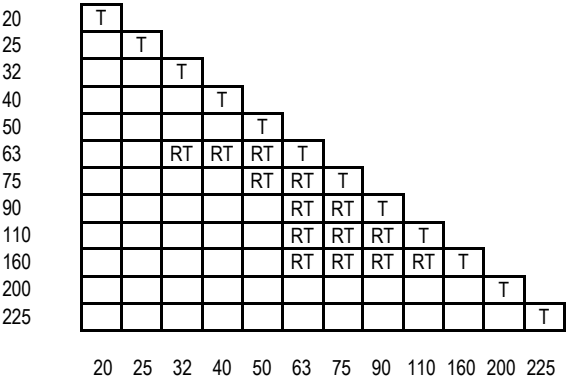
ITEMS	Nominal outside diameter (dn)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	20 – 32 40 – 50 63 – 225	S6.3 S8 S10	PE PE PE	PVDF, ISO 10931, SMLS PVDF, ISO 10931, SMLS PVDF, ISO 10931, SMLS	175,176	
CAP	20 – 110 125 – 225	AS PIPE AS PIPE	SOCW BW	PVDF, ISO 10931, SMLS, THICKNESS AS PIPE PVDF, ISO 10931, SMLS, THICKNESS AS PIPE	175,176	
90° ELBOW	20 – 110 125 – 225	MFR AS PIPE	SOCW BW	PVDF, ISO 10931, SMLS PVDF, ISO 10931, SMLS, THICKNESS AS PIPE	175,176,801	
45° ELBOW	20 – 110 125 – 225	MFR AS PIPE	SOCW BW	PVDF, ISO 10931, SMLS PVDF, ISO 10931, SMLS, THICKNESS AS PIPE	175,176,801	
COUPLING	20 – 110	MFR	SOCW	PVDF, ISO 10931, THICKNESS MIN. AS PIPE	801	
TEE	20 – 110 125 – 225	AS PIPE AS PIPE	SOCW BW	PVDF, ISO 10931, SMLS, THICKNESS AS PIPE PVDF, ISO 10931, SMLS, THICKNESS AS PIPE	175,176	
RED TEE	63 – 160	AS PIPE	BW	PVDF, ISO 10931, SMLS, THICKNESS AS PIPE		
REDUCER CON	25 – 110 160 – 225	AS PIPE AS PIPE	SOCW BW	PVDF, ISO 10931, SMLS, THICKNESS AS PIPE PVDF, ISO 10931, SMLS, THICKNESS AS PIPE	175,176	

ITEMS	Nominal outside diameter (d _n)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
FLANGE						
FLANGE ADAPTOR LOOSE BACKING FLG	20 – 225 20 – 225	MFR MFR	FF FF	PVDF, ISO 10931, THICKNESS AS PIPE PVDF, ISO 10931	35,175,176 801	
VALVES					277,801	
GLOBE	20 – 63	MFR	FF	PVDF BODY	TAG	
BALL	20 – 110	MFR	FF	PVDF BODY WITH PVDF BALL & STEM, FULL PORT		
DIAPHRAGM	20 – 63	MFR	FF	PVDF BODY WITH PTFE-EPDM DIAPHRAGM		
BOLTING	ALL	MFR		MFR	801,802	
GASKET	20 – 225	MFR		MFR	801,802	

	NOTES				REV
JOINT CONNECTS MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES SOCKETWELD BUTTWELD			
			SIZES		
			d _n 20 – 225		
			d _n 20 – 110		
			d _n 20 – 225		

NOTE	SPECIFIC PIPING NOTES	REV
35	All flanged joints and valve bonnets require safety shields after erection of the piping system.	
175	The manufacturer's recommendations and specifications must be followed during shipping, storing, handling, installation, erection, testing, final painting/insulating and for removal / maintenance requirements.	
176	Field Installation Note for plastic piping systems. Shop spools made to the required length with flanged ends are preferred and shall be used whenever possible. Field cutting and molding may be performed only when approved by the project and shall be done in accordance with the manufacturer's techniques, equipment and recommendations.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
801	MFR means by manufacturer.	
802	Vendor shall supply in complete set the Pipes, Fittings, Valves, Flanges, Bolts and Gaskets. That should be noted in Materials List and Order Contract.	

HEADER SIZE



BRANCH SIZE

BRANCH LEGEND		NOTES	REV
T	Tee		
RT	Reducing Tee		

PIPING BRANCH NOTES		REV

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
A03T	R00

				REV
SERVICES	Sodium hypochlorite pipeline	Flocculation pipeline	Coagulant pipeline	
	Corrosion inhibitor pipeline	Reducing agent pipeline	Scale inhibitor pipeline	
	Ammonia Phosphate(AP)			

PIPING MATERIAL	PVC-U	VALVE MATERIAL	PVC-U	WELD EXAMINATION	HOLD	REV
RATING / FACING	- / -	TRIM MATERIAL	EPDM	STRESS RELIEF	NOT REQUIRED	
CORROSION ALLOWANCE	0.0 mm (0.0")	CODE	PER PROJECT	BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (MPa)*

REV

SIZE

TEMP P °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
38																
93																

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES RTFE SEAT RATINGS (Mpa)

TEMP(C)	< NPS 2	REV
-28 to 38		
93		
149		
204		

BUTTERFLY RTFE SEAT RATINGS(Mpa)

TEMP(C)	ALL SIZES	REV
-28 to 38		
93		
149		
204		

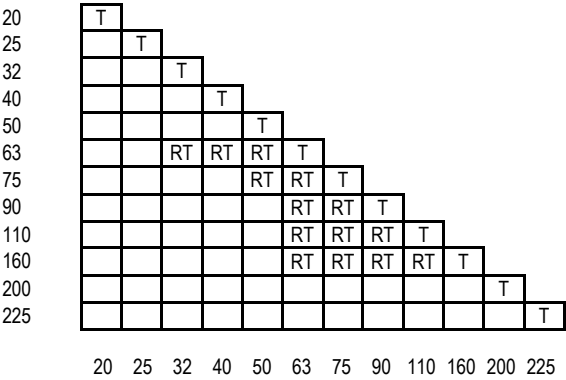
ITEMS	Nominal outside diameter (dn)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	20 – 32 40 – 50 63 – 225	S6.3 S8 S10	PE PE PE	PVC-U, ISO 15493, SMLS PVC-U, ISO 15493, SMLS PVC-U, ISO 15493, SMLS	175,176	
CAP	20 – 110 125 – 225	AS PIPE AS PIPE	SOCW BW	PVC-U, ISO 15493, SMLS, THICKNESS AS PIPE PVC-U, ISO 15493, SMLS, THICKNESS AS PIPE	175,176	
90° ELBOW	20 – 110 125 – 225	MFR AS PIPE	SOCW BW	PVC-U, ISO 15493, SMLS PVC-U, ISO 15493, SMLS, THICKNESS AS PIPE	175,176,801	
45° ELBOW	20 – 110 125 – 225	MFR AS PIPE	SOCW BW	PVC-U, ISO 15493, SMLS PVC-U, ISO 15493, SMLS, THICKNESS AS PIPE	175,176,801	
COUPLING	20 – 110	MFR	SOCW	PVC-U, ISO 15493, THICKNESS MIN. AS PIPE	801	
TEE	20 – 110 125 – 225	AS PIPE AS PIPE	SOCW BW	PVC-U, ISO 15493, SMLS, THICKNESS AS PIPE PVC-U, ISO 15493, SMLS, THICKNESS AS PIPE	175,176	
RED TEE	110 – 160	AS PIPE	BW	PVC-U, ISO 15493, SMLS, THICKNESS AS PIPE		
REDUCER CON	25 – 110 160 – 225	AS PIPE AS PIPE	SOCW BW	PVC-U, ISO 15493, SMLS, THICKNESS AS PIPE PVC-U, ISO 15493, SMLS, THICKNESS AS PIPE	175,176	

ITEMS	Nominal outside diameter (d _n)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
FLANGE FLANGE ADAPTOR LOOSE BACKING FLG	20 – 225 20 – 225	MFR MFR	FF FF	PVC-U, ISO 15493, THICKNESS AS PIPE PVC-U, ISO 15493	35,175,176 801	
VALVES GLOBE BALL DIAPHRAGM	20 – 63 20 – 110 20 – 63	MFR MFR MFR	FF FF FF	PVC-U BODY PVC-U BODY WITH PVC-U BALL & STEM, FULL PORT PVC-U BODY WITH PTFE-EPDM DIAPHRAGM	277,801	
BOLTING	ALL	MFR		MFR	801,802	
GASKET	20 – 225	MFR		MFR	801,802	

	NOTES		REV
JOINT CONNECTS MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES SOCKETWELD BUTTWELD	SIZES d _n 20 – 225 d _n 20 – 110 d _n 20 – 225

NOTE	SPECIFIC PIPING NOTES	REV
35	All flanged joints and valve bonnets require safety shields after erection of the piping system.	
175	The manufacturer's recommendations and specifications must be followed during shipping, storing, handling, installation, erection, testing, final painting/insulating and for removal / maintenance requirements.	
176	Field Installation Note for plastic piping systems. Shop spools made to the required length with flanged ends are preferred and shall be used whenever possible. Field cutting and molding may be performed only when approved by the project and shall be done in accordance with the manufacturer's techniques, equipment and recommendations.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
801	MFR means by manufacturer.	
802	Vendor shall supply in complete set the Pipes, Fittings, Valves, Flanges, Bolts and Gaskets. That should be noted in Materials List and Order Contract.	

HEADER SIZE



BRANCH SIZE

BRANCH LEGEND		NOTES	REV
T	Tee		
RT	Reducing Tee		

PIPING BRANCH NOTES		REV

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
B1A	R00

					REV
SERVICES	MP Condensate	Natural Gas	MP Steam		

PIPING MATERIAL	CARBON STEEL	VALVE MATERIAL	CARBON STEEL	WELD EXAMINATION		REV
RATING / FACING	ASME CLASS 300 / RF	TRIM MATERIAL	13 CR/HF (API #8)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.5mm	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)*

SIZE																
TEMP. °C	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20	24
-28 to 38	51.02	51.02	51.02	51.02	51.02	51.02	51.02	51.02	51.02	51.02	51.02	51.02	44.09	39.12	35.18	29.21
93	46.88	46.88	46.88	46.88	46.88	46.88	46.88	46.88	46.88	46.88	46.88	46.88	44.09	39.12	35.18	29.21
149	45.16	45.16	45.16	45.16	45.16	45.16	45.16	45.16	45.16	45.16	45.16	45.16	44.09	39.12	35.18	29.21
204	43.78	43.78	43.78	43.78	43.78	43.78	43.78	43.78	43.78	43.78	43.78	43.78	43.78	39.12	35.18	29.21
260	41.71	41.71	41.71	41.71	41.71	41.71	41.71	41.71	41.71	41.71	41.71	41.71	41.71	36.96	33.21	27.61
316	39.3	39.3	39.3	39.3	39.3	39.3	39.3	39.3	39.3	39.3	39.3	39.3	38.12	33.85	30.42	25.28
343	37.92	37.92	37.92	37.92	37.92	37.92	37.92	37.92	37.92	37.92	37.92	37.92	37.46	33.28	29.86	24.85
371	36.54	36.54	36.54	36.54	36.54	36.54	36.54	36.54	36.54	36.54	36.54	36.54	36.38	32.31	29.03	24.12

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES

RTFE SEAT RATINGS (Bar)

TEMP(C)	< NPS 2	REV
-28 to 38	51.02	
93	46.88	
149	28.93	
204	5.49	

BUTTERFLY

RTFE SEAT RATINGS(Bar)

TEMP(C)	ALL SIZES	REV
-28 to 38	51.02	
93	46.54	
149	26.87	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	½" - 1½"	80	PE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	2" - 16"	STD	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	18" - 24"	STD	BE	CS, API 5L GRADE B, EFW, ASME B36.10M, Ej=0.95		
NIPPLE	½" - 1½"	160	TOE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	½" - 1½"	160	TBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	½" - 1½"	80	PBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
SWAGE	½" - 1½"	160	TSE	CS, ASTM A234 GRADE WPB, MSS-SP-95	1	
	½" - 1½"	80	BBE	CS, ASTM A234 GRADE WPB, MSS-SP-95		
	½" - 1½"	160	TSE	CS, ASTM A234 GRADE WPB, MSS-SP-95		
	½" - 1½"	80	BBE	CS, ASTM A234 GRADE WPB, MSS-SP-95		
COUPLING	½" - 1½"	CL3000	SOCW	CS, ASTM A105, ASME B16.11		
FULL COUPLING	½" - 2"	CL3000	THRD	CS, ASTM A105, ASME B16.11	279	
OLET	½" - 1½"	CL3000	WELD	CS, ASTM A105, MSS-SP-97		
THREDOLET	½" - 2"	CL3000	WELD	CS, ASTM A105, MSS-SP-97	279	
WELDOLET	2" - 8"	AS PIPE	WELD	CS, ASTM A105, MSS-SP-97		
PLUG	½" - 2"		THRD	CS, ASTM A105, ASME B16.11, ROUND HEAD		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
CAP	½" - 1½" ½" - 2" 2" - 24"	CL3000 CL3000 AS PIPE	SOCW THRD BW	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9		279	
90° ELBOW SW LR	½" - 1½" 2" - 14" 16" - 24"	CL3000 AS PIPE AS PIPE	SOCW BW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9			
45° ELBOW	½" - 1½" 2" - 14" 16" - 24"	CL3000 AS PIPE AS PIPE	SOCW BW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9			
TEE RED TEE	½" - 1½" 2" - 14" 16" - 24" ½" - 1½" 2" - 14" 16" - 24"	CL3000 AS PIPE AS PIPE CL3000 AS PIPE AS PIPE	SOCW BW BW SOCW BW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9			
REDUCER CON ECC	2" - 14" 16" - 24" 2" - 14" 16" - 24"	AS PIPE AS PIPE AS PIPE AS PIPE	BW BW BW BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9			
FLANGE SOCKETWELD WELDNECK BLIND FLG DRIP RING FIGURE 8 BLANK+SPACER	½" - 1½" 2" - 24" ½" - 24" 1" - 12" 1" - 12" 14" - 24"	CL300 CL300 CL300 CL300 CL300 CL300	RF RF RF FF FF FF	CS, ASTM A105, ASME B16.5 CS, ASTM A105, ASME B16.5 THICKNESS AS PIPE CS, ASTM A105, ASME B16.5 CS, ASTM A516 GRADE 70, NORMALIZED, 1-1/2" THICK WITH ¾" SOCW TAPS (STANDARD DRAWINGS) CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48 CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48			
VALVES GATE	½" - 1½" ½" - 1½" ½" - 1½" ½" - 1½" ½" - 2" 2" 3" - 10" 12" - 24"	CL800 CL800 CL800 CL800 CL300 CL300 CL300 CL300	SOCW SOCW T/SW T/SW RF RF RF RF	CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT, GEAR OP.	TAG PVVGB68S0108 ACAD0ZZ PVVGF68S0108 ACAD0ZZ PVVGB68D0108 ACAD0ZZ PVVGF68D0108 ACAD0ZZ PVVGB63R0108 ACAD0ZZ PVVGF63R0308 BCAC0ZZ PVVGF63R0308 BCAC0ZZ PVVGF63R0308 BCAC5ZZ	41, 277 275 4 4, 275 24	
GLOBE	½" - 1½" 2" - 8" 10" - 12"	CL800 CL300 CL300	SOCW RF RF	CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8)	PVVLB68S0108A CAD0ZZ PVVLB63R0308 BCAC0ZZ PVVLB63R0308 BCAC5ZZ		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
LIFT CHECK	½" - 1½"	CL800	SOCW	CS BODY WITH 13 CR TRIM (API # 1)	PVVCL68S0101 NCAD0ZZ		
WAFER DUAL PLATER CHECK	2" - 24"	CL300	RF	CS BODY WITH 13 CR PLATES, METAL SEATS	PVVCD63W0301 NCAA0ZZ		
BALL	½" - 1½"	CL600	BW	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS	PVVB66B0171 ACAF0B2	27,55,43, 326	
	½" - 1½"	CL600	BW	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT	PVVBF66B0171 ACAF0B2	27,55,43, 326	
	½" - 1½"	CL600	T/SW	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS	PVVBF66D0171 ACAF0S2	27,56,43, 326	
	½" - 2"	CL300	RF	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT, SHORT PATTERN	PVVBF63R0171 ACAF0ZZ	27,43, 326	
WAF BUTTERFLY	3" - 4"	CL300	RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT	PVVUB63W0371 BCAG0ZZ	27,43, 326	
	6" - 24"	CL300	RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT, GEAR OP.	PVVUB63W0371 BCAG3ZZ	27,43, 326	
LUG BUTTERFLY	3" - 4"	CL300	RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT	PVVUB63L0371 BCAG0ZZ	27,43, 326	
	6" - 24"	CL300	RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT, GEAR OP.	PVVUB63L0371 BCAG3ZZ	27,43, 326	
BOLT/NUTS	ALL	CL300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEX NUTS			
GASKETS SPIRAL WOUND	½" - 24"	CL300		SPIRAL WOUND GASKET, 304 SS W/GRAPHITE FILLER, CS CENTERING RING AND 304SS INNER RING, ASME B16.20 / ASME B16.5			

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES	
		FLANGES	
		SOCKETWELD	
		BUTTWELD	
		SIZES	
		½" - 2"	
		3" - 24"	
		½" - 1 ½"	
		2" - 24"	

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
4	Socket-weld x Threaded gate valve shall be used for valved pressure, instrument, vent and drain connections only. Piping system end shall be socketweld and instrument end shall be threaded.	
24	Use when required at flanged equipment nozzles, or where noted on P&IDs or Modules.	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
43	Only to be used as indicated on the P&IDs.	
55	Supplier shall weld equal length of line class seamless plain end nipples on each buttweld end of the valve, overall length to be 250 mm.	
56	Supplier shall weld only a piece of line class seamless plain end nipple on the socketweld end of the socketweld x threaded valve, overall length to be 200 mm.	
59	The valve shall be installed in tandem with angle valve upstream as indicated on the P&IDs.	
200	The straightway (Y-type) or angle type blow-off valves may be used in vertical pipes or they may be used in horizontal runs of piping provided they are so constructed or installed that the lowest edge of the opening through the seat is at least 25 percent of the inside diameter below the center line of the valve.	
275	Only to be used where full port gate valve is required as indicated on P&ID's.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drains requiring plugs, connections to tube fittings and for connections to threaded proprietary items.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	
275	Only to be used where full port gate valve is required as indicated on P&ID's.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
B1B	R00

					REV
SERVICES	Ethylene Refrigerant	Process			

PIPING MATERIAL	CARBON STEEL	VALVE MATERIAL	CARBON STEEL	WELDEXAMINATION		REV
RATING / FACING	ASME CLASS 300 / RF	TRIM MATERIAL	13 CR/HF (API #8)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.5 mm	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)*

SIZE																	
TEMP. °C	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20	24	30
-28 to 38	51.02	51.02	51.02	51.02	51.02	51.02	51.02	51.02	51.02	51.02	49.62	45.16	39.41	35.00	44.60	37.05	37.88
93	46.88	46.88	46.88	46.88	46.88	46.88	46.88	46.88	46.88	46.88	46.88	45.16	39.41	35.00	44.60	37.05	37.88
149	45.16	45.16	45.16	45.16	45.16	45.16	45.16	45.16	45.16	45.16	45.16	45.16	39.41	35.00	44.60	37.05	37.88

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES RTFE SEAT RATINGS (Bar)

TEMP(C)	< NPS 2	REV
-28 to 38	51.02	
93	46.88	
149	28.93	
204	5.49	

BUTTERFLY RTFE SEAT RATINGS(Bar)

TEMP(C)	ALL SIZES	REV
-28 to 38	51.02	
93	46.54	
149	26.87	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	½" - 1½"	80	PE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	2" - 16"	STD	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	18"	STD	BE	CS, API 5L GRADE B, EFW, ASME B36.10M Ej=0.95		
	20" - 24"	12.7mm	BE	CS, API 5L GRADE B, EFW, ASME B36.10M Ej=0.95		
	30"	14.27mm	BE	CS, API 5L GRADE B, DSAW, ASME B36.10M Ej=1		
NIPPLE	½" - 1½"	160	TOE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	½" - 1½"	160	TBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	½" - 1½"	80	PBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
SWAGE	½" - 1½"	160	TSE	CS, ASTM A234 GRADE WPB, MSS-SP-95	1	
	½" - 1½"	80	BBE	CS, ASTM A234 GRADE WPB, MSS-SP-95		
	½" - 1½"	160	TSE	CS, ASTM A234 GRADE WPB, MSS-SP-95		
	½" - 1½"	80	BBE	CS, ASTM A234 GRADE WPB, MSS-SP-95		
COUPLING	½" - 1½"	CL3000	SOCW	CS, ASTM A105, ASME B16.11		
FULL COUPLING	½" - 2"	CL3000	THRD	CS, ASTM A105, ASME B16.11	279	
OLET	½" - 1½"	CL3000	WELD	CS, ASTM A105, MSS-SP-97		
THREDOLET	½" - 2"	CL3000	WELD	CS, ASTM A105, MSS-SP-97	279	
WELDOLET	2" - 8"	AS PIPE	WELD	CS, ASTM A105, MSS-SP-97		
PLUG						
ROUND HEAD	½" - 2"		THRD	CS, ASTM A105, ASME B16.11, ROUND HEAD		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
CAP	½" - 1½" ½" - 2" 2" - 30"	CL3000 CL3000 AS PIPE	SOCW THRD BW	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9		279	
90° ELBOW SW LR	½" - 1½" 2" - 16" 18" - 30"	CL3000 AS PIPE AS PIPE	SOCW BW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9			
45° ELBOW	½" - 1½" 2" - 16" 18" - 30"	CL3000 AS PIPE AS PIPE	SOCW BW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9			
TEE RED TEE	½" - 1½" 2" - 16" 18" - 30" ½" - 1½" 2" - 16" 18" - 30"	CL3000 AS PIPE AS PIPE CL3000 AS PIPE AS PIPE	SOCW BW BW SOCW BW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9			
REDUCER CON ECC	2" - 16" 18" - 30" 2" - 16" 18" - 30"	AS PIPE AS PIPE AS PIPE AS PIPE	BW BW BW BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9			
FLANGE SOCKETWELD WELDNECK BLIND FLG DRIP RING FIGURE 8 BLANK+SPACER	½" - 1½" 2" - 24" 30" ½" - 24" 30" 1" - 12" 1" - 12" 14" - 24"	CL300 CL300 CL300 CL300 CL300 CL300 CL300 CL300	RF RF RF RF RF FF FF FF	CS, ASTM A105, ASME B16.5 CS, ASTM A105, ASME B16.5 THICKNESS AS PIPE CS, ASTM A105, ASME B16.47 SERIES A THICKNESS AS PIPE CS, ASTM A105, ASME B16.5 CS, ASTM A105, ASME B16.47 SERIES A THICKNESS AS PIPE CS, ASTM A516 GRADE 70, NORMALIZED, 1-1/2" THICK WITH ¾" SOCW TAPS (STANDARD DRAWINGS) CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48 CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48			
VALVES GATE	½" - 1½" ½" - 1½" ½" - 1½" ½" - 1½" ½" - 2" 2" 3" - 10" 12" - 24" 30"	CL800 CL800 CL800 CL800 CL300 CL300 CL300 CL300 CL300	SOCW SOCW T/SW T/SW RF RF RF RF RF	CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT, GEAR OP. CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT, GEAR OP.	TAG PVVGB68S0108 ACAD0ZZ PVVGF68S0108 ACAD0ZZ PVVGB68D0108 ACAD0ZZ PVVGF68D0108 ACAD0ZZ PVVGB63R0108 ACAD0ZZ PVVGF63R0308 BCAC0ZZ PVVGF63R0308 BCAC0ZZ PVVGF63R0308 BCAC5ZZ PVVGF63K0308 BCAC5ZZ	41, 277 275 4 4, 275 24 43 33	
GLOBE	½" - 1½" 2" - 8" 10" - 12"	CL800 CL300 CL300	SOCW RF RF	CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8)	PVVLB68S0108A CAD0ZZ PVVLB63R0308 BCAC0ZZ PVVLB63R0308 BCAC5ZZ		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
LIFT CHECK	½" - 1½"	CL800	SOCW	CS BODY WITH 13 CR TRIM (API # 1)	PVVCL68S0101 NCAD0ZZ		
WAFER DUAL PLATE CHECK	2" - 24"	CL300	RF	CS BODY WITH 13 CR PLATES, METAL SEATS	PVVCD63W0301 NCAA0ZZ		
BALL	½" - 1½"	CL600	BW	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS	PVVBF66B0171 ACAF0B2	27,55,43, 326	
	½" - 1½"	CL600	BW	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT	PVVBF66B0171 ACAF0B2	27,55,43, 326	
	½" - 1½"	CL600	T/SW	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS	PVVBF66D0171 ACAF0S2	27,56,43, 326	
	½" - 2"	CL300	RF	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT, SHORT PATTERN	PVVBF63R0171 ACAF0ZZ	27,43,326	
WAF BUTTERFLY	3" - 4"	CL300	RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT	PVVUT63W0371 BCAG0ZZ	27,43,326	
	6" - 24"	CL300	RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT, GEAR OP.	PVVUT63W0371 BCAG3ZZ	27,43,326	
LUG BUTTERFLY	3" - 4"	CL300	RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT	PVVUT63L0371B CAG0ZZ	27,43,326	
	6" - 24"	CL300	RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT, GEAR OP.	PVVUT63L0371B CAG3ZZ	27,43,326	
BOLT/NUTS	ALL	CL300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEX NUTS			
GASKETS SPIRAL WOUND	½" - 24"	CL300		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, CS CENTERING RING AND 304SS INNER RING, ASME B16.20 / ASME B16.5			
	30"	CL300		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, CS CENTERING RING AND 304SS INNER RING, ASME B16.20 / ASME B16.47 SERIES A			

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD	

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
4	Socket-weld x Threaded gate valve shall be used for valved pressure, instrument, vent and drain connections only. Piping system end shall be socketweld and instrument end shall be threaded.	
24	Use when required at flanged equipment nozzles, or where noted on P&IDs or Modules.	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
33.	Flanged valves and equipment, NPS 30 and larger, shall have flange drilling and bolting patterns conforming to series A of ASMEB16.47.	
43	Only to be used as indicated on the P&IDs.	
55	Supplier shall weld equal length of line class seamless plain end nipples on each butt weld end of the valve, overall length to be 250 mm.	
56	Supplier shall weld only a piece of line class seamless plain end nipple on the socket weld end of the socket weld x threaded valve, overall length to be 200 mm.	
275	Only to be used where full port gate valve is required as indicated on P&ID's.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drains requiring plugs, connections to tube fittings and for connections to threaded proprietary items.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	

HEADER SIZE

BRANCH

BRANCH LEGEND		NOTES	REV
T	Tee(B.W. for sizes three inch and larger, S.W. for two inch and smaller)		
RT	Reducing Tee (B.W. for three inch and larger, S.W. for two inch and smaller)		
W	Weldolet (PWHT required at branch attachment welds when the thickness through the weld in any plane through the branch is greater than 1.5" for P1 material per ASME B31.3.)		
A	Socket		
S	Stub-in with or without Reinforcing Pad shall be set-on type as shown in detail "1" or "3" of Figure 328.5.4D of ASME B31.3		
P	Reinforcing Pad (Note d.)		

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
c.	Weldolet may be used in lieu of a stub-in Branch Weld if verified and approved by pressure design and flexibility analysis and shown on the piping design drawing	
e	PWHT required at all branch attachment welds when the thickness through the weld in any plane through the branch is greater than 38.1 mm (1.5") for P1 material per ASME B31.3 and greater than 19 mm (0.75") for P1 materials per ASME B31.1.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
B1E	R00

SERVICES	Hydrogen (from PSA)	Hydrogen				REV
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PIPING MATERIAL	CARBON STEEL	VALVE MATERIAL	CARBON STEEL	WELD EXAMINATION		REV
RATING / FACING	ASME CL300 / RF	TRIM MATERIAL	13 CR (API #1)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE(mm)	1.50 mm (0.059")	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
-28 to 38	51.02	51.02	51.02	51.02	51.02	51.02	51.02	50.69	50.86	50.60	44.50	41.89	35.37	31.41	28.24	23.50
93	46.54	46.54	46.54	46.54	46.54	46.54	46.54	46.54	46.54	46.54	44.50	41.89	35.37	31.41	28.24	23.50
149	45.16	45.16	45.16	45.16	45.16	45.16	45.16	45.16	45.16	45.16	44.50	41.89	35.37	31.41	28.24	23.50
204	43.78	43.78	43.78	43.78	43.78	43.78	43.78	43.78	43.78	43.78	43.78	41.89	35.37	31.41	28.24	23.50

*SHADED VALVES REPRESENTS THE LIMITS OF THE FLANGE RATING.

BALL VALVES RTFE SEAT RATINGS (bar)

TEMP(C)	< NPS 2	REV
-28 to 38	51.02	
93	46.54	
149	28.96	
204	5.52	

BUTTERFLY RTFE SEAT RATINGS(bar)

TEMP(C)	ALL SIZES	REV
-28 to 38	51.02	
93	46.54	
149	26.87	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" - 3/4" 1" - 1 1/2" 2" 3" - 18" 20" - 24"	160 80 80 STD 20	PE PE BE BE BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M	292	
NIPPLES	1/2" - 1" 1 1/2" 1/2" - 3/4" 1" - 1 1/2"	XXS 160 160 80	TBE TBE PBE PBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, ASME B36.10M	292	
SWAGES					1	
CONC&ECC	1/2" - 3/4" 1" - 1 1/2"	160 80	BBE BBE	CS, ASTM A234 GRADE WPB-S, SMLS, MSS-SP-95 CS, ASTM A234 GRADE WPB-S, SMLS, MSS-SP-95		
OLET						
SOCKOLET	1/2" - 3/4" 1" - 1 1/2"	6000 LB 3000 LB	WELD WELD	CS, ASTM A105, MSS-SP-97 CS, ASTM A105, MSS-SP-97		
SOCWELBOLET	1/2" - 3/4" 1" - 1 1/2"	6000 LB 3000 LB	WELD WELD	CS, ASTM A105, MSS-SP-97 CS, ASTM A105, MSS-SP-97		
WELDOLET	3" - 8"	AS PIPE	WELD	CS, ASTM A105, MSS-SP-97		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PLUG	½" – 2"		THRD	CS, ASTM A105, ASME B16.11, ROUND HEAD		
CAP	½" – ¾" 1" – 1 ½" 2" – 18" 20" – 24"	6000 LB 3000 LB AS PIPE AS PIPE	SOCW SOCW WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9		
90°ELBOW	½" – ¾" 1" – 1 ½" 2" – 18" 20" – 24"	6000 LB 3000 LB AS PIPE AS PIPE	SOCW SOCW WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9		
45°ELBOW	½" – ¾" 1" – 1 ½" 2" – 18" 20" – 24"	6000 LB 3000 LB AS PIPE AS PIPE	SOCW SOCW WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9		
COUPLING	½" – ¾" 1" – 1 ½"	6000 LB 3000 LB	SOCW SOCW	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11		
COUPLING(RE D)	½" – ¾" 1" – 1 ½"	6000 LB 3000 LB	SOCW SOCW	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11		
TEE	½" – ¾" 1" – 1 ½" 2" – 18" 20" – 24"	6000 LB 3000 LB AS PIPE AS PIPE	SOCW SOCW WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9		
RED TEE	½" – ¾" 1" – 1 ½" 2" – 18" 20" – 24"	6000 LB 3000 LB AS PIPE AS PIPE	SOCW SOCW WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9		
REDUCER CON	2" – 18" 20" – 24"	AS PIPE AS PIPE	WELD WELD	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9		
ECC	2" – 18" 20" – 24"	AS PIPE AS PIPE	WELD WELD	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9		
FLANGE						
SOCKETWELD	½" – 1 ½"	CL300	RF	CS, ASTM A105, ASME B16.5		
WELDNECK	2" – 24"	CL300	RF	CS, ASTM A105, ASME B16.5		
BLIND	½" – 24"	CL300	RF	CS, ASTM A105, ASME B16.5		
MISCELLANEOUS						
DRIP RING	1" – 12"	CL300	FF	CS, ASTM A516 GRADE 70, NORMALIZED, 38.100 mm (1-1/2") THICK WITH ¾" SOCKETWELD TAPS		
FIGURE 8 BLANK	1" – 12"	CL300	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48		
PADDLE BLANK	14"	CL300	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48		
PADDLE BLANK	16" – 24"	CL300	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48		
PADDLE SPACER	14"	CL300	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48		
PADDLE SPACER	16" – 24"	CL300	FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
VALVES						41,277	
GATE	½" – 1 ½"	CL800	SOCW	CS BODY WITH 13 CR TRIM (API #1)	PVVGB68S0101 ACAD0BZ	275	
GATE	½" – 1 ½"	CL800	SOCW	CS BODY WITH 13 CR TRIM (API #1), FULL PORT	PVVGf68S0101 ACAD0BZ	24	
GATE	½" – 2"	CL300	RF	CS BODY WITH 13 CR TRIM (API #1)	PVVGB63R0101 ACAD0BZ	43,275	
GATE	2"	CL300	RF	CS BODY WITH 13 CR TRIM (API #1), FULL PORT	PVVGf63R0301 ACAC0BZ		
GATE	2" – 10"	CL300	RF	CS BODY WITH 13 CR TRIM (API #1), FULL PORT	PVVGf63R0301 BCAC0BZ		
GATE	12" – 24"	CL300	RF	CS BODY WITH 13 CR TRIM (API #1), FULL PORT, GEAR OP.	PVVGf63R0301 BCAC5BZ		
GATE	2" – 10"	CL300	WELD	CS BODY WITH 13 CR TRIM (API #1), FULL PORT	PVVGf63B0301 BCAC0BZ		
GATE	12" – 24"	CL300	WELD	CS BODY WITH 13 CR TRIM (API #1), FULL PORT, GEAR OP.	PVVGf63B0301 BCAC5BZ		
GLOBE	½" – 1 ½"	CL800	SOCW	CS BODY WITH 13 CR TRIM (API #1)	PVVLB68S0101A CAD0BZ		
GLOBE	2" – 8"	CL300	RF	CS BODY WITH 13 CR TRIM (API #1)	PVVLB63R0301 BCAC0BZ		
GLOBE	10" – 12"	CL300	RF	CS BODY WITH 13 CR TRIM (API #1), GEAR OP.	PVVLB63R0301 BCAC5BZ		
GLOBE	2" – 8"	CL300	WELD	CS BODY WITH 13 CR TRIM (API #1)	PVVLB63B0301B CAC0BZ		
GLOBE	10" – 12"	CL300	WELD	CS BODY WITH 13 CR TRIM (API #1), GEAR OP.	PVVLB63B0301B CAC5BZ		
LIFT CHECK	½" – 1 ½"	CL800	SOCW	CS BODY WITH 13 CR TRIM (API #1)	PVVCL68S0101 NCAD0BZ		
WAFER CHECK	2" – 24"	CL300	WAF- RF	CS BODY WITH 13 CR PLATES, METAL SEATS	PVVCD63W0301 XNZA0BZ		
WAF BUTTERFLY	3–4"	CL300	WAF- RF	CS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUB63W0399 BCAG1BZ	27,43,326	
WAF BUTTERFLY	6" – 24"	CL300	WAF- RF	CS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP	PVVUB63W0399 BCAG3BZ	27,43,326	
LUG BUTTERFLY	3–4"	CL300	LUG- RF	CS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUB63L0399 BCAG1BZ	27,43,326	
LUG BUTTERFLY	6" – 24"	CL300	LUG- RF	CS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUB63L0399 BCAG3BZ	27,43,326	
BALL	½" – 1 ½"	CL600	BW	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT	PVVBF66E0171 ACAF0B3	27,55,326	
BALL	½" – 1 ½"	CL600	T/SW	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT	PVVBF66D0171 ACAF0S3	4,27,56, 326	
BALL	½" – 2"	CL300	RF	CS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT, SHORT PATTERN	PVVBK63R0171 ACAF0BZ	5,27,326	
BOLTING	ALL	CL300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A194 GRADE 2H HEAVY HEX NUTS			
GASKET	½" – 24"	CL300		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, SS INNER & CS OUTER RINGS, ASME B16.20 / ASME B16.5			

	NOTES		REV
JOINT CONNECTS			
MAINTENANCE		FLANGES	SIZES
MAINTENANCE		FLANGES	½" – 2"
NON-MAINTENANCE		SOCKETWELD	3" – 6"
NON-MAINTENANCE		BUTTWELD	½" – 1 ½"
			2" – 6"

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
4	Socket-weld x Threaded valve shall be used for valved pressure, and instrument connections only .Piping system end shall be socketwelded and instrument end shall be threaded.	
24	Use when required at flanged equipment nozzles, or where noted on P&IDs or Modules.	
27	Teflon seated valves are limited to maximum 204 °C. For pressure and temperatures outside table consideration should be given to alternative seat materials in consultation with manufacturers.	
41	Valve shall be suitable for hydrogen service.	
43	Only to be used as indicated on the P&IDs.	
55	Supplier shall weld equal length of line class seamless plain end nipples on each butt weld end of the valve, overall length to be 250mm.	
56	Supplier shall weld one piece of line class seamless plain end nipple on the butt weld end of the butt weld x threaded valve,overall length to be 200mm.	
275	Only to be used where full port gate valve is required as indicated on P&IDs .	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	

BRANCH CONNECTION CHART

[illegible]

BRANCH

BRANCH LEGEND

NOTES

REV

PIPING BRANCH NOTES

RFV

PROJECT NUMBER	01000
CLIENT	01000-MA-SPC-0002
LOCATION	

PIPING CLASS	REV
B2A	R00

						REV
SERVICES	Cooling Water Return	Cooling Water Supply	Dry Flare	Cold Liquid Drain	Process	
	Process (Propane)	Process (Recycle Propylene)	Propylene Refrigerant	Ammonia	Propane	
	Liquid Propane	Propane Gas	Propane Steam	Ethylene	Natural gas	
	Instrument air	Propylene				

						REV
PIPING MATERIAL	LT CARBON STEEL (IMPACT TESTED)	VALVE MATERIAL	LT CARBON STEEL	WELDE EXAMINATION	PER 108184-0000-PI-SPC-0006	
RATING / FACING	ASME CL300 / RF	TRIM MATERIAL	316 SS (API #10)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE(mm)	1.50 mm (0.059")	CODE	ASME B31.3	BENDING	NOT ALLOWE	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
-48 to 38	51.03	51.03	51.03	51.03	51.03	51.03	51.03	51.03	51.03	51.03	51.03	51.03	47.12	50.48	51.03	50.44
93	46.9	46.9	46.9	46.9	46.9	46.9	46.9	46.9	46.9	46.9	46.9	46.9	46.9	46.9	46.9	46.9
149	45.17	45.17	45.17	45.17	45.17	45.17	45.17	45.17	45.17	45.17	45.17	45.17	45.17	45.17	45.17	45.17
204	43.79	43.79	43.79	43.79	43.79	43.79	43.79	43.79	43.79	43.79	43.79	43.79	43.79	43.79	43.79	43.79
260	41.72	41.72	41.72	41.72	41.72	41.72	41.72	41.72	41.72	41.72	41.72	41.72	41.72	41.72	41.72	41.72
316	39.31	39.31	39.31	39.31	39.31	39.31	39.31	39.31	39.31	39.31	39.31	39.31	39.31	39.31	39.31	39.31

*SHADED VALVES REPRESENTS THE LIMITS OF THE FLANGE RATING.

BALL VALVES RTFE SEAT RATINGS (bar)

TEMP(C)	< NPS 2	>NPS2; < NPS 6	> NPS 6	REV
-48 to 38	47.92	47.92	19.65	
93	45.51	35.50	13.79	
149	28.96	18.96	7.58	
204	5.52	3.34	1.38	

BUTTERFLY RTFE SEAT RATINGS(bar)

TEMP(C)	ALL SIZES	REV
-48 to 38	47.92	
93	45.51	
149	26.87	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" - 1 1/2" 2" - 16" 18" 20" 24"	80 STD 30 30 XS	PE BE BE BE BE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M	143	
NIPPLES	1/2" - 1 1/2" 1/2" - 1 1/2" 1/2" - 1 1/2"	160 80 160	TOE PBE TOE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M	143	
SWAGES (CONC&ECC)	1/2" - 1 1/2" 1/2" - 1 1/2"	80 160	BBE TSE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95 LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95	1,143	
OLET SOCKOLET SOCW ELBOLET WELDOLET 45° LATROLET	1/2" - 1 1/2" 1/2" - 1 1/2" 2" - 8" 1/2" - 1 1/2"	3000 LB 3000 LB AS PIPE 3000 LB	SOCW SOCW WELD SOCW	LTCS, ASTM A350 GRADE LF2 CL1, MSS-SP-97 LTCS, ASTM A350 GRADE LF2 CL1, MSS-SP-97 LTCS, ASTM A350 GRADE LF2 CL1, MSS-SP-97 LTCS, ASTM A350 GRADE LF2 CL1, MSS-SP-97		
PLUG	1/2" - 2"		THRD	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11, ROUND HEAD		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
CAP	½" – 1 ½" 2" – 16" 18" – 24"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9	143	
90° ELBOW	½" – 1 ½" 2" – 16" 18" – 24"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9	143	
45° ELBOW	½" – 1 ½" 2" – 16" 18" – 24"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9	143	
COUPLING COUPLING(RED)	½" – 1 ½" ½" – 1 ½"	3000 LB 3000 LB	SOCW SOCW	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11 LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11	143	
45° S.W. LATERAL 45° LATERAL 45° SW LATERAL(RED) 45° LATERAL (RED)	½" – 1 ½" 2" – 16" 18" – 20" ½" – 1 ½" 2" – 16" 18" – 20"	3000 LB AS PIPE AS PIPE 3000 LB AS PIPE AS PIPE	SOCW WELD WELD SOCW WELD WELD	LTCS, ASTM A350 GRADE LF2 CL1, MSS-SP-97, MFR STD LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9, MFR STD LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9, MFR STD LTCS, ASTM A350 GRADE LF2 CL1, MSS-SP-97, MFR STD LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9, MFR STD LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9, MFR STD		
TEE RED TEE	½" – 1 ½" 2" – 16" 18" – 24" ½" – 1 ½" 2" – 16" 18" – 24"	3000 LB AS PIPE AS PIPE 3000 LB AS PIPE AS PIPE	SOCW WELD WELD SOCW WELD WELD	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9 LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9	143	
REDUCER CON ECC	 2" – 16" 18" – 24" 2" – 16" 18" – 24"	 AS PIPE AS PIPE AS PIPE AS PIPE	 WELD WELD WELD WELD	 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9	143	
FLANGE SOCKETWELD WELDNECK BLIND	 ½" – 1 ½" 2" – 24" ½" – 24"	 CL300 CL300 CL300	 RF RF RF	 LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5 LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5 LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5	143	
MISCELLANEOUS DRIP RING INSULATING JOINT FIGURE 8 BLANK	 1" – 12" 3" – 12" 1" – 4"	 CL300 CL300 CL300	 FF MFR FF	 LTCS, ASTM A516 GRADE 70 S-5, NORMALIZED AND IMPACT TESTED AT –46 °C, 38.100 mm (1-1/2") THICK WITH ¾" SOCKETWELD TAPS MFR LTCS, ASTM A516 GRADE 70 S-5, NORMALIZED, ASME B16.48	901	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
FIGURE 8 BLANK	6" - 12"	CL300	FF	SS, ASTM A240 GRADE 304, SOLUTION HEAT TREATED, ASME B16.48		
PADDLE BLANK	14" - 24"	CL300	FF	SS, ASTM A240 GRADE 304, SOLUTION HEAT TREATED, ASME B16.48		
PADDLE SPACER	14" - 24"	CL300	FF	SS, ASTM A240 GRADE 304, SOLUTION HEAT TREATED, ASME B16.48		
VALVES						
GATE	½" - 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API #10)	TAG PVVGB68S0710 ACLD0ZZ	143, 277
GATE	½" - 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVGFB68S0710 ACLD0ZZ	275 4
GATE	½" - 1 ½"	CL800	T/SW	LTCS BODY WITH 316 SS TRIM (API #10)	PVVGFB68D0710 ACLD0ZZ	4,275
GATE	½" - 1 ½"	CL800	T/SW	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVGFB68D0710 ACLD0ZZ	
GATE	½" - 1 ½"	CL300	RF	LTCS BODY WITH 316 SS TRIM (API #10)	PVVLB63R0710 ACLD0ZZ	24
GATE	½" - 1 ½"	CL300	RF	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVGFB63R0710 ACLD0ZZ	43, 275
GATE	2" - 10"	CL300	RF	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVGFB63R1110 BCLC0ZZ	
GATE	12" - 20"	CL300	RF	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT, GEAR OP.	PVVGFB63R1110 BCLC5ZZ	
GATE	12" - 20"	CL300	BW	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT, GEAR OP.	PVVGFB63B1110 BCLC5ZZ	
GLOBE	½" - 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API #10)	PVVLB68S0710A CLD0ZZ	
GLOBE	2" - 8"	CL300	RF	LTCS BODY WITH 316 SS TRIM (API #10)	PVVLB63R1110 BCLC0ZZ	
GLOBE	10" - 14"	CL300	RF	LTCS BODY WITH 316 SS TRIM (API #10), GEAR OP.	PVVLB63R1110 BCLC5ZZ	
LIFT CHECK	½" - 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API # 10) (HORIZ. ONLY)	PVVCL68S0710 NCLD0ZZ	
SWING CHECK	2" - 20"	CL300	RF	LTCS BODY WITH 316 SS TRIM (API # 10)	PVVCS63R1110 NCLC0ZZ	325
WAFER CHECK	2" - 20"	CL300	WAF- RF	LTCS BODY WITH 316 SS PLATES, METAL SEATS	PVWCD63W1110 XNZA0ZZ	328
WAF BUTTERFLY	3" - 4"	CL300	WAF- RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUB63W1199 BCLG1ZZ	27, 43, 326
WAF BUTTERFLY	6" - 20"	CL300	WAF- RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUB63W1199 BCLG3ZZ	27, 43, 326
LUG BUTTERFLY	3" - 4"	CL300	LUG- RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUB63L1199 BCLG1ZZ	27, 43, 326, 328
LUG BUTTERFLY	6" - 20"	CL300	LUG- RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUB63L1199 BCLG3ZZ	27, 43, 326, 328
BALL	½" - 1 ½"	CL600	BW	LTCS BODY WITH 316SS BALL&STEM,RTFE SEATS	PVVBH66E0771 ACLF0B2	27, 43, 55, 326
BALL	2" - 5"	CL300	RF	LTCS BODY WITH 316SS BALL&STEM,RTFE SEATS,SHORT PATTERN	PVVBH63R1171 BCLF1ZZ	27, 43, 326
BALL	6" - 20"	CL300	RF	LTCS BODY WITH 316SS BALL&STEM,RTFE SEATS,SHORT PATTERN,GEAR OP.	PVVBH63R1171 BCLF3ZZ	27, 43, 326
BOLTING	ALL	CL300		ALLOY STEEL, ASTM A320 GR.L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS		
GASKET	½" - 20"	CL300		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, SS INNER & SS OUTER RINGS, ASME B16.20 / ASME B16.5		

	NOTE			REV
JOINT CONNECTS			SIZES	
MAINTENANCE	144	FLANGES	1/2" – 1 1/2"	
MAINTENANCE NON-	144	FLANGES	2" – 20"	
MAINTENANCE NON-		SOCKETWELD	1/2" – 1 1/2"	
MAINTENANCE		BUTTWELD	2" – 20"	

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
4	Socket-weld x Threaded valve shall be used for valved pressure, and instrument connections only .Piping system end shall be socketwelded and instrument end shall be threaded.	
24	Use when required at flanged equipment nozzles, or where noted on P&IDs or Modules.	
27	Teflon seated valves are limited to maximum 204 °C. For pressure and temperatures outside table consideration should be given to alternative seat materials in consultation with manufacturers.	
33	Flanged valves and equipment, larger than NPS 24, shall have flange drilling and bolting patterns conforming to series B of ASME B16.47.	
43	Only to be used as indicated on the P&IDs .	
55	Supplier shall weld equal length of line class seamless plain end nipples on each butt weld end of the valve,overall length to be 250mm.	
143	In shop and/or field fabrication welds (girth, branch and fillet), all weld metal deposits and heat affected zone (HAZ) shall meet impact test requirements for low temperature service (at -48°C for L.T.C.S., at -105°C for semi-cryogenic S.S. "K" specs and for cryogenic S.S. "L" specs) per Paragraph 323 of ASME B31.3.	
275	Only to be used where full port gate valve is required as indicated on P&IDs .	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
325	Wafer and lug type valves shall not be used when the design temperature is greater than 600° F(315°C).	
326	This resilient seated valve does not in all cases meet full flange rating,See valve MAWP in this specification.	
328	Only for safety valve.	
901	Only to be used for the connection of Propane Pipeline and upground Propane Pipe.	

BRANCH CONNECTION CHART 45 DEGREE BRANCH CONNECTION CHART		
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HEADER SIZE

1/2	L																		
3/4	RL	L																	
1	RL	RL	L																
1 1/2	RL	RL	RL	L															
2	Ay	RL	RL	RL	L														
3	Ay	Ay	Ay	Ay	RL	L													
4	Ay	Ay	Ay	Ay	RL	RL	L												
6	Ay	Ay	Ay	Ay	P	RL	RL	L											
8	Ay	Ay	Ay	Ay	P	RL	RL	RL	L										
10	Ay	Ay	Ay	Ay	P	P	P	RL	RL	L									
12	Ay	Ay	Ay	Ay	P	P	P	RL	RL	RL	L								
14	Ay	Ay	Ay	Ay	P	P	P	P	RL	RL	RL	L							
16	Ay	Ay	Ay	Ay	P	P	P	P	P	RL	RL	RL	L						
18	Ay	Ay	Ay	Ay	P	P	P	P	P	P	P	RL	RL	L					
20	Ay	Ay	Ay	Ay	P	P	P	P	P	P	P	P	RL	RL	RL	L			
	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20				

BRANCH SIZE

BRANCH LEGEND		NOTES	REV
S	45 degree stub-in		
Ay	Socket-weld latrolet		
L	Lateral (B.W. for sizes two inch and larger, S.W. for 1 ½" and smaller)		
RL	Reducing lateral (B.W. for sizes two inch and larger, S.W. for 1 ½" and smaller)		
P	Stub-in with Reinforcing Pad shall be set-in type as shown in detail "2" or "4" of Figure 328.5.4D of ASME B31.3		

PIPING BRANCH NOTES

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
b.	For 45° branch connections, except that the branches from TLE's heater wash cone lines where 45° lateral stub-in is acceptable for atmospheric drain conditions.	
c.	Weldolet may be used in lieu of stub-in if required and verified by flexibility analysis and shown on the piping design drawing.	
d.	A PAD has been selected to provide reinforcement for this branch connection subject to the most severe coincident condition of internal pressure and fluid temperature seen in this specification. However, the specific pressure-temperature condition for this connection may eliminate the need for reinforcement. In addition, when the diameter-to-thickness ratio is restricted by paragraph 304.3.1 (3) (b) of ASME B31.3, finite element analysis (FEA) shall be performed .	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
B2B	R00

					REV
SERVICES	Ethylene Refrigerant	Process			

						REV
PIPING MATERIAL	L.T. CARBON STEEL (IMPACT TESTED)	VALVE MATERIAL	L.T. CARBON STEEL	WELDEXAMINATION		
RATING / FACING	ASME CLASS 300 / RF	TRIM MATERIAL	316SS (API #10)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.5 mm	CODE	ASME B31.3	BENDING	PER PROJECT	

**FLANGE RATING &
MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)***

SIZE																
TEMP. °C	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20	24
-48 to 38	47.92	47.92	47.92	47.92	47.92	47.92	47.92	47.92	47.92	47.92	47.92	47.92	46.38	41.20	37.00	30.74
93	45.16	45.16	45.16	45.16	45.16	45.16	45.16	45.16	45.16	45.16	45.16	45.16	46.38	41.20	37.00	30.74
149	44.13	44.13	44.13	44.13	44.13	44.13	44.13	44.13	44.13	44.13	44.13	44.13	46.38	41.20	37.00	30.74
204	42.75	42.75	42.75	42.75	42.75	42.75	42.75	42.75	42.75	42.75	42.75	42.75	46.38	41.20	37.00	30.74

**BUTTERFLY
RTFE SEAT RATINGS(Bar)**

TEMP(C)	ALL SIZES	REV
-48 to 38	47.92	
93	45.16	
149	44.13	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2 " - 2" 2" - 24"	80 STD	PE BE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M	143	
NIPPLE	TOE 1/2 " - 2" TBE 1/2 " - 2" PBE 1/2 " - 2"	160 160 80	TOE TBE PBE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M	143	
SWAGE	CON 1/2 " - 2" 1/2 " - 2" ECC 1/2 " - 2" 1/2 " - 2"	160 80 160 80	TSE BBE TSE BBE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95 LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95 LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95 LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95	1, 143	
COUPLING	FULL COUPLING 1/2 " - 1 1/2" FULL COUPLING 1/2 " - 2"	CL3000 CL3000	SOCW THRD	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11	143 279	
OLET	SOCKOLET 1/2 " - 1 1/2" THREDOLET 1/2 " - 2" WELDOLET 2" - 8"	CL3000 CL3000 AS PIPE	WELD WELD WELD	LTCS, ASTM A350 GRADE LF2 CLASS 1, MSS-SP-97 LTCS, ASTM A350 GRADE LF2 CLASS 1, MSS-SP-97 LTCS, ASTM A350 GRADE LF2 CLASS 1, MSS-SP-97	143 279	
PLUG	ROUND HEAD 1/2 " - 2"		THRD	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11, ROUND HEAD	143	
CAP	1/2 " - 1 1/2" 1/2 " - 2" 2" - 24"	CL3000 CL3000 AS PIPE	SOCW THRD BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9	143 279	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
90° ELBOW SW LR	½" - 1 ½" 2" – 24"	CL3000 AS PIPE	SOCW BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		143	
45° ELBOW	½" - 1 ½" 2" – 24"	CL3000 AS PIPE	SOCW BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		143	
TEE	½" - 1 ½" 2" – 24"	CL3000 AS PIPE	SOCW BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		143	
RED TEE	½" - 1 ½" 2" – 24"	CL3000 AS PIPE	SOCW BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
REDUCER CON ECC	2" – 24" 2" – 24"	AS PIPE AS PIPE	BW BW	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		143	
FLANGE SOCKETWELD	½" - 1 ½" ½" - 1 ½"	CL300 CL600	RF RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5 LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5		143,144 278	
WELDNECK	2" – 24" 2" – 24"	CL300 CL600	RF RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5 THICKNESS AS PIPE LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5 THICKNESS AS PIPE		278	
BLIND FLG	½" - 24" ½" - 24"	CL300 CL600	RF RF	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5 LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5		278	
DRIP RING	1" –12"	CL300	FF	CS, ASTM A516 GRADE 70 S-5, NORMALIZED, 1-1/2" THICK WITH ¾" SOCW TAPS (STANDARD DRAWINGS)			
FIGURE 8 BLANK+SPACER	1" –12" 14" – 24"	CL300 CL300	FF FF	CS, ASTM A516 GRADE 70 S-5, NORMALIZED, ASME B16.48 CS, ASTM A516 GRADE 70 S-5, NORMALIZED, ASME B16.48			
VALVES GATE	½" - 1 ½" ½" - 1 ½" ½" - 1 ½" ½" - 1 ½" ½" - 2" 2" 3" – 10" 12" – 24"	CL800 CL800 CL800 CL800 CL300 CL300 CL300 CL300	SOCW SOCW T/SW T/SW RF RF RF RF	LTCS BODY WITH 316 SS TRIM (API #10) LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT LTCS BODY WITH 316 SS TRIM (API #10) LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT LTCS BODY WITH 316 SS TRIM (API #10) LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT, GEAR OP.	TAG PVVGB68S0710 ACLD0ZZ PVVGF68S0710 ACLD0ZZ PVVGB68D0710 ACLD0ZZ PVVGF68D0710 ACLD0ZZ PVVGB63R1110 ACLD0ZZ PVVGF63R1110 BCLC0ZZ PVVGF63R1110 BCLC0ZZ PVVGF63R1110 BCLC5ZZ	143, 277 275 4 4, 275 24 43	
GLOBE	½" - 1 ½" 2" –8" 10" – 14"	CL800 CL300 CL300	SOCW RF RF	LTCS BODY WITH 316 SS TRIM (API #10) LTCS BODY WITH 316 SS TRIM (API #10) LTCS BODY WITH 316 SS TRIM (API #10), GEAR OP.	PVVLB68S0710A CLD0ZZ PVVLB62R1110 BCLC0ZZ PVVLB62R1110 BCLC5ZZ		
LIFT CHECK	½" - 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API # 10) (HORIZ. ONLY)	PVVCL68S0710 NCLD0ZZ		
WAFER DUAL PLATE CHECK	2" – 24"	CL300	RF	LTCS BODY WITH 316 SS PLATES, METAL SEATS	PVVCS62W1110 NCLA0ZZ		
WAF BUTTERFLY	3" –4" 6" – 24"	CL300 CL300	RF RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUT62W1171 BCLG0ZZ PVVUT62W1171 BCLG3ZZ	27,43,326 27,43,326	
LUG BUTTERFLY	3" –4"	CL300	RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUT62L1171B CLG0ZZ	27,43,326	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
	6" - 24"	CL300	RF	LTCS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUT62L1171B CLG3ZZ	27,43,326
BOLT/NUTS	ALL	CL300		ALLOY STEEL, ASTM A320 GRADE L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS		
	ALL	CL600		ALLOY STEEL, ASTM A320 GRADE L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS		
GASKETS SPIRAL WOUND	1/2" - 24"	CL300		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, 304SS INTER RING AND CENTERING RING, ASME B16.20 / ASME B16.5	144	
	1/2" - 24"	CL600		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, 304SS INTER RING AND CENTERING RING, ASME B16.20 / ASME B16.5		

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD	SIZES 1/2" - 2" 3" - 24" 1/2" - 1 1/2" 2" - 24"

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
4	Socket-weld x Threaded gate valve shall be used for valved pressure, instrument, vent and drain connections only. Piping system end shall be socketweld and instrument end shall be threaded.	
24	Use when required at flanged equipment nozzles, or where noted on P&IDs or Modules.	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
43	Only to be used as indicated on the P&IDs.	
143	In shop and/or field fabrication welds (girth, branch and fillet), all weld metal deposits and heat affected zone (HAZ) shall meet impact test requirements for low temperature service (at -48°C for semi-cryogenic S.S. "K" specs and -105°C for cryogenic S.S. "L" specs) per Paragraph 323 of ASME B31.3.	
144	Flanges mated to equipment with aluminum flanges shall use 1/8 in. soft Durabla (Durlon 9000) gaskets and one piece of 1/8 in. thick NEMA G-10 Gr. H22033 glass cloth epoxy resin isolation washer with 1/32 in. thick full length NEMA G-10 isolation sleeve and 1/8 in. thick steel washer between the nut and isolation washer on the aluminum flanges. Flange bolt torque per Manufacturer's recommendations (the G-10 material is limited to 121°C maximum).	
275	Only to be used where full port gate valve is required as indicated on P&ID's.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drains requiring plugs, connections to tube fittings and for connections to threaded proprietary items. See referenced Piping Modules (Section VI) and Standard Drawings.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
B2E	R00

SERVICES	Hydrogen					REV

PIPING MATERIAL	L.T. CARBON STEEL (IMPACT TESTED)	VALVE MATERIAL	L.T. CARBON STEEL	WELDEXAMINATION		REV
RATING / FACING	ASME CLASS 300 / RF	TRIM MATERIAL	316 SS (API# 10)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	3.0 mm. (0.118")	CODE	ASME B31.3	BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (MPa)*

REV

SIZE

TEM P °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
-38 to 38	5.102	5.102	5.102	5.102	5.102	5.102	5.102	5.102	5.102	5.102	5.102					
93	4.654	4.654	4.654	4.654	4.654	4.654	4.654	4.654	4.654	4.654	4.654					

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES
RTFE SEAT RATINGS (Mpa)

BUTTERFLY
RTFE SEAT RATINGS(Mpa)

TEMP(C)	≤ NPS 2	>NPS2; ≤ NPS4	REV	TEMP(C)	ALL SIZES	REV
-38 to 38	51.02	51.0		-38 to 38		
93	46.54	35.5		93		
149				149		
204				204		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" – 1 1/2" 2" 3" – 12"	160 80 STD	BE BE BE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M	143	
NIPPLES PBE	1/2" – 1 1/2"	160	BBE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M	143	
SWAGES CON&ECC CON&ECC	1/2" – 1 1/2" 1/2" – 4"	160 AS PIPE	BBE BBE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95 LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95	143 702	
OLET WELDOLET	1/2" – 8"		BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, MSS-SP-97	1001	
CAP	1/2" – 3/4" 1" – 1 1/2" 2" – 12"	AS PIPE AS PIPE AS PIPE	BW BW BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.9 LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		
90° ELBOW LR	1/2" 3/4" – 1 1/2" 2" – 12"	AS PIPE AS PIPE AS PIPE	BW BW BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.9 LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		

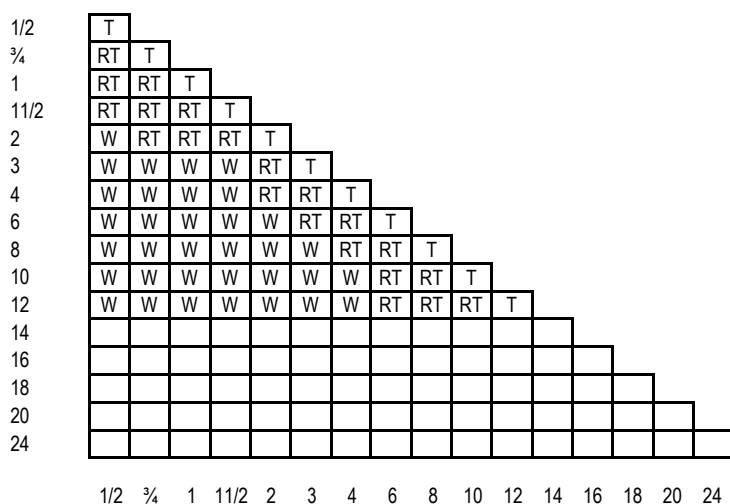
ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
45° ELBOW	1/2" 3/4" – 1 1/2" 2" – 12"	AS PIPE AS PIPE AS PIPE	BW BW BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.9 LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
TEE	1/2" 3/4" – 1 1/2" 2" – 12"	AS PIPE AS PIPE AS PIPE	BW BW BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.9 LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
RED TEE	3/4" – 1 1/2" 2" – 12"	AS PIPE AS PIPE	BW BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
REDUCER	CON ECC	2" – 12" 2" – 12"	AS PIPE AS PIPE	BW BW	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		
FLANGE						143	
WELDNECK	1/2" – 12"	CL300	M / FM	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5 THICKNESS AS PIPE		278	
BLIND FLG	1/2"- 12" 1/2"- 12"	CL300 CL600	FM FM	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5 LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.5		278	
MISCELLANEOUS FIGURE 8 BLANK	1" – 12"	CL 300	M&FM	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48			
VALVES						143,277	
GLOBE	1/2" – 1 1/2" 2" – 8" 10" – 12"	CL800 CL300 CL300	BW FM FM	LTCS BODY WITH 316 SS TRIM (API #10) LTCS BODY WITH 316 SS TRIM (API #10) LTCS BODY WITH 316 SS TRIM (API #10), GEAR OP.	TAG PVVLB68B0710F CED0BZ PVVLB63F1110F CEK0BZ PVVLB63F1110F CEK5BZ PVVCL68B0710 NCED0BZ PVVCS63F1110 NCEL0BZ		
LIFT CHECK	1/2" – 1 1/2"	CL800	BW	LTCS BODY WITH 316 SS TRIM (API # 10) (HORIZ. ONLY)			
SWING CHECK	2" – 12"	CL300	FM	LTCS BODY WITH 316 SS TRIM (API # 10)			
BALL	1/2" – 1 1/2" 2"	CL300 CL300	FM FM	LTCS BODY WITH 316 SS BALL & STEM, TRIM (API #10), FULL PORT LTCS BODY WITH 316 SS BALL & STEM, TRIM (API #10), FULL PORT	PVVBA63F0710F CEX0BZ PVVBA63F1110F CEX0BZ		
BOLTING	ALL ALL	CL300 CL600		ALLOY STEEL, ASTM A320 GRADE L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS ALLOY STEEL, ASTM A320 GRADE L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS			
GASKET SPIRAL WOUND	1/2"-8" 1/2"-8"	CL300 CL600		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, SS INNER RING, ASME B16.20 / ASME B16.5 SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, SS INNER RING, ASME B16.20 / ASME B16.5		144	

	NOTES			REV
JOINT CONNECTS			SIZES	
MAINTENANCE		FLANGES	1/2" – 1 1/2"	
MAINTENANCE		FLANGES	2" – 12"	
NON-MAINTENANCE				
NON-MAINTENANCE		BUTTWELD	1/2" – 12"	

NOTE	SPECIFIC PIPING NOTES	REV
143	In shop and/or field fabrication welds (girth, branch and fillet), all weld metal deposits and heat affected zone (HAZ) shall meet impact test requirements for low temperature service (at -48°C for L.T.C.S., at -105°C for semi-cryogenic S.S. "K" specs and for cryogenic S.S. "L" specs) per Paragraph 323 of ASME B31.3.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME Class 600 flanges shall only be used to match higher rated flange connections at control valves and special equipment.	
701	The max speed of hydrogen should not exceed 10/s.	
702	Applied in the case where the small end is 1 1/2 inch and smaller, and the lager end is two inch and lager.	
1001	The nominal size, schedule of main pipes and branches must be indicated when order is placed.	

BRANCH CONNECTION CHART	REV
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HEADER SIZE



BRANCH SIZE

BRANCH LEGEND		NOTES	REV
T	Tee (B.W.)		
RT	Reducing Tee (B.W.)		
W	Weldolet (PWHT required at branch attachment welds when the thickness through the weld in any plane through the branch is greater than 1.5" for P1 material per ASME B31.3.)		
P	Stub-in with Reinforcing Pad shall be sit-in type as shown in detail "2" or "4" of Figure 328.5.4D of ASME B31.3 (Note d.)		

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
b.	For 45° branch connections, see 45° Branch Table. The exception is the branches from heater wash cone lines on 'TLE' type heaters where 45° lateral stub-in Branch Weld is acceptable for atmospheric drain conditions.	
c.	Weldolet may be used in lieu of a stub-in Branch Weld if verified and approved by pressure design and flexibility analysis and shown on the piping design drawing	
d.	A pad has been selected to provide reinforcement for this branch connection subject to the most severe coincident condition of internal pressure and fluid temperature seen in this specification. However, the specific pressure temperature condition for this connection may eliminate the need for reinforcement. In addition, when the diameter to-thickness ratio is restricted by paragraph 304.3.1 (3) (b) of ASME B31.3, finite element analysis of the Pad design must be performed by the Contractor Stress Engineer.	
e.	PWHT required at all branch attachment welds when the thickness through the weld in any plane through the branch is greater than 38.1 mm (1.5") for P1 material per ASME B31.3 and greater than 19 mm (0.75") for P1 materials per ASME B31.1.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
B21A	R00

SERVICES					REV
	Filter outlet water				
	Oxygen Scavenger				

PIPING MATERIAL	304 S.S.	VALVE MATERIAL	316 SS	WELD EXAMINATION		REV
RATING / FACING	ASME CLASS 300 / RF	TRIM MATERIAL	316 SS (API# 10)	STRESS RELIEF	NOT REQUIRED	
CORROSION ALLOWANCE	0.500 mm (0.02")	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*

REV

SIZE

TEMP P °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
-28 to 38	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60
93	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES
 RTFE SEAT RATINGS (bar)

TEMP(C)	≤ NPS 2	>NPS2; ≤ NPS4	REV
-28 to 38	49.60	49.60	
93	41.86	35.14	
149			
204			

BUTTERFLY
 RTFE SEAT RATINGS(bar)

TEMP(C)	ALL SIZES	REV
-28 to 38		
93		
149		
204		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" – 1 1/2" 2" 3" – 12" 14" – 24"	40S 40S 40S 40S	PE BE BE BE	SS, ASTM A312 GRADE TP 304, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304, EFW, ASME B36.19M Ej=0.85		
NIPPLES	1/2" – 1 1/2"	40S	PBE	SS, ASTM A312 GRADE TP 304, SMLS, ASME B36.19M		
SWAGES CON&ECC CON&ECC	1/2" – 1 1/2" 1/2" – 4"	40S AS PIPE	PBE BLE/P SE	SS, ASTM A403 GRADE WP304-S, SMLS, MSS-SP-95 SS, ASTM A403 GRADE WP304-S, SMLS, MSS-SP-95	702	
OLET SOCKOLET	1/2" – 1 1/2"	CL3000	SOCW	SS, ASTM A182 GRADE F304, MSS-SP-97		
REINFORCING PAD	2" – 12" 14" – 24"	AS PIPE AS PIPE		SS, ASTM A312 GRADE TP 304, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304, EFW, ASME B36.19M Ej=0.85		
PLUG	1/2" – 2"	THRD		SS, ASTM A182 GRADE F304, ASME B16.11, ROUND HEAD	279	
CAP	1/2" – 1 1/2" 2" – 12"	CL3000 AS PIPE	SOCW BW	SS, ASTM A182 GRADE F304, ASME B16.11 SS, ASTM A403 GRADE WP304-S, ASME B16.9		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
90° ELBOW	1/2" – 1 1/2" 2" – 12" 14" – 24"	CL3000 AS PIPE AS PIPE	SOCW BW BW	SS, ASTM A182 GRADE F304, ASME B16.11 SS, ASTM A403 GRADE WP304-S, ASME B16.9 SS, ASTM A403 GRADE WP304-W, ASME B16.9		
45° ELBOW	1/2" – 1 1/2" 2" – 12" 14" – 24"	CL3000 AS PIPE AS PIPE	SOCW BW BW	SS, ASTM A182 GRADE F304, ASME B16.11 SS, ASTM A403 GRADE WP304-S, ASME B16.9 SS, ASTM A403 GRADE WP304-W, ASME B16.9		
COUPLING COUPLING COUPLING(RED)	1/2" – 1 1/2" 1/2" – 1 1/2"	CL3000 CL3000	SOCW SOCW	SS, ASTM A182 GRADE F304, ASME B16.11 SS, ASTM A182 GRADE F304, ASME B16.11		
TEE	1/2" – 1 1/2" 2" – 12" 14" – 24"	CL3000 AS PIPE AS PIPE	SOCW BW BW	SS, ASTM A182 GRADE F304, ASME B16.11 SS, ASTM A403 GRADE WP304-S, ASME B16.9 SS, ASTM A403 GRADE WP304-W, ASME B16.9		
RED TEE	1/2" – 1 1/2" 2" – 12" 14" – 24"	CL3000 AS PIPE AS PIPE	SOCW BW BW	SS, ASTM A182 GRADE F304, ASME B16.11 SS, ASTM A403 GRADE WP304-S, ASME B16.9 SS, ASTM A403 GRADE WP304-W, ASME B16.9		
REDUCER						
CON	2" – 12" 14" – 24"	AS PIPE AS PIPE	BW BW	SS, ASTM A403 GRADE WP304-S, ASME B16.9 SS, ASTM A403 GRADE WP304-W, ASME B16.9		
ECC	2" – 12" 14" – 24"	AS PIPE AS PIPE	BW BW	SS, ASTM A403 GRADE WP304-S, ASME B16.9 SS, ASTM A403 GRADE WP304-W, ASME B16.9		
FLANGE						
SOCKETWELD	1/2" – 1 1/2" 1/2" – 1 1/2"	CL300 CL600	RF RF	SS, ASTM A182 GRADE F304, ASME B16.5 SS, ASTM A182 GRADE F304, ASME B16.5	278	
WELDNECK	2" – 24" 2" – 24"	CL300 CL600	RF RF	SS, ASTM A182 GRADE F304, ASME B16.5 THICKNESS AS PIPE SS, ASTM A182 GRADE F304, ASME B16.5 THICKNESS AS PIPE	278	
BLIND FLG	1/2" – 24" 1/2" – 24"	CL300 CL600	RF RF	SS, ASTM A182 GRADE F304, ASME B16.5 SS, ASTM A182 GRADE F304, ASME B16.5	278	
MISCELLANEOUS						
FIGURE 8 BLANK	1" – 12"	CL300	FF	SS, ASTM A240 GRADE 304, ASME B16.48		
PADDLE BLANK	14" – 24"	CL300	FF	SS, ASTM A240 GRADE 304, ASME B16.48		
PADDLE SPACER	14" – 24"	CL300	FF	SS, ASTM A240 GRADE 304, ASME B16.48		
VALVES						
GATE	1/2" – 1 1/2"	CL800	SOCW	316 SS BODY WITH 316 SS TRIM (API #10)	TAG PVVGB68S3810 FCED0ZZ	277
	1/2" – 1 1/2"	CL300	RF	316 SS BODY WITH 316 SS TRIM (API #10)	PVVGB63R3810 FCED0ZZ	24
	2" – 10"	CL300	RF	316 SS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVGB63R5410 FCEC0ZZ	
	12" – 24"	CL300	RF	316 SS BODY WITH 316 SS TRIM (API #10), FULL PORT, GEAR OP.	PVVGB63R5410 FCEC5ZZ	
GLOBE	1/2" – 1 1/2"	CL800	SOCW	316 SS BODY WITH 316 SS TRIM (API #10)	PVVLB68S3810F CED0ZZ	
	2" – 8"	CL300	RF	316 SS BODY WITH 316 SS TRIM (API #10)	PVVLB63R5410F CEK0ZZ	
	10" – 12"	CL300	RF	316 SS BODY WITH 316 SS TRIM (API #10), GEAR OP.	PVVLB63R5410F CEK5ZZ	
LIFT CHECK	1/2" – 1 1/2"	CL800	SOCW	316 SS BODY WITH 316 SS TRIM (API #10) (HORIZ. ONLY)	PVVCL68S3810 NCED0ZZ	
SWING CHECK	1/2" – 1 1/2"	CL800	SOCW	316 SS BODY WITH 316 SS TRIM (API #10)	PVVCS68S3810 NCEL0ZZ	
SWING CHECK	2" – 24"	CL300	RF	316 SS BODY WITH 316 SS TRIM (API #10)	PVVCS63R5410 NCEL0ZZ	
BALL	1/2" – 4"	CL300	RF	316 SS BODY WITH 316 SS BALL & STEM, RTFE SEATS, FULL PORT	PVVA63R5410 FCEX0ZZ	27,326

BRANCH LEGEND		NOTES	REV
T	Tee (B.W. for two inch and larger, S.W. for 1 11/2" and smaller)		
A	Sockolet		
RT	Reducing Tee (B.W. for two inch and larger, S.W. for 1 11/2" and smaller)		
S	Stub-in without Reinforcing Pad shall be set -in type as shown in detail "2" or "4" of Figure 328.5.4D of ASME B31.3		D01
P	Stub-in with Reinforcing Pad shall be sit-in type as shown in detail "2" or "4" of Figure 328.5.4D of ASME B31.3 (Note d.)		

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
b.	For 45° branch connections, except that the branches from TLE's heater wash cone lines where 45° lateral stub-in is acceptable for atmospheric drain conditions.	
c.	Weldolet may be used in lieu of stub-in if required and verified by flexibility analysis and shown on the piping design drawing.	
d.	A PAD has been selected to provide reinforcement for this branch connection subject to the most severe coincident condition of internal pressure and fluid temperature seen in this specification. However, the specific pressure-temperature condition for this connection may eliminate the need for reinforcement. In addition, when the diameter-to-thickness ratio is restricted by paragraph 304.3.1 (3) (b) of ASME B31.3, finite element analysis (FEA) shall be performed .	

PROJECT NUMBER	01000	PIPING CLASS	REV
DOCUM. NO.	01000-MA-SPC-0002	B21D	R00
CONTR. DOCUM. NO.			

SERVICES	Lube oil(LO)					REV

PIPING MATERIAL	304 SS	VALVE MATERIAL	316 SS	WELD EXAMINATION		REV
RATING / FACING	ASME CLASS 300 / RF	TRIM MATERIAL	316 SS (API# 10)	STRESS RELIEF	NOT REQUIRED	
CORROSION ALLOWANCE	0.500 mm (0.02")	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*

REV

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
-28 to 38	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60					
93	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86					

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES
 RTFE SEAT RATINGS (bar)

TEMP(C)	≤ NPS 2	>NPS 2; ≤NPS4	REV
-28 to 38	49.60	49.60	
93	41.86	35.14	
149			
204			

BUTTERFLY
 RTFE SEAT RATINGS(bar)

TEMP(C)	ALL SIZES	REV
-28 to 38		
93		
149		
204		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" – 1 1/2" 2" 3" – 12"	40S 40S 40S	BE BE BE	SS, ASTM A312 GRADE TP 304, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304, SMLS, ASME B36.19M		
SWAGES CON&ECC CON&ECC	1/2" – 1 1/2" 1/2" – 4"	40S AS PIPE	BBE BBE	SS, ASTM A403 GRADE WP304-S, SMLS, MSS-SP-95 SS, ASTM A403 GRADE WP304-S, SMLS, MSS-SP-95	702	
OLET WELDOLET	1/2" – 1 1/2"		BW	SS, ASTM A182 GRADE F304, MSS-SP-97	1001	
REINFORCING PAD	2" – 12"	AS PIPE		SS, ASTM A312 GRADE TP 304, SMLS, ASME B36.19M		
CAP	1/2" – 1 1/2" 2" – 12"	AS PIPE AS PIPE	BW BW	SS, ASTM A182 GRADE F304, ASME B16.9 SS, ASTM A403 GRADE WP304-S, ASME B16.9		
90° ELBOW	1/2" – 1 1/2" 2" – 12"	AS PIPE AS PIPE	BW BW	SS, ASTM A182 GRADE F304, ASME B16.9 SS, ASTM A403 GRADE WP304-S, ASME B16.9		
45° ELBOW	1/2" – 1 1/2" 2" – 12"	AS PIPE AS PIPE	BW BW	SS, ASTM A182 GRADE F304, ASME B16.9 SS, ASTM A403 GRADE WP304-S, ASME B16.9		
TEE	1/2" – 1 1/2" 2" – 12"	AS PIPE AS PIPE	BW BW	SS, ASTM A182 GRADE F304, ASME B16.9 SS, ASTM A403 GRADE WP304-S, ASME B16.9		

NOTE	SPECIFIC PIPING NOTES	REV
27	TFE,PTFE seated valves are limited to maximum 204 °C .	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME Class 600 flanges shall only be used to match higher rated flange connections at control valves and special equipment.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	
702	Applied in the case where the small end is 1 1/2 inch and smaller, and the lager end is two inch and lager.	
1001	The nominal size, schedule of main pipes and branches must be indicated when order is placed.	

HEADER SIZE

BRANCH SIZE

PIPING BRANCH NOTES

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
B21K	R00

SERVICES	Propylene	Ethylene		REV

PIPING MATERIAL	304/304L SS (IMPACT TESTED)	VALVE MATERIAL	316 SS	WELD EXAMINATION		REV
RATING / FACING	ASME CLASS 300/ RF	TRIM MATERIAL	316SS (TRIM#10)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	0.5mm	CODE	ASME B31.3	BENDING	PER PROJECT	

**FLANGE RATING &
MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)*
SIZE**

TEMP (°C)	½	¾	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20	24
-48 TO 38	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60
93	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86
149	37.07	37.07	37.07	37.07	37.07	37.07	37.07	37.07	37.07	37.07	37.07	37.07	37.07	37.07	37.07	37.07
204	34.34	34.34	34.34	34.34	34.34	34.34	34.34	34.34	34.34	34.34	34.34	34.34	34.34	34.34	34.34	34.34

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

蝶阀 BUTTERFLY
RTFE 阀座温压表 RTFE SEAT RATINGS(Bar)

TEMP(C)	ALL SIZES	REV
-48 to 38	49.60	
93	41.86	
149	26.87	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	½" – 1½" 2" 3" – 4" 6" – 12" 14" – 18" 20" – 24"	40S 40S 10S 40S STD XS	PE BE BE BE BE BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, EFW, ASME B36.10M (X-RAY LONG SEAM, Ej=1) SS, ASTM A312 GRADE TP 304/304L, EFW, ASME B36.10M (X-RAY LONG SEAM Ej=1)	143	
NIPPLE	TOE ½" – 1½" TBE ½" – 1½" PBE ½" – 1½"	40S 40S 40S	TOE TBE PBE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M	143	
SWAGE	CON ½" – 1½" ½" – 1½" ECC ½" – 1½" ½" – 1½"	40S 40S 40S 40S	TSE BBE TSE BBE	SS, ASTM A403 GRADE WP304/304L-S, MSS-SP-95 SS, ASTM A403 GRADE WP304/304L-S, MSS-SP-95 SS, ASTM A403 GRADE WP304/304L-S, MSS-SP-95 SS, ASTM A403 GRADE WP304/304L-S, MSS-SP-95	1, 143	
COUPLING	FULL COUPLING ½" – 1½" FULL COUPLING ½" – 2"	CL3000 CL3000	SOCW THRD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A182 GRADE F304/304L, ASME B16.11	143 279	
OLET	SOCKOLET ½" – 1½" THREDOLET ½" – 2" WELDOLET 2" – 8"	CL3000 CL3000 AS PIPE	WELD WELD WELD	SS, ASTM A182 GRADE F304/304L, MSS-SP-97 SS, ASTM A182 GRADE F304/304L, MSS-SP-97 SS, ASTM A182 GRADE F304/304L, MSS-SP-97	143 279	
PLUG	ROUND HEAD ½" – 2"		THRD	SS, ASTM A182 GRADE F304/304L, ASME B16.11, ROUND HEAD	279	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
CAP	½" – 1½"	CL3000	SOCW	SS, ASTM A182 GRADE F304/304L, ASME B16.11		143	
	½" – 2"	CL3000	THRD	SS, ASTM A182 GRADE F304/304L, ASME B16.11		279	
	2" – 24"	AS PIPE	BW	SS,ASTM A403 GRADE WP 304/304L-S,ASME B16.9			
90° ELBOW SW LR	½" – 1½"	CL3000	SOCW	SS, ASTM A182 GRADE F304/304L, ASME B16.11		143	
	2" – 12"	AS PIPE	BW	SS,ASTM A403 GRADE WP304/304L-S,ASME B16.9			
	14" – 24"	AS PIPE	BW	SS,ASTM A403 GRADE WP304/304L-W,ASME B16.9			
45° ELBOW	½" – 1½"	CL3000	SOCW	SS, ASTM A182 GRADE F304/304L, ASME B16.11		143	
	2" – 12"	AS PIPE	BW	SS,ASTM A403 GRADE WP304/304L-S,ASME B16.9			
	14" – 24"	AS PIPE	BW	SS,ASTM A403 GRADE WP304/304L-W,ASME B16.9			
TEE	½" – 1½"	CL3000	SOCW	SS, ASTM A182 GRADE F304/304L, ASME B16.11		143	
	2" – 12"	AS PIPE	BW	SS,ASTM A403 GRADE WP304/304L-S,ASME B16.9			
	14" – 24"	AS PIPE	BW	SS,ASTM A403 GRADE WP304/304L-W,ASME B16.9			
RED TEE	½" – 1½"	CL3000	SOCW	SS, ASTM A182 GRADE F304/304L, ASME B16.11			
	2" – 12"	AS PIPE	BW	SS,ASTM A403 GRADE WP304/304L-S,ASME B16.9			
	14" – 24"	AS PIPE	BW	SS,ASTM A403 GRADE WP304/304L-W,ASME B16.9			
REDUCER CON ECC	2" – 12"	AS PIPE	BW	SS,ASTM A403 GRADE WP304/304L-S,ASME B16.9		143	
	14" – 24"	AS PIPE	BW	SS,ASTM A403 GRADE WP304/304L-W,ASME B16.9			
	2" – 12"	AS PIPE	BW	SS,ASTM A403 GRADE WP304/304L-S,ASME B16.9			
	14" – 24"	AS PIPE	BW	SS,ASTM A403 GRADE WP304/304L-W,ASME B16.9			
FLANGE SOCKETWELD WELDNECK BLIND FLG DRIP RING BLANK+SPACER	½" – 1½"	CL300	RF	SS,ASTM A182 GRADE F304/304L,ASME B16.5		143	
	2" – 24"	CL300	RF	SS,ASTM A182 GRADE F304/304L,ASME B16.5 THICKNESS AS PIPE			
	½" – 24"	CL300	RF	SS,ASTM A182 GRADE F304/304L,ASME B16.5			
	1" – 12"	CL300	FF	SS,ASTM A240 GRADE 304/304L(SOLUTION HEAT TREATED),1-1/2"THICK W/3/4"SOCW T&S(STANDARD DRWGS)			
	1" – 24"	CL300	FF	SS,ASTM A240 GRADE 304/304L (SOLUTION HEAT TREATED) ASME B16.48		266	
VALVES GATE GLOBE SWING CHECK	½" – 1½"	CL800	SOCW	SS BODY WITH 316 SS TRIM(API#10)	TAG PVVGB68S3810 ACND0ZZ	143,277 146,148	
	½" – 1½"	CL800	SOCW	SS BODY WITH 316 SS TRIM(API#10),FULL PORT	PVVG68S3810 ACND0ZZ	146,148, 275	
	½" – 1½"	CL800	T/SW	SS BODY WITH 316 SS TRIM(API#10)	PVVGB68D3810 ACNM0ZZ	4, 146,148, 279	
	½" – 1½"	CL800	T/SW	SS BODY WITH 316 SS TRIM(API#10),FULL PORT	PVVG68D3810 ACNM0ZZ	4,275,279, 146,148	
	½" – 2"	CL300	RF	SS BODY WITH 316 SS TRIM(API#10)	PVVGB63R3810 ACND0ZZ	24, 146,148	
	2"	CL300	RF	SS BODY WITH 316 SS TRIM(API#10),FULL PORT	PVVG63R5410 BCND0ZZ	146,148	
	3" – 10"	CL300	RF	SS BODY WITH 316 SS TRIM(API#10),FULL PORT	PVVG63R5410 BCNC0ZZ	146,148	
	12" – 24"	CL300	RF	SS BODY WITH 316 SS TRIM(API#10),FULL PORT , GEAR OP.	PVVG63R5410 BCNC5ZZ	146,148	
	½" – 1½"	CL800	SOCW	SS BODY WITH 316 SS TRIM(API#10)	PVVL68S3810A CND0ZZ	148	
	2" – 8"	CL300	RF	SS BODY WITH 316 SS TRIM(API#10)	PVVL63R5410 BCNC0ZZ	148	
	10" – 14"	CL300	RF	SS BODY WITH 316 SS TRIM(API#10) , GEAR OP.	PVVL63R5410 BCNC5ZZ	148	
	½" – 1½"	CL800	SOCW	SS BODY WITH 316 SS TRIM(API#10)	PVVC68S3810 NCND0ZZ		

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
4	Socket-weld x Threaded gate valve shall be used for valved pressure, instrument, vent and drain connections only. Piping system end shall be socketweld and instrument end shall be threaded.	
24	Use when required at flanged equipment nozzles, or where noted on P&IDs or Modules.	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
43	Only to be used as indicated on the P&IDs.	
143	In shop and/or field fabrication welds (girth, branch and fillet), all weld metal deposits and heat affected zone (HAZ) shall meet impact test requirements for low temperature service (at -48°C for L.T.C.S., at -48°C for semi-cryogenic S.S. "K" specs and -105°C for cryogenic S.S. "L" specs) per Paragraph 323 of ASME B31.3.	
144	Flanges mated to equipment with aluminum flanges shall use 1/8 in. soft Durabla (Durlon 9000) gaskets and one piece of 1/8 in. thick NEMA G-10 Gr. H22033 glass cloth epoxy resin isolation washer with 1/32 in. thick full length NEMA G-10 isolation sleeve and 1/8 in. thick steel washer between the nut and isolation washer on the aluminum flanges. Flange bolt torque per Manufacturer's recommendations (the G-10 material is limited to 121°C maximum).	
146	A 3.175 mm (1/8") diameter bleed hole shall be provided in the disc from the cavity of the valve to the "vent" side of the valve only. An arrow with "vent" shall be stamped on the outside of the valve body indicating the direction of venting. (See P&ID's for direction of venting for various valve installations of cryogenic gate valves)	
147	Extended bonnet valves shall have the minimum extended bonnet lengths from the centerline of the valve bore to the bottom of the packing (stuffing box) as stated in 3.3 of the Valve Specification Notes.	
148	Unless otherwise noted on the modules, valves shall not be installed in vertical lines. Valves in horizontal lines shall be installed with stem in vertical position or at least 45 degrees vertically from the horizontal centerline of the pipe.	
275	Only to be used where full port gate valve is required as indicated on P&ID's.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drains requiring plugs, connections to tube fittings and for connections to threaded proprietary items.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
B21L	R00

SERVICES	Ethylene Refrigerant (Liquid)	Cold Liquid Drain		REV
	Process	Ethylene	Ethylene flare	

PIPING MATERIAL	304/304L SS (IMPACT TESTED)	VALVE MATERIAL	316 SS	WELDEXAMINATION		REV
RATING / FACING	ASME CLASS 300/ RF	TRIM MATERIAL	316SS (TRIM#10)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	0.5mm	CODE	ASME B31.3	BENDING	PER PROJECT	

**FLANGE RATING &
MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)*
SIZE**

TEMP (°C)	½	¾	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20	24
-105 TO 38	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60	49.60
93	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86	41.86
149	37.07	37.07	37.07	37.07	37.07	37.07	37.07	37.07	37.07	37.07	37.07	37.07	37.07	37.07	37.07	37.07
204	34.34	34.34	34.34	34.34	34.34	34.34	34.34	34.34	34.34	34.34	34.34	34.34	34.34	34.34	34.34	34.34

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

蝶阀 BUTTERFLY
RTFE 阀座温压表 RTFE SEAT RATINGS(Bar)

TEMP(C)	ALL SIZES	REV
-105 to 38	49.60	
93	41.86	
149	26.87	
204	6.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	½" – 1½" 2" 3" – 4" 6" – 12" 14" – 18" 20" – 24"	40S 40S 10S 40S STD XS	PE BE BE BE BE BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, EFW, ASME B36.10M(X-RAY LONG SEAM, E _j =1) SS, ASTM A312 GRADE TP 304/304L, EFW, ASME B36.10M (X-RAY LONG SEAM E _j =1)	143	
NIPPLE	TOE ½" – 1½" TBE ½" – 1½" PBE ½" – 1½"	80S 80S 40S	TOE TBE PBE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M	143	
SWAGE	CON ½" – 1½" ½" – 1½" ECC ½" – 1½" ½" – 1½"	80S 40S 80S 40S	TSE BBE TSE BBE	SS, ASTM A403 GRADE WP304/304L-S, MSS-SP-95 SS, ASTM A403 GRADE WP304/304L-S, MSS-SP-95 SS, ASTM A403 GRADE WP304/304L-S, MSS-SP-95 SS, ASTM A403 GRADE WP304/304L-S, MSS-SP-95	1, 143	
COUPLING	FULL COUPLING ½" – 1½" FULL COUPLING ½" – 2"	CL3000 CL3000	SOCW THRD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A182 GRADE F304/304L, ASME B16.11	143 279	
OLET	SOCKOLET ½" – 1½" THREDOLET ½" – 2" WELDOLET 2" – 8"	CL3000 CL3000 AS PIPE	WELD WELD WELD	SS, ASTM A182 GRADE F304/304L, MSS-SP-97 SS, ASTM A182 GRADE F304/304L, MSS-SP-97 SS, ASTM A182 GRADE F304/304L, MSS-SP-97	143 279	
PLUG	ROUND HEAD ½" – 2"		THRD	SS, ASTM A182 GRADE F304/304L, ASME B16.11, ROUND HEAD	143	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
SWING CHECK	2" – 24"	CL300	RF	SS BODY WITH 316 SS TRIM(API#10)	PVVC63R5410 NCNC0ZZ		
WAF BUTTERFLY	3" – 4"	CL300	RF	SS BODY WITH 316 SS DISC.RTFE/316 SS SEAT, EXTENDED BONNET	PVVUT63R5464 BCNG0EV	27,43,147, 148,326	
	6" – 24"	CL300	RF	SS BODY WITH 316 SS DISC.RTFE/316 SS SEAT, EXTENDED BONNET, GEAR OP.	PVVUT63R5464 BCNG3EV	27,43,147, 148,326	
LUG BUTTERFLY	3" – 4"	CL300	RF	SS BODY WITH 316 SS DISC.RTFE/316 SS SEAT, EXTENDED BONNET	PVVUT63R5464 BCNG0EV	27,43,147, 148,326	
	6" – 24"	CL300	RF	SS BODY WITH 316 SS DISC.RTFE/316 SS SEAT, EXTENDED BONNET, GEAR OP.	PVVUT63R5464 BCNG3EV	27,43,147, 148,326	
BOLT/NUTS	ALL	CL300		ALLOY STEEL, ASTM A320 GRADE B8 CL2 WITH ASTM A194 GRADE 8 HEAVY HEX NUTS			
GASKETS SPIRAL WOUND	1/2" – 24"	CL300		SPIRAL WOUND, 304/304L SS W/GRAPHITE FILLER, 304/304L SS CENTERING RING AND INTER RING, ASME B16.20 / ASME B16.5		144	

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE			
		FLANGES	
		FLANGES	
		SOCKETWELD	
		BUTTWELD	

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
4	Socket-weld x Threaded gate valve shall be used for valved pressure, instrument, vent and drain connections only. Piping system end shall be socketweld and instrument end shall be threaded.	
24	Use when required at flanged equipment nozzles, or where noted on P&IDs or Modules.	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
43	Only to be used as indicated on the P&IDs.	
143	In shop and/or field fabrication welds (girth, branch and fillet), all weld metal deposits and heat affected zone (HAZ) shall meet impact test requirements for low temperature service (at -48°C for L.T.C.S., at -48°C for semi-cryogenic S.S. "K" specs and -105°C for cryogenic S.S. "L" specs) per Paragraph 323 of ASME B31.3.	
144	Flanges mated to equipment with aluminum flanges shall use 1/8 in. soft Durabla (Durlon 9000) gaskets and one piece of 1/8 in. thick NEMA G-10 Gr. H22033 glass cloth epoxy resin isolation washer with 1/32 in. thick full length NEMA G-10 isolation sleeve and 1/8 in. thick steel washer between the nut and isolation washer on the aluminum flanges. Flange bolt torque per Manufacturer's recommendations (the G-10 material is limited to 121°C maximum).	
146	A 3.175 mm (1/8") diameter bleed hole shall be provided in the disc from the cavity of the valve to the "vent" side of the valve only. An arrow with "vent" shall be stamped on the outside of the valve body indicating the direction of venting. (See P&ID's for direction of venting for various valve installations of cryogenic gate valves)	
147	Extended bonnet valves shall have the minimum extended bonnet lengths from the centerline of the valve bore to the bottom of the packing (stuffing box) as stated in 3.3 of the Valve Specification Notes.	
148	Unless otherwise noted on the modules, valves shall not be installed in vertical lines. Valves in horizontal lines shall be installed with stem in vertical position or at least 45 degrees vertically from the horizontal centerline of the pipe.	
149	No extension bonnet required for valved vents, drains and not used in normal service.	
266	Blind and spacer shall be used instead of Figure 8 in cryogenic services.	
275	Only to be used where full port gate valve is required as indicated on P&ID's.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drains requiring plugs, connections to tube fittings and for connections to threaded proprietary items.	
326	This resilient seated valve does not in all cases meet full flange rating.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCU. NO.	

PIPING CLASS	REV
B27A	R00

SERVICES	High Temperature Process HC (with Vacuum)		REV

PIPING MATERIAL	TP 321H Stainless Steel (Note 508)	VALVE MATERIAL	347HSS	WELD EXAMINATION	Class C Per Tech Spec (Note 531)	REV
RATING / FACING	ASME CLASS300 / RTJ	TRIM MATERIAL	347HSS/FHF	STRESS RELIEF	As per code	
CORROSION ALLOWANCE	0.5mm	CODE	ASME B31.3	BENDING		

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)

REV

TEMP. °C	SIZE																					
1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20	24	30	36	42	48	54	66	78
-38 to 38	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76
93	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76
149	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76
204	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76
260	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76
316	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76
343	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76
371	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76
399	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76
426	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76
454	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76
482	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76
510	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76
538	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76
565	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76
593	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76
649	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76	2.76

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

ITEM	SIZE RANGE	ENDS	SCH/RAT	MATERIAL DESCRIPTION	NOTES	REV
PIPE	1/2" - 1.5" 2" 3" - 12" 14" - 16" 18" - 24" 30" - 36" 42" 48" 54" 66" 78"	PE BE BE BE BE BE BE BE BE BE	SCH40S SCH40S SCH10S SCH10S SCH40S 9.53 MM 11.13 MM 12.7 MM 14.27 MM 17.48 MM 20.62 MM	SS, ASTM A312 GRADE TP 321H, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 321H, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 321H, SMLS, ASME B36.19M SS, ASTM A358 GRADE TP 321H CL 3 EFW, STR SM, Ej=1.0, RT = 100%, ASME B36.19M SS, ASTM A358 GRADE TP 321H CL 3 EFW, STR SM, Ej=1.0, RT = 100%, ASME B36.19M SS, ASTM A358 GRADE TP 321H CL 3 EFW, STR SM, Ej=1.0, RT = 100%, ASME B36.10M SS, ASTM A358 GRADE TP 321H CL 1, EFW, STR SM, Ej=1.0, RT = 100%, ASME B36.10M SS, ASTM A358 GRADE TP 321H CL 1, EFW, STR SM, Ej=1.0, RT = 100%, ASME B36.10M SS, ASTM A358 GRADE TP 321H CL 1, EFW, STR SM, Ej=1.0, RT = 100%, ASME B36.10M SS, ASTM A358 GRADE TP 321H CL 1, EFW, STR SM, Ej=1.0, RT = 100%, ASME B36.10M SS, ASTM A358 GRADE TP 321H CL 1, EFW, STR SM, Ej=1.0, RT = 100%, ASME B36.10M	705 705 705 705 705 705	
NIPPLES	1/2" - 1.5" 1/2" - 1.5" 1/2" - 1.5"	PBE PBE TOE	CALC CALC CALC	SS, ASTM A312 GRADE TP 321H, SMLS, ASME B36.19M, 3" (75mm) & 6" (150mm) LONG SS, ASTM A312 GRADE TP 321H, SMLS, ASME B36.19M, 3" (75mm) & 6" (150mm) LONG SS, ASTM A312 GRADE TP 321H, SMLS, ASME B36.19M, 3" (75mm) & 6" (150mm) LONG	503 03	

ITEM	SIZE RANGE	ENDS	SCH/RAT	MATERIAL DESCRIPTION		NOTES	REV
CONC. SWAGES	½" – 1.5" ½" – 2" 3" – 4" (½" – 2")	TSE PBE BLE x PSE	CALC CALC CALC	SS, ASTM A403 GRADE WP321H, SMLS, MSS-SP-95 SS, ASTM A403 GRADE WP321H, SMLS, MSS-SP-95 SS, ASTM A403 GRADE WP321H, SMLS, MSS-SP-95		03	
FITTINGS						02	
90 ELL	½" – 1.5"	SW	3000 LB	SS, ASTM A182 GRADE F321H, ASME B16.11			
45 ELL	½" – 1.5"	SW	3000 LB	SS, ASTM A182 GRADE F321H, ASME B16.11			
TEE	½" – 1.5"	SW	3000 LB	SS, ASTM A182 GRADE F321H, ASME B16.11			
TEE (RED)	½" – 1.5"	SW	3000 LB	SS, ASTM A182 GRADE F321H, ASME B16.11			
PLUG	½" – 1.5"	THRD		SS, ASTM A182 GRADE F321H, ASME B16.11, ROUND HEAD		03	
PLUG	½" - ½"	PE		SS, ASTM A182 GRADE F321H, ASME B16.11, BARSTOCK, ROUND HEAD		53	
COUPLING	½" – 1.5"	SW	3000 LB	SS, ASTM A182 GRADE F321H, ASME B16.11			
CAP	½" – 1.5"	SW	3000 LB	SS, ASTM A182 GRADE F321H, ASME B16.11			
CAP	½" – 1.5"	THRD	3000 LB	SS, ASTM A182 GRADE F321H, ASME B16.11		03	
SOCKOLET	½" – 1.5"	WELD	3000 LB	SS, ASTM A182 GRADE F321H, MSS-SP-97			
THREDOLET	½" – 1.5"	WELD	3000 LB	SS, ASTM A182 GRADE F321H, MSS-SP-97		03	
WELDOLET	2" – 78"	WELD		SS, ASTM A182 GRADE F321H, MSS-SP-97		701, 704	
90 ELL	2" – 6"	WELD		SS, ASTM A403 GRADE WP321H-S, ASME B16.9			
90 ELL	8" – 78"	WELD		SS, ASTM A403 GRADE WP321H-WX, ASME B16.9		701, 704	
45 ELL	2" – 6"	WELD		SS, ASTM A403 GRADE WP321H-S, ASME B16.9			
45 ELL	8" – 78"	WELD		SS, ASTM A403 GRADE WP321H-WX, ASME B16.9		701, 704	
TEE	2" – 6"	WELD		SS, ASTM A403 GRADE WP321H-S, ASME B16.9			
TEE	8" – 78"	WELD		SS, ASTM A403 GRADE WP321H-WX, ASME B16.9		701, 704	
TEE (RED.)	2" – 6"	WELD		SS, ASTM A403 GRADE WP321H-S, ASME B16.9			
TEE (RED.)	8" – 78"	WELD		SS, ASTM A403 GRADE WP321H-WX, ASME B16.9		701, 704	
CAP	2" – 78"	WELD		SS, ASTM A403 GRADE WP321H-S, ASME B16.9			
REDUCER (CC)	2" – 6"	WELD		SS, ASTM A403 GRADE WP321H-S, ASME B16.9			
REDUCER (CC)	8" – 78"	WELD		SS, ASTM A403 GRADE WP321H-WX, ASME B16.9		701, 704	
REDUCER (EC)	2" – 6"	WELD		SS, ASTM A403 GRADE WP321H-S, ASME B16.9			
REDUCER (EC)	8" – 78"	WELD		SS, ASTM A403 GRADE WP321H-WX, ASME B16.9		701, 704	
VALVES					TAG.		
GATE	½" – 1.5"	SW	CL 800	347H SS Body W/ 347H SS TRIM, Full HF SEAT, High Temp Service	PVVB68S44 38H11D0HV	708	
GATE	½" – 1.5"	THRD/S W	CL 800	347H SS Body W/ 347H SS TRIM, Full HF SEAT, High Temp Service	PVVB68D44 38H11D0HV	06, 03, 708	
GATE	½" – 1.5"	RTJ	CL 300	347H SS Body W/ 347H SS TRIM, Full HF SEAT, Full Port, High Temp Service	PVVB63J443 8H11D0HV	15, 708	
GATE	2" – 10"	RTJ	CL 300	347H SS Body W/ 347H SS TRIM, Full HF SEAT, Full Port High Temp Service	PVVB63J553 8H11C0HV	708	
GATE	12" – 24"	RTJ	CL 300	347H SS Body W/ 347H SS TRIM, Full HF SEAT, Full Port GO, High Temp Service	PVVB63J553 8H11C5HV	708	
GLOBE	½" – 1.5"	SW	CL 800	347H SS Body W/ 347H SS TRIM, Full HF SEAT, High Temp Service	PVVB68S443 8H11D0HV	708	
GLOBE	2" – 6"	RTJ	CL 300	347H SS Body W/ 347H SS TRIM, Full HF SEAT, High Temp Service	PVVB63J553 8H11C0HV	708	
GLOBE	8" – 14"	RTJ	CL 300	347H SS Body W/ 347H SS TRIM, Full HF SEAT, GO, High Temp Service	PVVB63J553 8H11C5HV	708	
LIFT CHECK	½" – 1.5"	SW	CL 800	347H SS Body W/ 347H SS TRIM, Full HF SEAT, Piston Type, High Temp Service	PVVB68S440 8N11D0HV	61, 708	
SWING CHECK	2" – 24"	RTJ	CL 300	347H SS Body W/ 347H SS TRIM, Full HF SEAT, High Temp Service	PVVB63J550 8N11C0HV	62, 708	
FLANGES							
SOCKETWELD	½" – 1.5"	RTJ	CL 300	SS, ASTM A182 GRADE F321H, ASME B16.5			
WELDNECK	2" – 24"	RTJ	CL 300	SS, ASTM A182 GRADE F321H, ASME B16.5			
WELDNECK	30" – 36"	RTJ	CL 300	SS, ASTM A182 GRADE F321H, ASME B16.47, Series A			
WELDNECK	42" – 60"	RTJ	CL 300	SS, ASTM A182 GRADE F321H, ASME B16.47, Series A, (GROOVE DIMENSIONS AS PER DETAIL DRAWING)		702	
WELDNECK	62" – 78"	RTJ	CL 300	SS, ASTM A182 GRADE F321H, (AS PER DETAIL DRAWING)			
BLIND	½" – 24"	RTJ	CL 300	SS, ASTM A182 GRADE F321H, ASME B16.5			
BLIND	30" – 60"	RTJ	CL 300	SS, ASTM A182 GRADE F321H, ASME B16.47, Series A, (AS PER DETAIL DRAWING)			
BLIND	62" – 78"	RTJ	CL 300	SS, ASTM A182 GRADE F321H, (AS PER DETAIL DRAWING)		702	
PR WN ORIFICE	1" – 24"	RTJ	CL 300	SS, ASTM A182 GRADE F321H W/ ½" SOCKETWELD TAPS, ASME B16.36			
MISCELLANEOUS							
PADDLE BLANK	½" – 24"	RTJ	CL 300	SS, ASTM A240 GRADE 321H, ASME B16.48			
PADDLE SPACER	½" – 24"	RTJ	CL 300	SS, ASTM A240 GRADE 321H, ASME B16.48			

ITEM	SIZE RANGE	ENDS	SCH/RAT	MATERIAL DESCRIPTION	NOTES	REV
Y-STRAINER	½" – 2"	SW	CL 800	SS, ASTM A182 GRADE F321H Body, TP 321H SS Screen W / 1/16" Dia. Hole @ 3/32" Center, High Temp Service	553	
Y-STRAINER	½" – 2"	SW	CL 800	SS, ASTM A182 GRADE F321H Body, TP 321H SS Screen W / 1/32" Dia. Hole @ 3/64" Center, High Temp Service,	554	
CONE STRAINER – TRUNCATED TYPE	3" – 24"	RTJ	CL 300	SS, ASTM A240 GRADE TP 321H SS, Perforated Plate with 1/8" Dia. Hole @ 3/16" Center, ASME B16.5	555	
GASKETS	½" – 24" 30" – 36" 42" – 60" 62" – 78"		CL 300 CL 300 CL 300 CL300	SOLID OVAL RING, ASME B16.20, RING TYPE JOINT, ASME B16.5, 347 SS SOLID OVAL RING,, ASME B16.20, RING TYPE JOINT, ASME B16. 47, Series A, SOLID OVAL RING,, ASME B16.20, RING TYPE JOINT, ASME B16. 47, Series A, (DIMENSIONS AS PER DETAIL DRAWING) SOLID OVAL RING, ASME B16.20 RING TYPE JOINT, (AS PER DETAIL DRAWING), 347 SS		
BOLTING						
STUD (1100°F/593°C MAX)	ALL			ALLOY STEEL, ASTM A193 GRADE B16, ASTM A 194 GRADE 4 HEAVY HEX NUTS		
STUD (1500°F/816°C MAX)	ALL			STUD BOLT / HEAVY HEX NUTS HASTELLOY-X, ASME SB572 UNS N06002, DIMENSIONS PER ASTM A193 / A194	527	

	NOTES			REV
JOINT CONNECTS			SIZES	
MAINTENANCE		FLANGES	1/2" – 2"	
MAINTENANCE		FLANGES	3" – 78"	
NON-MAINTENANCE		SOCKETWELD	1/2" – 1 1/2"	
NON-MAINTENANCE		BUTTWELD	2" – 78"	

NOTE	SPECIFIC PIPING NOTES	REV
02	All butt-welded component thicknesses shall match the pipe thickness.	
03	Threaded joints are permitted only at the outlet of vent and drain valves, hydrostatic connections, outlet instrument take-off valves, and to match equipment..	
06	These valves shall be used only for vent, drain, and instrument connections.	
09	Pressure and temperature rating can be limited by certain components permitted by this Practice. Manufacturer's recommended pressure-temperature restrictions shall be consulted.	
15	These valves shall be used if mating to flanged nozzles.	
53	This item shall be used for orifice flange tap connections only.	
61	These check valves shall be installed in a horizontal position with cover up.	
62	These check valves shall be installed in a horizontal position with cover up or in a vertical position with upward flow.	
503	Use this item for Branch Nipple which terminate at the valve or continue as tubing. (eg. For instrument connections, valved vents & drains, etc.).	
508	For use over design temperatures of 1000°F (538°C) all SS material shall have a 0.04% minimum Carbon content.	
527	Use Hastelloy X bolt for temperature above 1100°F (593°C) up to 1500°F (816°C).	
531	Examination shall be in accordance with requirements in "Standard for Piping Welding", refer to document number: 108184-0000-PI-SPC-0006.	
553	These check valves shall be installed in a horizontal position with hinge pin vertical or in a vertical position with upward flow.	
554	Use this strainer for Vapor / Gas Service, blow off connection requirements per "Technical Specification for Pipe, Fittings, and Flanges ".	
555	Cone Strainer shall be installed between the 2 flanges with the small end pointed upstream.	
701	Manufacturer of fittings above 48" shall provide proof that fittings supplied meet requirements of para 2.2 of ASME B16.9 for review by Buyer.	
702	These flanges have been calculated based on process design conditions on an individual basis.	
705	For straight piping without branches; with branches thickness to be calculated by DEC.	
708	Valves to be suitable for FV at 704°C.	

90° BRANCH CONNECTION CHART

BRANCH SIZE	78	T																								
	66	B	T	[R03]																						
	54	B	B	T	[R03]																					
	48	B	B	B	T	[R03]																				
	42	B	B	B	RT	T	[R03]																			
	36	B	B	B	RT	RT	T	[R03]																		
	30	B	B	B	RT	RT	RT	T	[R03]																	
	24	B	B	B	RT	RT	RT	RT	T	[R03]																
	20	B	B	B	B	B	RT	RT	RT	T	[R03]															
	18	B	B	B	B	RT	RT	RT	RT	RT	T															
	16	B	B	B	B	RT	RT	RT	RT	RT	RT	T														
	14	B	B	B	B	B	B	RT	RT	RT	RT	RT	T													
	12	B	B	B	B	B	B	RT	RT	RT	RT	RT	RT	T												
	10	B	B	B	B	B	B	RT	RT	RT	RT	RT	RT	RT	T											
	8	W	W	W	W	W	W	W	W	W	W	W	W	W	RT	T										
	6	W	W	W	W	W	W	W	W	W	W	W	W	W	W	RT	T									
	4	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	RT	T								
	3	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	RT	T							
	2	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	RT	T						
	1½	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	RT	T					
	1	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	RT	RT	T			
	¾	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	RT	RT	T		
	½	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	RT	RT	RT	T	
		78	66	54	48	42	36	30	24	20	18	16	14	12	10	8	6	4	3	2	1½	1	¾	½		
	HEADER SIZE																									

LEGEND:

RT = Reducing Tee (BW for 2" and larger, SW for 1.5" and smaller)

B = Branch Weld with Integral Reinforcing

A = Sockolet

T = Straight Tee (BW for 2" and larger, SW for 1.5" and smaller)

W = Weldolet

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
B27B	R00

SERVICES	Regeneration Air			REV

PIPING MATERIAL	TP 321H Stainless Steel (Note 508)	VALVE MATERIAL	347HSS	WELDEXAMINATION	Class C Per Tech Spec (Note 531)	REV
RATING / FACING	ASME CLASS300 / RTJ	TRIM MATERIAL	347HSS/FHF	STRESS RELIEF	As per code	
CORROSION ALLOWANCE	0.5mm	CODE	ASME B31.3	BENDING		

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)

REV

SIZE

TEMP. °C	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20	24	30	42	66	72	78	90	96
-28 to 38	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
93	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
149	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
204	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
260	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
316	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
343	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
371	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
399	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
426	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
454	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
482	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
510	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
538	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
565	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
593	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
649	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
704	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

ITEM	SIZE RANGE	ENDS	SCH/RAT	MATERIAL DESCRIPTION	NOTES	REV
PIPE	1/2" - 2"	PE	SCH40S	SS, ASTM A312 GRADE TP 321H, SMLS, ASME B36.19M		PIPE
	3" - 12"	BE	SCH10S	SS, ASTM A312 GRADE TP 321H, SMLS, ASME B36.19M		
	14" - 16"	BE	SCH10S	SS, ASTM A358 GRADE TP 321H CL 3 EFW, STR SM, Ej=1.0, RT = 100%, ASME B36.19M		
	18" - 30"	BE	SCH40S	SS, ASTM A358 GRADE TP 321H CL 3 EFW, STR SM, Ej=1.0, RT = 100%, ASME B36.19M		
	42"	BE	11.13 MM	SS, ASTM A358 GRADE TP 321H CL 1, EFW, STR SM, Ej=1.0, RT = 100%, ASME B36.10M	705	
	66"	BE	17.48 MM	SS, ASTM A358 GRADE TP 321H CL 1, EFW, STR SM, Ej=1.0, RT = 100%, ASME B36.10M	705	
	72"	BE	19.05 MM	SS, ASTM A358 GRADE TP 321H CL 1, EFW, STR SM, Ej=1.0, RT = 100%, ASME B36.10M	705	
	78"	BE	20.62 MM	SS, ASTM A358 GRADE TP 321H CL 1, EFW, STR SM, Ej=1.0, RT = 100%, ASME B36.10M	705	
	78"	BE	23.83 MM	SS, ASTM A358 GRADE TP 321H CL 1, EFW, STR SM, Ej=1.0, RT = 100%, ASME B36.10M	706	
	90"	BE	22.23 MM	SS, ASTM A358 GRADE TP 321H CL 1, EFW, STR SM, Ej=1.0, RT = 100%, ASME B36.10M	705	
	90"	BE	26.97 MM	SS, ASTM A358 GRADE TP 321H CL 1, EFW, STR SM, Ej=1.0, RT = 100%, ASME B36.10M	706	
	96"	BE	25.4 MM	SS, ASTM A358 GRADE TP 321H CL 1, EFW, STR SM, Ej=1.0, RT = 100%, ASME B36.10M	705	

ITEM	SIZE RANGE	ENDS	SCH/RAT	MATERIAL DESCRIPTION		NOTES	REV
NIPPLES	½" – 1.5"	PBE	CALC	SS, ASTM A312 GRADE TP 321H, SMLS, ASME B36.19M, 3" (75mm) & 6" (150mm) LONG		503	
	½" – 1.5"	PBE	CALC	SS, ASTM A312 GRADE TP 321H, SMLS, ASME B36.19M, 3" (75mm) & 6" (150mm) LONG			
	½" – 1.5"	TOE	CAL	SS, ASTM A312 GRADE TP 321H, SMLS, ASME B36.19M, 3" (75mm) & 6" (150mm) LONG		03	
CONC. SWAGES	½" – 1.5"	TSE	CALC	SS, ASTM A403 GRADE WP321H, SMLS, MSS-SP-95		03	
	½" – 2"	PBE	CALC	SS, ASTM A403 GRADE WP321H, SMLS, MSS-SP-95			
	3" – 4" (½" – 2")	BLE x PSE	CALC	SS, ASTM A403 GRADE WP321H, SMLS, MSS-SP-95			
FITTINGS	90 ELL ½" – 1.5"	SW	3000 LB	SS, ASTM A182 GRADE F321H, ASME B16.11		02	
	45 ELL ½" – 1.5"	SW	3000 LB	SS, ASTM A182 GRADE F321H, ASME B16.11			
	TEE ½" – 1.5"	SW	3000 LB	SS, ASTM A182 GRADE F321H, ASME B16.11		03	
	TEE (RED) ½" – 1.5"	SW	3000 LB	SS, ASTM A182 GRADE F321H, ASME B16.11			
	PLUG ½" – 1.5"	THRD		SS, ASTM A182 GRADE F321H, ASME B16.11, ROUND HEAD		53	
	PLUG ½" - ½"	PE		SS, ASTM A182 GRADE F321H, ASME B16.11, BARSTOCK, ROUND HEAD			
	COUPLING ½" – 1.5"	SW	3000 LB	SS, ASTM A182 GRADE F321H, ASME B16.11		03	
	CAP ½" – 1.5"	SW	3000 LB	SS, ASTM A182 GRADE F321H, ASME B16.11			
	CAP ½" – 1.5"	THRD	3000 LB	SS, ASTM A182 GRADE F321H, ASME B16.11		03	
	SOCKOLET ½" – 1.5"	WELD	3000 LB	SS, ASTM A182 GRADE F321H, MSS-SP-97			
	THREDOLET ½" – 1.5"	WELD	3000 LB	SS, ASTM A182 GRADE F321H, MSS-SP-97		03	
	WELDOLET 2" – 96"	WELD		SS, ASTM A182 GRADE F321H, MSS-SP-97			
	90 ELL 2" – 6"	WELD		SS, ASTM A403 GRADE WP321H-S, ASME B16.9		701, 704	
	90 ELL 8" – 96"	WELD		SS, ASTM A403 GRADE WP321H-WX, ASME B16.9			
	45 ELL 2" – 6"	WELD		SS, ASTM A403 GRADE WP321H-S, ASME B16.9		701, 704	
	45 ELL 8" – 96"	WELD		SS, ASTM A403 GRADE WP321H-WX, ASME B16.9			
	TEE 2" – 6"	WELD		SS, ASTM A403 GRADE WP321H-S, ASME B16.9		701, 704	
	TEE 8" – 96"	WELD		SS, ASTM A403 GRADE WP321H-WX, ASME B16.9			
	TEE (RED.) 2" – 6"	WELD		SS, ASTM A403 GRADE WP321H-S, ASME B16.9		701, 704	
	TEE (RED.) 8" – 96"	WELD		SS, ASTM A403 GRADE WP321H-WX, ASME B16.9			
	CAP 2" – 96"	WELD		SS, ASTM A403 GRADE WP321H-S, ASME B16.9		701, 704	
	REDUCER (CC) 2" – 6"	WELD		SS, ASTM A403 GRADE WP321H-S, ASME B16.9			
	REDUCER (CC) 8" –96"	WELD		SS, ASTM A403 GRADE WP321H-WX, ASME B16.9		701, 704	
	REDUCER (EC) 2" – 6"	WELD		SS, ASTM A403 GRADE WP321H-S, ASME B16.9			
	REDUCER (EC) 8" – 96"	WELD		SS, ASTM A403 GRADE WP321H-WX, ASME B16.9		701, 704	
VALVES					TAG.		
	GATE ½" – 1.5"	SW	CL 800	347H SS Body W/ 347H SS TRIM, Full HF SEAT, High Temp Service	PVVGB68S44 38H11D0HT	06, 03	
	GATE ½" – 1.5"	THRD/S W	CL 800	347H SS Body W/ 347H SS TRIM, Full HF SEAT, High Temp Service	PVVG68D44 38H11D0HT		
	GATE ½" – 1.5"	RTJ	CL 300	347H SS Body W/ 347H SS TRIM, Full HF SEAT, Full Port, High Temp Service	PVVG63J443 8H11D0HT	15	
	GATE 2" – 10"	RTJ	CL 300	347H SS Body W/ 347H SS TRIM, Full HF SEAT, Full Port High Temp Service	PVVG63J553 8H11C0HT		
	GATE 12" – 24"	RTJ	CL 300	347H SS Body W/ 347H SS TRIM, Full HF SEAT, Full Port GO, High Temp Service	PVVG63J553 8H11C5HT	61	
	GLOBE ½" – 1.5"	SW	CL 800	347H SS Body W/ 347H SS TRIM, Full HF SEAT, High Temp Service	PVVL68S443 8H11D0HT		
	GLOBE 2" – 6"	RTJ	CL 300	347H SS Body W/ 347H SS TRIM, Full HF SEAT, High Temp Service	PVVL63J553 8H11C0HT	62	
	GLOBE 8" – 14"	RTJ	CL 300	347H SS Body W/ 347H SS TRIM, Full HF SEAT, GO, High Temp Service	PVVL63J553 8H11C5HT		
	LIFT CHECK ½" – 1.5"	SW	CL 800	347H SS Body W/ 347H SS TRIM, Full HF SEAT, Piston Type, High Temp Service	PVVCL68S440 8N11D0HT		
SWING CHECK 2" – 24"	RTJ	CL 300	347H SS Body W/ 347H SS TRIM, Full HF SEAT, High Temp Service	PVVCS63J550 8N11C0HT			
FLANGES							
	SOCKETWELD ½" – 1.5"	RTJ	CL 300	SS, ASTM A182 GRADE F321H, ASME B16.5		702	
	WELDNECK 2" – 24"	RTJ	CL 300	SS, ASTM A182 GRADE F321H, ASME B16.5			
	WELDNECK 30" – 36"	RTJ	CL 300	SS, ASTM A182 GRADE F321H, ASME B16.47, Series A		702	
	WELDNECK 42" – 60"	RTJ	CL 300	SS, ASTM A182 GRADE F321H, ASME B16.47, Series A (GROOVE DIMENSIONS AS PER DETAIL DRAWING)			
	WELDNECK 62" – 96"	RTJ	CL 300	SS, ASTM A182 GRADE F321H, (AS PER DETAIL DRAWING)		702	
	BLIND ½" – 24"	RTJ	CL 300	SS, ASTM A182 GRADE F321H, ASME B16.5			
	BLIND 30" – 60"	RTJ	CL 300	SS, ASTM A182 GRADE F321H, ASME B16.47, Series A		702	
	BLIND 62" – 96"	RTJ	CL 300	SS, ASTM A182 GRADE F321H, (AS PER DETAIL DRAWING)			
	PR WN ORIFICE 1" – 24"	RTJ	CL 300	SS, ASTM A182 GRADE F321H W/ ½" SOCKETWELD TAPS, ASME B16.36			

NOTE	SPECIFIC PIPING NOTES	REV
02	All butt-welded component thicknesses shall match the pipe thickness.	
03	Threaded joints are permitted only at the outlet of vent and drain valves, hydrostatic connections, outlet instrument take-off valves, and to match equipment..	
06	These valves shall be used only for vent, drain, and instrument connections.	
09	Pressure and temperature rating can be limited by certain components permitted by this Practice. Manufacturer's recommended pressure-temperature restrictions shall be consulted.	
15	These valves shall be used if mating to flanged nozzles.	
53	This item shall be used for orifice flange tap connections only.	
61	These check valves shall be installed in a horizontal position with cover up.	
62	These check valves shall be installed in a horizontal position with cover up or in a vertical position with upward flow.	
503	Use this item for Branch Nipple which terminate at the valve or continue as tubing. (eg. For instrument connections, valved vents & drains, etc.).	
508	For use over design temperatures of 1000°F (538°C) all SS material shall have a 0.04% minimum Carbon content.	
527	Use Hastelloy X bolt for temperature above 1100°F (593°C) up to 1500°F (816°C).	
531	Examination shall be in accordance with requirements in "Standard for Piping Welding", refer to document number: 108184-0000-PI-SPC-0006.	
553	These check valves shall be installed in a horizontal position with hinge pin vertical or in a vertical position with upward flow.	
554	Use this strainer for Vapor / Gas Service, blow off connection requirements per "Technical Specification for Pipe, Fittings, and Flanges ".	
555	Cone Strainer shall be installed between the 2 flanges with the small end pointed upstream.	
701	Manufacturer of fittings above 48" shall provide proof that fittings supplied meet requirements of para 2.2 of ASME B16.9 for review by Buyer.	
702	These flanges have been calculated based on process design conditions on an individual basis.	
705	For straight piping without branches; with branches thickness to be calculated by DEC.	
706	For air inlet piping with branch connection 72" ID thicker wall is required as per ASME B31.3, paragraph 304.3.1.	

90° BRANCH CONNECTION CHART

BRANCH SIZE	96	T																						
	90	B	T	[R03]																				
	78	B	B	T	[R03]																			
	72	B	B ¹	B ¹	T	[R03]																		
	66	B	B	B ¹	B	T	[R03]																	
	42	B	B	B	B	B	T	[R03]																
	30	B	B	B	B	B	B	T	[R03]															
	24	B	B	B	B	B	B	RT	T	[R03]														
	20	B	B	B	B	B	B	RT	RT	T	[R03]													
	18	B	B	B	B	B	B	RT	RT	RT	T													
	16	B	B	B	B	B	B	RT	RT	RT	RT	T												
	14	B	B	B	B	B	B	RT	RT	RT	RT	RT	T											
	12	B	B	B	B	B	B	RT	RT	RT	RT	RT	RT	T										
	10	B	B	B	B	B	B	RT	RT	RT	RT	RT	RT	RT	T									
	8	W	W	W	W	W	W	W	W	W	W	W	W	W	RT	T								
	6	W	W	W	W	W	W	W	W	W	W	W	W	W	W	RT	T							
	4	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	RT	T						
	3	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	RT	T					
	2	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	RT	T				
	1½	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	RT	T				
1	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	RT	RT	T				
¾	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	RT	RT	T			
½	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	RT	RT	RT	T		
	96	90	78	72	66	42	30	24	20	18	16	14	12	10	8	6	4	3	2	1½	1	¾	½	
	HEADER SIZE																							

LEGEND:

RT = Reducing Tee (BW for 2" and larger, SW for 1.5" and smaller)

B = Branch Weld with Integral Reinforcing

A = Sockolet

T = Straight Tee (BW for 2" and larger, SW for 1.5" and smaller)

W = Weldolet

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
B54H	R00

SERVICES	LP Steam @ PDH Reactor	Reduction Gas @ PDH Reactor		REV

PIPING MATERIAL	1-1/4 CR - 1/2 MO	VALVE MATERIAL	1-1/4 CR - 1/2 MO	WELDEXAMINATION		REV
RATING / FACING	ASME CLASS300 / RTJ	TRIM MATERIAL	13Cr/HF (TRIM#8)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	3.0mm	CODE	ASME B31.3	BENDING		

**FLANGE RATING &
MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)***

TEMP. °C	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20	24
-28 to 38	51.70	51.70	51.70	51.70	51.70	51.70	51.70	51.70	51.70	51.70	46.11	41.91	36.59	49.66	44.62	37.07
93	51.52	51.52	51.52	51.52	51.52	51.52	51.52	50.05	49.65	48.95	43.11	39.21	34.21	46.41	41.68	34.64
149	49.73	49.73	49.73	49.73	49.73	49.73	49.73	48.18	47.86	47.12	41.46	37.71	32.93	44.71	40.17	33.35
204	47.86	47.86	47.86	47.86	47.86	47.86	47.86	46.88	46.45	45.82	40.25	36.68	32.02	43.45	39.02	32.41
260	45.62	45.62	45.62	45.62	45.62	45.62	45.62	45.62	45.62	45.05	39.58	36.05	31.48	42.68	38.34	31.84
316	42.30	42.30	42.30	42.30	42.30	42.30	42.30	42.30	42.30	42.30	38.41	34.98	30.56	41.42	37.21	30.92
343	40.60	40.60	40.60	40.60	40.60	40.60	40.60	40.60	40.60	40.60	37.31	33.92	29.62	40.19	36.12	30.02
371	39.12	39.12	39.12	39.12	39.12	39.12	39.12	39.12	39.12	39.12	35.91	32.66	28.56	38.72	34.78	28.91
399	36.59	36.59	36.59	36.59	36.59	36.59	36.59	36.59	36.59	36.59	35.00	31.87	27.82	36.59	33.89	28.16
426	35.14	35.14	35.14	35.14	35.14	35.14	35.14	35.14	35.14	35.14	34.55	31.41	27.43	35.14	33.45	27.81

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" - 3/4"	160	PE	1-1/4 CR - 1/2 MO, ASTM A335-P11 SMLS, ASME B36.10M		
	1" - 1 1/2"	80	PE	1-1/4 CR - 1/2 MO, ASTM A335-P11 SMLS, ASME B36.10M		
	2"	80	BE	1-1/4 CR - 1/2 MO, ASTM A335-P11 SMLS, ASME B36.10M		
	3" - 16"	STD	BE	1-1/4 CR - 1/2 MO, ASTM A335-P11 SMLS, ASME B36.10M		
	18" - 24"	XS	BE	1-1/4 CR - 1/2 MO, ASTM A335-P11 SMLS, ASME B36.10M		
NIPPLE	PBE 1/2" - 2"	160	PBE	1-1/4 CR - 1/2 MO, ASTM A335-P11 SMLS, ASME B36.10		
SWAGE	CON 1/2" - 2"	160	PBE	1-1/4 CR - 1/2 MO, ASTM A234-WP11 CL1-S, MSS SP-95	1,	
	ECC 1/2" - 2"	160	PBE	1-1/4 CR - 1/2 MO, ASTM A234-WP11 CL1-S, MSS SP-95		
COUPLING	1/2" - 3/4"	CL6000	SOCW	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.11		
FULL COUPLING	1" - 1 1/2"	CL3000	SOCW	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.11		
	1/2" - 1"	CL6000	THRD	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.11	279	
	1 1/2" - 2"	CL3000	THRD	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.11	279	
OLET	1/2" - 3/4"	CL6000	WELD	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, MSS SP-97		
SOCKOLET	1" - 1 1/2"	CL3000	WELD	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, MSS SP-97		
THREDOLET	1/2" - 1"	CL6000	WELD	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, MSS SP-97	279	
	1 1/2" - 2"	CL3000	WELD	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, MSS SP-97	279	
WELDOLET	2" - 8"	AS PIPE	WELD	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, MSS SP-97		
PLUG	ROUND HEAD 1/2" - 2"		THRD	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.11, ROUND HEAD		
CAP	1/2" - 3/4"	CL6000	SOCW	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.11		
	1" - 1 1/2"	CL3000	SOCW	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.11		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
	1/2" - 1"	CL6000	THRD	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.11		279	
	1 1/2" - 2"	CL3000	THRD	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.11		279	
	2" - 24"	AS PIPE	BW	1-1/4 CR - 1/2 MO, ASTM A234-WP11 CL1-S, ASME B16.9			
90° ELBOW							
SW	1/2" - 3/4"	CL6000	SOCW	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.11			
	1" - 1 1/2"	CL3000	SOCW	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.11			
LR	2" - 24"	AS PIPE	BW	1-1/4 CR - 1/2 MO, ASTM A234-WP11 CL1-S, ASME B16.9			
45° ELBOW							
SW	1/2" - 3/4"	CL6000	SOCW	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.11			
	1" - 1 1/2"	CL3000	SOCW	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.11			
	2" - 24"	AS PIPE	BW	1-1/4 CR - 1/2 MO, ASTM A234-WP11 CL1-S, ASME B16.9			
TEE							
	1/2" - 3/4"	CL6000	SOCW	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.11			
	1" - 1 1/2"	CL3000	SOCW	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.11			
	2" - 24"	AS PIPE	BW	1-1/4 CR - 1/2 MO, ASTM A234-WP11 CL1-S, ASME B16.9			
RED TEE							
	1/2" - 3/4"	CL6000	SOCW	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.11			
	1" - 1 1/2"	CL3000	SOCW	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.11			
	2" - 24"	AS PIPE	BW	1-1/4 CR - 1/2 MO, ASTM A234-WP11 CL1-S, ASME B16.9			
REDUCER							
CON	2" - 24"	AS PIPE	BW	1-1/4 CR - 1/2 MO, ASTM A234-WP11 CL1-S, ASME B16.9			
ECC	2" - 24"	AS PIPE	BW	1-1/4 CR - 1/2 MO, ASTM A234-WP11 CL1-S, ASME B16.9			
FLANGE							
SOCKETWELD	1/2" - 1 1/2"	CL300	RTJ	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.5			
WELDNECK	2" - 24"	CL300	RTJ	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.5, THICKNESS AS PIPE			
BLIND FLG	1/2" - 24"	CL300	RTJ	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.5			
DRIP RING	1" - 24"	CL300	RTJ	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2 (SOLUTION HEAT TREATED), 1-1/2" THICK W/3/4" SOCW TAs (STANDARD DRWGS)			
FIGURE 8	1" - 12"	CL300	RTJ	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.48			
BLANK & SPACER	14" - 24"	CL300	RTJ	1-1/4 CR - 1/2 MO, ASTM A182-F11 CL2, ASME B16.48			
VALVES							
GATE	1/2" - 1 1/2"	CL800	SOCW	1-1/4 CR - 1/2 MO BODY WITH 13 CR/HF TRIM (API #8)	TAG PVVGB68S1808 BQCD0ZZ	24	
	1/2" - 2"	CL300	RTJ	1-1/4 CR - 1/2 MO BODY WITH 13 CR/HF TRIM (API #8)	PVVG63J1808 BQCD0ZZ	43	
	2"	CL300	RTJ	1-1/4 CR - 1/2 MO BODY WITH 13 CR/HF TRIM (API #8) FULL PORT	PVVG63J1808B QCD0ZZ		
	3" - 10"	CL300	RTJ	1-1/4 CR - 1/2 MO BODY WITH 13 CR/HF TRIM (API #8) FULL PORT	PVVG63J2708B QCC0ZZ		
	12" - 24"	CL300	RTJ	1-1/4 CR - 1/2 MO BODY WITH 13 CR/HF TRIM (API #8), FULL PORT, GEAR OP.	PVVG63J2708B QCD5ZZ		
GLOBE	1/2" - 1 1/2"	CL800	SOCW	1-1/4 CR - 1/2 MO BODY WITH 13 CR/HF TRIM (API #8)	PVVL68S1808B QCD0ZZ		
	2" - 6"	CL300	RTJ	1-1/4 CR - 1/2 MO BODY WITH 13 CR/HF TRIM (API #8)	PVVL63J2708B QCC0ZZ		
	8" - 12"	CL300	RTJ	1-1/4 CR - 1/2 MO BODY WITH 13 CR/HF TRIM (API #8), GEAR OP.	PVVL63J2708B QCC5ZZ		
LIFT CHECK	1/2" - 1 1/2"	CL800	SOCW	1-1/4 CR - 1/2 MO BODY WITH 13 CR/HF TRIM (API #8), Piston Type	PVVC168S1808 NQCD0ZZ	61	
SWING CHECK	2" - 24"	CL300	RTJ	1-1/4 CR - 1/2 MO BODY WITH 13 CR PLATES, METAL SEATS	PVVC63J2708 NQCC0ZZ	62	
BOLT/NUTS							
	ALL	CL300		ALLOY STEEL, ASTM A193 GRADE B16 WITH ASTM A 194 GRADE 4 HEAVY HEX NUTS			

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
GASKETS	½" - 24"	CL300		OVAL RING , 5 CR-1/2 MO, MAX HARDNESS BHN 130, ASME B16.20 / ASME B16.5		

	NOTES			REV
JOINT CONNECTS			SIZES	
MAINTENANCE		FLANGES	1/2" - 2"	
MAINTENANCE		FLANGES	3" - 24"	
NON-MAINTENANCE		SOCKETWELD	1/2" - 1 1/2"	
NON-MAINTENANCE		BUTTWELD	2" - 24"	

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
5	See Piping Design Guidelines Standard Drawings or module sheets for instrument applications and dimensional data.	
6	The flanges, gasket and bolting materials on all vents, drains, pressure, instrument, or flow connections shall be furnished by Piping/BOM.	
24	Use when required at flanged equipment nozzles, or where noted on P&IDs or Modules.	
43	Only to be used as indicated on the P&IDs.	
47	See preamble section of Piping Material Specification, Piping Notes.	
61	These check valves shall be installed in a horizontal position with cover up.	
62	These check valves shall be installed in a horizontal position with cover up or in a vertical position with upward flow.	
275	Only to be used where full port gate valve is required for instrument as indicated on P&ID's.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drains requiring plugs, connections to tube fittings and for connections to threaded proprietary items.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
B54K	R00

SERVICES			REV
Gas Turbine Exhaust	DEC to confirm based on selected gas turbine.		

PIPING MATERIAL	1-1/4 CR - 1/2 MO	VALVE MATERIAL	1-1/4 CR - 1/2 MO	WELDEXAMINATION	Class B Per Tech Spec (Note 531)	REV
RATING / FACING	ASME CLASS300 / RTJ	TRIM MATERIAL	13Cr/HF (TRIM#8)	STRESS RELIEF	ASME B31.3 (W.T. > 1/2"/13 mm)	
CORROSION ALLOWANCE	1.5mm	CODE	ASME B31.3	BENDING		

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)

SIZE

TEMP. °C	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20	24	48	72	84	90
-28 to 38	5.170	5.170	5.170	5.170	5.170	5.128	5.116	5.065	5.085	5.055	4.456	4.051	3.539	3.142	2.826	2.351	0.5	0.5	0.5	0.5
93	5.152	5.152	5.152	5.152	5.152	4.915	4.802	4.735	4.755	4.725	4.165	3.786	3.312	2.938	2.641	2.197	0.5	0.5	0.5	0.5
149	4.973	4.973	4.973	4.973	4.973	4.725	4.615	4.558	4.576	4.552	4.011	3.647	3.186	2.832	2.546	2.116	0.5	0.5	0.5	0.5
204	4.786	4.786	4.786	4.786	4.786	4.588	4.492	4.428	4.445	4.422	3.896	3.541	3.096	2.751	2.473	2.054	0.5	0.5	0.5	0.5
260	4.562	4.562	4.562	4.562	4.562	4.523	4.412	4.356	4.376	4.352	3.828	3.482	3.041	2.702	2.429	2.019	0.5	0.5	0.5	0.5
316	4.230	4.230	4.230	4.230	4.230	4.230	4.230	4.230	4.230	4.215	3.716	3.378	2.952	2.624	2.356	1.959	0.5	0.5	0.5	0.5
343	4.060	4.060	4.060	4.060	4.060	4.060	4.060	4.060	4.060	4.060	3.604	3.279	2.866	2.546	2.286	1.902	0.5	0.5	0.5	0.5
371	3.912	3.912	3.912	3.912	3.912	3.912	3.912	3.912	3.912	3.912	3.471	3.161	2.761	2.451	2.204	1.831	0.5	0.5	0.5	0.5
399	3.659	3.659	3.659	3.659	3.659	3.659	3.659	3.659	3.659	3.659	3.382	3.081	2.691	2.391	2.146	1.788	0.5	0.5	0.5	0.5
426	5.170	5.170	5.170	5.170	5.170	5.128	5.116	5.065	5.085	5.055	4.456	4.051	3.539	3.142	2.826	2.351	0.5	0.5	0.5	0.5
454	5.152	5.152	5.152	5.152	5.152	4.915	4.802	4.735	4.755	4.725	4.165	3.786	3.312	2.938	2.641	2.197	0.5	0.5	0.5	0.5
482	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
510	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
538	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
565	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
593	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

ITEM	SIZE RANGE	ENDS	SCH/RAT	MATERIAL DESCRIPTION	NOTES	REV
PIPE	1/2" - 2"	PE	XS	1 1/4 CR - 1/2 MO, ASTM A335 GRADE P11, SMLS, ASME B36.10		
	3" - 12"	BE	STD	1 1/4 CR - 1/2 MO, ASTM A335 GRADE P11, SMLS, ASME B36.10		
	14" - 18"	BE	XS	1 1/4 CR - 1/2 MO, ASTM A335 GRADE P11, SMLS, ASME B36.10		
	20" - 24"	BE	CALC	1 1/4 CR - 1/2 MO, ASTM A335 GRADE P11, SMLS, ASME B36.10		
	48"	BE	10.31 MM	1 1/4 CR - 1/2 MO, ASTM A691 GRADE 1 1/4 CR CL 22, ASME B36.10, E _j =1.0	705	
	72"	BE	15.88 MM	1 1/4 CR - 1/2 MO, ASTM A691 GRADE 1 1/4 CR CL 22, ASME B36.10, E _j =1.0	705	
	84"	BE	19.05 MM	1 1/4 CR - 1/2 MO, ASTM A691 GRADE 1 1/4 CR CL 22, ASME B36.10, E _j =1.0	705	
	84"	BE	22.23 MM	1 1/4 CR - 1/2 MO, ASTM A691 GRADE 1 1/4 CR CL 22, ASME B36.10, E _j =1.0	706	
	90"	BE	23.83 MM	1 1/4 CR - 1/2 MO, ASTM A691 GRADE 1 1/4 CR CL 22, ASME B36.10, E _j =1.0	705	
	90"	BE	30.18 MM	1 1/4 CR - 1/2 MO, ASTM A691 GRADE 1 1/4 CR CL 22, ASME B36.10, E _j =1.0	706	
NIPPLES	1/2" - 1.5"	PBE	XS	1 1/4 CR - 1/2 MO, ASTM A335 GRADE P11, SMLS, ASME B36.10, 3" (75mm) & 6" (150mm) LONG		
	1/2" - 1.5"	PBE	160	1 1/4 CR - 1/2 MO, ASTM A335 GRADE P11, SMLS, ASME B36.10, 3" (75mm) & 6" (150mm) LONG	503	

ITEM	SIZE RANGE	ENDS	SCH/RAT	MATERIAL DESCRIPTION		NOTES	REV
	½" – 1.5"	TOE	160	1 ¼ CR – ½ MO, ASTM A335 GRADE P11, SMLS, ASME B36.10, 3" (75mm) & 6" (150mm) LONG		03	
CONC. SWAGES	½" – 1.5" ½" – 2" 3" – 4" (½" – 2")	TSE PBE BLE x PSE	160 XS STD x XS	1 ¼ CR – ½ MO, ASTM A234 GRADE WP11 CL 1, SMLS, MSS-SP-95 1 ¼ CR – ½ MO, ASTM A234 GRADE WP11 CL 1, SMLS, MSS-SP-95 1 ¼ CR – ½ MO, ASTM A234 GRADE WP11 CL 1, SMLS, MSS-SP-95		03	
FITTINGS						02	
90 ELL	½" – 1.5"	SW	3000 LB	1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, ASME B16.11			
45 ELL	½" – 1.5"	SW	3000 LB	1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, ASME B16.11			
TEE	½" – 1.5"	SW	3000 LB	1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, ASME B16.11			
TEE (RED)	½" – 1.5"	SW	3000 LB	1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, ASME B16.11			
PLUG	½" – 1.5"	THRD		1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, ASME B16.11, ROUND HEAD		03	
PLUG	½" – ½"	PE		1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, ASME B16.11, BARSTOCK, ROUND HEAD		53	
COUPLING	½" – 1.5"	SW	3000 LB	1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, ASME B16.11			
CAP	½" – 1.5"	SW	3000 LB	1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, ASME B16.11			
CAP	½" – 1.5"	THRD	3000 LB	1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, ASME B16.11		03	
SOCKOLET	½" – 1.5"	WELD	3000 LB	1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, MSS-SP-97			
THREDOLET	½" – 1.5"	WELD	3000 LB	1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, MSS-SP-97		03	
WELDOLET	2" – 24"	WELD		1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, MSS-SP-97			
WELDOLET	72" – 90"	WELD		1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, MSS-SP-97			
90 ELL	2" – 24"	WELD		1 ¼ CR – ½ MO, ASTM A234 GRADE WP11 CL 1-S, ASME B16.9			
90 ELL	72" – 90"	WELD		1 ¼ CR – ½ MO, ASTM A234 GRADE WP11 CL 1-W, ASME B16.9 / PER DETAIL DRAWING		701	
45 ELL	2" – 24"	WELD		1 ¼ CR – ½ MO, ASTM A234 GRADE WP11 CL 1-S, ASME B16.9			
45 ELL	72" – 84"	WELD		1 ¼ CR – ½ MO, ASTM A234 GRADE WP11 CL 1-W, ASME B16.9 / PER DETAIL DRAWING		701	
TEE	2" – 24"	WELD		1 ¼ CR – ½ MO, ASTM A234 GRADE WP11 CL 1-S, ASME B16.9			
TEE	72" – 84"	WELD		1 ¼ CR – ½ MO, ASTM A234 GRADE WP11 CL 1-W, ASME B16.9 / PER DETAIL DRAWING		701	
TEE (RED.)	2" – 24"	WELD		1 ¼ CR – ½ MO, ASTM A234 GRADE WP11 CL 1-S, ASME B16.9			
TEE (RED.)	72" – 84"	WELD		1 ¼ CR – ½ MO, ASTM A234 GRADE WP11 CL 1-W, ASME B16.9 / PER DETAIL DRAWING		701	
CAP	2" – 24"	WELD		1 ¼ CR – ½ MO, ASTM A234 GRADE WP11 CL 1-S, ASME B16.9			
CAP	72" – 84"	WELD		1 ¼ CR – ½ MO, ASTM A234 GRADE WP11 CL 1-S, ASME B16.9 / PER DETAIL DRAWING		701	
REDUCER (CC)	2" – 24"	WELD		1 ¼ CR – ½ MO, ASTM A234 GRADE WP11 CL 1-S, ASME B16.9			
REDUCER (CC)	72" – 90"	WELD		1 ¼ CR – ½ MO, ASTM A234 GRADE WP11 CL 1-S, ASME B16.9 / PER DETAIL DRAWING		701	
REDUCER (EC)	2" – 24"	WELD		1 ¼ CR – ½ MO, ASTM A234 GRADE WP11 CL 1-S, ASME B16.9			
VALVES							
GATE	½" – 1.5"	SW	CL 800	1 ¼ CR – ½ MO Body W/ 13 CR TRIM, HF SEAT (API #8)	PVVB68S1808 1QCD0ZZ	06, 03	
GATE	½" – 1.5"	THRD/SW	CL 800	1 ¼ CR – ½ MO Body W/ 13 CR TRIM, HF SEAT (API #8)	PVVB68D180 81QCD0ZZ		
GATE	½" – 1.5"	RTJ	CL 300	1 ¼ CR – ½ MO Body W/ 13 CR TRIM, HF SEAT (API #8), Conventional Port	PVVB63J1808 1QCD0ZZ	15	
GATE	2" – 10"	RTJ	CL 300	1 ¼ CR – ½ MO Body W/ 13 CR TRIM, HF SEAT (API #8), Full Bore	PVVG63J2708 1QCC0ZZ		
GATE	12" – 24"	RTJ	CL 300	1 ¼ CR – ½ MO Body W/ 13 CR TRIM, HF SEAT (API #8), Full Bore G.O.	PVVG63J2708 1QCC5ZZ		
GLOBE	½" – 1.5"	SW	CL 800	1 ¼ CR – ½ MO Body W/ 13 CR TRIM, HF SEAT (API #8)	PVVL68S1808 1QCD0ZZ		
GLOBE	2" – 6"	RTJ	CL 300	1 ¼ CR – ½ MO Body W/ 13 CR TRIM, HF SEAT (API #8)	PVVL63J2708 1QCC0ZZ		
GLOBE	8" – 14"	RTJ	CL 300	1 ¼ CR – ½ MO Body W/ 13 CR TRIM, HF SEAT (API #8), GO	PVVL63J2708 1QCC5ZZ		
LIFT CHECK	½" – 1.5"	SW	CL 800	1 ¼ CR – ½ MO Body W/ 13 CR TRIM, HF SEAT (API #8), Piston Type	PVVC68S1808 NQCD0ZZ	61	
SWING CHECK	2" – 24"	RTJ	CL 300	1 ¼ CR – ½ MO Body W/ 13 CR TRIM, HF SEAT (API #8)	PVVC63J2708 NQCC0ZZ	62	
FLANGES							
SOCKETWELD	½" – 1.5"	RTJ	CL 300	1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, ASME B16.5			
WELDNECK	2" – 24"	RTJ	CL 300	1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, ASME B16.5		702	
WELDNECK	48"	RTJ	CL 300	1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, (AS PER DETAIL DRAWING)		702	
WELDNECK	72" – 84"	RTJ	CL 300	1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, (AS PER DETAIL DRAWING)		702	
BLIND	½" – 24"	RTJ	CL 300	1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, ASME B16.5			
BLIND	72" – 84"	RTJ	CL 300	1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2, (AS PER DETAIL DRAWING)			
PR WN ORIFICE	1" – 24"	RTJ	CL 300	1 ¼ CR – ½ MO, ASTM A182 GRADE F11 CL 2 W/ ½" SW TAPS, ASME B16.36			
MISCELLANEOUS							
FIGURE 8 BLANK	½" – 12"	RTJ	CL 300	1 ¼ CR – ½ MO, ASTM A387 GRADE 11 CL 2, ASME B16.48			
PADDLE BLANK	14" – 24"	RTJ	CL 300	1 ¼ CR – ½ MO, ASTM A387 GRADE 11 CL 2, ASME B16.48			
PADDLE SPACER	14" – 24"	RTJ	CL 300	1 ¼ CR – ½ MO, ASTM A387 GRADE 11 CL 2, ASME B16.48			

ITEM	SIZE RANGE	ENDS	SCH/RAT	MATERIAL DESCRIPTION	NOTES	REV
PADDLE SPACER	72"	RTJ	CL 300	1 ¼ CR – ½ MO, ASTM A387 GRADE 11 CL 2, ASME B16.48 (AS PER DETAIL DRAWING)		
Y-STRAINER	½" – 1.5"	SW	CL 600	1 ¼ CR – ½ MO, ASTM A217 GRADE WC6 Body, TP 304SS Screen W / 1/16" Dia. Hole @ 3/32" Centre	553	
Y-STRAINER	½" – 1.5"	SW	CL 600	1 ¼ CR – ½ MO, ASTM A217 GRADE WC6 Body, TP 304SS Screen W / 1/32" Dia. Hole @ 3/64" Center	554	
CONE STRAINER – TRUNCATED TYPE	2" – 24"	RTJ	CL 300	SS, ASTM A240 GRADE TP 304SS, Perforated Plate with 1/8" Dia. Hole @ 3/16" Center, ASME B16.5	555	
GASKETS	½" – 24" 48" 72" – 84"		CL 300 CL 300 CL 300	SOLID OVAL RING,, ASME B16.20, RING TYPE JOINT, ASME B16.5, 5 Chromium SOLID OVAL RING,, ASME B16.20, RING TYPE JOINT, (AS PER DETAIL DRAWING), 5 Chromium SOLID OVAL RING,, ASME B16.20, RING TYPE JOINT, (AS PER DETAIL DRAWING), 5 Chromium		
BOLTING						
STUD	ALL		CL 300	ALLOY STEEL, ASTM A193 GRADE B16, ASTM A194 GRADE 4 HEAVY HEX NUTS		

	NOTES		REV
JOINT CONNECTS			
MAINTENANCE		FLANGES	
MAINTENANCE		FLANGES	
NON-MAINTENANCE		SOCKETWELD	
NON-MAINTENANCE		BUTTWELD	
		SIZES	
		½" – 2"	
		3" – 90"	
		½" – 1 ½"	
		2" – 90"	

NOTE	SPECIFIC PIPING NOTES	REV
02	All butt-welded component thicknesses shall match the pipe thickness.	
03	Threaded joints are permitted only at the outlet of vent and drain valves, hydrostatic connections, outlet instrument take-off valves, and to match equipment..	
06	These valves shall be used only for vent, drain, and instrument connections.	
09	Pressure and temperature rating can be limited by certain components permitted by this Practice. Manufacturer's recommended pressure-temperature restrictions shall be consulted.	
12	These flanges shall be used only to match Class 300 flange connections at control valves and special equipment.	
15	These valves shall be used if mating to flanged nozzles.	
53	This item shall be used for orifice flange tap connections only.	
61	These check valves shall be installed in a horizontal position with cover up.	
62	These check valves shall be installed in a horizontal position with cover up or in a vertical position with upward flow.	
503	Use this item for Branch Nipple which terminate at the valve or continue as tubing. (eg. For instrument connections, valved vents & drains, etc.).	
531	Examination shall be in accordance with requirements in "Standard for Piping Welding", refer to document number: 108184-0000-PI-SPC-0006.	
553	These check valves shall be installed in a horizontal position with hinge pin vertical or in a vertical position with upward flow.	
554	Use this strainer for Vapor / Gas Service, blow off connection requirements per "Technical Specification for Pipe, Fittings, and Flanges ".	
555	Cone Strainer shall be installed between the 2 flanges with the small end pointed upstream.	
701	Manufacturer of fittings above 48" shall provide proof that fittings supplied meet requirements of para 2.2 of ASME B16.9 for review by Buyer.	
702	These flanges have been calculated based on process design conditions on an individual basis.	
705	For straight piping without branches; with branches thickness to be calculated by DEC.	
706	For branch connection 90" or 84" header with 72" branch a thicker wall is required as per ASME B31.3, paragraph 304.3.1.	

90° BRANCH CONNECTION CHART

BRANCH SIZE	90	T																			
	84	B	T																		
	72	B	B	T																	
	48	B	B	B	T																
	24	B	B	B	RT	T															
	20	B	B	B	RT	RT	T														
	18	B	B	B	B	RT	RT	T													
	16	B	B	B	B	RT	RT	RT	T												
	14	B	B	B	B	RT	RT	RT	RT	T											
	12	B	B	B	B	RT	RT	RT	RT	RT	T										
	10	B	B	B	B	RT	RT	RT	RT	RT	RT	T									
	8	W	W	W	W	W	W	W	W	W	W	RT	T								
	6	W	W	W	W	W	W	W	W	W	W	W	RT	T							
	4	W	W	W	W	W	W	W	W	W	W	W	W	RT	T						
	3	W	W	W	W	W	W	W	W	W	W	W	W	W	RT	T					
	2	W	W	W	W	W	W	W	W	W	W	W	W	W	W	RT	T				
	1½	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	RT	T			
	1	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	RT	RT	T	
	¾	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	RT	RT	T	
	½	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	RT	RT	RT	T
		90	84	72	48	24	20	18	16	14	12	10	8	6	4	3	2	1½	1	¾	½
HEADER SIZE																					

LEGEND:

E = Reducing Tee (BW for 2" and larger, SW for 1.5" and smaller)
 B = Branch Weld (No Reinforcing Pads) (Note 705/706)
 A = Sockolet

T = Straight Tee (BW for 2" and larger, SW for 1.5" and smaller)
 W = Weldolet

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCU. NO.	

PIPING CLASS	REV
B09A	R00

SERVICES	Propane(P)					REV

PIPING MATERIAL	CARBON STEEL	VALVE MATERIAL	CARBON STEEL	WELDEXAMINATION	HOLD	REV
RATING / FACING	ASME CLASS 300 / RF	TRIM MATERIAL	316 SS (API#10)	STRESS RELIEF	NOT REQUIRED	
CORROSION ALLOWANCE	1.9 mm (0.075")	CODE	ASME B31.4/SNIP 2.05.06-85	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (MPa)*

REV

SIZE

TEMP P °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
-28 to 121	5.10	5.10	5.10	5.10	5.10	5.10	5.10	5.10	5.10	5.10	5.10					
150	4.92	4.92	4.92	4.92	4.92	4.92	4.92	4.92	4.92	4.92	4.92					

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES RTFE SEAT RATINGS (Mpa)

TEMP(C)	< NPS 2	REV
-28 to 38		
93		
149		
204		

BUTTERFLY RTFE SEAT RATINGS(Mpa)

TEMP(C)	ALL SIZES	REV
-28 to 38		
93		
149		
204		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	12"		PE	CS, API 5L PSL2 Gr. X56N, SMLS, 5.6mm	903	
HOT INDUCTION BENDING	12"		PE	CS, API 5L PSL2 Gr. X56N, SMLS, 6.3mm	903	
BARRED TEE	12" 12" × 6"		BW BW	CS, WPHY-56, MSS-SP-75, 6.3mm CS, WPHY-56, MSS-SP-75, 6.3mm	903 703,704 703,704	
REDUCER CON ECC	12" 12"		BW BW	CS, WPHY-56, ASME B16.9, 6.3mm CS, WPHY-56, ASME B16.9, 6.3mm		
FLANGE WELDNECK BLIND FLG	12" 12"	CL300 CL300	RF RF	CS, ASTM A694 Gr. F56, ASME B16.5, THICKNESS AS PIPE CS, ASTM A694 Gr. F56, ASME B16.5		
INSULATING JOINT	3"-12"		MFR	MFR	903 801,901	
VALVES SWING CHECK	12"	CL300	BW	CS BODY WITH 316 SS TRIM (API # 10)	TAG PVVCS63B0310 NCAL0ZZ 277 705	
BOLTING	ALL	CL300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEXNUTS		

PIPING BRANCH NOTES		REV

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
D1A	R00

					REV
SERVICES	HP Condensate	Boiler Feed Water	HP Boiler Blowdown		
	HP Boiler Continuous Blowdown	HP Steam			

PIPING MATERIAL	CARBON STEEL	VALVE MATERIAL	CARBON STEEL	WELDEXAMINATION		REV
RATING / FACING	ASME CLASS 600 / RF	TRIM MATERIAL	13 CR/HF (API #8)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.5mm	CODE	ASME B31.3	BENDING	PER PROJECT	

**FLANGE RATING &
MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)***

	SIZE																REV
TEMP. °C	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20	24	
-28 to 38	102.04	102.04	102.04	102.04	102.04	102.04	102.04	102.04	96.05	98.83	95.31	92.39	95.49	93.36	99.7	95.85	
93	93.77	93.77	93.77	93.77	93.77	93.77	93.77	93.77	93.77	93.77	93.77	93.77	93.77	93.36	93.77	93.77	
149	90.32	90.32	90.32	90.32	90.32	90.32	90.32	90.32	90.32	90.32	90.32	90.32	90.32	90.32	90.32	90.32	
204	87.22	87.22	87.22	87.22	87.22	87.22	87.22	87.22	87.22	87.22	87.22	87.22	87.22	87.22	87.22	87.22	
260	83.08	83.08	83.08	83.08	83.08	83.08	83.08	83.08	83.08	83.08	83.08	83.08	83.08	83.08	83.08	83.08	
316	78.26	78.26	78.26	78.26	78.26	78.26	78.26	78.26	78.26	78.26	78.26	78.26	78.26	78.26	78.26	78.26	
343	75.84	75.84	75.84	75.84	75.84	75.84	75.84	75.84	75.84	75.84	75.84	75.84	75.84	75.84	75.84	75.84	
371	73.08	73.08	73.08	73.08	73.08	73.08	73.08	73.08	73.08	73.08	73.08	73.08	73.08	73.08	73.08	73.08	
399	69.88	69.88	69.88	69.88	69.88	66.45	69.88	69.88	62.43	64.24	61.93	63.73	62.09	60.68	64.82	62.31	
427	56.88	56.88	56.88	56.88	56.88	55.2	56.88	56.88	51.87	53.37	51.45	52.95	51.59	50.41	53.86	51.75	

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2 " - 1 1/2 "	80	PE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	2"	80	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	3"	STD	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	4" - 6"	80	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	8"	60	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	10"	XS	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	12"	60	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M	280	
	14"	15.88mm	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M	280	
	16"	17.48mm	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M	280	
	18"	60	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M	280	
	20"	22.23mm	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M	280	
	24"	25.40mm	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M	280	
NIPPLE						
TOE	1/2 " - 2"	160	TOE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
TBE	1/2 " - 2"	160	TBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
PBE	1/2 " - 2"	80	PBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
SWAGE					1	
CON	1/2 " - 2"	160	TSE	CS, ASTM A234 GRADE WPB, MSS-SP-95		
	1/2 " - 2"	80	BBE	CS, ASTM A234 GRADE WPB, MSS-SP-95		
ECC	1/2 " - 2"	160	TSE	CS, ASTM A234 GRADE WPB, MSS-SP-95		
	1/2 " - 2"	80	BBE	CS, ASTM A234 GRADE WPB, MSS-SP-95		
COUPLING						
FULL COUPLING	1/2 " - 1 1/2 "	CL3000	SOCW	CS, ASTM A105, ASME B16.11		
FULL COUPLING	1/2 " - 2"	CL3000	THRD	CS, ASTM A105, ASME B16.11	279	
OLET						
SOCKOLET	1/2 " - 1 1/2 "	CL3000	WELD	CS, ASTM A105, MSS-SP-97		
THREDOLET	1/2 " - 2"	CL3000	WELD	CS, ASTM A105, MSS-SP-97	279	
WELDOLET	2" - 8"	AS PIPE	WELD	CS, ASTM A105, MSS-SP-97		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
PLUG ROUND HEAD	½" - 2"		THRD	CS, ASTM A105, ASME B16.11, ROUND HEAD			
CAP	½" - 1½" ½" - 2" 2" - 24"	CL3000 CL3000 AS PIPE	SOCW THRD BW	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9		279	
90° ELBOW SW LR	½" - 1½" 2" - 24"	CL3000 AS PIPE	SOCW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
45° ELBOW SW	½" - 1½" 2" - 24"	CL3000 AS PIPE	SOCW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
TEE	½" - 1½" 2" - 24"	CL3000 AS PIPE	SOCW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
RED TEE	½" - 1½" 2" - 24"	CL3000 AS PIPE	SOCW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
REDUCER CON ECC	2" - 24" 2" - 24"	AS PIPE AS PIPE	BW BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
FLANGE SOCKETWELD	½" - 1½"	CL600	RF	CS, ASTM A105, ASME B16.5			
WELDNECK	2" - 24"	CL600	RF	CS, ASTM A105, ASME B16.5 THICKNESS AS PIPE			
BLIND FLG	½" - 24"	CL600	RF	CS, ASTM A105, ASME B16.5			
DRIP RING	1" - 12"	CL600	FF	CS, ASTM A516 GRADE 70, NORMALIZED, 1-1/2" THICK WITH ¾" SOCW TAPS (STANDARD DRAWINGS)			
FIGURE 8 BLANK+SPACER	1" - 12" 14" - 24"	CL600 CL600	FF FF	CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48 CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48			
VALVES GATE	½" - 1½" ½" - 1½" ½" - 1½" ½" - 1½" ½" - 2" 2" 3" - 8" 10" - 24"	CL800 CL800 CL800 CL800 CL600 CL600 CL600 CL600	SOCW SOCW T/SW T/SW RF RF RF RF	CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT, GEAR OP.	TAG PVVGB68S0108 ACAD0ZZ PVVGF68S0108 ACAD0ZZ PVVGB68D0108 ACAD0ZZ PVVGF68D0108 ACAD0ZZ PVVGB66R0108 ACAD0ZZ PVVGF66R0308 BCAC0ZZ PVVGF66R0308 BCAC0ZZ PVVGF66R0308 BCAC5ZZ	277 275 4 4, 275 24 43	
GLOBE	½" - 1½" 2" - 6" 8" - 12"	CL800 CL600 CL600	SOCW RF RF	CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8) GEAR OP.	PVVLB68S0108A CAD0ZZ PVVLB66R0308 BCAC0ZZ PVVLB66R0308 BCAC5ZZ		
LIFT CHECK	½" - 1½"	CL800	SOCW	CS BODY WITH 13 CR TRIM (API # 1)	PVVCL68S0101 NCAD0ZZ		
WAFER DUAL PLATE CHECK	2" - 24"	CL600	RF	CS BODY WITH 13 CR PLATES, METAL SEATS	PVVCD66R0301 NCAA0ZZ		
TITLING DISC CHECK	2" - 24"	CL600	RF	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVC166R0308 NCAA0ZZ	43	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
STOP	½" - 1½"	CL800	SOCW	CS BODY WITH 13 CR / HF TRIM (API #8)	PVV168S0108A CAD0ZZ	43	
STOP-CHECK	½" - 1½"	CL800	SOCW	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVS68S0108 ACAN0ZZ	43, 281	
	2" - 8"	CL600	RF	CS BODY WITH 13 CR / HF TRIM (API #8)	PVVS66R0308 BCAN0ZZ	43, 281	
	10" - 14"	CL600	RF	CS BODY WITH 13 CR / HF TRIM (API #8) GEAR OP.	PVVS66R0308 BCAN5ZZ		
BOLT/NUTS	ALL	CL600		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEX NUTS			
GASKETS SPIRAL WOUND	½" - 24"	CL600		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, CS CENTERING RING AND 304 INTER RING, ASME B16.20 / ASME B16.5			

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES	SIZES ½" - 2" 3" - 24" ½" - 1 ½" 2" - 24"
		FLANGES	
		SOCKETWELD	
		BUTTWELD	

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
4	Socket-weld x Threaded gate valve shall be used for valved pressure, instrument, vent and drain connections only. Piping system end shall be socketweld and instrument end shall be threaded.	
24	Use when required at flanged equipment nozzles, or where noted on P&IDs or Modules.	
43	Only to be used as indicated on the P&IDs.	
275	Only to be used where full port gate valve is required as indicated on P&ID's.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drains requiring plugs, connections to tube fittings and for connections to threaded proprietary items.	
280	Impact testing of each product form (pipe & fittings) of material, welds (shop and field) and HAZ shall be in accordance with ASME B31.3 paragraph 323.3 (test temperature - 28°C)	
281	All stop-check valves shall have standard installations that have their flow axes horizontal and their stems in the vertical plane. Piping layouts may require non-standard installations of these valves that may result in different orientations. Only if written approval is obtained from the selected valve manufacturer may these valves have non-standard installations.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
D1B	R00

					REV
SERVICES	HP Boiler Blow down	Process			

PIPING MATERIAL	CARBON STEEL	VALVE MATERIAL	CARBON STEEL	WELDEXAMINATION		REV
RATING / FACING	ASME CLASS 600 / RF	TRIM MATERIAL	13 CR/HF (API #8)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	3.0mm	CODE	ASME B31.3	BENDING	PER PROJECT	

**FLANGE RATING &
MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)***

	SIZE												REV
TEMP. °C	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	
-28 to 38	102.04	102.04	102.04	102.04	102.04	102.04	102.04	102.04	102.04	102.04	102.04	102.04	
93	93.77	93.77	93.77	93.77	93.77	93.77	93.77	93.77	93.77	93.77	93.77	93.77	
149	90.32	90.32	90.32	90.32	90.32	90.32	90.32	90.32	90.32	90.32	90.32	90.32	
204	87.22	87.22	87.22	87.22	87.22	87.22	87.22	87.22	87.22	87.22	87.22	87.22	
260	83.08	83.08	83.08	83.08	83.08	83.08	83.08	83.08	83.08	83.08	83.08	83.08	
316	78.26	78.26	78.26	78.26	78.26	78.26	78.26	78.26	78.26	78.26	78.26	78.26	
343	75.84	75.84	75.84	75.84	75.84	75.84	75.84	75.84	75.84	75.84	75.84	75.84	
371	73.08	73.08	73.08	73.08	73.08	73.08	73.08	73.08	73.08	73.08	73.08	73.08	
399	69.88	69.88	69.88	69.88	69.88	69.88	69.88	69.88	66.86	67.84	69.12	69.88	
427	56.88	56.88	56.88	56.88	56.88	56.88	56.88	56.88	55.55	56.37	56.88	56.88	

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	½" - 1½"	160	PE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	2"	160	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	3" - 14"	80	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
NIPPLE	½" - 1½"	XXS	TOE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	½" - 1½"	XXS	TBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	½" - 1½"	160	PBE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
SWAGE	½" - 1½"	XXS	TSE	CS, ASTM A234 GRADE WPB, MSS-SP-95	1	
	½" - 1½"	160	BBE	CS, ASTM A234 GRADE WPB, MSS-SP-95		
	½" - 1½"	XXS	TSE	CS, ASTM A234 GRADE WPB, MSS-SP-95		
	½" - 1½"	160	BBE	CS, ASTM A234 GRADE WPB, MSS-SP-95		
COUPLING	½" - 1½"	CL6000	SOCW	CS, ASTM A105, ASME B16.11		
FULL COUPLING	½" - 2"	CL6000	THRD	CS, ASTM A105, ASME B16.11	279	
OLET	½" - 1½"	CL6000	WELD	CS, ASTM A105, MSS-SP-97		
SOCKOLET	½" - 2"	CL6000	WELD	CS, ASTM A105, MSS-SP-97	279	
THREDOLET	2" - 8"	AS PIPE	WELD	CS, ASTM A105, MSS-SP-97		
WELDOLET						
PLUG	½" - 2"		THRD	CS, ASTM A105, ASME B16.11, ROUND HEAD		
ROUND HEAD						
CAP	½" - 1½"	CL6000	SOCW	CS, ASTM A105, ASME B16.11		
	½" - 2"	CL6000	THRD	CS, ASTM A105, ASME B16.11	279	
	2" - 14"	AS PIPE	BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
90° ELBOW SW LR	½" - 1½" 2" - 14"	CL6000 AS PIPE	SOCW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
45° ELBOW SW	½" - 1½" 2" - 14"	CL6000 AS PIPE	SOCW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
TEE RED TEE	½" - 1½" 2" - 14" ½" - 1½" 2" - 14"	CL6000 AS PIPE CL6000 AS PIPE	SOCW BW SOCW BW	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
REDUCER CON ECC	2" - 14" 2" - 14"	AS PIPE AS PIPE	BW BW	CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9			
FLANGE SOCKETWELD WELDNECK BLIND FLG DRIP RING FIGURE 8 BLANK+SPACER	½" - 1½" 2" - 14" ½" - 14" 1" - 12" 1" - 10" 12" - 14"	CL600 CL600 CL600 CL600 CL600	RF RF RF FF FF FF	CS, ASTM A105, ASME B16.5 CS, ASTM A105, ASME B16.5 THICKNESS AS PIPE CS, ASTM A105, ASME B16.5 CS, ASTM A516 GRADE 70, NORMALIZED, 1-1/2" THICK WITH ¾" SOCW TAPS (STANDARD DRAWINGS) CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48 CS, ASTM A516 GRADE 70, NORMALIZED, ASME B16.48			
VALVES GATE	½" - 1½" ½" - 1½" ½" - 1½" ½" - 1½" ½" - 2" 2" 3" - 8" 10" - 14"	CL800 CL800 CL800 CL800 CL600 CL600 CL600 CL600	SOCW SOCW T/SW T/SW RF RF RF RF	CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT CS BODY WITH 13 CR / HF TRIM (API #8), FULL PORT, GEAR OP.	TAG PVVGB68S0108 ACAD0ZZ PVVGF68S0108 ACAD0ZZ PVVGB68D0108 ACAD0ZZ PVVGF68D0108 ACAD0ZZ PVVGB66R0108 ACAD0ZZ PVVGF66R0308 BCAC0ZZ PVVGF66R0308 BCAC0ZZ PVVGF66R0308 BCAC5ZZ	277 275 4 4, 275 24 43	
GLOBE	½" - 1½" 2" - 6" 8" - 12"	CL800 CL600 CL600	SOCW RF RF	CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8) GEAR OP.	PVVLB68S0108A CAD0ZZ PVVLB66R0308 BCAC0ZZ PVVLB66R0308 BCAC5ZZ		
CHECK WAFER CHECK TITLING DISC CHECK STOP STOP-CHECK	½" - 1½" 2" - 14" 2" - 14" ½" - 1½" ½" - 1½" 2" - 8" 10" - 14"	CL800 CL600 CL600 CL800 CL800 CL600 CL600	SOCW RF RF SOCW SOCW RF RF	CS BODY WITH 13 CR TRIM (API #1) CS BODY WITH 13 CR PLATES, METAL SEATS CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8) CS BODY WITH 13 CR / HF TRIM (API #8) GEAR OP.	PVVL68S0101 NCAD0ZZ PVVCD66R0301 NCAA0ZZ PVVC166R0308 NCAA0ZZ PVV168S0108A CAD0ZZ PVVSB68S0108 ACAN0ZZ PVVSB66R0308 BCAN0ZZ PVVSB66R0308 BCAN5ZZ	43 43 43, 281 43, 281	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
BOLT/NUTS	ALL	CL600		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEX NUTS		
GASKETS SPIRAL WOUND	½" - 14"	CL600		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, CS CENTERING RING AND 304 INTER RING, ASME B16.20 / ASME B16.5		

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD	SIZES ½" - 2" 3" - 14" ½" - 1 ½" 2" - 14"

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
4	Socket-weld x Threaded gate valve shall be used for valved pressure, instrument, vent and drain connections only. Piping system end shall be socketweld and instrument end shall be threaded.	
24	Use when required at flanged equipment nozzles, or where noted on P&IDs or Modules.	
43	Only to be used as indicated on the P&IDs.	
275	Only to be used where full port gate valve is required as indicated on P&ID's.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drains requiring plugs, connections to tube fittings and for connections to threaded proprietary items.	
280	Impact testing of each product form (pipe & fittings) of material, welds (shop and field) and HAZ shall be in accordance with ASME B31.3 paragraph 323.3 (test temperature – 28°C)	
281	All stop-check valves shall have standard installations that have their flow axes horizontal and their stems in the vertical plane. Piping layouts may require non-standard installations of these valves that may result in different orientations. Only if written approval is obtained from the selected valve manufacturer may these valves have non-standard installations.	

	REV
BRANCH CONNECTION CHART	

HEADER SIZE

0.5	T												
0.75	RT	T											
1	RT	RT	T										
1.5	RT	RT	RT	T									
2	A	A	A	A	T								
3	A	A	A	A	RT	T							
4	A	A	A	A	RT	RT	T						
6	A	A	A	A	W	RT	RT	T					
8	A	A	A	A	W	W	RT	RT	T				
10	A	A	A	A	W	W	W	RT	RT	RT	T		
12	A	A	A	A	W	W	W	W	RT	RT	RT	T	
14													
	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	

BRANCH

BRANCH LEGEND		NOTES	REV
T	Tee(B.W. for sizes three inch and larger, S.W. for two inch and smaller)		
RT	Reducing Tee (B.W. for three inch and larger, S.W. for two inch and smaller)		
W	Weldolet (PWHT required at branch attachment welds when the thickness through the weld in any plane through the branch is greater than 1.5" for P1 material per ASME B31.3.)		
A	Sockolet		
S	Stub-in with or without Reinforcing Pad shall be set-on type as shown in detail "1" or "3" of Figure 328.5.4D of ASME B31.3		
P	Reinforcing Pad (Note d.)		

PIPING BRANCH NOTES

		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
c.	Weldolet may be used in lieu of a stub-in Branch Weld if verified and approved by pressure design and flexibility analysis and shown on the piping design drawing	
e	PWHT required at all branch attachment welds when the thickness through the weld in any plane through the branch is greater than 38.1 mm (1.5") for P1 material per ASME B31.3 and greater than 19 mm (0.75") for P1 materials per ASME B31.1.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
D2A	R00

SERVICES	Process	Natural gas	REV

PIPING MATERIAL	LT CARBON STEEL (IMPACT TESTED)	VALVE MATERIAL	LT CARBON STEEL	WELD EXAMINATION		REV
RATING / FACING	ASME CL600 / RF	TRIM MATERIAL	316 SS (API #10)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.50 mm (0.059")	CODE	ASME B31.3	ASME B31.3	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (MPa)*

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12
-48 to 38	96.18	96.18	96.18	96.18	96.18	96.18	93.40	79.21	72.73	68.12	83.56
93	91.01	91.01	91.01	91.01	91.01	91.01	91.01	79.21	72.73	68.12	83.56
149	87.90	87.90	87.90	87.90	87.90	87.90	87.90	79.21	72.73	68.12	83.56
204	84.80	84.80	84.80	84.80	84.80	84.80	84.80	79.21	72.73	68.12	83.56
260	81.01	81.01	81.01	81.01	81.01	81.01	81.01	74.85	68.73	64.37	78.94
316	76.18	76.18	76.18	76.18	76.18	76.18	76.18	68.52	62.91	58.92	72.88

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" - 1-1/2" 2" 3" - 10" 12"	80 80 STD XS	PE BE BE BE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M	143	
NIPPLES	1/2" - 1-1/2" 1/2" - 1-1/2"	160 80	TOE PBE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M	143	
SWAGES BRANCH HEADER	1/2" - 1-1/2" 1/2" - 1-1/2" 1/2" - 1-1/2"	160 160 80	TSE BBE BBE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95 LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95 LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95	1,143	
OLET SOCKOLET WELDOLET	1/2" - 1-1/2" 2" - 4"	3000LB AS PIPE	SOCW WELD	LTCS, ASTM A350 GRADE LF2 CL1, MSS-SP-97 LTCS, ASTM A350 GRADE LF2 CL1, MSS-SP-97	143	
PLUG	1/2" - 1-1/2"		THRD	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11, ROUND HEAD	143	
CAP	1/2" - 1-1/2" 1/2" - 1-1/2" 2" - 12"	3000LB 3000LB AS PIPE	SOCW THRD WELD	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11 LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9	143 279	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
90° ELBOW	1/2" -1-1/2" 2" -12"	3000LB AS PIPE	SOCW WELD	LTCS, ASTM A350 GRADE LF2 CL1 , ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S , ASME B16.9		143	
45° ELBOW	1/2" -1-1/2" 2" -12"	3000LB AS PIPE	SOCW WELD	LTCS, ASTM A350 GRADE LF2 CL1 , ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S , ASME B16.9		143	
TEE	1/2" -1-1/2" 2" -12"	3000LB AS PIPE	SOCW WELD	LTCS, ASTM A350 GRADE LF2 CL1 , ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S , ASME B16.9		143	
TEE (RED)	1/2" -1-1/2" 2" -12"	3000LB AS PIPE	SOCW WELD	LTCS, ASTM A350 GRADE LF2 CL1 , ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S , ASME B16.9		143	
COUPLING	1/2" -1-1/2"	3000LB	SOCW	LTCS, ASTM A350 GRADE LF2 CL1 , ASME B16.11		143	
COUPLING(RED)	1/2" -1-1/2"	3000LB	SOCW	LTCS, ASTM A350 GRADE LF2 CL1 , ASME B16.11			
COUPLING	1/2" -1-1/2"	3000LB	THRD	LTCS, ASTM A350 GRADE LF2 CL1 , ASME B16.11		279	
COUPLING(RED)	1/2" -1-1/2"	3000LB	THRD	LTCS, ASTM A350 GRADE LF2 CL1 , ASME B16.11		279	
REDUCER	2" - 12"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
CON	2" - 12"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
FLANGE						143	
SOCKETWELD	1/2" -1-1/2"	CL600	RF	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5			
WELDNECK	2" -12"	CL600	RF	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5			
BLIND	1/2" -12"	CL600	RF	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5			
MISCELLANEOUS							
FIGURE 8 BLANK	1" - 12"	CL600	FF	LTCS, ASTM A516 GRADE 70 S-5, NORMALIZED, ASME B16.48			
VALVES						143,277	
GATE	1/2" -1-1/2"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM(API #10)	TAG PVVGB68S0710 ACLD0ZZ		
	1/2" -1-1/2"	CL800	T/SW	LTCS BODY WITH 316 SS TRIM(API #10)	PVVGB68D0710 ACLD0ZZ	4	
	1/2" -1-1/2"	CL600	RF	LTCS BODY WITH 316 SS TRIM(API #10)	PVVGB66R0710 ACLD0ZZ	24	
	2" -8"	CL600	RF	LTCS BODY WITH 316 SS TRIM(API #10), FULL PORT.	PVVG66R1110 BCLC0ZZ		
	10" -12"	CL600	RF	LTCS BODY WITH 316 SS TRIM(API #10), FULL PORT. GEAR OP	PVVG66R1110 BCLC5ZZ		
GLOBE	1/2" -1-1/2"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM(API #10)	PVVL68S0710A CLD0ZZ		
	2" -6"	CL600	RF	LTCS BODY WITH 316 SS TRIM(API #10)	PVVL66R1110 BCLC0ZZ		
	8" -10"	CL600	RF	LTCS BODY WITH 316 SS TRIM(API #10).GEAR OP	PVVL66R1110 BCLC5ZZ		
LIFT CHECK	1/2" -1-1/2"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM(API #10) (HORIZ. ONLY)	PVVL68S0710 NCLD0ZZ		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
LUG CHECK	2" –10"	CL600	LUG- RF	LTCS BODY WITH 316 SS PLATES,METAL SEATS.	PVVCD66L1110 XNZA0ZZ	
BALL	1/2" –1-1/2"	CL600	RF	LTCS BODY WITH 316 SS BALL&STEM,RTFE SEATS,FULL PORT	PVVBA66R0771 ACLF0ZZ	27,43,326
BALL	2" –3"	CL600	RF	LTCS BODY WITH 316 SS BALL&STEM,RTFE SEATS,FULL PORT	PVVBB66R1171 BCLF0ZZ	27,43,326
BALL	4" –8"	CL600	RF	LTCS BODY WITH 316 SS BALL&STEM,RTFE SEATS,FULL PORT,GEAR OP	PVVBB66R1171 BCLF3ZZ	27,43,326
BOLTING STUD	ALL	CL600		ALLOY STEEL, ASTM A320 GRADE L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS S-3, ASME B18.2.1/ASME B18.2.2		
GASKET	1/2" –4"	CL600		SPIRAL WOUND,304SS W/ GRAPHITE FILLER,304 SS GENTERING RING AND 304 SS INNER RING, ASME B16.20/ ASME B16.5		
	6" –12"	CL600		SPIRAL WOUND,304SS W/ GRAPHITE FILLER,304 SS GENTERING RING AND 304 SS INNER RING ,ASME B16.20 /ASME B16.5		
MISCELLANEOUS FIGURE 8 BLANK	1" – 10"	CL600	FF	CS,ASTM A516 GRADE 70,NORMALIZED,ASME B16.48		

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD	SIZES 1/2" –1-1/2" 2" – 10" 1/2" – 1-1/2" 2" – 10"

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
4	Socket-weld x Threaded valve shall be used for valved pressure, and instrument connections only .Piping system end shall be socketwelded and instrument end shall be threaded.	
6	The flanges, gasket and bolting materials on all vents, drains, pressure, instrument, or flow connections shall be furnished by Piping/BOM.	
24	Use when required at flanged equipment nozzles,or where noted on P&IDs or modules in piping standard drawings.	
27	TFE,PTFE seated valves are limited to maximum 204 °C.	
43	Only to be used as indicated on the P&IDs.	
143	In shop and/or field fabrication 9girth,branch and fillet),all weld metal deposits and affected zone(HAZ) shall meet impact test requirements for low temperature service(at -48 °C for L.T.C.S.,at -105 °C for cryogenic S.S "L"specs)per Paragraph 323 of ASME B31.3.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections(when applicable),gage glass piping, vents and drain requiring plugs, ,connections to tube fittings and for connections to threaded proprietary items. See referenced Modules in Piping Standard Drawings.	
326	This resilient seated valve does not in all cases meet full flange rating,See valve MAWP in this specification.	

	REV
BRANCH CONNECTION CHART	

HEADER SIZE

0.5	T											
0.75	RT	T										
1	RT	RT	T									
1.5	RT	RT	RT	T								
2	A	A	A	A	T							
3	A	A	A	A	RT	T						
4	A	A	A	A	W	RT	T					
6	A	A	A	A	W	RT	RT	T				
8	A	A	A	A	W	W	RT	RT	T			
10	A	A	A	A	W	W	W	RT	RT	T		
12	A	A	A	A	W	W	W	RT	RT	RT	T	
	0.5	0.75	1	1.5	2	3	4	6	8	10	12	
BRANCH												

BRANCH LEGEND		NOTES	REV
T	Tee (B.W. for two inch and larger, S.W. for 1 1½" and smaller)		
W	Weldolet (PWHT required at branch attachment welds when the thickness through the weld in any plane through the branch is greater than 38.100 mm (1.5") for P1 material per ASME B31.3.		
A	Sockolet		
RT	Reducing Tee (B.W. for two inch and larger, S.W. for 1 1½" and smaller)		

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
c.	Weldolet may be used in lieu of stub-in if required and verified by flexibility analysis and shown on the piping design drawing.	
d.	A PAD has been selected to provide reinforcement for this branch connection subject to the most severe coincident condition of internal pressure and fluid temperature seen in this specification. However, the specific pressure-temperature condition for this connection may eliminate the need for reinforcement. In addition, when the diameter-to-thickness ratio is restricted by paragraph 304.3.1 (3) (b) of ASME B31.3, finite element analysis (FEA) shall be performed .	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
D21A	R00

SERVICES	Phosphate					REV

PIPING MATERIAL	304 S.S.	VALVE MATERIAL	316 SS	WELD EXAMINATION	HOLD	REV
RATING / FACING	ASME CLASS 600 / RF	TRIM MATERIAL	316 SS (API# 10)	STRESS RELIEF	NOT REQUIRED	
CORROSION ALLOWANCE	0.500 mm (0.02")	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (MPa)*

REV

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
-28 to 38	8.433	8.433	8.433	8.433	8.433											
93	7.128	7.128	7.128	7.128	7.128											

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES
 RTFE SEAT RATINGS (bar)

TEMP(C)	≤ NPS 2	>NPS2; ≤ NPS4	REV
-28 to 38	75.90		
93	52.40		
149			
204			

BUTTERFLY
 RTFE SEAT RATINGS(bar)

TEMP(C)	ALL SIZES	REV
-28 to 38		
93		
149		
204		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" – 1 1/2" 2"	40S 40S	PE BE	SS, ASTM A312 GRADE TP 304, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304, SMLS, ASME B36.19M		
NIPPLES	1/2" – 1 1/2"	40S	PBE	SS, ASTM A312 GRADE TP 304, SMLS, ASME B36.19M		
SWAGES CON&ECC	1/2" – 1 1/2"	40S	PBE	SS, ASTM A403 GRADE WP304-S, SMLS, MSS-SP-95		
OLET SOCKOLET	1/2" – 1 1/2"	CL3000	SOCW	SS, ASTM A182 GRADE F304, MSS-SP-97		
CAP	1/2" – 1 1/2" 2"	CL3000 AS PIPE	SOCW BW	SS, ASTM A182 GRADE F304, ASME B16.11 SS, ASTM A403 GRADE WP304-S, ASME B16.9		
90° ELBOW	1/2" – 1 1/2" 2"	CL3000 AS PIPE	SOCW BW	SS, ASTM A182 GRADE F304, ASME B16.11 SS, ASTM A403 GRADE WP304-S, ASME B16.9		
45° ELBOW	1/2" – 1 1/2" 2"	CL3000 AS PIPE	SOCW BW	SS, ASTM A182 GRADE F304, ASME B16.11 SS, ASTM A403 GRADE WP304-S, ASME B16.9		
COUPLING COUPLING COUPLING(RD)	1/2" – 1 1/2" 1/2" – 1 1/2"	CL3000 CL3000	SOCW SOCW	SS, ASTM A182 GRADE F304, ASME B16.11 SS, ASTM A182 GRADE F304, ASME B16.11		

	NOTES			REV
JOINT CONNECTS			SIZES	
MAINTENANCE		FLANGES	1/2" - 1 1/2"	
MAINTENANCE		FLANGES	2"	
NON-MAINTENANCE		SOCKETWELD	1/2" - 1 1/2"	
NON-MAINTENANCE		BUTTWELD	2"	

HEADER SIZE



PIPING BRANCH NOTES

a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
b.	For 45° branch connections, except that the branches from TLE's heater wash cone lines where 45° lateral stub-in is acceptable for atmospheric drain conditions.	
c.	Weldolet may be used in lieu of stub-in if required and verified by flexibility analysis and shown on the piping design drawing.	
d.	A PAD has been selected to provide reinforcement for this branch connection subject to the most severe coincident condition of internal pressure and fluid temperature seen in this specification. However, the specific pressure-temperature condition for this connection may eliminate the need for reinforcement. In addition, when the diameter-to-thickness ratio is restricted by paragraph 304.3.1 (3) (b) of ASME B31.3, finite element analysis (FEA) shall be performed.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
D21K	R00

SERVICES	PP Catalyst	High Pressure Nitrogen—40.0 barg	Isopropanol	Cold Liquid Drain	REV
	Process	Low Pressure Nitrogen—4.0 barg	Powder	Silane	
	Tea	Washing Oil	Propylene	Recycle Gas	

PIPING MATERIAL	304/304L SS DUAL GRADE	VALVE MATERIAL	316SS STAINLESS STEEL	WELD EXAMINATION	PER 108184-0000-PI-SPC-0006	REV
RATING / FACING	ASME CL600 / RF	TRIM MATERIAL	316 SS (API #10)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	0.50 mm (0.02")	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (MPa)*

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12
-48 to 38	99.28	99.28	99.28	99.28	99.28	99.28	99.28	96.62	73.9	78.7	92.54
93	82.74	82.74	82.74	82.74	82.74	82.74	82.74	82.74	82.74	78.7	82.74
149	74.12	74.12	74.12	74.12	74.12	74.12	74.12	74.12	74.12	74.12	74.12
204	68.61	68.61	68.61	68.61	68.61	68.61	68.61	68.61	68.61	68.61	68.61

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES RTFE SEAT RATINGS (bar)

TEMP(C)	< NPS 2	>NPS2; < NPS 6	> NPS 6	REV
-48 to 38	75.84	51.02	19.65	
93	52.4	35.50	13.79	
149	28.95	18.96	7.58	
204	5.52	3.34	1.38	

BUTTERFLY RTFE SEAT RATINGS(bar)

TEMP(C)	ALL SIZES	REV
-48 to 38	99.28	
93	62.1	
149	37.9	
204	17.2	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	½" – 1.1/2" 2" – 10" 12" 14" – 16"	40S 40S 80S 60	PE BE BE BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M		
NIPPLES	½" – 1.1/2" ½" – 1.1/2"	40S 80S	PBE TOE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M		
SWAGES	½" – 1.1/2" ½" – 1.1/2"	40S 80S	BBE TSE	SS, ASTM A403 GRADE WP304/304L-S, MSS-SP-95 SS, ASTM A403 GRADE WP304/304L-S, MSS-SP-95	1	
OLET SOCKOLET SOCWELBOLET WELDOLET THREDOLET	½" – 1.1/2" ½" – 1.1/2" 2" – 8" ½" – 1.1/2"	3000 LB 3000 LB AS PIPE 3000 LB	SOCW SOCW WELD THRD	SS, ASTM A182 GRADE F304/304L, MSS-SP-97 SS, ASTM A182 GRADE F304/304L, MSS-SP-97 SS, ASTM A182 GRADE F304/304L, MSS-SP-97 SS, ASTM A182 GRADE F304/304L, MSS-SP-97	279	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PLUG	½" – 1.1/2"		THRD	SS, ASTM A182 GRADE F304/304L, ASME B16.11, ROUND HEAD		
CAP	½" – 1.1/2" ½" – 1.1/2" 14" – 16"	3000 LB 3000 LB AS PIPE	SOCW THRD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9	279	
90° ELBOW	½" – 1.1/2" 2" – 12" 14" – 16"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9		
45° ELBOW	½" – 1.1/2" 2" – 12" 14" – 16"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9		
COUPLING	½" – 1.1/2" ½" – 1.1/2"	3000 LB 3000 LB	SOCW SOCW	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A182 GRADE F304/304L, ASME B16.11		
COUPLING(RED)	½" – 1.1/2" ½" – 1.1/2"	3000 LB 3000 LB	THRD THRD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A182 GRADE F304/304L, ASME B16.11	279 279	
45° S.W. LATERAL	½" – 1 ½"	3000 LB	SOCW	SS, ASTM A182 GRADE F304/304L, MSS-SP-97, MFR STD		
45° LATERAL	2" – 12"	AS PIPE	WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9, MFR STD		
45° SW LATERAL(RED)	½" – 1 ½"	3000 LB	SOCW	SS, ASTM A182 GRADE F304/304L, MSS-SP-97, MFR STD		
45° LATERAL (RED)	2" – 12"	AS PIPE	WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9, MFR STD		
TEE	½" – 1.1/2" 2" – 12" 14" – 16"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9		
RED TEE	½" – 1.1/2" 2" – 12" 14" – 16"	3000 LB AS PIPE AS PIPE	SOCW WELD WELD	SS, ASTM A182 GRADE F304/304L, ASME B16.11 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9		
REDUCER						
CON	2" – 12" 14" – 16"	AS PIPE AS PIPE	WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9		
ECC	2" – 12" 14" – 16"	AS PIPE AS PIPE	WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9		
FLANGE						
SOCKETWELD	½" – 1.1/2"	CL600	RF	SS, ASTM A182 GRADE F304/304L, ASME B16.5		
WELDNECK	2" – 12" 14" – 16"	CL600 CL600	RF RF	SS, ASTM A182 GRADE F304/304L, ASME B16.5 SS, ASTM A182 GRADE F304/304L, ASME B16.5		
BLIND	½" – 12"	CL600	RF	SS, ASTM A182 GRADE F304/304L, ASME B16.5		
MISCELLANEOUS						
PADDLE BLANK	1" – 12"	CL600	FF	SS, ASTM A240 GRADE 304/304L (SOLUTION HEAT TREATED), ASME B16.48	266	
PADDLE SPACER	1" – 12"	CL600	FF	SS, ASTM A240 GRADE 304/304L (SOLUTION HEAT TREATED), ASME B16.48	266	
VALVES						
GATE	½" – 1.1/2"	CL800	SOCW	SS BODY WITH 316 SS TRIM (API #10), VACUUM	143, 277 146, 148	
GATE	½" – 1.1/2"	CL800	SOCW	SS BODY WITH 316 SS TRIM (API #10), FULL PORT, VACUUM	146, 148, 275	
GATE	½" – 1.1/2"	CL800	T/SW	SS BODY WITH 316 SS TRIM (API #10)	4, 146, 148	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
GATE	½" – 1.1/2"	CL800	T/SW	SS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVG68D3810 ACND0ZZ	4,146, 148, 275	
GATE	½" – 1.1/2"	CL600	RF	SS BODY WITH 316 SS TRIM (API #10)	PVVG66R3810 ACND0ZZ	24, 146, 148	
GATE	2" – 8"	CL600	RF	SS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVG66R5410 BCNC0ZZ	146, 148	
GATE	10" – 12"	CL600	RF	SS BODY WITH 316 SS TRIM (API #10), FULL PORT, GEAR OP	PVVG66R5410 BCNC5ZZ	146, 148	
GLOBE	½" – 1.1/2"	CL800	SOCW	SS BODY WITH 316 SS TRIM (API #10), VACUUM	PVVL68S3810A CND0VV	148	
GLOBE	2" – 6"	CL600	RF	SS BODY WITH 316 SS TRIM (API #10)	PVVL66R5410 BCNC0ZZ	148	
GLOBE	8" – 10"	CL600	RF	SS BODY WITH 316 SS TRIM (API #10), GEAR OP	PVVL66R5410 BCNC5ZZ	148	
SWING CHECK	½" – 1.1/2"	CL800	SOCW	SS BODY WITH 316 SS TRIM (API #10)	PVVC68S3810 NCND0ZZ		
WAFER CHECK	2" – 12"	CL600	WAF- RF	SS BODY WITH 316 SS PLATES, METAL SEATS	PVVC66W5410 XNZA0ZZ		
WAF BUTTERFLY	3"	CL600	WAF- RF	SS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUB66W5499 BCNG1ZZ	27, 43, 148, 326	
WAF BUTTERFLY	4" – 24"	CL600	WAF- RF	SS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUB66W5499 BCNG3ZZ	27, 43, 148, 326	
LUG BUTTERFLY	3"	CL600	LUG- RF	SS BODY WITH 316 SS DISC, RTFE/316 SS SEAT	PVVUB66L5499 BCNG1ZZ	27, 43, 148, 326, 328	
LUG BUTTERFLY	4" – 24"	CL600	LUG- RF	SS BODY WITH 316 SS DISC, RTFE/316 SS SEAT, GEAR OP.	PVVUB66L5499 BCNG3ZZ	27, 43, 148, 326, 328	
BALL	½" – 1.1/2"	CL600	BW	SS BODY WITH 316SS BALL&STEM, RTFE SEATS, VACUUM	PVVBH66E3871 ACNF0B4	27, 55, 43, 326	
BALL	½" – 1.1/2"	CL600	BW	SS BODY WITH 316SS BALL&STEM, RTFE SEATS, FULL PORT, VACUUM	PVVBH66E3871 ACNF0B4	27, 55, 43, 326	
BALL	½" – 1.1/2"	CL600	T/SW	SS BODY WITH 316SS BALL&STEM, RTFE SEATS	PVVBH66D3871 ACNF0S2	27, 56, 43, 326	
BALL	½" – 1.1/2"	CL600	RF	SS BODY WITH 316SS BALL&STEM, RTFE SEATS, FULL PORT	PVVBH66R3871 BCNF0ZZ	27, 43, 326	
BALL	2" – 3"	CL600	RF	SS BODY WITH 316SS BALL&STEM, RTFE SEATS	PVVB66R5471 BCNF1ZZ	27, 43, 326	
BALL	4" – 24"	CL600	RF	SS BODY WITH 316SS BALL&STEM, RTFE SEATS	PVVB66R5471 BCNF3ZZ	27, 43, 326	
NEEDLE	½" – 1.1/2"	CL800	SOCW	SS BODY WITH 316 SS TRIM (API #10)	PVVB68S3810 ACNN0ZZ	148	
BOLTING	ALL	CL600		ALLOY STEEL, ASTM A320 GRADE B8 CL2 WITH ASTM A194 GRADE 8 HEAVY HEX NUTS			
GASKET	½" – 4"	CL600		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, 304 SS INNER RING & CENTERING RING, ASME B16.20 / ASME B16.5			
	6" – 16"	CL600		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER, 304 SS INNER RING & CENTERING RING, ASME B16.20 / ASME B16.5			

	NOTES		REV
JOINT CONNECTS			
MAINTENANCE		FLANGES	
MAINTENANCE		FLANGES	
NON-MAINTENANCE		SOCKETWELD	
NON-MAINTENANCE		BUTT WELD	
		SIZES	
		½" – 1.1/2"	
		2" – 12"	
		½" – 1 ½"	
		2" – 12"	

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
4	Socket-weld x Threaded valve shall be used for valved pressure, and instrument connections only .Piping system end shall be socketwelded and instrument end shall be threaded.	
24	Use when required at flanged equipment nozzle,or where noted on P&IDs or Modules in Piping Standard Drawings.	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
43	Only to be used as indicated on the P&IDs .	
47	See Modules section of Piping Standard Drawings.	
55	Supplier shall weld equal length of line class seamless plain end nipples on each butt weld end of the valve, overall length to be 250 mm.	
56	Supplier shall weld one piece of line class samless plain end nipple on the socket weld end of the socket weld x threaded valve,overall length to be 200mm.	
143	In shop and/or field fabrication 9girth,branch and fillet),all weld metal deposits and affected zone(HAZ) shall meet impact test requirements for low temperature service(at -48°C for L.T.C.S.,at -105°C for cryogenic S.S "L"specs)per Paragraph 323 of ASME B31.3.	
146	A 1/8 in. diameter bleed hole shall be provided in the disc from the cavity of the valve to the "vent" side of the valve only. An arrow with "vent" shall be stamped on the outside of the valve body indicating the direction of venting. (See P&ID's for direction of venting for various valve installations of cryogenic gate valves).	
148	Unless otherwise noted on the modules, valves shall not be installed in vertical lines. Valves in horizontal lines shall be installed with stem in vertical position or at least 45 degrees vertically from the horizontal centerline of the pipe.	
275	Only to be used where full port gate valve is required as indicated on P&IDs .	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME Class 300 flanges shall only be used to match higher rated flange connections at control valves and special equipment.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections(when applicable),gage glass piping, vents and drain requiring plugs, ,connections to tube fittings and for connections to threaded proprietary items. See referenced Modules in Piping Standard Drawings.	
326	This resilient seated valve does not in all cases meet full flange rating,See valve MAWP in this specification.	

	REV
90 DEGREE BRANCH CONNECTION CHART	

HEADER SIZE

1/2	T																
3/4	RT	T															
1	RT	RT	T														
1½	RT	RT	RT	T													
2	A	A	A	A	T												
3	A	A	A	A	RT												
4	A	A	A	A	W	RT											
6	A	A	A	A	W	W	RT	T									
8	A	A	A	A	W	W	W	RT	T								
10	A	A	A	A	W	W	W	W	RT	T							
12	A	A	A	A	W	W	W	W	W	RT	RT	T					
14	A	A	A	A	W	W	W	W	W	W	RT	RT	T				
16	A	A	A	A	W	W	W	W	W	W	RT	RT	RT	T			
	½	¾	1	1½	2	3	4	6	8	10	12	14	16				

BRANCH SIZE

45 DEGREE BRANCH CONNECTION CHART	REV
-----------------------------------	-----

HEADER SIZE

1/2	L											
3/4	RL	L										
1	RL	RL	L									
1 1/2	RL	RL	RL	L								
2	Ay	Ay	Ay	Ay	L							
3	Ay	Ay	Ay	Ay	P	L						
4	Ay	Ay	Ay	Ay	P	P	L					
6	Ay	Ay	Ay	Ay	P	P	P	L				
8	Ay	Ay	Ay	Ay	P	P	P	P	L			
10	Ay	Ay	Ay	Ay	P	P	P	P	P	L		
12	Ay	Ay	Ay	Ay	P	P	P	P	P	RL	L	
	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	

BRANCH SIZE

BRANCH LEGEND		NOTES	REV
S	45 degree stub-in		
Ay	Socket-weld latrolet		
L	Lateral (B.W. for sizes two inch and larger, S.W. for 1 ½" and smaller)		
RL	Reducing lateral (B.W. for sizes two inch and larger, S.W. for 1 ½" and smaller)		
Y	Butt-weld latrolet(PWHT required at branch attachment welds when the thickness through the weld in any plane through the branch is greater than 38.100 mm (1.5") for P1 material per ASME B31.3.)		

		REV
PIPING BRANCH NOTES		
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
b.	For 45° branch connections, except that the branches from TLE's heater wash cone lines where 45° lateral stub-in is acceptable for atmospheric drain conditions.	
c.	Weldolet may be used in lieu of stub-in if required and verified by flexibility analysis and shown on the piping design drawing.	
d.	A PAD has been selected to provide reinforcement for this branch connection subject to the most severe coincident condition of internal pressure and fluid temperature seen in this specification. However, the specific pressure-temperature condition for this connection may eliminate the need for reinforcement. In addition, when the diameter-to-thickness ratio is restricted by paragraph 304.3.1 (3) (b) of ASME B31.3, finite element analysis (FEA) shall be performed .	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
D23B	R00

SERVICES	Chemical Injection (Phosphate)			REV

PIPING MATERIAL	316/316L SS DUAL MARKED STAINLESS STEEL	VALVE MATERIAL	316 SS	WELDEXAMINATION		REV
RATING / FACING	ASME CLASS 600 / RF	TRIM MATERIAL	316 SS (TRIM#16)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	0.5mm	CODE	ASME B31.3	BENDING	PER PROJECT	

**FLANGE RATING &
MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)*
SIZE**

TEMP (C)	½	¾	1	1 1/2	2	3	4	6	8
-48 TO 38	99.3	99.3	99.3	99.3	99.3	99.3	90.89	77.44	71.45
93	86.05	86.05	86.05	86.05	86.05	86.05	86.05	77.44	71.45
149	77.14	77.14	77.14	77.14	77.14	77.14	77.14	77.14	71.45
204	70.94	70.94	70.94	70.94	70.94	70.94	70.94	70.94	69.20

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	½" - 1½"	80S	PE	SS, ASTM A312 GRADE TP 316/316L, SMLS, ASME B36.19M		
	2"	80S	BE	SS, ASTM A312 GRADE TP 316/316L, SMLS, ASME B36.19M		
	3" - 8"	40S	BE	SS, ASTM A312 GRADE TP 316/316L, SMLS, ASME B36.19M		
NIPPLE	½" - 1½"	80S	PBE	SS, ASTM A312 GRADE TP 316/316L, SMLS, ASME B36.19M		
SWAGE	½" - 1½"	80S	PBE	SS, ASTM A403 GRADE WP 316/316L -S, MSS-SP-95	1,	
	½" - 1½"	80S	PBE	SS, ASTM A403 GRADE WP 316/316L -S, MSS-SP-95		
COUPLING	½" - 1½"	CL3000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
FULL COUPLING	½" - 2"	CL3000	THRD	SS, ASTM A182 GRADE F 316/316L, ASME B16.11	279	
OLET	½" - 1½"	CL3000	WELD	SS, ASTM A182 GRADE F 316/316L, MSS-SP-97		
SOCKOLET	½" - 2"	CL3000	WELD	SS, ASTM A182 GRADE F 316/316L, MSS-SP-97	279	
THREDOLET	2" - 8"	AS PIPE	WELD	SS, ASTM A182 GRADE F 316/316L, MSS-SP-97		
WELDOLET						
PLUG	½" - 2"		THRD	SS, ASTM A182 GRADE F 316/316L, ASME B16.11, ROUND HEAD		
ROUND HEAD						
CAP	½" - 1½"	CL3000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
	½" - 2"	CL3000	THRD	SS, ASTM A182 GRADE F 316/316L, ASME B16.11	279	
	2" - 8"	AS PIPE	BW	SS, ASTM A403 GRADE F 316/316L-S, ASME B16.9		
90° ELBOW	½" - 1½"	CL3000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
SW	2" - 8"	AS PIPE	BW	SS, ASTM A403 GRADE F 316/316L-S, ASME B16.9		
LR						
45° ELBOW	½" - 1½"	CL3000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
SW	2" - 8"	AS PIPE	BW	SS, ASTM A403 GRADE F 316/316L-S, ASME B16.9		
TEE	½" - 1½"	CL3000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
	2" - 8"	AS PIPE	BW	SS, ASTM A403 GRADE F 316/316L-S, ASME B16.9		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
RED TEE	½" - 1½" 2" - 8"	CL3000 AS PIPE	SOCW BW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11 SS, ASTM A403 GRADE F 316/316L-S, ASME B16.9			
REDUCER CON ECC	2" - 8" 2" - 8"	AS PIPE AS PIPE	BW BW	SS, ASTM A403 GRADE F 316/316L-S, ASME B16.9 SS, ASTM A403 GRADE F 316/316L-S, ASME B16.9			
FLANGE SOCKETWELD WELDNECK BLIND FLG	½" - 1½" 2" - 8" ½" - 8"	CL600 CL600 CL600	RF RF RF	SS, ASTM A182 GRADE F 316/316L, ASME B16.5 SS, ASTM A182 GRADE F 316/316L, ASME B16.5 THICKNESS AS PIPE SS, ASTM A182 GRADE F 316/316L, ASME B16.5		54	
DRIP RING FIGURE 8	1" - 8" 1" - 8"	CL600 CL600	FF FF	SS, ASTM A240 GRADE 316/316L (SOLUTION HEAT TREATED), 1-1/2" THICK W/3/4" SOCW T&S (STANDARD DRWG) SS, ASTM A240 GRADE 316/316L, ASME B16.48			
VALVES GATE	½" - 1½" ½" - 1½" ½" - 2" 2" 3" - 8"	CL800 CL800 CL600 CL600 CL600	SOCW SOCW RF RF RF	SS BODY WITH 316SSTRIM(API#16) SS BODY WITH 316SSTRIM(API#16), FULL PORT SS BODY WITH 316SSTRIM(API#16) SS BODY WITH 316SSTRIM(API#16), FULL PORT SS BODY WITH 316SSTRIM(API#16), FULL PORT	TAG PVVGB68S3816 AJND0ZZ PVVGF68S3816 AJND0ZZ PVVGB62R3816 AJND0ZZ PVVGF62R5416 BJNC0ZZ PVVGF62R5416 BJNC0ZZ	277 275 24 43,275,	
GLOBE	½" - 1½" 2" - 8"	CL800 CL600	SOCW RF	SS BODY WITH 316SSTRIM(API#16) SS BODY WITH 316SSTRIM(API#16)	PVVLB68S3816A JND0ZZ PVVLB62R5416 BJNC0ZZ		
SWING CHECK	½" - 1½" 2" - 12"	CL600 CL600	RF RF	SS BODY WITH 316SSTRIM(API#16) SS BODY WITH 316SSTRIM(API#16)	PVVCS62R5416 NJND0ZZ PVVCS62R5416 NJNC0ZZ		
BOLT/NUTS	ALL	CL600		ALLOY STEEL, ASTM A320 GRADE B8M WITH ASTM A 194 GRADE 8 HEAVY HEX NUTS			
GASKETS SPIRAL WOUND	½" - 12"	CL600		SPIRAL WOUND, 316L SS W/ PTFE FILLER, 316L CENTERING RING AND 316L INTER RING, ASME B16.20 / ASME B16.5		54	

	NOTES				REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD		SIZES ½" - 2" 3" - 8" ½" - 1 ½" 2" - 8"	

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
4	Socket-weld x Threaded gate valve shall be used for valved pressure, instrument, vent and drain connections only. Piping system end shall be socketweld and instrument end shall be threaded.	
24	Use when required at flanged equipment nozzles, or where noted on P&IDs or Modules.	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
43	Only to be used as indicated on the P&IDs.	
54	Flat face line class flanges may be used to AS PIPE equipment connections as indicated on P&IDs only, and full face line class gaskets shall be used.	
275	Only to be used where full port gate valve is required as indicated on P&ID's.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drains requiring plugs, connections to tube fittings and for connections to threaded proprietary items.	

REV
BRANCH CONNECTION CHART

HEADER SIZE

0.5	T								
0.75	RT	T							
1	RT	RT	T						
1.5	RT	RT	RT	T					
2	A	A	A	A	T				
3	A	A	A	A	RT	T			
4	A	A	A	A	RT	RT	T		
6	A	A	A	A	W	W	RT	RT	T
8									
0.5	0.75	1	1.5	2	3	4	6	8	

BRANCH

BRANCH LEGEND	NOTES	REV
T	Tee(B.W. for sizes three inch and larger, S.W. for two inch and smaller)	
RT	Reducing Tee (B.W. for three inch and larger, S.W. for two inch and smaller)	
W	Weldolet (PWHT required at branch attachment welds when the thickness through the weld in any plane through the branch is greater than 1.5" for P1 material per ASME B31.3.)	
A	Sockolet	
S	Stub-in with or without Reinforcing Pad shall be set-on type as shown in detail "1" or "3" of Figure 328.5.4D of ASME B31.3	
P	Reinforcing Pad (Note d.)	

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
c.	Weldolet may be used in lieu of a stub-in Branch Weld if verified and approved by pressure design and flexibility analysis and shown on the piping design drawing	
e	PWHT required at all branch attachment welds when the thickness through the weld in any plane through the branch is greater than 38.1 mm (1.5") for P1 material per ASME B31.3 and greater than 19 mm (0.75") for P1 materials per ASME B31.1.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
D24L	R00

SERVICES	Catalyst	Cold Liquid Drain		REV

PIPING MATERIAL	316L SS STAINLESS STEEL	VALVE MATERIAL	316L SS STAINLESS STEEL	WELDEXAMINATION		REV
RATING / FACING	ASME CL600 / RF	TRIM MATERIAL	316 SS (API# 10)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	0.50 mm (0.02")	CODE	ASME B31.3	BENDING	PER PROJECT	

**FLANGE RATING &
MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*
SIZE**

TEMP (C)	1	1-1/2	2	3
-48 TO 38	82.73	82.73	82.73	82.73
93	70.32	70.32	70.32	70.32
149	62.74	62.74	62.74	62.74

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

**BALL VALVES
RTFE SEAT RATINGS (bar)**

TEMP(C)	≤NPS2	>NPS2	REV
-29 to 38	75.84	51.02	
93	52.4	35.50	
149	28.95	18.96	
204	5.52	3.34	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1" - 1-1/2" 2" - 3"	40S 40S	PE BE	SS, ASTM A312 GRADE TP 316L, SMLS, ASME B36.19M SS, ASTM A312 GRADE TP 316L, SMLS, ASME B36.19M		
NIPPLE	1" - 1-1/2"	80S	PBE	SS, ASTM A312 GRADE TP 316L, SMLS, ASME B36.19M		
SWAGE CON ECC	1" - 3" 1" - 3"	80S 80S	PBE PBE	SS, ASTM A403 GRADE WP 316L-S, MSS-SP-95 SS, ASTM A403 GRADE WP 316L-S, MSS-SP-95	1	
COUPLING	1" - 1-1/2" 2" - 3"	3000 LB 3000 LB	SOCW THRD	SS, ASTM A182 GRADE F316L, ASME B16.11 SS, ASTM A182 GRADE F316L, ASME B16.11	279	
COUPLING (RED)	1" - 1-1/2" 2" - 3"	3000 LB 3000 LB	SOCW THRD	SS, ASTM A182 GRADE F316L, ASME B16.11 SS, ASTM A182 GRADE F316L, ASME B16.11	279	
OLET SOCKOLET THREDOLET WELDOLET	1" - 1-1/2" 1/2" - 2" 2"	3000 LB 3000 LB AS PIPE	WELD WELD WELD	SS, ASTM A182 GRADE F316L, MSS-SP-97 SS, ASTM A182 GRADE F316L, MSS-SP-97 SS, ASTM A182 GRADE F316L, MSS-SP-97	279	
PLUG ROUND HEAD	1" - 2"		THRD	SS, ASTM A182 GRADE F316L, ASME B16.11, ROUND HEAD		
CAP	1" - 1-1/2" 1/2" - 2" 2" - 3"	3000 LB 3000 LB AS PIPE	SOCW THRD BW	SS, ASTM A182 GRADE F316L, ASME B16.11 SS, ASTM A182 GRADE F316L, ASME B16.11 SS, ASTM A403 GRADE WP316L-S, ASME B16.9	279	

90° ELBOW	1"- 1-1/2" 2" - 3"	3000 LB AS PIPE	SOCW BW	SS, ASTM A182 GRADE F316L, ASME B16.11 SS,ASTM A403 GRADE WP316L-S,ASME B16.9		
45° ELBOW	1"- 1-1/2" 2" - 3"	3000 LB AS PIPE	SOCW BW	SS, ASTM A182 GRADE F316L, ASME B16.11 SS,ASTM A403 GRADE WP316L-S,ASME B16.9		
90° BENDS	1"- 1-1/2" 2" - 3"	AS PIPE AS PIPE	BW BW	SS, ASTM A312 GRADE TP 316L, SMLS, ASME B36.19M, 15X DIAMETER SS, ASTM A312 GRADE TP 316L, SMLS, ASME B36.19M, 15X DIAMETER		
45° BENDS	1"- 1-1/2" 2" - 3"	AS PIPE AS PIPE	BW BW	SS, ASTM A312 GRADE TP 316L, SMLS, ASME B36.19M, 15X DIAMETER SS, ASTM A312 GRADE TP 316L, SMLS, ASME B36.19M, 15X DIAMETER		
TEE	1"- 1-1/2" 2" - 3"	3000 LB AS PIPE	SOCW BW	SS, ASTM A182 GRADE F316L, ASME B16.11 SS,ASTM A403 GRADE WP316L-S,ASME B16.9		
TEE (RED)	1"- 1-1/2" 2" - 3"	3000 LB AS PIPE	SOCW BW	SS, ASTM A182 GRADE F316L, ASME B16.11 SS,ASTM A403 GRADE WP 316L-S,ASME B16.9		
FLANGE						
SOCKETWELD WELDNECK	1"- 1-1/2" 2" - 3"	CL600 CL600	RF RF	SS,ASTM A182 GRADE F316L,ASME B16.5 SS,ASTM A182 GRADE F316L,ASME B16.5		
BLIND FLG	1" - 3"	CL600	RF	SS,ASTM A182 GRADE F316L,ASME B16.5		
MISCELLANEOUS						
FIGURE 8 BLANK	1" - 3"	CL600	FF	SS,ASTM A240 GRADE 316L,ASME B16.48		
VALVES					277	
GLOBE	1"- 1-1/2" 2" - 3"	CL800 CL600	SOCW RF	316L SS BODY WITH 316 SS TRIM(API#10) 316L SS BODY WITH 316 SS TRIM(API#10)	43	
SWING CHECK WAFER CHECK	1"- 1-1/2" 2" - 3"	CL800 CL600	SOCW WAF-RF	316L SS BODY WITH 316 SS TRIM(API#10) 316L SS BODY WITH 316 SS PLATES, METAL SEATS		
BALL	1"- 1-1/2"	CL600	BW	316L SS BODY WITH 316 SS BALL&STEM,FULL PORT, RTFE SEATS		27,55, 43,326
BALL	2" - 3"	CL600	RF	316L SS BODY WITH 316 SS BALL&STEM,FULL PORT, RTFE SEATS,		27,43,326
BOLT/NUTS	ALL	CL600		ALLOY STEEL, ASTM A320 GRADE B8 CL2 WITH ASTM A194 GRADE 8 HEAVY HEX NUTS		
GASKETS	1" - 3"	CL600		SPIRAL WOUND, 316LSS W/GRAPHITE FILLER, 316LSS CENTERING RING, AND CS OUTER RINGS, ASME B16.20 / ASME B16.5		

	NOTES		REV
JOINT CONNECTS			
MAINTENANCE			
NON-MAINTENANCE			
		FLANGES SOCKETWELD	
			SIZES 1/2" - 2" 1/2" - 1 1/2"

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
27	TFE,PTFE seated valves are limited to maximum 204°C.	
43	Only to be used as indicated on the P&IDs.	
55	Supplier shall weld equal length of line class seamless plain end nipples on each buttweld end of the valve, overall length to be 250 mm.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drains requiring plugs, connections to tube fittings and for connections to threaded proprietary items.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	

REV
BRANCH CONNECTION CHART

1 T
 1½ RT T
 2 A A T
 3 A A W T

 1 1½ 2 3

BRANCH

BRANCH LEGEND		NOTES	REV
T	Tee(B.W. for sizes three inch and larger, S.W. for two inch and smaller)		
RT	Reducing Tee (B.W. for three inch and larger, S.W. for two inch and smaller)		
W	Weldolet (PWHT required at branch attachment welds when the thickness through the weld in any plane through the branch is greater than 1.5" for P1 material per ASME B31.3.)		
A	Sockolet		

PROJECT NUMBER	01000
CLIENT	01000-MA-SPC-0002
LOCATION	

PIPING CLASS	REV
D24P	R00

SERVICES	High Pressure Nitrogen—40.0 barg	Powder	Propylene	REV

PIPING MATERIAL	304/304L SS DUAL GRADE	VALVE MATERIAL	316SS STAINLESS STEEL	WELD EXAMINATION	REV
RATING / FACING	ASME CL600 / RF	TRIM MATERIAL	316 SS/HF (API #12)	STRESS RELIEF	
CORROSION ALLOWANCE	0.50 mm(0.02")	CODE		BENDING	
				NOT REQUIRED	
				NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*

REV

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16
-48 to 38	84.33	84.33	84.33	84.33	84.33	84.33	84.33	84.33	84.33	84.33	84.33	84.33	84.33
93	71.28	71.28	71.28	71.28	71.28	71.28	71.28	71.28	71.28	71.28	71.28	71.28	71.28
149	63.94	63.94	63.94	63.94	63.94	63.94	63.94	63.94	63.94	63.94	63.94	63.94	63.94
204	57.99	57.99	57.99	57.99	57.99	57.99	57.99	57.99	57.99	57.99	57.99	57.99	57.99
260	53.74	53.74	53.74	53.74	53.74	53.74	53.74	53.74	53.74	53.74	53.74	53.74	53.74
316	50.58	50.58	50.58	50.58	50.58	50.58	50.58	50.58	50.58	50.58	50.58	50.58	50.58
343	49.15	49.15	49.15	49.15	49.15	49.15	49.15	49.15	49.15	49.15	49.15	49.15	49.15

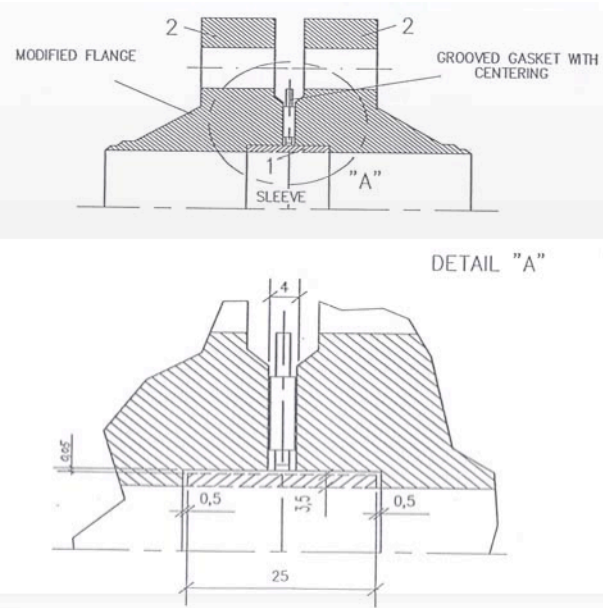
BALL VALVES
 RTFE SEAT RATINGS (bar)

TEMP(C)	≤NPS 2	>NPS2 ; ≤NPS 6	REV
-48 to 38	75.84	51.10	
93	52.40	35.50	
149	28.96	18.96	
204	5.51	3.44	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	½" –1-1/2"	40S	BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M		
	2" –3"	40S	BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M		
	4" –8"	80S	BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M		
	10" –12"	80S	BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M		
	14" –16"	60	BE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M		
NIPPLES	½" – 1-1/2"	80S	TOE	SS, ASTM A312 GRADE TP 304/304L, SMLS, ASME B36.19M		
SWAGES	½" –1-1/2"	80S	TSE	SS, ASTM A403 GRADE WP304/304L-S, SMLS, MSS-SP-95	1	
	½" –1-1/2"	40S	BBE	SS, ASTM A403 GRADE WP304/304L-S, SMLS, MSS-SP-95		
OLET WELDOLET	½" –8"	AS PIPE	WELD	SS, ASTM A182 GRADE F304/304L, MSS-SP-97	330	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
CAP	½" –10" 12" –16"	AS PIPE AS PIPE	WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9		330	
STUB END	½" – 10"	AS PIPE	WELD	(SPECIAL DESIGN)SS, ASTM A403 GRADE WP304/304L-S TYPE B, MSS-SP-43(SEE PIPING STANDARD DRWG Figure 2)		330	
SLEEVE	2" – 3"	AS PIPE	WELD	(SPECIAL DESIGN)SS, ASTM A312 GRADE TP 304/304L-S, ASME B36.19M(SEE PIPING STANDARD DRWG Figure 1)		330	
90° ELBOW	½" –1-1/2" 2" –10" 12" –16"	AS PIPE AS PIPE AS PIPE	WELD WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9		330	
90° BENDS	2" – 10" 2" – 10"	AS PIPE AS PIPE	WELD WELD	SS, ASTM A312 GRADE TP304/304L, SMLS,ASME B36.19M,10X DIAMETER SS, ASTM A312 GRADE TP304/304L, SMLS,ASME B36.19M,5X DIAMETER		330	
45° ELBOW	½" –1-1/2" 2" –10" 12" –16"	AS PIPE AS PIPE AS PIPE	WELD WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9		330	
45° BENDS	2" – 10"	AS PIPE	WELD	SS, ASTM A312 GRADE TP304/304L, SMLS,ASME B36.19M,10X DIAMETER		330	
TEE	½" –1-1/2" 2" –10"	AS PIPE AS PIPE	WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9		330	
RED	½" –1-1/2" 2" –10"	AS PIPE AS PIPE	WELD WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9 SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9			
REDUCER						330	
CON	2" –10"	AS PIPE	WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9			
ECC	2" –10"	AS PIPE	WELD	SS, ASTM A403 GRADE WP304/304L-S, ASME B16.9			
FLANGE						279 374 375	
LAP JOINT	½" –10"	CL600	RF	(SPECIAL DESIGN,FIG.2)CS,ASTM A105, ASME B16.5(SEE PIPING STANDARD DRWG Figure 2)			
THREADED	½" –1-1/2"	CL600	RF	SS, ASTM A182 GRADE F304/304L, ASME B16.5			
WELDNECK	½" –16"	CL600	RF	SS, ASTM A182 GRADE F304/304L, ASME B16.5			
	2" –6"	CL600	RF	(SPECIAL DESIGN)WITH LEEVE FIG.1,FOR REACTOR DISCHARGE LINES)SS, ASTM A182 GRADE F304/304L, ASME B16.5, (SEE PIPING STANDARD DRWG Figure 1)			
BLIND	½" –10"	CL600	RF	SS, ASTM A182 GRADE F304/304L, ASME B16.5			
FIGURE 8 BLANK	½" –2"	CL600	FF	SS, ASTM A240 GRADE F304/304L, ASME B16.48		277 147,148	
VALVES							
GATE	½" –1-1/2"	CL800	SOCW	SS BODY WITH 316 SS/HF TRIM (API #12),EXTENDED BONNET	TAG PVVGB68S3812 ACND0ZZ		

Flange Connection for Powder lines with Flanges (special Design)
Figure 1. Welding Neck Flange (special Design with Sleeve)



CONTINUATION

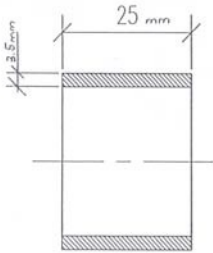
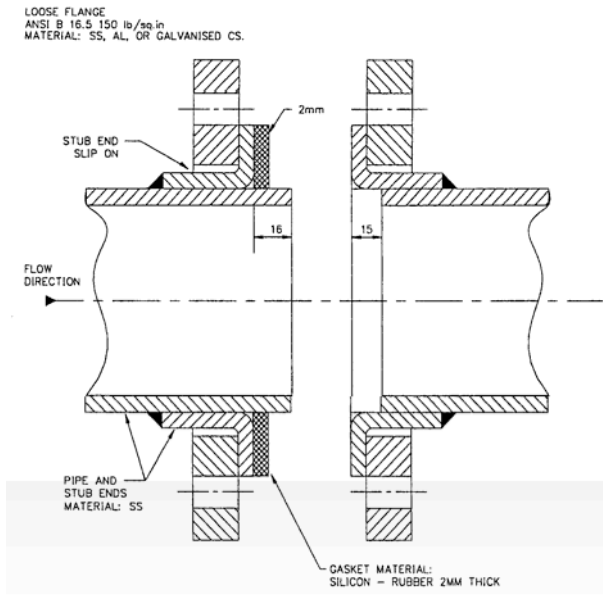


Figure 2: Slip On Flanges (special Design)



PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
E2A	R00

SERVICES	Natural gas(FG)	Propane				REV

PIPING MATERIAL	L.T. CARBON STEEL (IMPACT TESTED)	VALVE MATERIAL	L.T. CARBON STEEL	WELDEXAMINATION		REV
RATING / FACING	ASME CLASS 900 / RTJ	TRIM MATERIAL	316 SS (API# 16)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.50 mm (0.059")	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*

REV

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
-38 to 38	153.2	153.2	153.2	153.2	153.2	153.2	153.2	153.2	153.2	153.2	153.2					
50	150.4	150.4	150.4	150.4	150.4	150.4	150.4	150.4	150.4	150.4	150.4					
100	139.8	139.8	139.8	139.8	139.8	139.8	139.8	139.8	139.8	139.8	139.8					

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" – 1 1/2" 2" – 12"	80 XS	BE BE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M	143,951	
NIPPLES	1/2" – 1 1/2"	80	BBE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M	143,951	
SWAGES CON&ECC CON&ECC	1/2" – 1 1/2" 1/2" – 4"	80 AS PIPE	PBE BLE/P SE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95 LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95	143,951 702	
OLET SOCKOLET WELDOLET	1/2" – 1 1/2" 2" – 8"	CL3000 AS PIPE	SOCW BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, MSS-SP-97 LTCS, ASTM A350 GRADE LF2 CLASS 1, MSS-SP-97	143	
CAP	1/2" – 1 1/2" 2" – 12"	CL3000 AS PIPE	SOCW BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9	143	
90° ELBOW	1/2" – 1 1/2" 2" – 12"	CL3000 AS PIPE	SOCW BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9	143	
45° ELBOW	1/2" – 1 1/2" 2" – 12"	CL3000 AS PIPE	SOCW BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9	143	
TEE	1/2" – 1 1/2" 2" – 12"	CL3000 AS PIPE	SOCW BW	LTCS, ASTM A350 GRADE LF2 CLASS 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		

	NOTES				REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD		SIZES 1/2" – 11/2" 2" – 12" 1/2" – 11/2" 1/2" – 12"	

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
c.	Weldolet may be used in lieu of a stub-in Branch Weld if verified and approved by pressure design and flexibility analysis and shown on the piping design drawing	
d.	A pad has been selected to provide reinforcement for this branch connection subject to the most severe coincident condition of internal pressure and fluid temperature seen in this specification. However, the specific pressure temperature condition for this connection may eliminate the need for reinforcement. In addition, when the diameter to-thickness ratio is restricted by paragraph 304.3.1 (3) (b) of ASME B31.3, finite element analysis of the Pad design must be performed by the Contractor Stress Engineer.	
e.	PWHT required at all branch attachment welds when the thickness through the weld in any plane through the branch is greater than 38.1 mm (1.5") for P1 material per ASME B31.3 and greater than 19 mm (0.75") for P1 materials per ASME B31.1.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
E54K	R00

SERVICES	Primary high pressure steam	HP Clean Condensate			REV

PIPING MATERIAL	1-1/4Cr - 1/2Mo	VALVE MATERIAL	1-1/4Cr - 1/2Mo	WELD EXAMINATION	HOLD	REV
RATING / FACING	ASME CLASS 900 / RJ	TRIM MATERIAL	304/HF (API# 15)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.5 mm (0.059")	CODE	ASME B31.3	BENDING	PER PROJECT	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (MPa)*

REV

SIZE

TEMP P °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24
-28 to 38	15.51	15.51	15.51	15.51	15.51	15.51	15.51	15.51	15.51	15.51	15.51	15.51	15.51			
50	15.51	15.51	15.51	15.51	15.51	15.51	15.51	15.51	15.51	15.51	15.51	15.51	15.51			
100	15.44	15.44	15.44	15.44	15.44	15.44	15.44	15.44	15.44	15.44	15.44	15.44	15.44			
150	14.92	14.92	14.92	14.92	14.92	14.92	14.92	14.92	14.92	14.92	14.92	14.92	14.92			
200	14.39	14.39	14.39	14.39	14.39	14.39	14.39	14.39	14.39	14.39	14.39	14.39	14.39			
250	13.90	13.90	13.90	13.90	13.90	13.90	13.90	13.90	13.90	13.90	13.90	13.90	13.90			
300	12.86	12.86	12.86	12.86	12.86	12.86	12.86	12.86	12.86	12.86	12.86	12.86	12.86			
350	12.07	12.07	12.07	12.07	12.07	12.07	12.07	12.07	12.07	12.07	12.07	12.07	12.07			
400	10.98	10.98	10.98	10.98	10.98	10.98	10.98	10.98	10.98	10.98	10.98	10.98	10.98			
450	10.14	10.14	10.14	10.14	10.14	10.14	10.14	10.14	10.14	10.14	10.14	10.14	10.14			
475	9.51	9.51	9.51	9.51	9.51	9.51	9.51	9.51	9.51	9.51	9.51	9.51	9.51			
500	7.72	7.72	7.72	7.72	7.72	7.72	7.72	7.72	7.72	7.72	7.72	7.72	7.72			

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES
 RTFE SEAT RATINGS (Mpa)

TEMP(C)	ALL SIZES	REV
-28 to 38		
93		
149		
204		

BUTTERFLY
 RTFE SEAT RATINGS(Mpa)

TEMP(C)	ALL SIZES	REV
-28 to 38		
93		
149		
204		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" –1 1/2" 2" – 16"	80 80	BE BE	1-1/4Cr - 1/2Mo, ASTM A335-P11, SMLS, ASME B36.10M 1-1/4Cr - 1/2Mo, ASTM A335-P11, SMLS, ASME B36.10M	951	
NIPPLES	1/2" – 2"	80	BBE	1-1/4Cr - 1/2Mo, ASTM A335-P11, SMLS, ASME B36.10M	951	
SWAGES CON&ECC CON&ECC	1/2" –1 1/2" 1/2" – 4"	80 80	BBE BBE	1-1/4Cr - 1/2Mo, ASTM A234-WP11 CL1-S, SMLS, MSS-SP-95 1-1/4Cr - 1/2Mo, ASTM A234-WP11 CL1-S, SMLS, MSS-SP-95	951 702	
OLET WELDOLET	1/2" – 16"		BW	1-1/4Cr - 1/2Mo, ASTM A182-F11 CL2, MSS-SP-97	1001	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV	
CAP	1/2" – 1 1/2" 2" – 16"	AS PIPE AS PIPE	BW BW	1-1/4Cr - 1/2Mo, ASTM A182-F11 CL2, ASME B16.9 1-1/4Cr - 1/2Mo, ASTM A234-WP11 CL1-S, ASME B16.9				
90° ELBOW	1/2" – 1 1/2" 2" – 16"	AS PIPE AS PIPE	BW BW	1-1/4Cr - 1/2Mo, ASTM A182-F11 CL2, ASME B16.9 1-1/4Cr - 1/2Mo, ASTM A234-WP11 CL1-S, ASME B16.9				
45° ELBOW	1/2" – 1 1/2" 2" – 16"	AS PIPE AS PIPE	BW BW	1-1/4Cr - 1/2Mo, ASTM A182-F11 CL2, ASME B16.9 1-1/4Cr - 1/2Mo, ASTM A234-WP11 CL1-S, ASME B16.9				
TEE	1/2" – 1 1/2" 2" – 16"	AS PIPE AS PIPE	BW BW	1-1/4Cr - 1/2Mo, ASTM A182-F11 CL2, ASME B16.9 1-1/4Cr - 1/2Mo, ASTM A234-WP11 CL1-S, ASME B16.9				
RED TEE	1/2" – 1 1/2" 2" – 16"	AS PIPE AS PIPE	BW BW	1-1/4Cr - 1/2Mo, ASTM A182-F11 CL2, ASME B16.9 1-1/4Cr - 1/2Mo, ASTM A234-WP11 CL1-S, ASME B16.9				
REDUCER	CON ECC	2" – 16" 2" – 16"	AS PIPE AS PIPE	BW BW	1-1/4Cr - 1/2Mo, ASTM A234-WP11 CL1-S, ASME B16.9 1-1/4Cr - 1/2Mo, ASTM A234-WP11 CL1-S, ASME B16.9			
FLANGE	WELDNECK BLIND FLG	1/2"- 16" 1/2"- 16" 1/2"- 16" 1/2"- 16"	CL900 CL1500 CL900 CL1500	RTJ RTJ RTJ RTJ	1-1/4Cr - 1/2Mo, ASTM A182-F11 CL2, ASME B16.5 THICKNESS AS PIPE 1-1/4Cr - 1/2Mo, ASTM A182-F11 CL2, ASME B16.5 THICKNESS AS PIPE 1-1/4Cr - 1/2Mo, ASTM A182-F11 CL2, ASME B16.5 1-1/4Cr - 1/2Mo, ASTM A182-F11 CL2, ASME B16.5	278 278		
MISCELLANEOUS	FIGURE 8 BLANK PADDLE BLANK PADDLE SPACER	1" – 8" 10" – 16" 10" – 16"	CL900 CL900 CL900	RTJ RTJ RTJ	1-1/4Cr - 1/2Mo, ASTM A182-F11 CL2, ASME B16.48 1-1/4Cr - 1/2Mo, ASTM A182-F11 CL2, ASME B16.48 1-1/4Cr - 1/2Mo, ASTM A182-F11 CL2, ASME B16.48			
VALVES	GATE	1/2" – 1 1/2" 1/2" – 1 1/2" 2" – 6" 8" – 16" 2" – 6" 8" – 16"	CL1500 CL900 CL900 CL900 CL900 CL900	BW RTJ BW BW RTJ RTJ	1-1/4Cr - 1/2Mo BODY WITH 304 SS TRIM (API #15), FULL PORT 1-1/4Cr - 1/2Mo BODY WITH 304 SS TRIM (API #15), FULL PORT 1-1/4Cr - 1/2Mo BODY WITH 304 SS TRIM (API #15), FULL PORT 1-1/4Cr - 1/2Mo BODY WITH 304 SS TRIM (API #15), FULL PORT, GEAR OP. 1-1/4Cr - 1/2Mo BODY WITH 304 SS TRIM (API #15), FULL PORT 1-1/4Cr - 1/2Mo BODY WITH 304 SS TRIM (API #15), FULL PORT, GEAR OP.	TAG PVVGP75B1815 FMAD0ZZ PVVGP69J1815F MAD0ZZ PVVGP69B2715 FMAC0ZZ PVVGP69B2715 FMAC5ZZ PVVGP69J2715F MAC0ZZ PVVGP69J2715F MAC5ZZ	277 24,290 290 290	
	GLOBE	1/2" – 1 1/2" 2" – 4" 6" – 16" 2" – 4" 6" – 16"	CL1500 CL900 CL900 CL900 CL900	BW BW BW RTJ RTJ	1-1/4Cr - 1/2Mo BODY WITH 304 SS TRIM (API #15) 1-1/4Cr - 1/2Mo BODY WITH 304 SS TRIM (API #15) 1-1/4Cr - 1/2Mo BODY WITH 304 SS TRIM (API #15), GEAR OP. 1-1/4Cr - 1/2Mo BODY WITH 304 SS TRIM (API #15) 1-1/4Cr - 1/2Mo BODY WITH 304 SS TRIM (API #15), GEAR OP.	PVVLP75B1815F MAD0ZZ PVVLP69B2715F MAK0ZZ PVVLP69B2715F MAK5ZZ PVVLP69J2715F MAK0ZZ PVVLP69J2715F MAK5ZZ	290 290	
	LIFT CHECK	1/2" – 1 1/2"	CL1500	BW	1-1/4Cr - 1/2Mo BODY WITH 304 SS TRIM (API #15) (HORIZ. ONLY)	PVVCP75B1815 NMAD0ZZ		
	SWING CHECK	2" – 16"	CL900	BW	1-1/4Cr - 1/2Mo BODY WITH 304 SS TRIM (API #15)	PVVCQ69B2715 NMAL0ZZ		
BOLTING	ALL ALL	CL900 CL1500		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEXNUTS ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEXNUTS		278		

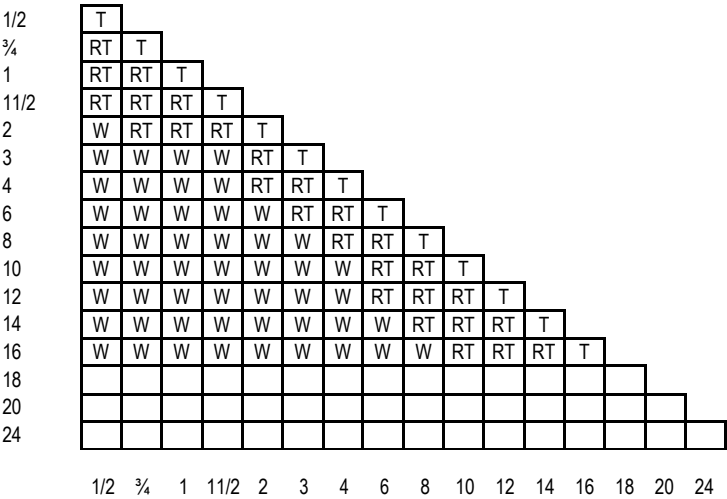
ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
GASKET OVAL RING	1/2"- 16" 1/2"- 16"	CL900 CL1500		OVAL RING, 316SS, MAX HARDNESS BHN 160, ASME B16.20 / ASME B16.5 OVAL RING, 316SS, MAX HARDNESS BHN 160, ASME B16.20 / ASME B16.5	278	

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES BUTTWELD	SIZES 1/2" – 11/2" 2" – 16" 1/2" – 16"

NOTE	SPECIFIC PIPING NOTES	REV
24	Use when required at flanged equipment nozzle, or where noted on P&IDs or Modules in Piping Standard Drawings.	
277	All valves shown below have detailed descriptions provided in the Valve Specification	
278	ASME Class 1500 flanges shall only be used to match higher rated flange connections at control valves and special equipment.	
290	Use when required for maintenance, flanged equipment nozzles, or where noted on the P&ID.	
702	Applied in the case where the small end is 1 1/2 inch and smaller, and the lager end is two inch and lager.	
951	The thickness does not in all cases meet full flange rating, it is based on the piping Design Condition.	
1001	The nominal size, schedule of main pipes and branches must be indicated when order is placed.	

	REV
BRANCH CONNECTION CHART	

HEADER SIZE



BRANCH SIZE

BRANCH LEGEND		NOTES	REV
T	Tee		
W	Weldolet		
RT	Reducing Tee		

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
c.	Weldolet may be used in lieu of stub-in if required and verified by flexibility analysis and shown on the piping design drawing.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
F2A	R00

					REV
SERVICES	Carbon Dioxide	Powder	High Pressure Nitrogen—40.0 barg		

						REV
PIPING MATERIAL	LT CARBON STEEL (IMPACT TESTED)	VALVE MATERIAL	LT CARBON STEEL	WELD EXAMINATION		
RATING / FACING	ASME CL1500 / BW	TRIM MATERIAL	316SS (API #10)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.50 mm (0.059")	CODE	ASME B31.3	BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)

REV

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10
-48 to 38	255.45	255.45	255.45	255.45	255.45	255.45	208.12	188.23	191.05	182.47
93	234.08	234.08	234.08	234.08	234.08	234.08	208.12	188.23	191.05	182.47
149	225.46	225.46	225.46	225.46	225.46	225.46	208.12	188.23	191.05	182.47
204	218.56	218.56	218.56	218.56	218.56	218.56	208.12	188.23	191.05	182.47

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES
 RTFE SEAT RATINGS (bar)

TEMP(C)	≤NPS2	>NPS2	REV
-29 to 38	75.84	51.02	
93	52.4	35.50	
149	28.95	18.96	
204	5.52	3.34	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	½" – 1 ½" 2" – 3" 4" – 10"	160 160 120	PE BE BE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M	143	
NIPPLES	½" – 1 ½"	160	PBE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M	143	
SWAGES CON & ECC	½" – 2"	160	BBE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95	1,143	
OLET WELDOLET	½" – 4"	AS PIPE	WELD	LTCS, ASTM A350 GRADE LF2 CL1, MSS-SP-97	143	
CAP	½" – 1 ½" 2" – 10"	AS PIPE AS PIPE	WELD WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9	143	
90° ELBOW	½" – 1 ½" 2" – 10"	AS PIPE AS PIPE	WELD WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9	143	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
45° ELBOW	½" – 1 ½" 2" – 10"	AS PIPE AS PIPE	WELD WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		143	
TEE	½" – 1 ½" 2" – 10"	AS PIPE AS PIPE	WELD WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
TEE (RED)	½" – 1 ½" 2" – 10"	AS PIPE AS PIPE	WELD WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
REDUCER							
CON	3" – 10"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9, ASTM A420 WPL6-S, ASME B16.9			
ECC	3" – 10"	AS PIPE	WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9, ASTM A420 WPL6-S, ASME B16.9			
FLANGE						143	
WELDNECK	0.5" – 10"	CL1500	RF	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5			
BLIND	½" – 10"	CL1500	RF	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5			
MISCELLANEOUS							
DRIP RING	1" – 10"	CL1500	RF	LTCS, ASTM A516 GRADE 70 S-5, NORMALIZED AND IMPACT TESTED AT -46 C, 38.1 mm (1-1/2") THICK WITH ¾" SOCKETWELD TAPS			
FIGURE 8 BLANK	1" – 8"	CL1500	RF	LTCS, ASTM A516 GRADE 70 S-5,, NORMALIZED, ASME B16.48			
PADDLE BLANK	10"	CL1500	RF	LTCS, ASTM A516 GRADE 70 S-5,, NORMALIZED, ASME B16.48			
PADDLE SPACE	10"	CL1500	RF	LTCS, ASTM A516 GRADE 70 S-5,, NORMALIZED, ASME B16.48			
VALVES						143, 277	
GATE	½" – 1 ½"	CL1500	BW	LTCS BODY WITH 316 SS TRIM (API #10)	TAG PVVGB75B0710 ACLD0ZZ		
GATE	2" – 4"	CL1500	BW	LTCS BODY WITH 316 SS TRIM (API #10)	PVVGB75B1110 BCLC0ZZ		
GATE	6" – 10"	CL1500	BW	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT, GEAR OP.	PVVG75B1110 BCLC5ZZ	275	
GLOBE	½" – 1 ½"	CL1500	BW	LTCS BODY WITH 316 SS TRIM (API #10)	PVVL75B0710A CLD0ZZ		
LIFT CHECK	½" – 1 ½"	CL1500	BW	LTCS BODY WITH 316 SS TRIM (API # 10)	PVVCL75B0710 NCLD0ZZ		
SWING CHECK	2" – 10"	CL1500	BW	LTCS BODY WITH 316 SS TRIM (API #10)	PVVCS75B1110 NCLC0ZZ		
BALL	½" – 1 ½"	CL1500	BW	LTCS BODY WITH 316 SS BALL&STEM, RTFE SEATS, FULL PORT, LONG PATTERN	PVVBA75B0710 ACLF0B2	27,55,326	
BALL	2" – 2 ½"	CL1500	BW	LTCS BODY WITH 316 SS BALL&STEM, RTFE SEATS, FULL PORT, LONG PATTERN	PVVBA75B1110 BCLF0B2	27,55,326	
STOP	1" – 2"	CL1500	RF	LTCS BODY WITH 316 SS TRIM (API #10), Y-PATTERN	PVV1X75R0710 ACLNOZZ	24, 209	
STOP	½" – 2 ½"	CL1500	BW	LTCS BODY WITH 316 SS TRIM (API #10)	PVV1B75B0710A CLNOZZ	209	
BOLTING	ALL	CL1500		ALLOY STEEL, ASTM A320 GR.L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS			
GASKET	ALL	CL1500		SPIRAL WOUND, 304SS W/ GRAPHITE FILLER, 304 SS GENTERING RING AND 304 SS INNER RING, ASME B16.20/ ASME B16.5			

BRANCH CONNECTION CHART		REV
-------------------------	--	-----

HEADER SIZE

$\frac{1}{2}$	T									
$\frac{3}{4}$	RT	T								
1	RT	RT	T							
$1\frac{1}{2}$	RT	RT	RT	T						
2	W	RT	RT	RT	T					
3	W	W	W	RT	RT	T				
4	W	W	W	W	RT	RT	T			
6	W	W	W	W	W	RT	RT	T		
8	W	W	W	W	W	W	RT	RT	T	
10	W	W	W	W	W	W	W	RT	RT	T
	$\frac{1}{2}$	$\frac{3}{4}$	1	$1\frac{1}{2}$	2	3	4	6	8	10

BRANCH SIZE

BRANCH

BRANCH LEGEND		NOTES	REV
T	Tee (B.W.)		
W	Weldolet (PWHT required at branch attachment welds when the thickness through the weld in any plane through the branch is greater than 38.100 mm (1.5") for P1 material per ASME B31.3.		
A	Sockolet		
RT	Reducing Tee (B.W.)		

PIPING BRANCH NOTES

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
F23A	R00

SERVICES	Hydraulic Oil	Seal Oil		REV

PIPING MATERIAL	316/316L SS DUAL MARKED STAINLESS STEEL	VALVE MATERIAL	316 SS	WELD EXAMINATION		REV
RATING / FACING	ASME CLASS 1500 / RF	TRIM MATERIAL	316 SS (TRIM #16)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	0.5mm	CODE	ASME B31.3	BENDING	PER PROJECT	

**FLANGE RATING &
MAXIMUM ALLOWABLE WORKING PRESSURE (Bar)*
SIZE**

TEMP (C)	½	¾	1	1 1/2	2	3	4
-28 TO 38	248.2	248.2	248.2	240.9	211.1	202.5	177.3
93	215.14	215.14	215.14	215.14	211.1	202.5	177.3
149	192.37	192.37	192.37	192.37	192.37	192.37	177.3
204	177.38	177.38	177.38	177.38	177.38	177.38	177.3

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	½" - 1½"	80S	PE	SS, ASTM A312 GRADE TP 316/316L, SMLS, ASME B36.19M		
	2" - 4"	80S	BE	SS, ASTM A312 GRADE TP 316/316L, SMLS, ASME B36.19M		
NIPPLE	½" - 1½"	80S	PBE	SS, ASTM A312 GRADE TP 316/316L, SMLS, ASME B36.19M		
SWAGE	½" - 1½"	160	PBE	SS, ASTM A403 GRADE WP 316/316L-S, MSS-SP-95	1,	
CON	½" - 1½"	160	PBE	SS, ASTM A403 GRADE WP 316/316L-S, MSS-SP-95		
ECC						
COUPLING FULL	½" - 1½"	CL6000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
COUPLING						
OLET	½" - 1½"	CL6000	WELD	SS, ASTM A182 GRADE F 316/316L, MSS-SP-97		
SOCKOLET	2" - 4"	AS PIPE	WELD	SS, ASTM A182 GRADE F 316/316L, MSS-SP-97		
WELDOLET						
PLUG	½" - 2"		THRD	SS, ASTM A182 GRADE F 316/316L, ASME B16.11, ROUND HEAD		
ROUND HEAD						
CAP	½" - 1½"	CL6000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
	2" - 4"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-S, ASME B16.9		
90° ELBOW	½" - 1½"	CL6000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
SW	2" - 4"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-S, ASME B16.9		
LR						
45° ELBOW	½" - 1½"	CL6000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
SW	2" - 4"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-S, ASME B16.9		
TEE	½" - 1½"	CL6000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
	2" - 4"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-S, ASME B16.9		
RED TEE	½" - 1½"	CL6000	SOCW	SS, ASTM A182 GRADE F 316/316L, ASME B16.11		
	2" - 4"	AS PIPE	BW	SS, ASTM A403 GRADE WP 316/316L-S, ASME B16.9		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
REDUCER CON ECC	2" - 4" 2" - 4"	AS PIPE AS PIPE	BW BW	SS,ASTM A403 GRADE WP 316/316L-S,ASME B16.9 SS,ASTM A403 GRADE WP 316/316L-S,ASME B16.9			
FLANGE WELDNECK BLIND FLG DRIP RING FIGURE 8	½" - 4" ½" - 4" 1" - 4" 1" - 4"	CL1500 CL1500 CL1500 CL1500	RF RF FF FF	SS,ASTM A182 GRADE F 316/316L,ASME B16.5 THICKNESS AS PIPE SS,ASTM A182 GRADE F 316/316L,ASME B16.5 SS,ASTM A240 GRADE 316/316L(SOLUTION HEAT TREATED),1-1/2"THICK W/3/4"SOCW T AS(STANDARD DRWGS) SS,ASTM A240 GRADE 316/316L,ASME B16.48		54	
VALVES GATE GLOBE LIFT CHECK	½" - 1½" ½" - 2" 2" 3" - 4" ½" - 1½" 2" - 4" ½" - 1½"	CL1500 CL1500 CL1500 CL1500 CL1500 CL1500 CL1500	SOCW BW RF RF SOCW BW SOCW	316 SS BODY WITH 316SS TRIM(API#16) 316SS BODY WITH 316SS TRIM(API#16) 316 SS BODY WITH 316SS TRIM(API#16),FULL PORT 316 SS BODY WITH 316SS TRIM(API#16),FULL PORT 316 SS BODY WITH 316SS TRIM(API#16) 316 SS BODY WITH 316SS TRIM(API#16) SS BODY WITH 316SS TRIM(API#16),PISTON	TAG PVVGB75S3816 ACED0ZZ PVVGB75B3816 ACED0ZZ PVVGF75B5416 BCEC0ZZ PVVGF75B5416 BCEC0ZZ PVVLB75S3816A CED0ZZ PVVLB75B5416B CEC0ZZ PVVCL75S3816 NCED0ZZ	277	
BOLT/NUTS	ALL	CL1500		ALLOY STEEL, ASTM A193 GRADE B8M WITH ASTM A 194 GRADE 8 HEAVY HEX NUTS			
GASKETS SPIRAL WOUND	½" - 4"	CL1500		SPIRAL WOUND,316L SS W/ PTFE FILLER, 316L CENTERING RING AND 316L INTER RING, ASME B16.20 / ASME B16.5		54	

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD	SIZES ½" - 2" 3" - 4" ½" - 1½" 2" - 4"

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
24	Use when required at flanged equipment nozzles, or where noted on P&IDs or Modules.	
43	Only to be used as indicated on the P&IDs.	
54	Flat face line class flanges may be used to AS PIPE equipment connections as indicated on P&IDs only, and full face line class gaskets shall be used.	
275	Only to be used where full port gate valve is required for instrument as indicated on P&ID's.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	

BRANCH CONNECTION CHART

0.5	T								
0.75	RT	T							
1	RT	RT	T						
1.5	RT	RT	RT	T					
2	A	A	A	A	T				
3	A	A	A	A	RT	T			
4	A	A	A	A	RT	RT	T		
6	A	A	A	A	W	RT	RT	T	
8	A	A	A	A	W	W	RT	RT	T
0.5									
0.75									
1									
1.5									
2									
3									
4									
6									
8									

BRANCH LEGEND		NOTES	REV
T	Tee(B.W. for sizes three inch and larger, S.W. for two inch and smaller)		
RT	Reducing Tee (B.W. for three inch and larger, S.W. for two inch and smaller)		
W	Weldolet (PWHT required at branch attachment welds when the thickness through the weld in any plane through the branch is greater than 1.5" for P1 material per ASME B31.3.)		
A	Sockolet		
S	Stub-in with or without Reinforcing Pad shall be set-on type as shown in detail "1" or "3" of Figure 328.5.4D of ASME B31.3		
P	Reinforcing Pad (Note d.)		

PIPING BRANCH NOTES

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
G82E	R00

SERVICES	Hydrogen					REV

PIPING MATERIAL	316 SS	VALVE MATERIAL	316 SS	WELD EXAMINATION	HOLD	REV
RATING / FACING	ASME CLASS 2500 / RJ	TRIM MATERIAL	HF (API 16#)	STRESS RELIEF	NOT REQUIRED	
CORROSION ALLOWANCE	0.500mm(0.02")	CODE	ASME B31.3	BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (MPa)*

REV

SIZE

TEMP P °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16
-38 to 38	41.37	41.37	41.37	41.37	41.37	41.37	41.37	41.37	41.37	41.37	41.37		
50	40.09	40.09	40.09	40.09	40.09	40.09	40.09	40.09	40.09	40.09	40.09		
100	35.16	35.16	35.16	35.16	35.16	35.16	35.16	35.16	35.16	35.16	35.16		
150	32.08	32.08	32.08	32.08	32.08	32.08	32.08	32.08	32.08	32.08	32.08		
200	29.72	29.72	29.72	29.72	29.72	29.72	29.72	29.72	29.72	29.72	29.72		
250	27.81	27.81	27.81	27.81	27.81	27.81	27.81	27.81	27.81	27.81	27.81		

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES
 RTFE SEAT RATINGS (Mpa)

TEMP(C)	ALL SIZES	REV
-38 to 38		
93		
149		
204		

BUTTERFLY
 RTFE SEAT RATINGS(Mpa)

TEMP(C)	ALL SIZES	REV
-38 to 38		
93		
149		
204		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" – 1 1/2"	160	BE	SS, ASTM A312 GRADE TP 316, SMLS, ASME B36.10M	951	
	2" – 6"	160	BE	SS, ASTM A312 GRADE TP 316, SMLS, ASME B36.10M		
	8" – 12"	140	BE	SS, ASTM A312 GRADE TP 316, SMLS, ASME B36.10M		
NIPPLES	1/2" – 1 1/2"	160	BBE	SS, ASTM A312 GRADE TP 316, SMLS, ASME B36.10M	951	
SWAGES CON&ECC	1/2" –1 1/2"	160	BBE	SS, ASTM A403 GRADE WP316-S, MSS-SP-95	951	
OLET WELDOLET	1/2" – 12"		BW	SS,ASTM A182 GRADE F316,MSS-SP-97	1001	
CAP	1/2" – 1 1/2" 2" – 12"	AS PIPE AS PIPE	BW BW	SS, ASTM A182 GRADE F316, ASME B16.9 SS,A403 GRADE WP 316-S,ASME B16.9		
90° ELBOW	1/2" – 1 1/2" 2" – 12"	AS PIPE AS PIPE	BW BW	SS, ASTM A182 GRADE F316, ASME B16.9 SS,A403 GRADE WP 316-S,ASME B16.9		
45° ELBOW	1/2" – 1 1/2" 2" – 12"	AS PIPE AS PIPE	BW BW	SS, ASTM A182 GRADE F316, ASME B16.9 SS,A403 GRADE WP 316-S,ASME B16.9		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
TEE	1/2" – 1 1/2" 2" – 12"	AS PIPE AS PIPE	BW BW	SS, ASTM A182 GRADE F316, ASME B16.9 SS,A403 GRADE WP 316-S,ASME B16.9		
RED TEE	1/2" – 1 1/2" 2" – 12"	AS PIPE AS PIPE	BW BW	SS, ASTM A182 GRADE F316, ASME B16.9 SS, ASTM A182 GRADE F316, ASME B16.9		
REDUCER						
CON	2" – 12"	AS PIPE	BW	SS,A403 GRADE WP 316-S,ASME B16.9		
ECC	2" – 12"	AS PIPE	BW	SS,A403 GRADE WP 316-S,ASME B16.9		
FLANGE						
WELDNECK	2" – 12"	CL2500	RTJ	SS,ASTM A182 GRADE F316,ASME B16.5 THICKNESS AS PIPE		
BLIND FLG	1/2"- 12"	CL2500	RTJ	SS,ASTM A182 GRADE F316,ASME B16.5		
VALVES					277	
GLOBE	1/2" – 1 1/2"	CL2500	BW	PSB,OS&Y, 316 SS BODY WITH TRIM(API#5), SCH160	TAG PVVLP85B3816F MLD0BZ PVVLP85B5416F MLC0BZ PVVLP85B5416F MLC5BZ PVVLP85B5416F MLC5BZ	
	2"	CL2500	BW	PSB,OS&Y, 316 SS BODY WITH TRIM(API#5), SCH160		
	3" – 6"	CL2500	BW	PSB,OS&Y, 316 SS BODY WITH TRIM(API#5), GEAR OP., SCH160		
	8"	CL2500	BW	PSB,OS&Y, 316 SS BODY WITH TRIM(API#5), GEAR OP., SCH140		
BOLTING	ALL	CL2500		ALLOY STEEL, ASTM A320 GR.L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS		
GASKET						
OVAl RING	ALL	CL2500		OVAl RING, 316SS, MAX HARDNESS BHN 160, ASME B16.20 / ASME B16.5		

	NOTES		REV
JOINT CONNECTS			
MAINTENANCE		FLANGES	
MAINTENANCE		FLANGES	
NON-MAINTENANCE			
NON-MAINTENANCE		BUTTWELD	

NOTE	SPECIFIC PIPING NOTES	REV
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
951	The thickness does not in all cases meet full flange rating, it is based on the piping Design Condition.	
1001	The nominal size, schedule of main pipes and branches must be indicated when order is placed.	

BRANCH CONNECTION CHART	REV
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HEADER SIZE

1/2	T																		
3/4	RT	T																	
1	RT	RT	T																
1 1/2	RT	RT	RT	T															
2	W	RT	RT	RT	T														
3	W	W	W	W	RT	T													
4	W	W	W	W	RT	RT	T												
6	W	W	W	W	W	RT	RT	RT	T										
8	W	W	W	W	W	W	RT	RT	RT	T									
10	W	W	W	W	W	W	W	RT	RT	RT	T								
12	W	W	W	W	W	W	W	RT	RT	RT	RT	T							
14																			
16																			
	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16						

BRANCH SIZE

BRANCH LEGEND		NOTES	REV
T	Tee		
W	Weldolet		
RT	Reducing Tee		

PIPING BRANCH NOTES

a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
c.	Weldolet may be used in lieu of stub-in if required and verified by flexibility analysis and shown on the piping design drawing.	
d.	A PAD has been selected to provide reinforcement for this branch connection subject to the most severe coincident condition of internal pressure and fluid temperature seen in this specification. However, the specific pressure-temperature condition for this connection may eliminate the need for reinforcement. In addition, when the diameter-to-thickness ratio is restricted by paragraph 304.3.1 (3) (b) of ASME B31.3, finite element analysis (FEA) shall be performed.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
H1A	R00

SERVICES	Rev
Cooling Water Supply	Cooling Water Return
Waste Water	Non-Contaminated Sewer
Hot water	High pressure fire water
Recovered water	Ro washing water
Oily wastewater	Sanitary wastewater
Off gas	Return sludge
Scum	Caustic liquid Urea(Hold)
	Safety Valve to Atmosphere
	Blow down
	Chemical Drain Service Water
	Oily Contaminated Rainy Water Drain
	Fire Water (Aboveground)
	Service water
	Clean surface water
	Wet Air
	Retentant
	Chemical Sewer
	Oily sludge
	Liquid slop oil
	Exces activated sludge
	Cooling Water (UG)

PIPING MATERIAL	CARBON STEEL	VALVE MATERIAL	CARBON STEEL	WELDEXAMINATION	REV
RATING / FACING	ASME CL150 / RF	TRIM MATERIAL	13CR (API #1)	STRESS RELIEF	PER CODE
CORROSION ALLOWANCE(mm)	3.00 mm (0.118")	CODE	ASME B31.3	BENDING	PER PROJECT

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)

SIZE

TEMP P °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16	18	20	24	30	36
-28-38	19.65	19.65	19.65	19.65	19.48	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65
93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93
149	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86	15.86
204	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79
260	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72
315	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65
343	8.61	8.61	8.61	8.61	8.61	8.61	8.61	8.61	8.61	8.61	8.61	8.61	8.61	8.61	8.61	8.61	8.61	8.61

TEMP P °C	40	48	72
-28-38	19.65	19.65	19.65
93	17.93	17.93	17.93
149	15.86	15.86	15.86
204	13.79	13.79	13.79
260	11.72	11.72	11.72
315	9.65	9.65	9.65
343	8.61	8.61	8.61

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVES RTFE SEAT RATINGS (bar)

TEMP(C)	< NPS 2	REV
-28 to 38	19.65	
93	17.93	
149	15.86	
204	5.49	

BUTTERFLY BUNA-N RTFE SEAT RATINGS (bar)

TEMP(C)	ALL SIZES	REV
-28 to 38	HOLD	
93	HOLD	
149	HOLD	
204	HOLD	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" - 1 1/2"	80	PE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	2" - 6"	STD	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	8" - 12"	30	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		
	14" - 16"	20	BE	CS, ASTM A106 GRADE B, SMLS, ASME B36.10M		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
	18" - 36" 40 48 54 64 72	20 XS 14.27mm 15.88mm 17.48mm 19.05mm	BE BE BE BE BE BE	CS, API 5L GRADE B, DSAW, ASME B36.10M, E _j =0.95 CS, API 5L GRADE B, DSAW, ASME B36.10M, E _j =0.95 CS, API 5L GRADE B, DSAW, ASME B36.10M, E _j =0.95 CS, API 5L GRADE B, DSAW, ASME B36.10M, E _j =0.95 CS, API 5L GRADE B, DSAW, ASME B36.10M, E _j =0.95 CS, API 5L GRADE B, DSAW, ASME B36.10M, E _j =0.95	548 548 548 548	
NIPPLES	½" - 1½" ½" - 1½"	160 160	TOE TBE	CS, ASTM A53 GRADE B, SMLS, ASME B36.10M CS, ASTM A53 GRADE B, SMLS, ASME B36.10M		
SWAGES CON ECC	½" - 1½" ½" - 1½"	80 80	BBE BBE	CS, ASTM A234 GRADE WPB-S, MSS-SP-95 CS, ASTM A234 GRADE WPB-S, MSS-SP-95	1	
OLET SOCKOLET THRD ELET WELD OLET	½" - 1½" ½" - 1½" 2" - 8"	3000 LB 3000 LB AS PIPE	SOCW THRD WELD	CS, ASTM A105, MSS-SP-97 CS, ASTM A105, MSS-SP-97 CS, ASTM A105, MSS-SP-97		
PLUG	½" - 2"		THRD	CS, ASTM A105, ASME B16.11, ROUND HEAD		
CAP	½" - 1½" ½" - 1½" 2" - 16" 18" - 36" 40" - 48" 54" - 72"	3000 LB 3000 LB AS PIPE AS PIPE AS PIPE AS PIPE	SOCW THRD WELD WELD WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 CS, ASTM A234 GRADE WPB-W, (GENERALLY ASME B16.9)	279 548 61,548	
90° ELBOW	½" - 1½" 2" - 6" 8" - 16" 18" - 36" 40" - 48" 54" - 72"	3000 LB AS PIPE AS PIPE AS PIPE AS PIPE AS PIPE	SOCW WELD WELD WELD WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 CS, ASTM A234 GRADE WPB-W, (GENERALLY ASME B16.9)	548 61,548	
45° ELBOW	½" - 1½" 2" - 6" 8" - 16" 18" - 36" 40" - 48" 54" - 72"	3000 LB AS PIPE AS PIPE AS PIPE AS PIPE AS PIPE	SOCW WELD WELD WELD WELD WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 CS, ASTM A234 GRADE WPB-W, (GENERALLY ASME B16.9)	548 61,548	
COUPLING COUPLING (RED)	½" - 1½" ½" - 1½"	3000 LB 3000 LB	SOCW SOCW	CS, ASTM A105, ASME B16.11 CS, ASTM A105, ASME B16.11		
UNION (GJ)	½" - 1½"	3000 LB	THRD	CS, ASTM A105, ASME B16.11		
TEE TEE (RED)	½" - 1½" 2" - 6" 8" - 16" 18" - 48" 64" - 72" ½" - 1½" 2" - 6"	3000 LB AS PIPE AS PIPE AS PIPE AS PIPE 3000 LB AS PIPE	SOCW WELD WELD WELD WELD SOCW WELD	CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-S, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 CS, ASTM A234 GRADE WPB-W, ASME B16.9 CS, ASTM A105, ASME B16.11 CS, ASTM A234 GRADE WPB-S, ASME B16.9	548 548	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
SWING CHECK	2" – 12"	CL150	RF	CS BODY WITH 13 CR TRIM (API #1)	PVVCS62R0301 NCAC0ZZ		
WAFER CHECK	14" – 24"	CL150	WAF- RF	CS BODY WITH 13 CR PLATES, METAL SEATS	PVVCD62W0301 XNZA0ZZ		
WAFER CHECK	26" – 36"	CL150	WAF- RF	CS BODY WITH 13 CR PLATES, METAL SEATS	PVVCD62I0301X NZA0ZZ		
WAF BUTTERFLY	2" – 6"	CL150	WAF- RF	CS BODY WITH 304 SS DISC. BUNA-N /304 SS SEAT	PVVUB62W0387 BCAG0ZZ	46,326	
WAF BUTTERFLY	8" – 24"	CL150	WAF- RF	CS BODY WITH 304 SS DISC. BUNA-N /304 SS SEAT, WORM GEAR	PVVUB62W0387 BCAG3ZZ	46,326	
WAF BUTTERFLY	26" – 48"	CL150	WAF- RF	CS BODY WITH 304 SS DISC. BUNA-N /304 SS SEAT, WORM GEAR	PVVUB62I0387B CAG3ZZ	46,326	
BALL	½" – 1½"	CL600	SOCW	CS BODY WITH 13Cr BALL & STEM, PTFE SEATS	PVVBD68S0161 ACAF0ZZ	16,326	
BALL	2" – 4"	CL150	RF	CS BODY WITH 13Cr BALL & STEM, PTFE SEATS	PVVBD62R0361 BCAF0ZZ	16,326	
BALL	6"	CL150	RF	CS BODY WITH 13Cr BALL & STEM, PTFE SEATS , WORM GEAR	PVVBD62R0361 BCAF3ZZ	16,326	
NEEDLE	½" – 1½"	CL800	SOCW	CS BODY WITH 13 CR TRIM (API #1)	PVVNB68S0101 ACAN0ZZ		
BOLTING	ALL	CL150		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A194 GRADE 2H HEAVY HEX NUTS			
	ALL	CL300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A194 GRADE 2H HEAVY HEX NUTS			
GASKET	½" – 14" 16" – 24" 26" – 48" ½" – 14" 16" – 24" 26" – 36"	CL150 CL150 CL150 CL300 CL300 CL300		1.5 mm COMPOSITE GASKET, 304+GRAF, ASME B16.21/ ASME B16.5 3.0 mm COMPOSITE GASKET, 304+GRAF, ASME B16.21/ ASME B16.5 3.0 mm COMPOSITE GASKET, 304+GRAF, ASME B16.21/ ASME B16.47A 1.5 mm COMPOSITE GASKET, 304+GRAF, ASME B16.21/ ASME B16.5 3.0 mm COMPOSITE GASKET, 304+GRAF, ASME B16.21/ ASME B16.5 3.0 mm COMPOSITE GASKET, 304+GRAF, ASME B16.21/ ASME B16.47A			

	NOTES			REV
JOINT CONNECTS		FLANGES FLANGES SOCW BUTTWELD	SIZES	
MAINTENANCE			½" – 1½"	
MAINTENANCE			2" – 36"	
NON-MAINTENANCE			½" – 1½"	
NON-MAINTENANCE			2" – 36"	

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size..	
14	For drainage only.	
15	Used as drain valve or root valve to connect instrument.	
16	Working temperature shall not exceed 150°C .	
20	To be installed at the horizontal pipeline only.	
46	Working temperature shall not exceed 120°C .	
61	Fittings and branch connections above NPS 48 diameter, outside the jurisdiction of the specified industry standard, shall be designed or approved by the Buyer's piping engineer. See piping design guidelines or specialty item data sheets.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME Class 300 flanges shall only be used to match higher rated flange connections at control valves and special equipment.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections(when applicable), gage glass piping, vents and drain requiring plugs, ,connections to tube fittings and for connections to threaded proprietary items. See referenced Modules in Piping Standard Drawings.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	
548	Pipe and pipe components thicker than 0.500" may require impact-tested materials. See ASME B31.3, Paragraph 323.2.2.	

BRANCH CONNECTION CHART		REV
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HEADER SIZE

1/2	T
¾	RT T
1	RT RT T
1½	RT RT RT T
2	A A A RT T
3	A A A A RT T
4	A A A A RT RT T
6	A A A A S RT RT T
8	A A A A P P RT RT T
10	A A A A P P P RT RT T
12	A A A A P P P P RT RT T
14	A A A A P P P P RT RT RT T
16	A A A A P P P P P P RT RT RT T
18	A A A A P P P P P P P RT RT RT T
20	A A A A P P P P P P P P RT RT RT T
24	A A A A P P P P P P P P P P RT RT RT T
30	A A A A P P P P P P P P P P P RT RT RT T
36	A A A A P P P P P P P P P P P P RT RT RT T
40	A A A A P P P P P P P P P P P P P RT RT RT T
48	A A A A P P P P P P P P P P P P P P P RT RT T
72	A A A A P P P P P P P P P P P P P P P P P RT RT T
	1/2 ¾ 1 1½ 2 3 4 6 8 10 12 14 16 18 20 24 30 36 40 48 72

BRANCH SIZE

BRBNCH

<u>BRBNCH LEGEND</u>		NOTES	REV
A	Sockolet		
P	Stub-in with Reinforcing Pad shall be set-in type as shown in detail "2" or "4" of Figure 328.5.4D of ASME B31.3		
T	Tee (B.W. for two inch and larger, S.W. for 1 11/2" and smaller)		
RT	Reducing Tee(B.W. for two inch and larger, S.W. for 1 11/2" and smaller)		
S	Stub-in without Reinforcing Pad shall be set-in type as shown in detail "2" or "4" of Figure 328.5.4D of BSME B31.3		

PIPING BRANCH NOTES

		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
H2A	R00

SERVICES	Service Water (City Water)	Waste Water	Blow down	Chemical Drain Service Water	REV
	Non-Contaminated Sewer	Oily Contaminated Rainy Water Drain	Fire Water (Aboveground)	Cooling Water Return	
	Cooling Water Supply	Saturation Air			

PIPING MATERIAL	LT CARBON STEEL (IMPACT TESTED)	VALVE MATERIAL	LT CARBON STEEL	WELD EXAMINATION	REV
RATING / FACING	ASME CL150 / RF	TRIM MATERIAL	316 SS (API #10)	STRESS RELIEF	PER CODE
CORROSION ALLOWANCE	3.00 mm (0.118")	CODE	ASME B31.3	BENDING	PER PROJECT

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16
-48-38	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65
93	17.92	17.92	17.92	17.92	17.92	17.92	17.92	17.92	17.92	17.92	17.92	17.92	17.92
149	15.85	15.85	15.85	15.85	15.85	15.85	15.85	15.85	15.85	15.85	15.85	15.85	15.85
204	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79	13.79
260	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72	11.72
315	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65	9.65
343	8.61	8.61	8.61	8.61	8.61	8.61	8.61	8.61	8.61	8.61	8.61	8.61	8.61

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	½" - 1 ½" 2" - 6" 8" - 16"	80 STD 30	PE BE BE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M	143	
NIPPLES	½" - 1 ½" ½" - 1 ½"	160 80	TBE PBE	LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M LTCS, ASTM A333 GRADE 6, SMLS, ASME B36.10M	143	
SWAGES CON & ECC	½" - 2"	80	BLE	LTCS, ASTM A420 GRADE WPL6-S, SMLS, MSS-SP-95	1,143	
OLET SOCKOLET WELDOLET	½" - 1 ½" 2" - 8"	3000 LB AS PIPE	SOCW WELD	LTCS, ASTM A350 GRADE LF2 CL 1, MSS-SP-97 LTCS, ASTM A350 GRADE LF2 CL 1, MSS-SP-97	143	
PLUG	½" - 2"	THRD		LTCS, ASTM A350 GRADE LF2 CL 1, ASME B16.11, ROUND HEAD	143 279	
CAP	½" - 1 ½" 2" - 16"	3000 LB AS PIPE	SOCW WELD	LTCS, ASTM A350 GRADE LF2 CL 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9	143	
90° ELBOW	½" - 1 ½" 2" - 16"	3000 LB AS PIPE	SOCW WELD	LTCS, ASTM A350 GRADE LF2 CL 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9	143	
45° ELBOW	½" - 1 ½" 2" - 16"	3000 LB AS PIPE	SOCW WELD	LTCS, ASTM A350 GRADE LF2 CL 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9	143	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
COUPLING COUPLING COUPLING(RED)	½" – 1 ½" ½" – 1 ½"	3000LB 3000LB	SOCW SOCW	LTCS, ASTM A350 GRADE LF2 CL 1, ASME B16.11 LTCS, ASTM A350 GRADE LF2 CL 1, ASME B16.11		143	
TEE	½" – 1 ½" 2" – 16"	3000LB AS PIPE	SOCW WELD	LTCS, ASTM A350 GRADE LF2 CL 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
RED	½" – 1 ½" 2" – 16"	3000LB AS PIPE	SOCW WELD	LTCS, ASTM A350 GRADE LF2 CL 1, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9			
REDUCER CON ECC	2" – 16" 2" – 16"	AS PIPE AS PIPE	WELD WELD	LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, ASME B16.9		143	
FLANGE						143	
SOCKETWELD	½" – 1 ½" ½ "– 1 ½"	CL150 CL300	RF RF	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5 LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5		278	
WELDNECK	2" – 16"	CL150	RF	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5			
WELDNECK	2" – 12"	CL300	RF	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5		278	
BLIND FLG	½ "– 12" ½ "– 12"	CL150 CL300	RF RF	LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5 LTCS, ASTM A350 GRADE LF2 CL1, ASME B16.5		278	
MISCELLANEOUS							
DRIP RING	1" – 12"	CL150	FF	LTCS, ASTM A516 GRADE 70 S-5,, NORMALIZED, 38.100 mm (1-1/2") THICK WITH ¾" SOCW TAPS (A-400 DRAWINGS)			
FIGURE 8 BLANK	1" – 8"	CL150	FF	LTCS, ASTM A516 GRADE 70 S-5,, NORMALIZED, ASME B16.48			
FIGURE 8 BLANK	10" – 12"	CL150	FF	LTCS, ASTM A516 GRADE 70 S-5,, NORMALIZED, ASME B16.48			
VALVES					TAG	143,277	
GATE	½" – 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM(API #10)	PVVGB68S0710 ACLD0ZZ		
	½" – 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVGF68S0710 ACLD0ZZ	275	
	½" – 2"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10)	PVVGB62R0710 ACLD0ZZ	24	
	2"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVGF62R1110 BCLC0ZZ	275	
	3"– 14"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT	PVVGF62R1110 BCLC0ZZ		
	16"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10), FULL PORT,GEAR OP	PVVGF62R1110 BCLC5ZZ		
GLOBE	½" – 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API #10)	PVVLB68S0710A CLD0ZZ		
	2" – 12"	CL150	RF	LTCS BODY WITH 316 SS TRIM (API #10)	PVVLB62R1110 BCLC0ZZ		
LIFT CHECK	½" – 1 ½"	CL800	SOCW	LTCS BODY WITH 316 SS TRIM (API #10) (HORIZ. ONLY)	PVVCL68S0710 NCLD0ZZ		
WAFER CHECK	2" – 16"	CL150	WAF- RF	LTCS BODY WITH 316 SS PLATES, METAL SEATS	PVVCD62W1110 XNZA0ZZ		
WAF BUTTERFLY	2" – 6"	CL150	WAF- RF	LTCS BODY WITH 316 SS DISC, RTFE SEATS	PVVUB62W1199 BCLG1ZZ	27,43,326, 801	
WAF BUTTERFLY	8" – 16"	CL150	WAF- RF	LTCS BODY WITH 316 SS DISC, RTFE SEATS, GEAR OP	PVVUB62W1199 BCLG3ZZ	27,43,326, 801	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
BOLTING	ALL	CL150		ALLOY STEEL, ASTM A320 GR.L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS		
GASKET SPIRAL WOUND	½" – 16"	CL150		SPIRAL WOUND, 304 SS W/GRAPHITE FILLER,SS INNER & CS OUTER RING, ASME B16.20 / ASME B16.5		

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE		FLANGES FLANGES SOCKETWELD BUTTWELD	SIZES ½" – 1 ½" 2" – 12" ½" – 1 ½" 2" – 12"

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
24	Use when required at flanged equipment nozzle, or where noted on P&IDs or Modules in Piping Standard Drawings.	
27	TFE, PTFE seated valves are limited to maximum 204°C.	
43	Only to be used as indicated on the P&IDs.	
143	In shop and/or field fabrication welds (girth, branch and fillet), all weld metal deposits and heat affected zone (HAZ) shall meet impact test requirements for low temperature service (at -48°C for L.T.C.S., at -105°C for semi-cryogenic S.S. "K" specs and for cryogenic S.S. "L" specs) per Paragraph 323 of ASME B31.3.	
275	Only to be used where full port gate valve is required as indicated on P&IDs .	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME Class 300 flanges shall only be used to match higher rated flange connections at control valves and special equipment.	
326	This resilient seated valve does not in all cases meet full flange rating, See valve MAWP in this specification.	
510	The pad thickness equals run pipe thickness, the pad width equals 1/2 of branch O.D. These items can be replaced with a Latrolet or Lateral depending on engineering requirements, availability and economics.	
801	LCB body valves are limited to maximum 340°C.	

BRANCH CONNECTION CHART		REV
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HEADER SIZE

1/2	T																		
¾	RT	T																	
1	RT	RT	T																
1 1/2	RT	RT	RT	T															
2	A	A	A	RT	T														
3	A	A	A	A	RT	T													
4	A	A	A	A	RT	RT	T												
6	A	A	A	A	S	RT	RT	T											
8	A	A	A	A	S	S	RT	RT	T										
10	A	A	A	A	P	S	RT	RT	RT	T									
12	A	A	A	A	P	P	P	RT	RT	RT	T								
14	A	A	A	A	S	S	S	P	RT	RT	RT	T							
16	A	A	A	A	P	P	P	P	P	RT	RT	RT	T						
	1/2	¾	1	1 1/2	2	3	4	6	8	10	12	14	16						

BRANCH SIZE

BRANCH LEGEND		NOTES	REV
T	Tee (B.W. for two inch and larger, S.W. for 1 1½" and smaller)		
A	Sockolet		
RT	Reducing Tee (B.W. for two inch and larger, S.W. for 1 1½" and smaller)		
S	Stub-in without Reinforcing Pad shall be sit-in type as shown in detail "2" or "4" of Figure 328.5.4D of ASME B31.3		
P	Stub-in with Reinforcing Pad shall be sit-in type as shown in detail "2" or "4" of Figure 328.5.4D of ASME B31.3		

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
H3A	R00

					REV
SERVICES	Instrument Air	Potable Water	Instrument Air (unfiltered)	Fire Water	

						REV
PIPING MATERIAL	LT CARBON STEEL (GALV)	VALVE MATERIAL	LT CARBON STEEL	WELD EXAMINATION		
RATING / FACING	ASME CL150/ RF	TRIM MATERIAL	316 SS (API # 10)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	1.50 mm (0.059")	CODE	ASME B31.3	BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)

REV

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16
-38 to 38	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65
93	17.92	17.92	17.92	17.92	17.92	17.92	17.92	17.92	17.92	17.92	17.92	17.92	17.92
149	15.85	15.85	15.85	15.85	15.85	15.85	15.85	15.85	15.85	15.85	15.85	15.85	15.85

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVE
SEAT RATINGS (MPa)

TEMP(C)	NPS < 2	REV
-28 to 38	19.65	
93	17.92	
149	15.85	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	½" – 1½"	80	T&C	LTCS, ASTM A333 GRADE 6, SMLS, GALVANIZED, ASME B36.10M	83, 84, 85, 292	
	2" - 6"	STD	BE	LTCS, ASTM A333 GRADE 6, SMLS, GALVANIZED, ASME B36.10M		
	8" - 12"	30	BE	LTCS, ASTM A333 GRADE 6, SMLS, GALVANIZED, ASME B36.10M		
	14" - 16"	30	BE	LTCS, ASTM A333 GRADE 6, SMLS, GALVANIZED, ASME B36.10M		
NIPPLES	½" – 1½"	80	TOE	LTCS, ASTM A333 GRADE 6, SMLS, GALVANIZED, ASME B36.10M	83, 84, 85,	
	½" – 1½"	80	TBE	LTCS, ASTM A333 GRADE 6, SMLS, GALVANIZED, ASME B36.10M		
SWAGES	CON ½" – 1½"	80	TSE	LTCS, ASTM A420 GRADE WPL6-S, GALVANIZED, MSS-SP-95	1,83,84, 85	
		80	TBE	LTCS, ASTM A420 GRADE WPL6-S, GALVANIZED, MSS-SP-95		
	ECC ½" – 1½"	80	TSE	LTCS, ASTM A420 GRADE WPL6-S, GALVANIZED, MSS-SP-95		
		80	TBE	LTCS, ASTM A420 GRADE WPL6-S, GALVANIZED, MSS-SP-95		
OLET THREDOLET THRD ELBOLET WELDOLET	½" – 1½"	3000 LB	WELD	LTCS, ASTM A350 GRADE LF2 CL 1, GALVANIZED, MSS-SP-97	83,84,85	
	½" – 1½"	3000 LB	WELD	LTCS, ASTM A350 GRADE LF2 CL 1, GALVANIZED, MSS-SP-97		
	2" – 8"	AS PIPE	WELD	LTCS, ASTM A350 GRADE LF2 CL 1, GALVANIZED, MSS-SP-97		
PLUG	½" – 1½"		THRD	LTCS, ASTM A350 GRADE LF2 CL 1, GALVANIZED, ASME B16.11, ROUND HEAD	83,84,85	
CAP	½" – 1½" 2" – 16"	3000 LB AS PIPE	THRD WELD	LTCS, ASTM A350 GRADE LF2 CL 1, GALVANIZED, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, GALVANIZED, ASME B16.9	83,84,85	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
90° ELBOW	½" – 1½" 2" – 6" 8" – 16"	3000 LB AS PIPE AS PIPE	THRD WELD WELD	LTCS, ASTM A350 GRADE LF2 CL 1, GALVANIZED, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, GALVANIZED, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, GALVANIZED, ASME B16.9		83,84,85	
45° ELBOW	½" – 1½" 2" – 6" 8" – 16"	3000 LB AS PIPE AS PIPE	THRD WELD WELD	LTCS, ASTM A350 GRADE LF2 CL 1, GALVANIZED, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, GALVANIZED, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, GALVANIZED, ASME B16.9		83,84,85	
COUPLING	½" – 1½"	3000 LB	THRD	LTCS, ASTM A350 GRADE LF2 CL 1, GALVANIZED, ASME B16.11		83,84,85	
COUPLING(RED)	½" – 1½"	3000 LB	THRD	LTCS, ASTM A350 GRADE LF2 CL 1, GALVANIZED, ASME B16.11			
UNION (GJ)	½" – 1½"	3000 LB	THRD	LTCS, ASTM A350 GRADE LF2 CL 1, GALVANIZED, ASME B16.11		83,84,85	
TEE	½" – 1½" 2" – 6" 8" – 16"	3000 LB AS PIPE AS PIPE	THRD WELD WELD	LTCS, ASTM A350 GRADE LF2 CL 1, GALVANIZED, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, GALVANIZED, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, GALVANIZED, ASME B16.9		83,84,85	
TEE (RED)	½" – 1½" 2" – 6" 8" – 16"	3000 LB AS PIPE AS PIPE	THRD WELD WELD	LTCS, ASTM A350 GRADE LF2 CL 1, GALVANIZED, ASME B16.11 LTCS, ASTM A420 GRADE WPL6-S, GALVANIZED, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, GALVANIZED, ASME B16.9			
REDUCER CON ECC	2" – 6" 8" – 16" 2" – 6" 8" – 16"	AS PIPE AS PIPE AS PIPE AS PIPE	WELD WELD WELD WELD	LTCS, ASTM A420 GRADE WPL6-S, GALVANIZED, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, GALVANIZED, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, GALVANIZED, ASME B16.9 LTCS, ASTM A420 GRADE WPL6-S, GALVANIZED, ASME B16.9		83,84,85	
FLANGE THREADED	½" – 2"	CL150	RF	LTCS, ASTM A350 GRADE LF2 CL 1, GALVANIZED, ASME B16.5		72, 84, 85	
SLIP-ON	3" – 16"	CL150	RF	LTCS, ASTM A350 GRADE LF2 CL 1, ASME B16.5		71	
WELDNECK	3" – 16"	CL150	RF	LTCS, ASTM A350 GRADE LF2 CL 1, GALVANIZED, ASME B16.5			
BLIND	½" – 16"	CL150	RF	LTCS, ASTM A350 GRADE LF2 CL 1, GALVANIZED, ASME B16.5			
THREADED	½" – 2"	CL300	RF	LTCS, ASTM A350 GRADE LF2 CL 1, GALVANIZED, ASME B16.5		278	
WELDNECK	3" – 16"	CL300	RF	LTCS, ASTM A350 GRADE LF2 CL 1, GALVANIZED, ASME B16.5		71, 278	
BLIND	½" – 16"	CL300	RF	LTCS, ASTM A350 GRADE LF2 CL 1, GALVANIZED, ASME B16.5		278	
MISCELLANEOUS DRIP RING	1"–12"	CL150	FF	LTCS, ASTM A516 GRADE 70, NORMALIZED, GALVANIZED, 38.100 mm (1-1/2") THICK WITH ¾" THREADED TAPS		84,85	
FIGURE 8 BLANK BLANK&SPACER	1"–12" 14"–16"	CL150 CL150	FF FF	LTCS, ASTM A516 GRADE 70, NORMALIZED, GALVANIZED, ASME B16.48 LTCS, ASTM A516 GRADE 70 S-5, NORMALIZED, GALVANIZED, ASME B16.48			
VALVES					TAG	277	
GATE	½" – 1½"	CL800	THRD	LTCS BODY WITH 316 SS TRIM(API #10)	PVVB68N0710 EJLD0ZZ	401	
GATE	2" – 14"	CL150	RF	LTCS BODY WITH 316 SS TRIM(API #10)	PVVB62R1110 EJLC0ZZ		
GATE	16"	CL150	RF	LTCS BODY WITH 316 SS TRIM(API #10),GEAR OP	PVVB62R1110 EJLC5ZZ		
GATE	2" – 12"	CL150	RF	LTCS BODY WITH 316 SS TRIM(API #10), FM/UL	PVVB62R1110 EJLC0FU		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
GLOBE	½" – 1½"	CL800	THRD	LTCS BODY WITH 316 SS TRIM(API #10)	PVVLB68N0710 EJLD0ZZ	401	
GLOBE	2" – 12"	CL150	RF	LTCS BODY WITH 316 SS TRIM(API #10)	PVVLB62R1110 EJLC0ZZ		
SWING CHECK	½" – 1½"	CL800	THRD	LTCS BODY WITH 316 SS TRIM(API #10)	PVVCS68N0710 NJLD0ZZ		
LIFT CHECK	½" – 1½"	CL800	THRD	LTCS BODY WITH 316 SS TRIM(API #10), HORIZ. ONLY	PVVCL68N0710 NJLD0ZZ	43	
WAFER CHECK	2" – 16"	CL150	WAF- RF	LTCS BODY WITH 316 SS PLATES, PTFE SEAT	PVVCD62W1164 XNZAOZZ		
BALL	½" – 1½"	CL800	THRD	LTCS BODY WITH 316 SS BALL& STEM, PTFE SEAT	PVVVBH68N0764 EJLF0ZZ	43,326,401	
WAF BUTTERFLY	2" – 6"	CL150	WAF- RF	LTCS BODY WITH 304 SS DISC. BUNA-N /304 SS SEAT	PVVUB62W1187 EJLG0FU	46,326	
WAF BUTTERFLY	8" – 12"	CL150	WAF- RF	LTCS BODY WITH 304 SS DISC. BUNA-N /304 SS SEAT, WORM GEAR	PVVUB62W1187 EJLG3FU	46,326	
BOLTING	ALL	CL150		ALLOY STEEL, ASTM A320 GR.L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS		72	
	ALL	CL300		ALLOY STEEL, ASTM A320 GR.L7 WITH ASTM A194 GRADE 4 HEAVY HEX NUTS			
GASKET	½" – 12"	CL150	RF	FLAT RING, COMPRESSED NON –ASBESTOS, 1.6mm THK, ASME B16.21 / ASME B16.5		72	
	14" – 16"	CL150	FF	FLAT RING, COMPRESSED NON –ASBESTOS, 3.0mm THK, ASME B16.21 / ASME B16.5		72	
	1" – 12"	CL300	RF	FLAT RING, COMPRESSED NON –ASBESTOS, 1.6mm THK, ASME B16.21 / ASME B16.5			
	½" – 12"	CL300	FF	FLAT RING, COMPRESSED NON –ASBESTOS, 1.6mm THK, ASME B16.21 / ASME B16.5		72	

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE	84, 85	UNIONS FLANGES THREADED BUTTWELD	SIZES ½" – 1½" 2" – 12" ½" – 1½" 2" – 12"

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
43	Only to be used as indicated on the P&IDs.	
46	Working temperature shall not exceed 120°C.	
71	To be used against butt-weld fittings and/or when mating against wafer type check valves or resilient seated butterfly valves or flanged rotating/vibrating equipment.	
72	Use R.F. line flanges and flat ring line class gaskets (B16.21) against R.F.flanges with alloy steel stud bolts ASTM A193 GRADE B7 and ASTM A194 GRADE 2H heavy hex nuts. Where C.I. flange is used against R.F. flange,a rubber ring spacer shall be used to fill up the gap between the portion of the R.F. of the flange and outside diameter of flange.In addition, ASTM A193 GRADE B7 bolts with ASTM A194 GRADE 2H heavy hex nuts and line class full face non-asbestos gasket B16.21 shall be used.	
83	Rack piping and fittings shall be purchased galvanized.	
84	All shop fabricated pipe, flanges and fittings shall be hot-dipped galvanized after fabrication in accordance with ASTM A153.	
85	Surfaces where galvanizing has been burned-off during field welding, etc, shall be wire brushed and zinc coated or cold galvanized using ZRC compound.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME Class 300 flanges shall only be used to match higher rated flange connections at control valves and special equipment.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections(when applicable),gage glass piping, vents and drain requiring plugs, ,connections to tube fittings and for connections to threaded proprietary items. See referenced Modules in Piping Standard Drawings.	
326	This resilient seated valve does not in all cases meet full flange rating,See valve MAWP in this specification.	
401	Intermediate strength bolting. Flange joint sealing to be verified in accordance with ASME B16.5, table 1B, note 3.	

BRANCH CONNECTION CHART		REV
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HEADER SIZE 管座尺寸

1/2	T																
3/4	RT	T															
1	RT	RT	T														
1 1/2	RT	RT	RT	T													
2	B	RT	RT	RT	T												
3	B	B	B	B	RT	T											
4	B	B	B	B	S	RT	T										
6	B	B	B	B	S	S	RT	T									
8	B	B	B	B	S	S	S	RT	T								
10	B	B	B	B	S	S	S	S	RT	T							
12	B	B	B	B	S	S	S	S	S	RT	T						
14	B	B	B	B	S	S	S	S	S	S	RT	T					
16	B	B	B	B	S	S	S	S	S	S	S	RT	T				
	1/2	3/4	1	1 1/2	2	3	4	6	8	10	12	14	16				

BRANCH SIZE

BRANCH

BRANCH LEGEND		NOTES	REV
B	Thredolet		
T	Tee (B.W. for two inch and larger,THRD. for 1 11/2" and smaller)		
RT	Reducing Tee (B.W. for two inch and larger, THRD for 1 11/2" and smaller)		
S	Stub-in without Reinforcing Pad shall be set-in type as shown in detail "2" or "4" of Figure 328.5.4D of ASME B31.3		

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
c.	Weldolet may be used in lieu of stub-in if required and verified by flexibility analysis and shown on the piping design drawing.	
d.	A PAD has been selected to provide reinforcement for this branch connection subject to the most severe coincident condition of internal pressure and fluid temperature seen in this specification. However, the specific pressure-temperature condition for this connection may eliminate the need for reinforcement. In addition, when the diameter-to-thickness ratio is restricted by paragraph 304.3.1 (3) (b) of ASME B31.3, finite element analysis (FEA) shall be performed .	

PROJECT NUMBER	01000	PIPING CLASS	REV
DOCUM. NO.	01000-MA-SPC-0002	H3B	R00
CONTR. DOCU. NO.			

SERVICES	Fire Water				REV
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PIPING MATERIAL	CARBON STEEL (GALV)	VALVE MATERIAL	CARBON STEEL	WELD EXAMINATION		REV
RATING / FACING	ASME CL150/ RF	TRIM MATERIAL	13 CR/HF (API #8)	STRESS RELIEF	PER CODE	
CORROSION ALLOWANCE	0.000 mm (0.000")	CODE	ASME B31.3	BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)

REV

SIZE

TEMP °C	1/2	3/4	1	1-1/2	2	3	4	6	8	10	12	14	16
0 to 38	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65	19.65
93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93	17.93

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

BALL VALVE
SEAT RATINGS (MPa)

TEMP(C)	NPS < 2	REV
0 to 38	19.65	
93	17.93	
149	15.86	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" – 1 1/2" 2" - 6" 8" - 12"	80 STD 30	T&C BE BE	CS, ASTM A106 GRADE B, SMLS, GALVANIZED, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, GALVANIZED, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, GALVANIZED, ASME B36.10M	83, 84, 85, 292	
NIPPLES	1/2" – 1 1/2" 1/2" – 1 1/2"	80 80	TOE TBE	CS, ASTM A106 GRADE B, SMLS, GALVANIZED, ASME B36.10M CS, ASTM A106 GRADE B, SMLS, GALVANIZED, ASME B36.10M	83, 84, 85,	
SWAGES CON ECC	1/2" – 1 1/2" 1/2" – 1 1/2" 1/2" – 1 1/2" 1/2" – 1 1/2"	80 80 80 80	TSE TBE TSE TBE	CS, ASTM A234 GRADE WPB-S, GALVANIZED, MSS-SP-95 CS, ASTM A234 GRADE WPB-S, GALVANIZED, MSS-SP-95 CS, ASTM A234 GRADE WPB-S, GALVANIZED, MSS-SP-95 CS, ASTM A234 GRADE WPB-S, GALVANIZED, MSS-SP-95	1,83,84, 85	
OLET THREDOLET THRD ELBOLET	1/2" – 1 1/2" 1/2" – 1 1/2"	3000 LB 3000 LB	WELD WELD	CS, ASTM A105, GALVANIZED, MSS-SP-97 CS, ASTM A105, GALVANIZED, MSS-SP-97	83,84,85	
PLUG	1/2" – 1 1/2"		THRD	CS, ASTM A105, GALVANIZED, ASME B16.11, ROUND HEAD	83,84,85	
CAP	1/2" – 1 1/2" 2" – 12"	3000 LB AS PIPE	THRD WELD	CS, ASTM A105, GALVANIZED, ASME B16.11 CS, ASTM A234 GRADE WPB-S, GALVANIZED, ASME B16.9	83,84,85	
90° ELBOW	1/2" – 1 1/2" 2" – 6"	3000 LB AS PIPE	THRD WELD	CS, ASTM A105, GALVANIZED, ASME B16.11 CS, ASTM A234 GRADE WPB-S, GALVANIZED, ASME B16.9	83,84,85	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
	8" – 12"	AS PIPE	WELD	CS, ASTM A234 GRADE WPB-S, GALVANIZED, ASME B16.9			
45° ELBOW	½" – 1½" 2" – 6" 8" – 12"	3000 LB AS PIPE AS PIPE	THRD WELD WELD	CS, ASTM A105, GALVANIZED, ASME B16.11 CS, ASTM A234 GRADE WPB-S, GALVANIZED, ASME B16.9 CS, ASTM A234 GRADE WPB-S, GALVANIZED, ASME B16.9		83,84,85	
COUPLING	½" – 1½"	3000 LB	THRD	CS, ASTM A105, GALVANIZED, ASME B16.11		83,84,85	
COUPLING(RD)	½" – 1½"	3000 LB	THRD	CS, ASTM A105, GALVANIZED, ASME B16.11			
UNION (GJ)	½" – 1½"	3000 LB	THRD	CS, ASTM A105, GALVANIZED, ASME B16.11		83,84,85	
TEE	½" – 1½" 2" – 6" 8" – 12"	3000 LB AS PIPE AS PIPE	THRD WELD WELD	CS, ASTM A105, GALVANIZED, ASME B16.11 CS, ASTM A234 GRADE WPB-S, GALVANIZED, ASME B16.9 CS, ASTM A234 GRADE WPB-S, GALVANIZED, ASME B16.9		83,84,85	
TEE (RED)	½" – 1½" 2" – 6" 8" – 12"	3000 LB AS PIPE AS PIPE	THRD WELD WELD	CS, ASTM A105, GALVANIZED, ASME B16.11 CS, ASTM A234 GRADE WPB-S, GALVANIZED, ASME B16.9 CS, ASTM A234 GRADE WPB-S, GALVANIZED, ASME B16.9			
REDUCER						83,84,85	
CON	2" – 6" 8" – 12"	AS PIPE AS PIPE	WELD WELD	CS, ASTM A234 GRADE WPB-S, GALVANIZED, ASME B16.9 CS, ASTM A234 GRADE WPB-S, GALVANIZED, ASME B16.9			
ECC	2" – 6" 8" – 12"	AS PIPE AS PIPE	WELD WELD	CS, ASTM A234 GRADE WPB-S, GALVANIZED, ASME B16.9 CS, ASTM A234 GRADE WPB-S, GALVANIZED, ASME B16.9			
FLANGE						72, 84, 85	
THREADED	½" – 2"	CL150	RF	CS, ASTM A105, GALVANIZED, ASME B16.5			
SLIP-ON	3" – 12"	CL150	RF	CS, ASTM A105, GALVANIZED, ASME B16.5			
WELDNECK	3" – 12"	CL150	RF	CS, ASTM A105, GALVANIZED, ASME B16.5		71	
BLIND	½" – 12"	CL150	RF	CS, ASTM A105, GALVANIZED, ASME B16.5			
THREADED	½" – 2"	CL300	RF	CS, ASTM A105, GALVANIZED, ASME B16.5		278	
WELDNECK	3" – 12"	CL300	RF	CS, ASTM A105, GALVANIZED, ASME B16.5		71, 278	
BLIND	½" – 12"	CL300	RF	CS, ASTM A105, GALVANIZED, ASME B16.5		278	
MISCELLANEOUS						84,85	
DRIP RING	1"–12"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, GALVANIZED, 38.100 mm (1-1/2") THICK WITH ¾" THREADED TAPS			
FIGURE 8 BLANK	1"–12"	CL150	FF	CS, ASTM A516 GRADE 70, NORMALIZED, GALVANIZED, ASME B16.48			
VALVES						277	
GATE	½" – 1½"	CL800	THRD	CS BODY WITH 13 CR / HF TRIM (API #8), FM/UL	TAG PVVGB68N0101 ACAD0ZZ	401	
GATE	2" – 12"	CL150	RF	CS BODY WITH 13 CR / HF TRIM (API #8), FM/UL	PVVGB62R0301 BCAC0ZZ		
WAF BUTTERFLY	3" – 6"	CL150	WAF- RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT , FM/UL	PVVUB62W0399 BCAG1ZZ	27,43	
	8" – 16"	CL150	WAF- RF	CS BODY WITH 316 SS DISC. RTFE/316 SS SEAT, GEAR OP. , FM/UL	PVVUB62W0399 BCAG3ZZ	27,43	

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
BOLTING	ALL	CL150		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEX NUTS	72	
	ALL	CL300		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEX NUTS		
GASKET	½" – 12"	CL150	RF	FLAT RING, COMPRESSED NON –ASBESTOS, 1.6mm THK, ASME B16.21 / ASME B16.5	72	
	1" – 12"	CL300	RF	FLAT RING, COMPRESSED NON –ASBESTOS, 1.6mm THK, ASME B16.21 / ASME B16.5		
	½" – 12"	CL300	FF	FLAT RING, COMPRESSED NON –ASBESTOS, 1.6mm THK, ASME B16.21 / ASME B16.5	72	

	NOTES		REV
JOINT CONNECTS MAINTENANCE MAINTENANCE NON-MAINTENANCE NON-MAINTENANCE	84, 85	UNIONS FLANGES THREADED BUTTWELD	SIZES
			½" – 1½"
			2" – 12"
			½" – 1½"
			2" – 12"

NOTE	SPECIFIC PIPING NOTES	REV
1	Use swages where small end does not exceed two inch size and large end does not exceed three inch size.	
43	Only to be used as indicated on the P&IDs.	
71	To be used against butt-weld fittings and/or when mating against wafer type check valves or resilient seated butterfly valves or flanged rotating/vibrating equipment.	
72	Use R.F. line flanges and flat ring line class gaskets (B16.21) against R.F.flanges with alloy steel stud bolts ASTM A193 GRADE B7 and ASTM A194 GRADE 2H heavy hex nuts. Where C.I. flange is used against R.F. flange, a rubber ring spacer shall be used to fill up the gap between the portion of the R.F. of the flange and outside diameter of flange. In addition, ASTM A193 GRADE B7 bolts with ASTM A194 GRADE 2H heavy hex nuts and line class full face non-asbestos gasket B16.21 shall be used.	
83	Rack piping and fittings shall be purchased galvanized.	
84	All shop fabricated pipe, flanges and fittings shall be hot-dipped galvanized after fabrication in accordance with ASTM A153.	
85	Surfaces where galvanizing has been burned-off during field welding, etc, shall be wire brushed and zinc coated or cold galvanized using ZRC compound.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
278	ASME Class 300 flanges shall only be used to match higher rated flange connections at control valves and special equipment.	
279	Threaded piping components shall only be used for such items as pressure instrument connections, orifice flange instrument connections (when applicable), gage glass piping, vents and drain requiring plugs, connections to tube fittings and for connections to threaded proprietary items. See referenced Modules in Piping Standard Drawings.	
401	Intermediate strength bolting. Flange joint sealing to be verified in accordance with ASME B16.5, table 1B, note 3.	

BRANCH CONNECTION CHART		REV
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HEADER SIZE 管座尺寸

½	T															
¾	RT	T														
1	RT	RT	T													
1½	RT	RT	RT	T												
2	B	RT	RT	RT	T											
3	B	B	B	B	RT	T										
4	B	B	B	B	S	RT	T									
6	B	B	B	B	S	S	RT	T								
8	B	B	B	B	S	S	S	RT	T							
10	B	B	B	B	S	S	S	S	RT	T						
12	B	B	B	B	S	S	S	S	S	RT	T					
14	B	B	B	B	S	S	S	S	S	S	RT	T				
16	B	B	B	B	S	S	S	S	S	S	S	RT	T			
	1/2	¾	1	1½	2	3	4	6	8	10	12	14	16			

BRANCH SIZE

BRANCH

BRANCH LEGEND		NOTES	REV
B	Threadolet		
T	Tee (B.W. for two inch and larger, THRD. for 1 11/2" and smaller)		
RT	Reducing Tee (B.W. for two inch and larger, THRD for 1 11/2" and smaller)		
S	Stub-in without Reinforcing Pad shall be set-in type as shown in detail "2" or "4" of Figure 328.5.4D of ASME B31.3		

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
H61B	R00

					REV
SERVICES	Service Water (UG)	Oily Contaminated Rainy Water (UG)	Clean Surface Water (UG)	Sanitary Wastewater (UG)	
	High Salt Water	Clean Surface Water			

						REV
PIPING MATERIAL	FRP	VALVE MATERIAL	CARBON STEEL	WELD EXAMINATION	NOT REQUIRED	D01
RATING / FACING	ASME CL150/ FF	TRIM MATERIAL	13 CR (API #1)	STRESS RELIEF	NOT REQUIRED	
CORROSION ALLOWANCE	0.000 mm	CODE	ASME B31.3	BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)

REV

SIZE

TEMP °C	2	3	4	6	8	10	12	14	16	18	20	24	26~96
0 to 40	10	10	10	10	10	10	10	10	10	10	10	10	10

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	2" - 16"	10 Barg	O-ring	FRP, FILAMENT-WOUND, ASTM D2996 Type III, AWWA C950	259	D01
	18" - 80"	10 Barg	O-ring	FRP, FILAMENT-WOUND, ASTM D2996 Type III, AWWA C950		D01
90° ELBOW	2" - 16"	10 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950	259	D01
	18" - 80"	10 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950		D01
45° ELBOW	2" - 16"	10 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950	259	D01
	18" - 80"	10 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950		D01
TEE	2" - 16"	10 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950	259	D01
	18" - 24"	10 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950		D01
TEE (RED)	2" - 16"	10 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950		D01
	18" - 24"	10 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950		D01
REDUCER	CON 2" - 16" 18" - 48"	10 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950	259	D01
		10 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950		D01
	ECC 2" - 16" 18" - 48"	10 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950		D01
		10 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950		D01
FLANGE FLANGE	2" - 24"	CL150	FF	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950, DIMENSIONS PER ASME B16.5	259	

HEADER SIZE 管座尺寸

2	T
3	RT T
4	RT RT T
6	RT RT RT T
8	RT RT RT RT T
10	RT RT RT RT RT T
12	RT RT RT RT RT RT T
14	RT RT RT RT RT RT RT T
16	RT RT RT RT RT RT RT RT T
18	RT RT RT RT RT RT RT RT RT T
20	RT RT RT RT RT RT RT RT RT RT T
24	RT RT RT RT RT RT RT RT RT RT RT T
	2 3 4 6 8 10 12 14 16 18 20 24

BRANCH SIZE

BRANCH

BRANCH LEGEND		NOTES	REV
T	Tee		
RT	Reducing Tee		

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCU. NO.	

PIPING CLASS	REV
H61C	R00

					REV
SERVICES	RAW Water (UG)				

						REV
PIPING MATERIAL	FRP	VALVE MATERIAL	--	WELD EXAMINATION	NOT REQUIRED	D01
RATING / FACING	ASME CL300/ RF	TRIM MATERIAL	--	STRESS RELIEF	NOT REQUIRED	
CORROSION ALLOWANCE	0.000 mm	CODE	ASME B31.3	BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)

REV
D01

SIZE			
TEMP °C	16	18	20
0 to 40	18	18	18

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	16" - 20"	18 Barg	Key-Lock	FRP, FILAMENT-WOUND, ASTM D2996 Type III, AWWA C950	259	D01
90° ELBOW	16" - 20"	18 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950	259	D01
45° ELBOW	16" - 20"	18 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950	259	D01
TEE	16" - 20"	18 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950	259	D01
TEE (RED)	16" - 20"	18 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950		D01
REDUCER	CON 16" - 20"	18 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950	259	D01
	ECC 16" - 20"	18 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950		D01
FLANGE	FLANGE 16" – 20"	CL300	RF	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950, DIMENSIONS PER ASME B16.5	259	D01
	BLIND 16" – 20"	CL300	RF	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950, DIMENSIONS PER ASME B16.5		D01
BOLTING	ALL	CL150		ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEX NUTS		
GASKET	16" – 20"	CL300	FF	FLAT RING, COMPRESSED NON –ASBESTOS,3.0mm THK, ASME B16.21 / ASME B16.5		D01

NOTE	SPECIFIC PIPING NOTES	REV
259	Underground installation shall be as per the manufacturer's recommended procedures.	

	REV
BRANCH CONNECTION CHART	D01

HEADER SIZE
 管座尺寸

16

T

18

RT

T

20

RT

RT

T

16

18

20

BRANCH SIZE

BRANCH

BRANCH LEGEND		NOTES	REV
T	Tee		
RT	Reducing Tee		

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
H61D	R00

SERVICES					REV
	Potable Water(UG)				

PIPING MATERIAL	HDPE	VALVE MATERIAL	CARBON STEEL	WELD EXAMINATION	NOT REQUIRED	REV
RATING / FACING	ASME CL150/ FF	TRIM MATERIAL	13 CR (API #1)	STRESS RELIEF	NOT REQUIRED	D01
CORROSION ALLOWANCE	0.0 mm (0.0")	CODE	PER PROJECT	BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (MPa)*

REV

SIZE

TEMP °C	2	3	4	6	8
0 to 40	8	8	8	8	8

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

ITEMS	Nominal outside diameter (dn)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	20 – 32 40 – 160	8 bar 8 bar	PE PE	HDPE, EN 12201-2 HDPE, EN 12201-2	175	D01
CAP	20 – 160	8 bar	Socket	HDPE, EN 12201-3	175	D01
90° ELBOW	20 – 160	8 bar	Socket	HDPE, EN 12201-3	175	D01
45° ELBOW	20 – 160	8 bar	Socket	HDPE, EN 12201-3	175	D01
COUPLING	20 – 160	8 bar	Socket	HDPE, EN 12201-3	175	D01
TEE	20 – 160	8 bar	Socket	HDPE, EN 12201-3	175	D01
RED TEE	20 – 160	8 bar	Socket	HDPE, EN 12201-3		
REDUCER CON	20 – 160	8 bar	Socket	HDPE, EN 12201-3	175	D01
FLANGE FLANGE	20 – 160	CL150	FF	HDPE, EN 12201-3, ASME B16.5	35,175	D01
VALVES GATE	½" – 1½"	CL150	RF	CS BODY WITH 13 CR TRIM (API #1)	TAG PVVGB62R0101 ACAD0ZZ 277,805	D01

ITEMS	Nominal outside diameter (d _n)	Sch/RATING	ENDS	DESCRIPTION		NOTES	REV
GATE	2" – 6"	CL150	RF	CS BODY WITH 13 CR TRIM (API #1)	PVVGB62R0301 BCAC0ZZ		
SWING CHECK	2" – 6"	CL150	RF	CS BODY WITH 13 CR TRIM (API #1)	PVVCS62R0301 NCAC0ZZ		D01
BOLTING	ALL			ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEX NUTS		802	D01
GASKET	1/2" – 6"	CL150	FF	FLAT RING, COMPRESSED NON –ASBESTOS, 3.0mm THK, ASME B16.21 / ASME B16.5		802	D01

	NOTES		REV
JOINT CONNECTS MAINTENANCE NON-MAINTENANCE		FLANGES SOCKETWELD	SIZES d_n 20 – 160 d_n 20 – 160 D01

NOTE	SPECIFIC PIPING NOTES	REV
35	All flanged joints and valve bonnets require safety shields after erection of the piping system.	
175	The manufacturer's recommendations and specifications must be followed during shipping, storing, handling, installation, erection, testing, final painting/insulating and for removal / maintenance requirements.	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	
801	MFR means by manufacturer.	
802	Vendor shall supply in complete set the Pipes, Fittings, Valves, Flanges, Bolts and Gaskets. That should be noted in Materials List and Order Contract.	
805	These valve should only be installed in valve well, can't be buried.	D01

	REV
BRANCH CONNECTION CHART	D01

HEADER SIZE

20	T																		
25	RT	T																	
32	RT	RT	T																
40	RT	RT	RT	T															
50	RT	RT	RT	RT	T														
63	RT	RT	RT	RT	RT	T													
75	RT	RT	RT	RT	RT	RT	T												
90	RT	RT	RT	RT	RT	RT	RT	T											
110	RT	RT	RT	RT	RT	RT	RT	RT	T										
140	RT	RT	RT	RT	RT	RT	RT	RT	RT	T									
160	RT	RT	RT	RT	RT	RT	RT	RT	RT	RT	T								
	20	25	32	40	50	63	75	90	110	140	160								

BRANCH SIZE

<u>BRANCH LEGEND</u>		NOTES	REV
T	Tee		
RT	Reducing Tee		

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
H61E	R00

					REV
SERVICES	Fire Water & Foam (UG)				

						REV
PIPING MATERIAL	FRP	VALVE MATERIAL	Ductile Iron	WELDEXAMINATION	NOT REQUIRED	D01
RATING / FACING	ASME CL150/ FF	TRIM MATERIAL	BRONZE	STRESS RELIEF	NOT REQUIRED	
CORROSION ALLOWANCE	0.000 mm	CODE	ASME B31.3	BENDING	NOT ALLOWED	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)

REV
D01

SIZE

TEMP °C	2	3	4	6	8	10	12	14	16	18	20	24
0 to 40	16	16	16	16	16	16	16	16	16	16	16	16

*SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	4" - 16"	16 Barg	Key-Lock	FRP, FILAMENT-WOUND, ASTM D2996 Type III, AWWA C950	259	D01
	18" - 24"	16 Barg	Key-Lock	FRP, FILAMENT-WOUND, ASTM D2996 Type III, AWWA C950		D01
90° ELBOW	4" - 16"	16 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950	259	D01
	18" - 24"	16 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950		D01
45° ELBOW	4" - 16"	16 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950	259	D01
	18" - 24"	16 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950		D01
TEE	4" - 16"	16 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950	259	D01
	18" - 24"	16 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950		D01
TEE (RED)	4" - 16"	16 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950		D01
	18" - 24"	16 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950		D01
REDUCER	CON 4" - 16"	16 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950	259	D01
		16 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950		D01
	ECC 4" - 16"	16 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950		D01
		16 Barg	Socket	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950		D01

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
FLANGE					259	
FLANGE	4" – 24"	CL150	FF	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950, DIMENSIONS PER ASME B16.5		D01
BLIND	4" – 24"	CL150	FF	FRP, CONTACT MOLDED, ASTM D2996 Type III, AWWA C950, DIMENSIONS PER ASME B16.5		D01
VALVES					259	
GATE	4" – 14"	250psi	FF	DI BODY WITH BRONZE TRIM, POST INDICATOR, FM/UL	TAG PVVGB65R8B94 EUAQ0WV	D01
GATE	16" – 24"	250psi	FF	DI BODY WITH BRONZE TRIM, POST INDICATOR, FM/UL, GEAR OP.	PVVGB65R8B94 EUAQ3WV	D01
BOLTING	ALL			ALLOY STEEL, ASTM A193 GRADE B7 WITH ASTM A 194 GRADE 2H HEAVY HEX NUTS		D01
GASKET	4" – 24"	CL150	FF	FLAT RING, COMPRESSED NON –ASBESTOS,3.0mm THK, ASME B16.21 / ASME B16.5		D01

NOTE	SPECIFIC PIPING NOTES	REV
259	Underground installation shall be as per the manufacturer's recommended procedures.	

REV
BRANCH CONNECTION CHART

HEADER SIZE
 管座尺寸

2	T
3	RT T
4	RT RT T
6	RT RT RT T
8	RT RT RT RT T
10	RT RT RT RT RT T
12	RT RT RT RT RT RT T
14	RT RT RT RT RT RT RT T
16	RT RT RT RT RT RT RT RT T
18	RT RT RT RT RT RT RT RT RT T
20	RT RT RT RT RT RT RT RT RT RT T
24	RT RT RT RT RT RT RT RT RT RT RT T
	2 3 4 6 8 10 12 14 16 18 20 24

BRANCH SIZE

BRANCH

BRANCH LEGEND		NOTES	REV
T	Tee		
RT	Reducing Tee		

PROJECT NUMBER	01000
CLIENT	01000-MA-SPC-0002
LOCATION	

PIPING CLASS	REV
S82A	R00

SERVICES	Carbon Dioxide	Catalyst	Liquid/Melttable Additive	REV
	Cold Liquid Drain	Process		

PIPING MATERIAL	DUPLEX 2205	VALVE MATERIAL	316 SS STAINLESS STEEL	WELDEXAMINATION	NOT REQUIRED	REV
RATING / FACING	NOT APPLICABLE	TRIM MATERIAL	316 SS (API #10)	STRESS RELIEF	NOT REQUIRED	
CORROSION ALLOWANCE	0.50 mm (0.02")	CODE	ASME B31.3	BENDING	PERMITTED (Note 115)	

FLANGE RATING & MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*

REV

SIZE

TEMP °C	3	6	10	12	16	20	22	25
-48 to 38	217.91	217.91	217.91	217.91	217.91	217.91	217.91	217.91
93	217.91	217.91	217.91	217.91	217.91	217.91	217.91	217.91
120	210.88	210.88	210.88	210.88	210.88	210.88	210.88	210.88

ITEM	SIZE RANGE (O.D.)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
TUBING	1/4" – 3/8"	0.9mm (0,035")	TUBE END	TUBING, ASTM A789 S32205, SMLS	115 195	
	1/2"	1.2mm (0,047")	TUBE END	TUBING, ASTM A789 S32205, SMLS		
	5/8" – 3/4"	1.7mm (0,067")	TUBE END	TUBING, ASTM A789 S32205, SMLS		
	7/8" – 1"	2.1mm (0,083")	TUBE END	TUBING, ASTM A789 S32205, SMLS		
FITTINGS					118,195	
MALE CONNECTOR	¼" – 1"			COMPRESSION TYPE, ASTM A479 S32205, MANUFACTURER'S STANDARD		
FEMALE CONNECTOR	¼" – 1"			COMPRESSION TYPE, ASTM A479 S32205, MANUFACTURER'S STANDARD		
TUBE BH UNION	¼" – 1"			COMPRESSION TYPE, ASTM A479 S32205, MANUFACTURER'S STANDARD		
TUBE / MNPT 90 ELL	¼" – 1"			COMPRESSION TYPE, ASTM A479 S32205, MANUFACTURER'S STANDARD		
TUBE 90 ELL	¼" – 1"			COMPRESSION TYPE, ASTM A479 S32205, MANUFACTURER'S STANDARD		
TUBE 90 BEND	¼" – 1"		TUBE END	ASTM A789 S32205, SMLS, 15X DIAMETER , MANUFACTURER'S STANDARD		
TUBE 45 BEND	¼" – 1"		TUBE END	ASTM A789 S32205, SMLS, 15X DIAMETER , MANUFACTURER'S STANDARD		

UNION TEE (RD)	¼" – 1"			COMPRESSION TYPE, ASTM A182 F60, MANUFACTURER'S STANDARD		
FLANGES TUBE END	¼" – 1"		RF	COMPRESSION TYPE, ASTM A182 F60, MANUFACTURER'S STANDARD		
VALVES					TAG	118,277
BALL	¼"	CL1500	FNPT	316 SS BODY AND TRIM, PTFE SEAT AND PACKING, OVAL HW	PVVBW7 5N3864E MZM0FT	
BALL	3/8"	CL1500	FNPT	316 SS BODY AND TRIM, PTFE SEAT AND PACKING, OVAL HW	PVVBW7 5N3864E MZM0FT	
BALL	½"	CL1500	FNPT	316 SS BODY AND TRIM, PTFE SEAT AND PACKING, OVAL HW	PVVBW7 5N3864E MZM0FT	
BALL	¾"	CL1500	FNPT	316 SS BODY AND TRIM, PTFE SEAT AND PACKING, OVAL HW	PVVBW7 5N3864E MZM0FT	
GATE	½"	CL1500	FNPT	316 SS BODY AND TRIM, PTFE PACKING	PVVBW7 5N3864E MZM0FT	
GATE	¾"	CL1500	FNPT	316 SS BODY AND TRIM, PTFE PACKING	PVVBW7 5N3864E MZM0FT	
CHECK	¼"	CL1500	FNPT	316 SS BODY AND TRIM:MANUFACTURER	PVVBW7 5N3864E MZM0FT	
CHECK	3/8"	CL1500	FNPT	316 SS BODY AND TRIM:MANUFACTURER	PVVBW7 5N3864E MZM0FT	
CHECK	½"	CL1500	FNPT	316 SS BODY AND TRIM:MANUFACTURER	PVVBW7 5N3864E MZM0FT	
CHECK	1"	CL1500	FNPT	316 SS BODY AND TRIM:MANUFACTURER	PVVBW7 5N3864E MZM0FT	

NOTE	SPECIFIC PIPING NOTES	REV
115	Tubing bends made to the radii of standard tube benders are acceptable.	
118	Copper and copper alloys for internals shall not be permitted in contact with flowing medium.	
195	Tubing leads and returns to and from the tubing adjacent to the traced items shall be insulated and jacketed in accordance with Engineering Specification (HOLD), "Steam Tracing Piping Requirements"	
277	All valves shown below have detailed descriptions provided in the Valve Specification.	

	REV
BRANCH CONNECTION CHART	

1/4	T						
3/8	RT	T					
1/2	RT	RT	T				
5/8	RT	RT	RT	T			
3/4	RT	RT	RT	RT	T		
7/8	RT	RT	RT	RT	RT	T	
1	RT	RT	RT	RT	RT	RT	T
	1/4	3/8	1/2	5/8	3/4	7/8	1

BRANCH LEGEND		NOTES	REV
T	Tee		
RT	Reducing Tee		

PROJECT NUMBER	01000
DOCUM. NO.	01000-MA-SPC-0002
CONTR. DOCUM. NO.	

PIPING CLASS	REV
T62A	R00

SERVICES	NaClO	Oxidant		REV
	Reductant	Scale inhibitor	HCl	

PIPING MATERIAL	PVC	VALVE MATERIAL	PVC	WELD EXAMINATION		REV
RATING / FACING	PVC CLASS150 / FF	TRIM MATERIAL	PVC	STRESS RELIEF	NOT REQUIRED	
CORROSION ALLOWANCE	0.0mm	CODE	ASME B31.3	BENDING		

**FLANGE RATING &
MAXIMUM ALLOWABLE WORKING PRESSURE (bar)*
SIZE**

REV

TEMP (C)	1/2	3/4	1	1 1/2	2								
AMB	10.34	10.34	10.34	10.34	10.34								
38	6.2	6.2	6.2	6.2	6.2								

* SHADED VALUES REPRESENTS THE LIMITS OF THE FLANGE RATING

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
PIPE	1/2" - 2"	80	PE	PVC, ASTM D1784 CLASS 12454-B MANUFACTURED TO ASTM D1785 (UV PROTECTED)	121, 125, 126	
NIPPLES	1/2" - 2"	80	PE	PVC, ASTM D1784 CLASS 12454-B MANUFACTURED TO ASTM D1785 (UV PROTECTED)	121, 125, 126	
FITTINGS					121, 125, 126	
90 ELL	1/2" - 2"	80	SOCW	PVC, ASTM D1784 CLASS 12454-B MANUFACTURED TO ASTM D2467 (UV PROTECTED)		
45 ELL	1/2" - 2"	80	SOCW	PVC, ASTM D1784 CLASS 12454-B MANUFACTURED TO ASTM D2467 (UV PROTECTED)		
45 LATERAL	1/2" - 2"	80	SOCW	PVC, ASTM D1784 CLASS 12454-B MANUFACTURED TO ASTM D2467 (UV PROTECTED)		
45 LATERAL (RED)	1/2" - 2"	80	SOCW	PVC, ASTM D1784 CLASS 12454-B MANUFACTURED TO ASTM D2467 (UV PROTECTED)		
TEE	1/2" - 2"	80	SOCW	PVC, ASTM D1784 CLASS 12454-B MANUFACTURED TO ASTM D2467 (UV PROTECTED)		
TEE (RED)	1/2" - 2"	80	SOCW	PVC, ASTM D1784 CLASS 12454-B MANUFACTURED TO ASTM D2467 (UV PROTECTED)		
PLUG			THRD	PVC, ASTM D1784 CLASS 12454-B MANUFACTURED TO ASTM D2467 (UV PROTECTED)		
COUPLING	1/2" - 2"	80	SOCW	PVC, ASTM D1784 CLASS 12454-B MANUFACTURED TO ASTM D2467 (UV PROTECTED)		
COUPLING (RD)	1/2" - 2"	80	SOCW	PVC, ASTM D1784 CLASS 12454-B MANUFACTURED TO ASTM D2467 (UV PROTECTED)		
CAP	1/2" - 2"	80	SOCW	PVC, ASTM D1784 CLASS 12454-B MANUFACTURED TO ASTM D2467 (UV PROTECTED)		
BUSHING	1/2" - 2"	80	MALE/ THRD	PVC, ASTM D1784 CLASS 12454-B MANUFACTURED TO ASTM D2467 (UV PROTECTED)		
FLANGE ADAPTER UNION (GJ)	1/2" - 2"	80	SOCW	PVC, ASTM D1784 CLASS 12454-B MANUFACTURED TO ASTM D2467 (UV PROTECTED) (TO BE FURNISHED TOGETHER WITH VAN STONE FLANGE)		
VALVES					277	
BALL	1/2" - 2"	CL150	THRD	PVC BODY AND BALL & STEM, CLASS 12454-B, TFE SEATS, (UV PROTECTED), HANDWHEEL OPERATED		
FLANGES					124	
VAN STONE STYLE	1/2" - 2"	CL150 (80)	FF	PVC, ASTM D1784 CLASS 12454-B MANUFACTURED TO ASTM D2467 (UV PROTECTED) (TO BE FURNISHED TOGETHER WITH FLANGE ADAPTER)		
BLIND	1/2" - 2"	CL150	FF	PVC, ASTM D1784 CLASS 12454-B MANUFACTURED TO ASTM D2467 (UV PROTECTED)		

ITEMS	Size (IN)	Sch/RATING	ENDS	DESCRIPTION	NOTES	REV
GASKETS	½" – 2"	CL150	FF	3 mm THK. FULL FACE VITON (ASME B16.21), SHORE A HARDNESS 50 TO 70 DUROMETER	122	
BOLTING STUD	ALL	CL150		ALLOY STEEL, GB 3077 GRADE 35CrMoA (ASTM A193 GRADE B7) WITH GB 3077 GRADE 35 Q&T (ASTM A194 GRADE 2H) HEAVY HEX NUTS	123, , 292	
WASHERS	ALL			FLAT CUT WROUGHT STEEL WASHERS (ASTM F844), U.S. STANDARD SIZES (ANSI B18.22.1, TYPE A), TO BE USED ON FLANGED JOINTS UNDER NUT		
JOINT MATERIAL				SOLVENT CEMENT FOR PVC (ASTM D1784 CLASS 12454-B) PIPE AND FITTINGS PER ASTM D2564	126	

NOTE	SPECIFIC PIPING NOTES	REV
121	Where as determined by Civil Engineering Group, underground piping installation in heavy traffic areas shall be encased with C.S. pipe, ASTM A53 Gr. B ERW or API 5L Gr. B SAW, Std. Wt. The C.S. pipe sleeve shall extend one meter on either side of the road and shall be purchased with 90 mils exterior extruded polyethylene coating and fittings or field welds shall be coated in the field after welding with coal tar epoxy and then wrapped with tape per project spec	
122	Gaskets to be used at all flanged connections.	
123	All bolts shall be installed with washers under the nuts.	
124	Use flanges only when absolutely necessary. Maximum pressure rating for flange is 1.03 MPa. Flange connection below ground shall be avoided.	
125	One coupling (PVC) for every 6.1 meter length of pipe shall be provided	
126	Design and installation of the PVC gravity sewer and drain piping and fittings shall be in accordance with the manufacturer's recommendations.	
277	All valves shown below have detailed descriptions provided in Valve Specification.	
292	All primary onshore piping materials are listed with offshore ASTM equivalent materials listed in parenthesis	

	REV
BRANCH CONNECTION CHART	

HEADER SIZE

0.5	T				
0.75	RT	T			
1	RT	RT	T		
1.5	RT	RT	RT	T	
2	A	A	A	A	T
0.5	0.75	1	1.5	2	

BRANCH

BRANCH LEGEND		NOTES	REV
T	Tee(B.W. for sizes three inch and larger, S.W. for two inch and smaller)		
RT	Reducing Tee (B.W. for three inch and larger, S.W. for two inch and smaller)		

PIPING BRANCH NOTES		REV
a.	Where additional reinforcing is required for mechanical strength, use reinforcement as indicated on the piping drawing.	
b.	For 45° branch connections, see 45° Branch Table. The exception is the branches from heater wash cone lines on 'TLE' type heaters where 45° lateral stub-in Branch Weld is acceptable for atmospheric drain conditions.	
c.	Weldolet may be used in lieu of a stub-in Branch Weld if verified and approved by pressure design and flexibility analysis and shown on the piping design drawing	
d.	A pad has been selected to provide reinforcement for this branch connection subject to the most severe coincident condition of internal pressure and fluid temperature seen in this specification. However, the specific pressure-temperature condition for this connection may eliminate the need for reinforcement. In addition, when the diameter-to-thickness ratio is restricted by paragraph 304.3.1 (3) (b) of ASME B31.3, finite element analysis of the Pad design must be performed by the Contractor Stress Engineer.	
e	PWHT required at all branch attachment welds when the thickness through the weld in any plane through the branch is greater than 38.1 mm (1.5") for P1 material per ASME B31.3 and greater than 19 mm (0.75") for P1 materials per ASME B31.1.	

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ATTECHMENT1—VALVE SPECIFICAITION Discipline: Material							

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GENERAL NOTES

1.0 Scope

This Valve Specification is applicable for the design and selection of valve materials during the execution of Karabatan Power Plant Project .

2.1 Normative References

2.2 International Standards

API 594	Check Valves: Wafer, Wafer-Lug, and Double Flanged Type
API 598	Valve Inspection and Testing
API 599	Metal Plug Valves - Flanged and welding Ends
API 600	Bolted Bonnet Steel Gate Valves for Petroleum & Natural Gas Industries
API 602	Steel Gate, Globe and Check Valves for Sizes DN100 and Smaller for the Petroleum and Natural Gas Industries
API 603	Corrosion-Resistant, Bolted Bonnet Gate Valves--Flanged & Butt-Welding Ends
API 607	Fire Test for Soft-Seated Quarter-Turn Valves
API 608	Metal Ball Valves – Flanged, Threaded and Welding Ends
API 609	Butterfly Valves: Double Flanged, Lug and Wafer-Type
API 941	Steels for Hydrogen Service at Elevated Temperatures and Pressures in Petroleum Refineries and Petrochemical Plants
ASME B16.5	Pipe Flanges and Flanged Fittings (NPS 1/2 Through NPS 24)
ASME B16.10	Face-to-Face & End-to-End Dimensions of Valves
ASME B16.34	Valves–Flanged, Threaded, and Welding End
ASME B16.47	Large Diameter Steel Flanges (NPS 26 Through NPS 60)
BS EN ISO17292	Metal Ball Valves for the Petroleum, Petrochemical and Allied Industries
MSS SP-81	Stainless Steel, Bonnetless, Flanged Knife Gate Valves

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2.2 KPI Project Specification

108184-0000-MA-SPC-0001	Piping Material Specification
108184-0000-MA-SPC-0002	Piping Material Classification
108184-0000-MA-SPC-0007	Standard for Protective Coatings and Application Requirements
108184-0000-MA-SPC-0008	Color Coding
108184-0000-PI-SPC-0006	Standard For Piping Welding

3.1 Technical Requirement

3.2 Genaral

- 3.2.1 Unless otherwise specified, steel valves shall conform to ASME B16.34.
- 3.2.2 Face-to-face dimensions of valves shall conform to ASME B16.10 to the extent covered. For valves not covered in ASME B16.10, certified dimensions drawings shall be provided by the Selected Vendor.
- 3.2.3 All flanged valves shall have end flanges conforming dimensionally to ASME B16.5 or 16.47A.
- 3.2.4 Valves specified as needing to be ‘fire safe’ shall have been tested and certified per API 607.
- 3.2.5 Each individual valve shall be subjected to a shell and seat pressure test per API 598.
- 3.2.6 Antistatic devices shall be furnished according to EN ISO 17292.
- 3.2.7 Buttweld, socketweld and threaded end carbon steel valves shall have a 0.25% maximum carbon content.
- 3.2.8 All pressure testing of austenitic stainless steel valves shall be carried out with water having a maximum of 50 ppm chloride content.
- 3.2.9 The design and manufacture of all valves shall satisfy with the requirements of particular fluid services.

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3.1.10 End and bonnet flanges shall be cast or forged integrally with the valve body. Conversion of weld end valves to flanged by welding on flanges is not permitted.

3.1.11 Valves both ends with nipples shall be connected by buttwelding, sockekwelding is forbidden.

3.1.12 All valves with weld deposits shall be impact tested as per project requirements. All valves for low temperature service shall have weld deposits and heat affected zone (HAZ) meeting impact test requirements for low temperature service per Paragraph 323 of ASME B31.3.

3.1.13 Unless otherwise specified, all austenitic stainless steel materials (300 series) shall be furnished in a solution annealed condition. Approved repairs, if any, shall be made prior to solution annealing.

3.1.14 In case where PWHT is required, valve trims shall be assembled after final heat treatment.

3.1.15 The wadge of gate valve chose solid wadge from NPS 0.5" to NPS 1.5". For size larger than NPS 2" ,the wadge type is flexible wadge.

3.1.16 The disc type of globe valve is swivel.

3.2 Valve Operators

Unless otherwise noted, gate valves and globe valves shall be supplied with bevel gear operators as follows; quarter turn valves shall be supplied with worm gear operators as follows. However, the operating force at the half opening position by the handwheel or lever shall not exceed 35kgf.

ASME Class	Gate	Globe	Ball	Butterfly	Plug
150	≥16	≥14	≥6	≥8	≥6
300	≥12	≥10	≥6	≥6	≥6
600	≥10	≥8	≥4	≥4	
900	≥8	≥6			
1500	≥6	≥6			
2500	≥6	≥4			

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3.3 Extended Bonnet Lengths for Low Temperature Service

Extended bonnet valves for low temperature service shall have the minimum extended bonnet lengths stated in the table below, measured from the centerline of the valve bore to the bottom of the packing (stuffing box).

Valve	Centerline of Valve Bore to Bottom of Valve Packing (mm)	
Size	Rising Stem	Quarter-Turn
(NPS)	(i.e. Gate, Globe)	(e.g. Butterfly, Ball)
½,3/4,1	305	190
1½	355	216
2	406	254
3	457	330
4	558	355
6	610	432
8	685	457
10	889	635
12	1016	711
14	1118	787
16	1194	838
18	1295	889
20	1397	940
24	1600	990

GENERAL NOTES

4.0 Abbreviation List

Abbreviation	Description
BB	Bolted Bonnet
BW	Buttweld
CP	Column Plug
FB	Full Port
FLB	Floating Ball
FNPT	60°Female NPT Threads
GBP	Stainless steel wire&flexible graphite braided packing
GBPR	Stainless steel wire&flexible graphite braided packing+flexible
NLUB	Non-lubricated
OS&Y	Outside Screw And Yoke
STL	Stellite
S.S/FG SWG	Spiral-wound Stainless Steel gasket with
SW	Socket Welding
SEY	Side Entry
TRUB	Trunnion Ball
NBE	Nipple Both Ends
MFR STD	Manufacturer Standard
STD PATN	Standard Pattern
PWHT	Post welding heat treatment

GENERAL NOTES

5.0 An example explaining the valve code

PVVGB62R0108ACAD0ZZ

PV	VG	B	02	R	01	08	A	C	A	D	0	ZZ
阀门	阀门类型	结构形式	压力等级	连接形式	阀体材料	阀内件	填料/弹簧	垫片	螺栓螺母	标准	驱动型式	特殊要求
Valve	Valve Type	Structure Type	Rating	Connection Type	Body	Trim	Packing/Spring	Gasket	Bolt/Nut	Design Standard	Operation	Special Requirement

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NO.	VALVE NUMBER	NPS	DESCRIPTION					SPEC.	REV.	REMARK		
1	PVVB!Ball Valve											
1. 1	PVVBA62R0108AVAF1VV	1/2" - 1"	Ball Valve, FP, SEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket:Non-ASB Gasket, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 608, Vacuum					A1B				
1. 2	PVVBA62R0108AVAF1ZZ	1-1/2" - 2"	Ball Valve, FP, SEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket:Non-ASB Gasket, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 608,					A1B				
1. 3	PVVBA62R0169EJAXOZZ	1/2" - 1-1/2"	Ball Valve, FP, SEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A105, Trim:304/RTFE, Stem:A182 F304, Packing:PTFE, Gasket:S. S/PTFE SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, BS EN ISO17292,					A1R				
1. 4	PVVBA62R0316FCAXOZZ	1/2" - 3"	Ball Valve, FP, SEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:STL, Stem:A182 F316, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, BS EN ISO17292,					A1W				
1. 5	PVVBA62R0358FCAXOZZ	1/2" - 3"	Ball Valve, FP, SEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim: Alloy20/PTFE, Stem: B462 UNS N08020, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, BS EN ISO17292,					A1S				
1. 6	PVVBA62R0369EJAXOZZ	2" - 3"	Ball Valve, FP, SEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:304/RTFE, Stem:A182 F304, Packing:PTFE, Gasket:S. S/PTFE SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, BS EN ISO17292,					A1R				

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NO.	VALVE NUMBER	NPS	DESCRIPTION					SPEC.	REV.	REMARK		
1. 7	PVVBA62R9097EKAXOZZ	1" - 3"	Ball Valve, FP, SEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB+PTFE, Trim:304/PTFE LINED/RTFE, Stem:304/PTFE LINED, Packing:PTFE, Gasket:R-PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, BS EN ISO17292,					A01S				
1. 8	PVVBA63F5410FCEX0BZ	1/2" - 4"	Ball Valve, FP, SEY, FLB, CL300 (PN5. 0MPa), FE, ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, BS EN ISO17292, Hydrogen Service					B2E				
1. 9	PVVBA63R5410FCEXOZZ	1/2" - 4"	Ball Valve, FP, SEY, FLB, CL300 (PN5. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, BS EN ISO17292,					B21A, B21D				
1. 10	PVVBA66R0771ACLF0ZZ	1/2" - 1-1/2"	Ball Valve, FP, SEY, FLB, CL600 (PN11. 0MPa), RF (ASME B16. 5), ASTM A350 LF2 CL1, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 608,					D2A				
1. 11	PVVBA66R3871ECNF1ZZ	1/2" - 1-1/2"	Ball Valve, FP, SEY, FLB, CL600 (PN11. 0MPa), RF (ASME B16. 5), ASTM A182 F316, Trim:316/RTFE, Stem:A182 F316, Packing:PTFE, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, LEVER, API 608,					D24P				
1. 12	PVVBA66R5471ECNF1ZZ	2" - 3"	Ball Valve, FP, SEY, FLB, CL600 (PN11. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316/RTFE, Stem:A182 F316, Packing:PTFE, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, LEVER, API 608,					D24P				

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NO.	VALVE NUMBER	NPS	DESCRIPTION					SPEC.	REV.	REMARK	
1. 13	PVVBA66R5471ECNF3ZZ	6"	Ball Valve, FP, SEY, FLB, CL600 (PN11. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316/RTFE, Stem:A182 F316, Packing:PTFE, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Worm Gear, API 608,					D24P			
1. 14	PVVBB66R1171BCLF0ZZ	2" - 3"	Ball Valve, FP, SEY, TRUB, CL600 (PN11. OMPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 608,					D2A			
1. 15	PVVBB66R1171BCLF3ZZ	4" - 6"	Ball Valve, FP, SEY, TRUB, CL600 (PN11. OMPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Worm Gear, API 608,					D2A			
1. 16	PVVBD62R0361BCAF0ZZ	2" - 4"	Ball Valve, REP, SEY, FLB, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/PTFE, Stem: A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608,					H1A			
1. 17	PVVBD62R0361BCAF3ZZ	6"	Ball Valve, REP, SEY, FLB, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/PTFE, Stem: A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 608,					H1A			
1. 18	PVVBD66R5471BCNF1ZZ	2-1/2"	Ball Valve, REP, SEY, FLB, CL600 (PN11. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, LEVER, API 608,					D21K			

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NO.	VALVE NUMBER	NPS	DESCRIPTION					SPEC.	REV.	REMARK	
1. 19	PVVBD68S0161ACAF0ZZ	1/2" - 1-1/2"	Ball Valve, REP, SEY, FLB, CL800, SW, ASTM A105, Trim:13Cr/PTFE, Stem: A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608,					H1A			
1. 20	PVVBE62R037CECAF3VV	10" - 24"	Ball Valve, REP, SEY, TRUB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Ball:nickel, Seat:RPTEF, Packing:PTFE, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 608, Vacuum					A1DV			
1. 21	PVVBE62R037CECAF3ZZ	10" - 24"	Ball Valve, REP, SEY, TRUB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Ball:nickel, Seat:RPTEF, Packing:PTFE, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 608,					A1D			
1. 22	PVVBF63R0171ACAF0ZZ	1/2" - 2"	Ball Valve, FP, TEY, FLB, CL300 (PN5. 0MPa), RF (ASME B16. 5), ASTM A105, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608,					B1A, B1B			
1. 23	PVVBF63R5471BCNF0ZZ	1" - 4"	Ball Valve, FP, TEY, FLB, CL300 (PN5. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 608,					B21K			
1. 24	PVVBF63R5471BCNF5ZZ	6"	Ball Valve, FP, TEY, FLB, CL300 (PN5. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Bevel Gear, API 608,					B21K			
1. 25	PVVBF66B0171ACAF0B2	1/2" - 1-1/2"	Ball Valve, FP, TEY, FLB, CL600 (PN11. 0MPa), BW, ASTM A105, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608, both ends with nipple, overall lengh 250mm					B1A			

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NO.	VALVE NUMBER	NPS	DESCRIPTION					SPEC.	REV.	REMARK	
1. 26	PVVBF66B0171ACAF0ZZ	1/2" - 1-1/2"	Ball Valve, FP, TEY, FLB, CL600 (PN11. OMPa) , BW, ASTM A105, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608,					B1B			
1. 27	PVVBF66D0171ACAF0S2	1/2" - 1-1/2"	Ball Valve, FP, TEY, FLB, CL600 (PN11. OMPa) , SW/FNPT, ASTM A105, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608, SW end with nipple, overall lengh 200mm					B1A			
1. 28	PVVBF66D0171ACAF0S3	1/2" - 1-1/2"	Ball Valve, FP, TEY, FLB, CL600 (PN11. OMPa) , SW/FNPT, ASTM A105, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608, SW end with nipple, overall lengh 200mm, Hydrogen Service					B1E			
1. 29	PVVBF66D0171ACAF0ZZ	1/2" - 1-1/2"	Ball Valve, FP, TEY, FLB, CL600 (PN11. OMPa) , SW/FNPT, ASTM A105, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608,					B1B			
1. 30	PVVBF66D0710ACLF0S2	1/2" - 1-1/2"	Ball Valve, FP, TEY, FLB, CL600 (PN11. OMPa) , SW/FNPT, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 608, SW end with nipple, overall lengh 200mm					A2A			
1. 31	PVVBF66E0171ACAF0B3	1/2" - 1-1/2"	Ball Valve, FP, TEY, FLB, CL600 (PN11. OMPa) , NBE, ASTM A105, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608, both ends with nipple, overall lengh 250mm, Hydrogen Service					B1E			
1. 32	PVVBF66E0171ACAF0VV	1/2" - 1-1/2"	Ball Valve, FP, TEY, FLB, CL600 (PN11. OMPa) , NBE, ASTM A105, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608, Vacuum					A1AV			

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NO.	VALVE NUMBER	NPS	DESCRIPTION					SPEC.	REV.	REMARK	
1.33	PVVBF66E0171ACAF0ZZ	1/2" - 1-1/2"	Ball Valve, FP, TEY, FLB, CL600 (PN11. OMPa) , NBE, ASTM A105, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608,					A1A, A1E			
1.34	PVVBF66E3864ACAF0B2	1/2" - 1-1/2"	Ball Valve, FP, TEY, FLB, CL600 (PN11. OMPa) , NBE, ASTM A182 F316, Trim:316/PTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608, both ends with nipple, overall lengh 250mm					A21A			
1.35	PVVBF66E3971ACNF0B2	1" - 1-1/2"	Ball Valve, FP, TEY, FLB, CL600 (PN11. OMPa) , NBE, ASTM A182 F316L, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 608, both ends with nipple, overall lengh 250mm					D24L			
1.36	PVVBF66R5371BCNF0ZZ	2" - 3"	Ball Valve, FP, TEY, FLB, CL600 (PN11. OMPa) , RF (ASME B16.5) , ASTM A351 CF3M, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 608,					D24L			
1.37	PVVBF66S0710ACLF0ZZ	1/2" - 1-1/2"	Ball Valve, FP, TEY, FLB, CL600 (PN11. OMPa) , SW, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 608,					A2A			
1.38	PVVBH66D0171ACAF0VV	1/2" - 1-1/2"	Ball Valve, REP, TEY, FLB, CL600 (PN11. OMPa) , SW/FNPT, ASTM A105, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608, Vacuum					A1AV			
1.39	PVVBH66D3864ACAF0S2	1/2" - 1-1/2"	Ball Valve, REP, TEY, FLB, CL600 (PN11. OMPa) , SW/FNPT, ASTM A182 F316, Trim:316/PTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608, SW end with nipple, overall lengh 200mm					A21A			

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NO.	VALVE NUMBER	NPS	DESCRIPTION					SPEC.	REV.	REMARK	
1. 40	PVVBH66D3871ACNFOS2	1/2" - 1-1/2"	Ball Valve, REP, TEY, FLB, CL600 (PN11. OMPa), SW/FNPT, ASTM A182 F316, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 608, SW end with nipple, overall lengh 200mm					A21K			
1. 41	PVVBH66E0171ACAF0VV	1/2" - 1-1/2"	Ball Valve, REP, TEY, FLB, CL600 (PN11. OMPa), NBE, ASTM A105, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608, Vacuum					A1AV			
1. 42	PVVBH66E0171ACAF0ZZ	1/2" - 1-1/2"	Ball Valve, REP, TEY, FLB, CL600 (PN11. OMPa), NBE, ASTM A105, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608,					A1A, A1E			
1. 43	PVVBH66E0771ACLF0B2	1/2" - 1-1/2"	Ball Valve, REP, TEY, FLB, CL600 (PN11. OMPa), NBE, ASTM A350 LF2 CL1, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 608, both ends with nipple, overall lengh 250mm					B2A			
1. 44	PVVBH66E3864ACAF0B2	1/2" - 1-1/2"	Ball Valve, REP, TEY, FLB, CL600 (PN11. OMPa), NBE, ASTM A182 F316, Trim:316/PTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608, both ends with nipple, overall lengh 250mm					A21A			
1. 45	PVVBH66E3871ACNF0B2	1/2" - 1-1/2"	Ball Valve, REP, TEY, FLB, CL600 (PN11. OMPa), NBE, ASTM A182 F316, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 608, both ends with nipple, overall lengh 250mm					A21K			
1. 46	PVVBH68N0764EJLF0ZZ	1/2" - 1-1/2"	Ball Valve, REP, TEY, FLB, CL800, FNPT, ASTM A350 LF2 CL1, Trim:316/PTFE, Stem:A182 F316, Packing:PTFE, Gasket:S. S/PTFE SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 608,					H3A			

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NO.	VALVE NUMBER	NPS	DESCRIPTION					SPEC.	REV.	REMARK	
1. 47	PVVBK62R0171ACAF0ZZ	1/2" - 1-1/2"	Ball Valve, Short PATN, FP, SEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A105, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608,					A1A, A1E			
1. 48	PVVBK62R0171BCAF1ZZ	1/2" - 2"	Ball Valve, Short PATN, FP, SEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A105, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 608,					A1E			
1. 49	PVVBK62R0171EUAF3F1	2" - 6"	Ball Valve, Short PATN, FP, SEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A105, Trim:316/RTFE, Stem:A182 F316, Packing:PTFE, Gasket:PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 608, POST INDICATOR, FM/UL					H61B			
1. 50	PVVBK62R0371BCAF1VV	1/2" - 2"	Ball Valve, Short PATN, FP, SEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 608, Vacuum					A1AV			
1. 51	PVVBK62R0371BCAF1ZZ	2"	Ball Valve, Short PATN, FP, SEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 608,					A1A			
1. 52	PVVBK62R3864ACAF0ZZ	1/2" - 2"	Ball Valve, Short PATN, FP, SEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A182 F316, Trim:316/PTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608,					A21A			

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NO.	VALVE NUMBER	NPS	DESCRIPTION					SPEC.	REV.	REMARK	
1. 53	PVVBK62R3871BCNF0ZZ	1/2" - 1-1/2"	Ball Valve, Short PATN, FP, SEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A182 F316, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 608,					A21K			
1. 54	PVVBK62R3871ECAFOZZ	1/2" - 4"	Ball Valve, Short PATN, FP, SEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A182 F316, Trim:316/RTFE, Stem:A182 F316, Packing:PTFE, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608,					A24P, A24P1, A24P2			
1. 55	PVVBK62R3871ECAF3ZZ	6"	Ball Valve, Short PATN, FP, SEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A182 F316, Trim:316/RTFE, Stem:A182 F316, Packing:PTFE, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 608,					A24P1			
1. 56	PVVBK62R5464BCAF1ZZ	3" - 4"	Ball Valve, Short PATN, FP, SEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316/PTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 608,					A21A			
1. 57	PVVBK62R5471ECAF3ZZ	6"	Ball Valve, Short PATN, FP, SEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316/RTFE, Stem:A182 F316, Packing:PTFE, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 608,					A24P, A24P2			
1. 58	PVVBK63R0171ACAF0BZ	1/2" - 2"	Ball Valve, Short PATN, FP, SEY, FLB, CL300 (PN5. 0MPa), RF (ASME B16. 5), ASTM A105, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608, Hydrogen Service					B1E			

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1. 59	PVVBN62R0171ECAF1VV	3/4" - 1-1/2"	Ball Valve, Short PATN, REP, SEY, FLB, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A105, Trim:316/RTFE, Stem:A182 F316, Packing:PTFE, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 608, Vacuum					A1DV			
1. 60	PVVBN62R0171ECAF1ZZ	3/4" - 1-1/2"	Ball Valve, Short PATN, REP, SEY, FLB, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A105, Trim:316/RTFE, Stem:A182 F316, Packing:PTFE, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 608,					A1D			
1. 61	PVVBN62R0371ECAF1VV	2" - 6"	Ball Valve, Short PATN, REP, SEY, FLB, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:316/RTFE, Stem:A182 F316, Packing:PTFE, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 608, Vacuum					A1DV			
1. 62	PVVBN62R0371ECAF1ZZ	2" - 6"	Ball Valve, Short PATN, REP, SEY, FLB, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:316/RTFE, Stem:A182 F316, Packing:PTFE, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 608,					A1D			
1. 63	PVVBN62R0371ECAF3VV	8"	Ball Valve, Short PATN, REP, SEY, FLB, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:316/RTFE, Stem:A182 F316, Packing:PTFE, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 608, Vacuum					A1DV			
1. 64	PVVBN62R0371ECAF3ZZ	8"	Ball Valve, Short PATN, REP, SEY, FLB, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:316/RTFE, Stem:A182 F316, Packing:PTFE, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 608,					A1D			

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NO.	VALVE NUMBER	NPS	DESCRIPTION					SPEC.	REV.	REMARK	
1. 65	PVVBN62R1110BCLF1ZZ	2" - 4"	Ball Valve, Short PATN, REP, SEY, FLB, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, LEVER, API 608,					A2A			
1. 66	PVVBN62R5471BCNF1ZZ	2" - 4"	Ball Valve, Short PATN, REP, SEY, FLB, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, LEVER, API 608,					A21K			
1. 67	PVVBN62R5471BCNF3ZZ	6" - 24"	Ball Valve, Short PATN, REP, SEY, FLB, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Worm Gear, API 608,					A21K			
1. 68	PVVBN63R1171BCLF1ZZ	2" - 5"	Ball Valve, Short PATN, REP, SEY, FLB, CL300 (PN5. OMPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, LEVER, API 608,					B2A			
1. 69	PVVBP62R1110BCLF3ZZ	6" - 14"	Ball Valve, Short PATN, REP, SEY, TRUB, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Worm Gear, API 608,					A2A			
1. 70	PVVBP63R1171BCLF3ZZ	6" - 16"	Ball Valve, Short PATN, REP, SEY, TRUB, CL300 (PN5. OMPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Worm Gear, API 608,					B2A			

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NO.	VALVE NUMBER	NPS	DESCRIPTION					SPEC.	REV.	REMARK	
1. 71	PVVBQ62R3871ACAF0ZZ	1/2" - 2"	Ball Valve, Short PATN, FP, TEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A182 F316, Trim:316/RTFE, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608,					A21D			
1. 72	PVVBQ62R5471BCAF0ZZ	3" - 8"	Ball Valve, Short PATN, FP, TEY, FLB, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 608,					A21D			
1. 73	PVVBW75N3864EMZMOFT	1/4" - 3/4"	Ball Valve, Three pieces, FP, CL1500 (PN26. 0Mpa), FNPT, ASTM A182 F316, Trim:316/PTFE, Stem:A182 F316, Packing:PTFE, Gasket:MFR STD, MFR STD, For Tubing					S82A			
2	PVVC!Check Valve										
2. 1	PVVC166R0308NCAA0ZZ	2" - 24"	Check Valve, BC, TITLING DISC, CL600 (PN11. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 594,					D1A			
2. 2	PVVC275N381MMZMOFT	1/4" - 1"	-					S82A			
2. 3	PVVC62I0301XNZAOZZ	30" - 36"	Check Valve, Dual plate, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 47 A), ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Spring:INCONEL-X750, API 594,					A1K, H1A			
2. 4	PVVC62I1110XNZAOZZ	28" - 48"	Check Valve, Dual plate, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 47 A), ASTM A352 LCB, Trim:316, Stem:A182 F316, Spring:INCONEL-X750, API 594,					A2A			

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NO.	VALVE NUMBER	NPS	DESCRIPTION					SPEC.	REV.	REMARK	
2. 5	PVVCD62K032CXCAA0VV	28" - 48"	Check Valve, Dual plate, CL150 (PN2. 0MPa), RF (ASME B16. 47 A), ASTM A216 WCB, Trim:13Cr/STL, Disc:410 SS, Spring: INCONEL-X750, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 594, Vacuum					A1DV			
2. 6	PVVCD62K032CXCAA0ZZ	28" - 48"	Check Valve, Dual plate, CL150 (PN2. 0MPa), RF (ASME B16. 47 A), ASTM A216 WCB, Trim:13Cr/STL, Disc:410 SS, Spring: INCONEL-X750, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 594,					A1D			
2. 7	PVVCD62R032CXCAA0VV	12" - 24"	Check Valve, Dual plate, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Disc:410 SS, Spring: INCONEL-X750, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 594, Vacuum					A1DV			
2. 8	PVVCD62R032CXCAA0ZZ	12" - 24"	Check Valve, Dual plate, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Disc:410 SS, Spring: INCONEL-X750, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 594,					A1D			
2. 9	PVVCD62R5410NJNA0ZZ	2" - 16"	Check Valve, Dual plate, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Gasket:S. S/PTFE SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 594,					A23B			
2. 10	PVVCD62R5410NKNA0ZZ	2" - 16"	Check Valve, Dual plate, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Gasket:R-PTFE, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 594,					A23D			
2. 11	PVVCD62W0301XNZAOZZ	2" - 24"	Check Valve, Dual plate, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Spring: INCONEL-X750, API 594,					A1E, A1K, H1A			

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2. 12	PVVCD62W0308XNZA0VV	2" - 24"	Check Valve, Dual plate, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Spring:INCONEL-X750, API 594, Vacuum					A1AV			
2. 13	PVVCD62W0308XNZA0ZZ	2" - 16"	Check Valve, Dual plate, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Spring:INCONEL-X750, API 594,					A1A, A1B			
2. 14	PVVCD62W1110NCLA0VV	2" - 24"	Check Valve, Dual plate, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 594, Vacuum					A2DV			
2. 15	PVVCD62W1110NCLA0ZZ	2" - 24"	Check Valve, Dual plate, CL150 (PN2. 0MPa), Wafer (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 594,					A2D, A2E			
2. 16	PVVCD62W1110XNZA0ZZ	2" - 24"	Check Valve, Dual plate, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Spring:INCONEL-X750, API 594,					A2A, H2A			
2. 17	PVVCD62W1164XNZA0ZZ	2" - 16"	Check Valve, Dual plate, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A352 LCB, Trim:316/PTFE, Stem:A182 F316, Spring:INCONEL-X750, API 594,					H3A			
2. 18	PVVCD62W5410XNZA0ZZ	2" - 24"	Check Valve, Dual plate, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Spring:INCONEL-X750, API 594,					A21A			
2. 19	PVVCD63W0301NCAA0ZZ	2" - 24"	Check Valve, Dual plate, CL300 (PN5. 0MPa), Wafer (ASME B16. 5), ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 594,					B1A, B1B			

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2.20	PVVCD63W0301XNZAOBZ	2" - 24"	Check Valve,Dual plate,CL300(PN5.0MPa),Wafer,RF(ASME B16.5),ASTM A216 WCB,Trim:13Cr,Stem:A182 F6a CL.2, Spring:INCONEL-X750, API 594,Hydrogen Service					B1E			
2.21	PVVCD63W1110XNZAOZZ	2" - 20"	Check Valve,Dual plate,CL300(PN5.0MPa),Wafer,RF(ASME B16.5),ASTM A352 LCB,Trim:316, Stem:A182 F316, Spring:INCONEL-X750, API 594,					B2A			
2.22	PVVCD63W5410YNEAOZZ	2" - 12"	Check Valve,Dual plate,CL300(PN5.0MPa),Wafer,RF(ASME B16.5),ASTM A351 CF8M,Trim:316,Stem:A182 F316, Spring:316,Bolt/Nut:A193 Gr.B8M CLASS 1/A194 Gr.8M, API 594,					B21D			
2.23	PVVCD66L1110XNZAOZZ	2" - 10"	Check Valve,Dual plate,CL600(PN11.0MPa),Lug (ASME B16.5),ASTM A352 LCB,Trim:316, Stem:A182 F316, Spring:INCONEL-X750, API 594,					D2A			
2.24	PVVCD66R0301NCAA0ZZ	2" - 24"	Check Valve,Dual plate,CL600(PN11.0MPa),RF(ASME B16.5),ASTM A216 WCB,Trim:13Cr,Stem:A182 F6a CL.2,Gasket: S.S/FG SWG,Bolt/Nut:A193 Gr.B7/A194 Gr.2H, API 594,					D1A			
2.25	PVVCD66W5310XNZAOZZ	2" - 3"	Check Valve,Dual plate,CL600(PN11.0MPa),Wafer,RF(ASME B16.5),ASTM A351 CF3M,Trim:316,Stem:A182 F316, Spring:INCONEL-X750, API 594,					D24L			
2.26	PVVCD66W5410XNZAOZZ	2" - 12"	Check Valve,Dual plate,CL600(PN11.0MPa),Wafer,RF(ASME B16.5),ASTM A351 CF8M,Trim:316,Stem:A182 F316, Spring:INCONEL-X750, API 594,					D21K			
2.27	PVVCL62R0102NCAD0ZZ	1/2" - 1-1/2"	Check Valve,BC,Lift,CL150(PN2.0MPa),RF(ASME B16.5),ASTM A105,Trim:304,Stem:A182 F304,Gasket: S.S/FG SWG,Bolt/Nut:A193 Gr.B7/A194 Gr.2H, API 602,					A1R			

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2. 28	PVVCL62R0108NVAD0ZZ	1/2" - 1-1/2"	Check Valve, BC, Lift, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Gasket:Non-ASB Gasket, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A1B			
2. 29	PVVCL62R0113NCAD0ZZ	1/2" - 1-1/2"	Check Valve, BC, Lift, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A105, Trim:alloy 20, Stem:B462 US N08200, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A1S			
2. 30	PVVCL62R9097NKAL0ZZ	1" - 24"	Check Valve, BC, Lift, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB+PTFE, Trim:304/PTFE LINED/RTFE, Stem:304/PTFE LINED, Gasket:R-PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, BS 1868,					A01S			
2. 31	PVVCL66R3812NCND0ZZ	1/2" - 1-1/2"	Check Valve, BC, Lift, CL600 (PN11. OMPa), RF (ASME B16. 5), ASTM A182 F316, Trim:316/STL, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,					D24P			
2. 32	PVVCL68B0105NCAD0ZZ	1/2" - 1-1/2"	Check Valve, BC, Lift, CL800, BW, ASTM A105, Trim:STL, Stem:A182 F6a CL. 2, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A1W			
2. 33	PVVCL68B3810NCED0BZ	1/2" - 1-1/2"	Check Valve, BC, Lift, CL800, BW, ASTM A182 F316, Trim:316, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, API 602, Hydrogen Service					B2E			
2. 34	PVVCL68N0101NCAD0ZZ	1/2" - 1-1/2"	Check Valve, BC, Lift, CL800, FNPT, ASTM A105, Trim:13Cr, Stem:A182 F6a CL. 2, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					H3A			
2. 35	PVVCL68S0101NCAD0BZ	1/2" - 1-1/2"	Check Valve, BC, Lift, CL800, SW, ASTM A105, Trim:13Cr, Stem:A182 F6a CL. 2, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602, Hydrogen Service					B1E			

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2. 36	PVVCL68S0101NCAD0ZZ	1/2" - 1-1/2"	Check Valve, BC, Lift, CL800, SW, ASTM A105, Trim:13Cr, Stem:A182 F6a CL. 2, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A1K, B1A, B1B, D1A, H1A			
2. 37	PVVCL68S0108NCAD0VV	1/2" - 1-1/2"	Check Valve, BC, Lift, CL800, SW, ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602, Vacuum					A1AV, A1DV			
2. 38	PVVCL68S0108NCAD0ZZ	1/2" - 1-1/2"	Check Valve, BC, Lift, CL800, SW, ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A1A, A1D			
2. 39	PVVCL68S0710NCLD0VV	1/2" - 1-1/2"	Check Valve, BC, Lift, CL800, SW, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602, Vacuum					A2DV			
2. 40	PVVCL68S0710NCLD0ZZ	1/2" - 1-1/2"	Check Valve, BC, Lift, CL800, SW, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602,					A2A, A2D, A2E, B2A, B2B, D2A, H2A			
2. 41	PVVCL68S1808NQCD0ZZ	1/2" - 1-1/2"	Check Valve, BC, Lift, CL800, SW, ASTM A182 F11 CL2, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Gasket:12Cr5Mo(1Cr5Mo), Bolt/Nut:A193 Gr. B16/A194 Gr. 4, API 602,					B54H, B54K			
2. 42	PVVCL68S3810NCED0ZZ	1/2" - 1-1/2"	Check Valve, BC, Lift, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, API 602,					B21A, B21D			

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2. 43	PVVCL68S4408N11DOHT	1/2" - 1-1/2"	Check Valve, BC, Lift, CL800, SW, ASTM A182 F347H, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, API 602, HIGH TEMPERATURE SEVICE					B27B			
2. 44	PVVCL68S4408N11DOHV	1/2" - 1-1/2"	Check Valve, BC, Lift, CL800, SW, ASTM A182 F347H, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, API 602, FULL VACUUM AT HIGH TEMPERATURE SEVICE					B27A			
2. 45	PVVCL75B0710NCLDOZZ	1/2" - 1-1/2"	Check Valve, BC, Lift, CL1500 (PN26. 0Mpa), BW, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602,					F2A			
2. 46	PVVCL75S3816NCEDOZZ	1/2" - 1-1/2"	Check Valve, BC, Lift, CL1500 (PN26. 0MPa), SW, ASTM A182 F316, Trim:STL, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, API 602,					F23A			
2. 47	PVVCM62R9097NKALOZZ	1" - 24"	Check Valve, Vertical lift, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB+PTFE, Trim:304/PTFE LINED/RTFE, Stem:304/PTFE LINED, Gasket:R- PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, BS 1868,					A01S			
2. 48	PVVCP75B1815NMADOZZ	1/2" - 1-1/2"	Check Valve, PSC, Lift, CL1500 (PN26. 0Mpa), BW, ASTM A182 F11 CL2, Trim:STL, Stem:A182 F304, Gasket:MFR STD, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					E54K			
2. 49	PVVCQ69B2715NMALOZZ	2" - 16"	Check Valve, PSC, SWING, CL900 (PN15. 0MPa), BW, ASTM A217 WC6, Trim:STL, Stem:A182 F304, Gasket:MFR STD, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, BS 1868,					E54K			

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2. 50	PVVCS62R0301NCACOZZ	2" - 12"	Check Valve, BC, Swing, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,					H1A			
2. 51	PVVCS62R0302NCALOZZ	2" - 8"	Check Valve, BC, Swing, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:304, Stem:A182 F304, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, BS 1868,					A1R			
2. 52	PVVCS62R0305NCALOZZ	2" - 24"	Check Valve, BC, Swing, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:STL, Stem:A182 F6a CL. 2, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, BS 1868,					A1W			
2. 53	PVVCS62R0308NCACOVV	2" - 24"	Check Valve, BC, Swing, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600, Vacuum					A1DV			
2. 54	PVVCS62R0308YCACOZZ	2" - 24"	Check Valve, BC, Swing, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Spring:316, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,					A1D			
2. 55	PVVCS62R0313NCALOZZ	2" - 6"	Check Valve, BC, Swing, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:alloy 20, Stem:B462 US N08200, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, BS 1868,					A1S			
2. 56	PVVCS62R3810NCADOZZ	1/2" - 1-1/2"	Check Valve, BC, Swing, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A182 F316, Trim:316, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A24P, A24P1, A24P2			
2. 57	PVVCS62R3810NKNAOZZ	3/4" - 2"	Check Valve, BC, Swing, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A182 F316, Trim:316, Stem:A182 F316, Gasket:R-PTFE, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 594,					A21D			

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2. 58	PVVCS62R3810NKND0ZZ	1/2" - 16"	Check Valve, BC, Swing, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A182 F316, Trim:316, Stem:A182 F316, Gasket:R-PTFE, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,					A23D			
2. 59	PVVCS62R5410NCNC0ZZ	2" - 24"	Check Valve, BC, Swing, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600,					A21L			
2. 60	PVVCS62R5410NKNA0ZZ	3" - 12"	Check Valve, BC, Swing, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Gasket:R-PTFE, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 594,					A21D			
2. 61	PVVCS62R5416NJNC0ZZ	2" - 16"	Check Valve, BC, Swing, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:STL, Stem:A182 F316, Gasket:S. S/PTFE SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600,					A23B			
2. 62	PVVCS62R5416NJND0ZZ	1/2" - 1-1/2"	Check Valve, BC, Swing, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:STL, Stem:A182 F316, Gasket:S. S/PTFE SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,					A23B			
2. 63	PVVCS62W1110NCLA0ZZ	2" - 24"	Check Valve, BC, Swing, CL150 (PN2. OMPa), Wafer, RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 594,					B2B			
2. 64	PVVCS63F5410NCELOBZ	2" - 12"	Check Valve, BC, Swing, CL300 (PN5. OMPa), FE, ASTM A351 CF8M, Trim:316, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, BS 1868, Hydrogen Service					B2E			
2. 65	PVVCS63J2708NQCC0ZZ	2" - 24"	Check Valve, BC, Swing, CL300 (PN5. OMPa), RJ, ASTM A217 WC6, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Gasket:12Cr5Mo(1Cr5Mo), Bolt/Nut:A193 Gr. B16/A194 Gr. 4, API 600,					B54H, B54K			

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2. 66	PVVC63J5508N11COHT	2" - 24"	Check Valve, BC, Swing, CL300 (PN5. 0MPa), RJ, ASTM A351 CF8C, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, API 600, HIGH TEMPERATURE SEVICE				B27B				
2. 67	PVVC63J5508N11COHV	2" - 24"	Check Valve, BC, Swing, CL300 (PN5. 0MPa), RJ, ASTM A351 CF8C, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, API 600, FULL VACUUM AT HIGH TEMPERATURE SEVICE				B27A				
2. 68	PVVC63R5410NCELOZZ	2" - 24"	Check Valve, BC, Swing, CL300 (PN5. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, BS 1868,				B21A, B21D				
2. 69	PVVC63R5410NCNCOZZ	2" - 24"	Check Valve, BC, Swing, CL300 (PN5. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600,				B21L				
2. 70	PVVC63W5410NNZCOZZ	2" - 24"	Check Valve, BC, Swing, CL300 (PN5. 0MPa), Wafer, RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, API 600,				B21K				
2. 71	PVVC66R5412NCNCOZZ	3" - 6"	Check Valve, BC, Swing, CL600 (PN11. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316/STL, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600,				D24P				
2. 72	PVVC68N0710NJLDOZZ	1/2" - 1-1/2"	Check Valve, BC, Swing, CL800, FNPT, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Gasket:S. S/PTFE SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602,				H3A				
2. 73	PVVC68S3810NCAD0ZZ	1/2" - 1-1/2"	Check Valve, BC, Swing, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,				A21A				

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2. 74	PVVCS68S3810NCEL0ZZ	1/2" - 1-1/2"	Check Valve, BC, Swing, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, BS 1868,				B21A				
2. 75	PVVCS68S3810NCND0ZZ	1/2" - 1-1/2"	Check Valve, BC, Swing, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,				A21K, A21L, B21K, B21L, D21K				
2. 76	PVVCS68S3910NCND0ZZ	1" - 1-1/2"	Check Valve, BC, Swing, CL800, SW, ASTM A182 F316L, Trim:316, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,				D24L				
2. 77	PVVCS75B1110NCLC0ZZ	2" - 10"	Check Valve, BC, Swing, CL1500 (PN26. 0Mpa), BW, ASTM A352 LCB, Trim:316, Stem:A182 F316, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 600,				F2A				
2. 78	PVVCY68S0101NCAD0ZZ	1/2" - 1-1/2"	Check Valve, BC, Lift, Y PATN, CL800, SW, ASTM A105, Trim:13Cr, Stem:A182 F6a CL. 2, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,				A1E				
3	PVVD!Diaphragm Valve										
3. 1	PVVDW62R9097EKAMOZZ	1" - 4"	Diaphragm Valve, WB, Ball, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB+PTFE, Trim:304/PTFE LINED/RTFE, Stem:304/PTFE LINED, Packing:PTFE, Gasket:R-PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, MFR STD,				A01S				
4	PVVG!Gate Valve										

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4. 1	PVVGB62K0301BCAC5ZZ	30" - 36"	Gate Valve, BB, OS&Y, CL150 (PN2. OMPa) , RF (ASME B16. 47 A), ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,				H1A				
4. 2	PVVGB62K0308BCAC5ZZ	28" - 48"	Gate Valve, BB, OS&Y, CL150 (PN2. OMPa) , RF (ASME B16. 47 A), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,				A1D				
4. 3	PVVGB62R0101ACAD0ZZ	1/2" - 2"	Gate Valve, BB, OS&Y, CL150 (PN2. OMPa) , RF (ASME B16. 5), ASTM A105, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,				A1E, A1K, H1A				
4. 4	PVVGB62R0102FCAD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL150 (PN2. OMPa) , RF (ASME B16. 5), ASTM A105, Trim:304, Stem:A182 F304, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,				A1R				
4. 5	PVVGB62R0105FCAD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL150 (PN2. OMPa) , RF (ASME B16. 5), ASTM A105, Trim:STL, Stem:A182 F6a CL. 2, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,				A1W				
4. 6	PVVGB62R0108ACAD0VV	1/2" - 2"	Gate Valve, BB, OS&Y, CL150 (PN2. OMPa) , RF (ASME B16. 5), ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602, Vacuum				A1AV, A1DV				
4. 7	PVVGB62R0108ACAD0ZZ	1/2" - 2"	Gate Valve, BB, OS&Y, CL150 (PN2. OMPa) , RF (ASME B16. 5), ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,				A1A, A1D				
4. 8	PVVGB62R0113FCAD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL150 (PN2. OMPa) , RF (ASME B16. 5), ASTM A105, Trim:alloy 20, Stem:B462 US N08200, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,				A1S				

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4. 9	PVVGB62R0301BCAC0ZZ	2" - 14"	Gate Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,					H1A, H3B			
4. 10	PVVGB62R0301BCAC5ZZ	16" - 24"	Gate Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,					H1A			
4. 11	PVVGB62R0302FCAC0ZZ	2" - 8"	Gate Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:304, Stem:A182 F304, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,					A1R			
4. 12	PVVGB62R0305FCAC0ZZ	2" - 12"	Gate Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:STL, Stem:A182 F6a CL. 2, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,					A1W			
4. 13	PVVGB62R0305FCAC5ZZ	16" - 24"	Gate Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:STL, Stem:A182 F6a CL. 2, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,					A1W			
4. 14	PVVGB62R0308BCAC0ZZ	2" - 14"	Gate Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,					A1D			
4. 15	PVVGB62R0308BCAC5ZZ	16" - 24"	Gate Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,					A1D			
4. 16	PVVGB62R0313FCAC0ZZ	2" - 8"	Gate Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:alloy 20, Stem:B462 US N08200, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,					A1S			

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4. 17	PVVGB62R0710ACLD0VV	1/2" - 2"	Gate Valve, BB, OS&Y, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602, Vacuum					A2DV			
4. 18	PVVGB62R0710ACLD0ZZ	1/2" - 2"	Gate Valve, BB, OS&Y, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602,					A2A, A2D, A2E, H2A			
4. 19	PVVGB62R1110EJLC0ZZ	2" - 14"	Gate Valve, BB, OS&Y, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:PTFE, Gasket:S. S/PTFE SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 600,					H3A			
4. 20	PVVGB62R1110EJLC5ZZ	16"	Gate Valve, BB, OS&Y, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:PTFE, Gasket:S. S/PTFE SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Bevel Gear, API 600,					H3A			
4. 21	PVVGB62R3810ACAD0ZZ	1/2" - 2"	Gate Valve, BB, OS&Y, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A21A			
4. 22	PVVGB62R3810ACND0EV	1/2" - 2"	Gate Valve, BB, OS&Y, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602, Extented stem					A21L			
4. 23	PVVGB62R3810ACND0ZZ	1/2" - 2"	Gate Valve, BB, OS&Y, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,					A21K			
4. 24	PVVGB62R3810AKND0ZZ	1/2" - 2"	Gate Valve, BB, OS&Y, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket:R- PTFE, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,					A21D, A23D			

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4. 25	PVVGB62R3816AJND0ZZ	1/2" - 2"	Gate Valve, BB, OS&Y, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A182 F316, Trim:STL, Stem:A182 F316, Packing:GBP, Gasket:S. S/PTFE SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,					A23B			
4. 26	PVVGB62R8Z95EUAQ0F1	2" - 8"	Gate Valve, BB, OS&Y, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A126 GR. B, Trim:EPR Lined, Stem:A182 F6a CL. 2, Packing:PTFE, Gasket:PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, MSS SP-70, POST INDICATOR, FM/UL					H61B			
4. 27	PVVGB62R8Z95EUAQ0ZZ	2" - 14"	Gate Valve, BB, OS&Y, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A126 GR. B, Trim:EPR Lined, Stem:A182 F6a CL. 2, Packing:PTFE, Gasket:PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, MSS SP-70,					H61B			
4. 28	PVVGB62R8Z95EUAQ3ZZ	16" - 24"	Gate Valve, BB, OS&Y, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A126 GR. B, Trim:EPR Lined, Stem:A182 F6a CL. 2, Packing:PTFE, Gasket:PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, MSS SP-70,					H61B, H61C			
4. 29	PVVGB62R9097EKAC0ZZ	2" - 12"	Gate Valve, BB, OS&Y, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB+PTFE, Trim:304/PTFE LINED/RTFE, Stem:304/PTFE LINED, Packing:PTFE, Gasket:R-PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,					A01S			
4. 30	PVVGB62R9097EKAC5ZZ	14" - 24"	Gate Valve, BB, OS&Y, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB+PTFE, Trim:304/PTFE LINED/RTFE, Stem:304/PTFE LINED, Packing:PTFE, Gasket:R-PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,					A01S			

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4. 31	PVVGB62R9097EKAD0ZZ	1" - 1-1/2"	Gate Valve, BB, OS&Y, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB+PTFE, Trim:304/PTFE LINED/RTFE, Stem:304/PTFE LINED, Packing:PTFE, Gasket:R-PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A01S			
4. 32	PVVGB63J18081QCD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL300 (PN5. 0MPa), RJ, ASTM A182 F11 CL2, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:High Temperature Packing for 593℃ Service, Gasket:12Cr5Mo (1Cr5Mo), Bolt/Nut:A193 Gr. B16/A194 Gr. 4, API 602,					B54K			
4. 33	PVVGB63J1808BQCD0ZZ	1/2" - 2"	Gate Valve, BB, OS&Y, CL300 (PN5. 0MPa), RJ, ASTM A182 F11 CL2, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket:12Cr5Mo (1Cr5Mo), Bolt/Nut:A193 Gr. B16/A194 Gr. 4, API 602,					B54H			
4. 34	PVVGB63J27081QCC0ZZ	2" - 10"	Gate Valve, BB, OS&Y, CL300 (PN5. 0MPa), RJ, ASTM A217 WC6, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:High Temperature Packing for 593℃ Service, Gasket:12Cr5Mo (1Cr5Mo), Bolt/Nut:A193 Gr. B16/A194 Gr. 4, API 600,					B54K			
4. 35	PVVGB63J27081QCC5ZZ	12" - 24"	Gate Valve, BB, OS&Y, CL300 (PN5. 0MPa), RJ, ASTM A217 WC6, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:High Temperature Packing for 593℃ Service, Gasket:12Cr5Mo (1Cr5Mo), Bolt/Nut:A193 Gr. B16/A194 Gr. 4, Bevel Gear, API 600,					B54K			
4. 36	PVVGB63J4438H11DOHT	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL300 (PN5. 0MPa), RJ, ASTM A182 F347H, Trim:STL, Stem:A182 F347H, Packing:High Temperature Packing for 704℃ Service, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, API 602, HIGH TEMPERATURE SEVICE					B27B			

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4. 37	PVVGB63J4438H11D0HV	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL300 (PN5. 0MPa), RJ, ASTM A182 F347H, Trim:STL, Stem:A182 F347H, Packing:High Temperature Packing for 704℃ Service, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, API 602, FULL VACUUM AT HIGH TEMPERATURE SEVICE					B27A			
4. 38	PVVGB63J5538H11C0HT	2" - 10"	Gate Valve, BB, OS&Y, CL300 (PN5. 0MPa), RJ, ASTM A351 CF8C, Trim:STL, Stem:A182 F347H, Packing:High Temperature Packing for 704℃ Service, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, API 600, HIGH TEMPERATURE SEVICE					B27B			
4. 39	PVVGB63J5538H11C0HV	2" - 10"	Gate Valve, BB, OS&Y, CL300 (PN5. 0MPa), RJ, ASTM A351 CF8C, Trim:STL, Stem:A182 F347H, Packing:High Temperature Packing for 704℃ Service, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, API 600, FULL VACUUM AT HIGH TEMPERATURE SEVICE					B27A			
4. 40	PVVGB63J5538H11C5HT	12" - 24"	Gate Valve, BB, OS&Y, CL300 (PN5. 0MPa), RJ, ASTM A351 CF8C, Trim:STL, Stem:A182 F347H, Packing:High Temperature Packing for 704℃ Service, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, Bevel Gear, API 600, HIGH TEMPERATURE SEVICE					B27B			
4. 41	PVVGB63J5538H11C5HV	12" - 24"	Gate Valve, BB, OS&Y, CL300 (PN5. 0MPa), RJ, ASTM A351 CF8C, Trim:STL, Stem:A182 F347H, Packing:High Temperature Packing for 704℃ Service, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, Bevel Gear, API 600, FULL VACUUM AT HIGH TEMPERATURE SEVICE					B27A			
4. 42	PVVGB63R0101ACAD0BZ	1/2" - 2"	Gate Valve, BB, OS&Y, CL300 (PN5. 0MPa), RF (ASME B16. 5), ASTM A105, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602, Hydrogen Service					B1E			

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4. 43	PVVGB63R0108ACAD0VV	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL300 (PN5.0MPa), RF (ASME B16.5), ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S.S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602, Vacuum					A1DV			
4. 44	PVVGB63R0108ACAD0ZZ	1/2" - 2"	Gate Valve, BB, OS&Y, CL300 (PN5.0MPa), RF (ASME B16.5), ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S.S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A1D, B1A, B1B			
4. 45	PVVGB63R0710ACLD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL300 (PN5.0MPa), RF (ASME B16.5), ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S.S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602,					B2A			
4. 46	PVVGB63R1110ACLD0ZZ	1/2" - 2"	Gate Valve, BB, OS&Y, CL300 (PN5.0MPa), RF (ASME B16.5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S.S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602,					B2B			
4. 47	PVVGB63R3810ACND0EV	1/2" - 2"	Gate Valve, BB, OS&Y, CL300 (PN5.0MPa), RF (ASME B16.5), ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S.S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602, Extented stem					B21L			
4. 48	PVVGB63R3810ACND0ZZ	1/2" - 2"	Gate Valve, BB, OS&Y, CL300 (PN5.0MPa), RF (ASME B16.5), ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S.S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,					B21K			
4. 49	PVVGB63R3810FCED0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL300 (PN5.0MPa), RF (ASME B16.5), ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:FG, Gasket: S.S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, API 602,					B21A, B21D			
4. 50	PVVGB63R5410FCECOZZ	2" - 10"	Gate Valve, BB, OS&Y, CL300 (PN5.0MPa), RF (ASME B16.5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:FG, Gasket: S.S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, API 600,					B21A, B21D			

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4. 51	PVVGB63R5410FCEC5ZZ	12" - 24"	Gate Valve, BB, OS&Y, CL300 (PN5. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, Bevel Gear, API 600,					B21A, B21D			
4. 52	PVVGB66R0108ACAD0ZZ	1/2" - 2"	Gate Valve, BB, OS&Y, CL600 (PN11. 0MPa), RF (ASME B16. 5), ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					D1A			
4. 53	PVVGB66R0710ACLD0ZZ	1/2" - 2"	Gate Valve, BB, OS&Y, CL600 (PN11. 0MPa), RF (ASME B16. 5), ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602,					D2A			
4. 54	PVVGB66R3810ACND0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL600 (PN11. 0MPa), RF (ASME B16. 5), ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,					D21K			
4. 55	PVVGB68B0105FCAD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, BW, ASTM A105, Trim:STL, Stem:A182 F6a CL. 2, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A1W			
4. 56	PVVGB68D0101ACAD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW/FNPT, ASTM A105, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A1E, H1A			
4. 57	PVVGB68D0108ACAD0VV	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW/FNPT, ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602, Vacuum					A1AV, A1DV			
4. 58	PVVGB68D0108ACAD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW/FNPT, ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A1D, B1A, B1B, D1A			

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4. 59	PVVGB68D0710ACLD0VV	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW/FNPT, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602, Vacuum					A2DV			
4. 60	PVVGB68D0710ACLD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW/FNPT, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602,					A2D, A2E, B2A, B2B, D2A			
4. 61	PVVGB68D18081QCD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW/FNPT, ASTM A182 F11 CL2, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:High Temperature Packing for 593℃ Service, Gasket:12Cr5Mo(1Cr5Mo), Bolt/Nut:A193 Gr. B16/A194 Gr. 4, API 602,					B54K			
4. 62	PVVGB68D3810ACND0EV	1" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW/FNPT, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602, Extented stem					B21L			
4. 63	PVVGB68D3810ACND0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW/FNPT, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,					B21L, D21K			
4. 64	PVVGB68D3810ACNM0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW/FNPT, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, MFR STD,					A21K, B21K			
4. 65	PVVGB68D4438H11DOHT	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW/FNPT, ASTM A182 F347H, Trim:STL, Stem:A182 F347H, Packing:High Temperature Packing for 704℃ Service, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, API 602, HIGH TEMPERATURE SEVICE					B27B			

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4. 66	PVVGB68D4438H11D0HV	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW/FNPT, ASTM A182 F347H, Trim:STL, Stem:A182 F347H, Packing:High Temperature Packing for 704℃ Service, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, API 602, FULL VACUUM AT HIGH TEMPERATURE SEVICE				B27A				
4. 67	PVVGB68N0101ACAD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, FNPT, ASTM A105, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,				H3B				
4. 68	PVVGB68N0710EJLD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, FNPT, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:PTFE, Gasket:S. S/PTFE SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602,				H3A				
4. 69	PVVGB68S0101ACAD0BZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW, ASTM A105, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602, Hydrogen Service				B1E				
4. 70	PVVGB68S0101ACAD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW, ASTM A105, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,				A1E, A1K, H1A				
4. 71	PVVGB68S0108ACAD0VV	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW, ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602, Vacuum				A1AV, A1DV				
4. 72	PVVGB68S0108ACAD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW, ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,				A1A, A1D, B1A, B1B, D1A				
4. 73	PVVGB68S0710ACLD0VV	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602, Vacuum				A2DV				

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4. 74	PVVGB68S0710ACLD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602,					A2A, A2D, A2E, B2A, B2B, D2A, H2A			
4. 75	PVVGB68S18081QCD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW, ASTM A182 F11 CL2, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:High Temperature Packing for 593℃ Service, Gasket:12Cr5Mo (1Cr5Mo), Bolt/Nut:A193 Gr. B16/A194 Gr. 4, API 602,					B54K			
4. 76	PVVGB68S1808BQCD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW, ASTM A182 F11 CL2, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket:12Cr5Mo (1Cr5Mo), Bolt/Nut:A193 Gr. B16/A194 Gr. 4, API 602,					B54H			
4. 77	PVVGB68S3810ACAD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A21A			
4. 78	PVVGB68S3810ACND0EV	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602, Extented stem					A21L, B21L			
4. 79	PVVGB68S3810ACND0VV	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602, Vacuum					D21K			
4. 80	PVVGB68S3810ACND0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,					A21K, B21K			

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4. 81	PVVGB68S3810AKND0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket:R-PTFE, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,					A23D			
4. 82	PVVGB68S3810FCED0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, API 602,					B21A			
4. 83	PVVGB68S3812ACND0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW, ASTM A182 F316, Trim:316/STL, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,					D24P			
4. 84	PVVGB68S3816AJND0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW, ASTM A182 F316, Trim:STL, Stem:A182 F316, Packing:GBP, Gasket:S. S/PTFE SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,					A23B			
4. 85	PVVGB68S4438H11D0HT	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW, ASTM A182 F347H, Trim:STL, Stem:A182 F347H, Packing:High Temperature Packing for 704℃ Service, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, API 602, HIGH TEMPERATURE SEVICE					B27B			
4. 86	PVVGB68S4438H11D0HV	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL800, SW, ASTM A182 F347H, Trim:STL, Stem:A182 F347H, Packing:High Temperature Packing for 704℃ Service, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, API 602, FULL VACUUM AT HIGH TEMPERATURE SEVICE					B27A			
4. 87	PVVGB75B0710ACLD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL1500 (PN26. OMpa), BW, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602,					F2A			

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序号 NO.	阀 号 VALVE NUMBER	公 称 直 径 NPS	描 述 DESCRIPTION				管道等级 SPEC.	修改号 REV.	备 注 REMARK		
4. 88	PVVGB75B1110BCLC0ZZ	2" - 4"	Gate Valve, BB, OS&Y, CL1500 (PN26. 0Mpa), BW, ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 600,				F2A				
4. 89	PVVGB75N3864EMZM0FT	1/2" - 3/4"	Gate Valve, BB, OS&Y, CL1500 (PN26. 0Mpa), FNPT, ASTM A182 F316, Trim:316/PTFE, Stem:A182 F316, Packing:PTFE, Gasket:MFR STD, MFR STD, For Tubing				S82A				
4. 90	PVVGB75S3816ACED0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, CL1500 (PN26. 0MPa), SW, ASTM A182 F316, Trim:STL, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, API 602,				F23A				
4. 91	PVVG62K0301BCAC5ZZ	30"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 47 A), ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,				A1K				
4. 92	PVVG62K0308BCAC5VV	28" - 48"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 47 A), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600, Vacuum				A1DV				
4. 93	PVVG62K1110BCLC5ZZ	28" - 48"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 47 A), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Bevel Gear, API 600,				A2A				
4. 94	PVVG62R0301ACAC0ZZ	2"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,				A1K				
4. 95	PVVG62R0301BCAC0ZZ	2" - 14"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,				A1E, A1K				

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4. 96	PVVGF62R0301BCAC5ZZ	14" - 24"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,					A1E, A1K			
4. 97	PVVGF62R0308ACAC0ZZ	2"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,					A1A			
4. 98	PVVGF62R0308ACAD0VV	2"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602, Vacuum					A1AV			
4. 99	PVVGF62R0308BCAC0VV	2" - 14"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600, Vacuum					A1AV, A1DV			
4. 100	PVVGF62R0308BCAC0ZZ	3" - 14"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,					A1A			
4. 101	PVVGF62R0308BCAC5VV	16" - 24"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600, Vacuum					A1AV, A1DV			
4. 102	PVVGF62R0308BCAC5ZZ	16"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,					A1A			

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4. 103	PVVGF62R1110BCLCOVV	2" - 14"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 600, Vacuum					A2DV			
4. 104	PVVGF62R1110BCLCOZZ	2" - 14"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 600,					A2A, A2D, A2E, H2A			
4. 105	PVVGF62R1110BCLC5VV	16" - 24"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Bevel Gear, API 600, Vacuum					A2DV			
4. 106	PVVGF62R1110BCLC5ZZ	16" - 24"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Bevel Gear, API 600,					A2A, A2D, A2E, H2A			
4. 107	PVVGF62R5410BCAC0ZZ	2" - 14"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,					A21A			
4. 108	PVVGF62R5410BCAC5ZZ	16" - 24"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,					A21A			
4. 109	PVVGF62R5410BCNCOEV	2" - 14"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600, Extented stem					A21L			

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4. 110	PVVGF62R5410BCNC0ZZ	2" - 14"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600,					A21K			
4. 111	PVVGF62R5410BCNC5EV	16" - 24"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Bevel Gear, API 600, Extented stem					A21L			
4. 112	PVVGF62R5410BCNC5ZZ	16" - 24"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Bevel Gear, API 600,					A21K			
4. 113	PVVGF62R5410BKNC0ZZ	2" - 12"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket:R-PTFE, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600,					A21D, A23D			
4. 114	PVVGF62R5416BJNC0ZZ	2" - 14"	Gate Valve, BB, OS&Y, FP, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:STL, Stem:A182 F316, Packing:GBPR, Gasket:S. S/PTFE SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600,					A23B			
4. 115	PVVGF63B0301BCAC0BZ	2" - 10"	Gate Valve, BB, OS&Y, FP, CL300 (PN5. 0MPa), BW, ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600, Hydrogen Service					B1E			
4. 116	PVVGF63B0301BCAC5BZ	12" - 24"	Gate Valve, BB, OS&Y, FP, CL300 (PN5. 0MPa), BW, ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600, Hydrogen Service					B1E			

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4. 117	PVVGF63D3810ACND0EV	1" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL300 (PN5. 0MPa), SW/FNPT, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602, Extented stem					B21L			
4. 118	PVVGF63J1808BQCD0ZZ	2"	Gate Valve, BB, OS&Y, FP, CL300 (PN5. 0MPa), RJ, ASTM A182 F11 CL2, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket:12Cr5Mo(1Cr5Mo), Bolt/Nut:A193 Gr. B16/A194 Gr. 4, API 602,					B54H			
4. 119	PVVGF63J2708BQCC0ZZ	3" - 10"	Gate Valve, BB, OS&Y, FP, CL300 (PN5. 0MPa), RJ, ASTM A217 WC6, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket:12Cr5Mo(1Cr5Mo), Bolt/Nut:A193 Gr. B16/A194 Gr. 4, API 600,					B54H			
4. 120	PVVGF63J2708BQCD5ZZ	12" - 24"	Gate Valve, BB, OS&Y, FP, CL300 (PN5. 0MPa), RJ, ASTM A217 WC6, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket:12Cr5Mo(1Cr5Mo), Bolt/Nut:A193 Gr. B16/A194 Gr. 4, Bevel Gear, API 602,					B54H			
4. 121	PVVGF63K0308BCAC5ZZ	30"	Gate Valve, BB, OS&Y, FP, CL300 (PN5. 0MPa), RF (ASME B16. 47 A), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,					B1B			
4. 122	PVVGF63R0301BCAC0BZ	2" - 10"	Gate Valve, BB, OS&Y, FP, CL300 (PN5. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600, Hydrogen Service					B1E			

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4.123	PVVGF63R0301BCAC5BZ	12" - 24"	Gate Valve, BB, OS&Y, FP, CL300 (PN5.0MPa), RF (ASME B16.5), ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL.2, Packing:GBPR, Gasket: S.S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600, Hydrogen Service				B1E				
4.124	PVVGF63R0308BCAC0ZZ	2" - 10"	Gate Valve, BB, OS&Y, FP, CL300 (PN5.0MPa), RF (ASME B16.5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL.2, Packing:GBPR, Gasket: S.S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,				B1A, B1B				
4.125	PVVGF63R0308BCAC5ZZ	12" - 24"	Gate Valve, BB, OS&Y, FP, CL300 (PN5.0MPa), RF (ASME B16.5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL.2, Packing:GBPR, Gasket: S.S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,				B1A, B1B				
4.126	PVVGF63R1110BCLC0ZZ	2" - 10"	Gate Valve, BB, OS&Y, FP, CL300 (PN5.0MPa), RF (ASME B16.5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S.S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 600,				B2A, B2B				
4.127	PVVGF63R1110BCLC5ZZ	12" - 24"	Gate Valve, BB, OS&Y, FP, CL300 (PN5.0MPa), RF (ASME B16.5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S.S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Bevel Gear, API 600,				B2A, B2B				
4.128	PVVGF63R5410BCNC0EV	2" - 10"	Gate Valve, BB, OS&Y, FP, CL300 (PN5.0MPa), RF (ASME B16.5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S.S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600, Extented stem				B21L				
4.129	PVVGF63R5410BCNC0ZZ	3" - 10"	Gate Valve, BB, OS&Y, FP, CL300 (PN5.0MPa), RF (ASME B16.5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S.S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600,				B21K				

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4. 130	PVVGF63R5410BCNC5EV	12" - 24"	Gate Valve, BB, OS&Y, FP, CL300 (PN5. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Bevel Gear, API 600, Extented stem				B21L				
4. 131	PVVGF63R5410BCNC5ZZ	12" - 24"	Gate Valve, BB, OS&Y, FP, CL300 (PN5. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Bevel Gear, API 600,				B21K				
4. 132	PVVGF63R5410BCND0ZZ	2"	Gate Valve, BB, OS&Y, FP, CL300 (PN5. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,				B21K				
4. 133	PVVGF66R0308BCAC0ZZ	2" - 8"	Gate Valve, BB, OS&Y, FP, CL600 (PN11. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,				D1A				
4. 134	PVVGF66R0308BCAC5ZZ	10" - 24"	Gate Valve, BB, OS&Y, FP, CL600 (PN11. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,				D1A				
4. 135	PVVGF66R1110BCLC0ZZ	2" - 8"	Gate Valve, BB, OS&Y, FP, CL600 (PN11. 0MPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 600,				D2A				
4. 136	PVVGF66R1110BCLC5ZZ	10" - 12"	Gate Valve, BB, OS&Y, FP, CL600 (PN11. 0MPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Bevel Gear, API 600,				D2A				

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4. 137	PVVGF66R5410BCNC0ZZ	2" - 8"	Gate Valve, BB, OS&Y, FP, CL600 (PN11. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600,					D21K			
4. 138	PVVGF66R5410BCNC5ZZ	10" - 12"	Gate Valve, BB, OS&Y, FP, CL600 (PN11. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Bevel Gear, API 600,					D21K			
4. 139	PVVGF66R5412BCNC0ZZ	2" - 8"	Gate Valve, BB, OS&Y, FP, CL600 (PN11. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316/STL, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600,					D24P			
4. 140	PVVGF66R5412BCNC5ZZ	10" - 16"	Gate Valve, BB, OS&Y, FP, CL600 (PN11. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316/STL, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Bevel Gear, API 600,					D24P			
4. 141	PVVGF68D0101ACAD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW/FNPT, ASTM A105, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A1E			
4. 142	PVVGF68D0108ACAD0VV	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW/FNPT, ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602, Vacuum					A1AV			
4. 143	PVVGF68D0108ACAD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW/FNPT, ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					B1A, B1B, D1A			

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4. 144	PVVGF68D0710ACLD0VV	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW/FNPT, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602, Vacuum					A2DV			
4. 145	PVVGF68D0710ACLD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW/FNPT, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602,					A2D, A2E, B2A, B2B			
4. 146	PVVGF68D3810ACND0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW/FNPT, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,					A21K, B21L, D21K			
4. 147	PVVGF68D3810ACNM0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW/FNPT, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, MFR STD,					B21K			
4. 148	PVVGF68S0101ACAD0BZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW, ASTM A105, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602, Hydrogen Service					B1E			
4. 149	PVVGF68S0101ACAD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW, ASTM A105, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A1E			
4. 150	PVVGF68S0108ACAD0VV	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW, ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602, Vacuum					A1AV			
4. 151	PVVGF68S0108ACAD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW, ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A1A, B1A, B1B, D1A			

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4. 152	PVVGF68S0710ACLD0VV	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602, Vacuum				A2DV				
4. 153	PVVGF68S0710ACLD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602,				A2A, A2D, A2E, B2A, B2B, H2A				
4. 154	PVVGF68S3810ACAD0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,				A21A				
4. 155	PVVGF68S3810ACND0EV	1" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602, Extented stem				A21L				
4. 156	PVVGF68S3810ACND0VV	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602, Vacuum				D21K				
4. 157	PVVGF68S3810ACND0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,				A21K, A21L, B21K				
4. 158	PVVGF68S3810AKND0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket:R-PTFE, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,				A23D				
4. 159	PVVGF68S3816AJND0ZZ	1/2" - 1-1/2"	Gate Valve, BB, OS&Y, FP, CL800, SW, ASTM A182 F316, Trim:STL, Stem:A182 F316, Packing:GBP, Gasket:S. S/PTFE SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,				A23B				

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4. 160	PVVG75B1110BCLC5ZZ	6" - 10"	Gate Valve, BB, OS&Y, FP, CL1500 (PN26. 0Mpa) , BW, ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Bevel Gear, API 600,					F2A			
4. 161	PVVG75B5416BCECOZZ	2" - 8"	Gate Valve, BB, OS&Y, FP, CL1500 (PN26. 0Mpa) , BW, ASTM A351 CF8M, Trim:STL, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, API 600,					F23A			
4. 162	PVVG69B2715FMACOZZ	2" - 6"	Gate Valve, PSB, OS&Y, CL900 (PN15. 0MPa) , BW, ASTM A217 WC6, Trim:STL, Stem:A182 F304, Packing:FG, Gasket:MFR STD, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,					E54K			
4. 163	PVVG69B2715FMAC5ZZ	8" - 16"	Gate Valve, PSB, OS&Y, CL900 (PN15. 0MPa) , BW, ASTM A217 WC6, Trim:STL, Stem:A182 F304, Packing:FG, Gasket:MFR STD, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,					E54K			
4. 164	PVVG69J2715FMACOZZ	2" - 6"	Gate Valve, PSB, OS&Y, CL900 (PN15. 0MPa) , RJ, ASTM A217 WC6, Trim:STL, Stem:A182 F304, Packing:FG, Gasket:MFR STD, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,					E54K			
4. 165	PVVG69J2715FMAC5ZZ	8" - 16"	Gate Valve, PSB, OS&Y, CL900 (PN15. 0MPa) , RJ, ASTM A217 WC6, Trim:STL, Stem:A182 F304, Packing:FG, Gasket:MFR STD, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,					E54K			
4. 166	PVVG75B1815FMADOZZ	1/2" - 1-1/2"	Gate Valve, PSB, OS&Y, CL1500 (PN26. 0Mpa) , BW, ASTM A182 F11 CL2, Trim:STL, Stem:A182 F304, Packing:FG, Gasket:MFR STD, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					E54K			
5	PVVK!Knife Valve										

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5. 1	PVVKB62R9B2DBCAU0ZZ	3" - 14"	Knife Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), G-ALZN 10 SI 8 MG BODY, Trim:316, Plate:316, Stem:316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, MSS SP-81,					A24P			
5. 2	PVVKB62R9B2DBCAU5ZZ	16" - 24"	Knife Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), G-ALZN 10 SI 8 MG BODY, Trim:316, Plate:316, Stem:316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, MSS SP-81,					A24P			
6	PVVL!Globe Valve										
6. 1	PVVLB62R0102FCAD0ZZ	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A105, Trim:304, Stem:A182 F304, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A1R			
6. 2	PVVLB62R0108AVAD0VP	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket:Non-ASB Gasket, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602, Parabolic Disc For Vacuum					A1B			
6. 3	PVVLB62R0113FCAD0ZZ	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A105, Trim:alloy 20, Stem:B462 US N08200, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A1S			
6. 4	PVVLB62R0301BCAC0ZZ	2" - 12"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,					A1E, A1K, H1A			
6. 5	PVVLB62R0301BCAC5ZZ	14"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,					A1E, A1K			

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6. 6	PVVLB62R0302FCAK0ZZ	2" - 8"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:304, Stem:A182 F304, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, BS 1873,					A1R			
6. 7	PVVLB62R0305FCAK0ZZ	2" - 8"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:STL, Stem:A182 F6a CL. 2, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, BS 1873,					A1W			
6. 8	PVVLB62R0305FCAK5ZZ	14"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:STL, Stem:A182 F6a CL. 2, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, BS 1873,					A1W			
6. 9	PVVLB62R0308BCAC0VV	2" - 12"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600, Vacuum					A1AV, A1DV			
6. 10	PVVLB62R0308BCAC0ZZ	2" - 12"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,					A1A, A1D			
6. 11	PVVLB62R0308BCAC5VV	14" - 16"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600, Vacuum					A1AV, A1DV			
6. 12	PVVLB62R0308BCAC5ZZ	14"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,					A1A, A1D			

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6. 13	PVVLB62R0308BVACOPD	2"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket:Non-ASB Gasket, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600, Parabolic Disc					A1B			
6. 14	PVVLB62R0313FCAK0ZZ	2" - 8"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:alloy 20, Stem:B462 US N08200, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, BS 1873,					A1S			
6. 15	PVVLB62R1110BCLCOVV	2" - 12"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 600, Vacuum					A2DV			
6. 16	PVVLB62R1110BCLCOZZ	2" - 12"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 600,					A2A, A2D, A2E, B2B, H2A			
6. 17	PVVLB62R1110BCLC5VV	14"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Bevel Gear, API 600, Vacuum					A2DV			
6. 18	PVVLB62R1110BCLC5ZZ	10" - 14"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Bevel Gear, API 600,					A2A, A2D, A2E, B2B			
6. 19	PVVLB62R1110EJLC0ZZ	2" - 12"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:PTFE, Gasket:S. S/PTFE SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 600,					H3A			

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6. 20	PVVLB62R3810AKNC0ZZ	1/2" - 2"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket:R-PTFE, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600,					A21D			
6. 21	PVVLB62R5410BCAC0ZZ	2" - 12"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,					A21A, A24P			
6. 22	PVVLB62R5410BCAC5ZZ	14"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,					A21A			
6. 23	PVVLB62R5410BCNCOEV	2" - 12"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600, Extented stem					A21L			
6. 24	PVVLB62R5410BCNC0ZZ	2" - 12"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600,					A21K			
6. 25	PVVLB62R5410BCNC5EV	14"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Bevel Gear, API 600, Extented stem					A21L			
6. 26	PVVLB62R5410BCNC5ZZ	14"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Bevel Gear, API 600,					A21K			

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6. 27	PVVLB62R5410BKNC0ZZ	2" - 12"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket:R-PTFE, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600,				A21D, A23D				
6. 28	PVVLB62R5416BJNC0ZZ	2" - 8"	Globe Valve, BB, OS&Y, CL150 (PN2. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:STL, Stem:A182 F316, Packing:GBPR, Gasket:S. S/PTFE SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600,				A23B				
6. 29	PVVLB63B0301BCAC0BZ	2" - 8"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), BW, ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600, Hydrogen Service				B1E				
6. 30	PVVLB63B0301BCAC5BZ	10" - 12"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), BW, ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600, Hydrogen Service				B1E				
6. 31	PVVLB63F5410FCEK0BZ	2" - 8"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), FE, ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, BS 1873, Hydrogen Service				B2E				
6. 32	PVVLB63F5410FCEK5BZ	10" - 12"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), FE, ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, Bevel Gear, BS 1873, Hydrogen Service				B2E				
6. 33	PVVLB63J27081QCC0ZZ	2" - 6"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RJ, ASTM A217 WC6, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:High Temperature Packing for 593℃ Service, Gasket:12Cr5Mo (1Cr5Mo), Bolt/Nut:A193 Gr. B16/A194 Gr. 4, API 600,				B54K				

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6. 34	PVVLB63J27081QCC5ZZ	8" - 14"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RJ, ASTM A217 WC6, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:High Temperature Packing for 593℃ Service, Gasket:12Cr5Mo (1Cr5Mo), Bolt/Nut:A193 Gr. B16/A194 Gr. 4, Bevel Gear, API 600,					B54K			
6. 35	PVVLB63J2708BQCC0ZZ	2" - 6"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RJ, ASTM A217 WC6, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket:12Cr5Mo (1Cr5Mo), Bolt/Nut:A193 Gr. B16/A194 Gr. 4, API 600,					B54H			
6. 36	PVVLB63J2708BQCC5ZZ	8" - 12"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RJ, ASTM A217 WC6, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket:12Cr5Mo (1Cr5Mo), Bolt/Nut:A193 Gr. B16/A194 Gr. 4, Bevel Gear, API 600,					B54H			
6. 37	PVVLB63J5538H11COHT	2" - 6"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RJ, ASTM A351 CF8C, Trim:STL, Stem:A182 F347H, Packing:High Temperature Packing for 704℃ Service, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, API 600, HIGH TEMPERATURE SEVICE					B27B			
6. 38	PVVLB63J5538H11COHV	2" - 6"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RJ, ASTM A351 CF8C, Trim:STL, Stem:A182 F347H, Packing:High Temperature Packing for 704℃ Service, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, API 600, FULL VACUUM AT HIGH TEMPERATURE SEVICE					B27A			
6. 39	PVVLB63J5538H11C5HT	8" - 14"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RJ, ASTM A351 CF8C, Trim:STL, Stem:A182 F347H, Packing:High Temperature Packing for 704℃ Service, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, Bevel Gear, API 600, HIGH TEMPERATURE SEVICE					B27B			

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6. 40	PVVLB63J5538H11C5HV	8" - 14"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RJ, ASTM A351 CF8C, Trim:STL, Stem:A182 F347H, Packing:High Temperature Packing for 704℃ Service, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, Bevel Gear, API 600, FULL VACUUM AT HIGH TEMPERATURE SEVICE					B27A			
6. 41	PVVLB63R0301BCAC0BZ	2" - 8"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600, Hydrogen Service					B1E			
6. 42	PVVLB63R0301BCAC5BZ	10" - 12"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600, Hydrogen Service					B1E			
6. 43	PVVLB63R0308BCAC0ZZ	2" - 8"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,					B1A, B1B			
6. 44	PVVLB63R0308BCAC5ZZ	10" - 14"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,					B1A, B1B			
6. 45	PVVLB63R1110BCLC0ZZ	2" - 8"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 600,					B2A			
6. 46	PVVLB63R1110BCLC5ZZ	10" - 14"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RF (ASME B16. 5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Bevel Gear, API 600,					B2A			

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6. 47	PVVLB63R3810FCED0ZZ	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RF (ASME B16. 5), ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, API 602,					B21D			
6. 48	PVVLB63R5410BCNCOEV	2" - 8"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600, Extented stem					B21L			
6. 49	PVVLB63R5410BCNCOZZ	2" - 8"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600,					B21K			
6. 50	PVVLB63R5410BCNC5EV	10" - 14"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Bevel Gear, API 600, Extented stem					B21L			
6. 51	PVVLB63R5410BCNC5ZZ	10" - 14"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Bevel Gear, API 600,					B21K			
6. 52	PVVLB63R5410FCEK0ZZ	2" - 8"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, BS 1873,					B21A, B21D			
6. 53	PVVLB63R5410FCEK5ZZ	10" - 12"	Globe Valve, BB, OS&Y, CL300 (PN5. OMPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, Bevel Gear, BS 1873,					B21A, B21D			

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6. 54	PVVLB66R0308BCAC0ZZ	2" - 6"	Globe Valve, BB, OS&Y, CL600 (PN11.0MPa), RF (ASME B16.5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL.2, Packing:GBPR, Gasket: S.S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 600,					D1A			
6. 55	PVVLB66R0308BCAC5ZZ	8" - 12"	Globe Valve, BB, OS&Y, CL600 (PN11.0MPa), RF (ASME B16.5), ASTM A216 WCB, Trim:13Cr/STL, Stem:A182 F6a CL.2, Packing:GBPR, Gasket: S.S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 600,					D1A			
6. 56	PVVLB66R1110BCLC0ZZ	2" - 6"	Globe Valve, BB, OS&Y, CL600 (PN11.0MPa), RF (ASME B16.5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S.S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 600,					D2A			
6. 57	PVVLB66R1110BCLC5ZZ	8" - 10"	Globe Valve, BB, OS&Y, CL600 (PN11.0MPa), RF (ASME B16.5), ASTM A352 LCB, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S.S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Bevel Gear, API 600,					D2A			
6. 58	PVVLB66R5310BCNC0ZZ	2" - 3"	Globe Valve, BB, OS&Y, CL600 (PN11.0MPa), RF (ASME B16.5), ASTM A351 CF3M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S.S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600,					D24L			
6. 59	PVVLB66R5410BCNC0ZZ	2" - 6"	Globe Valve, BB, OS&Y, CL600 (PN11.0MPa), RF (ASME B16.5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S.S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600,					D21K			
6. 60	PVVLB66R5410BCNC5ZZ	8" - 10"	Globe Valve, BB, OS&Y, CL600 (PN11.0MPa), RF (ASME B16.5), ASTM A351 CF8M, Trim:316, Stem:A182 F316, Packing:GBPR, Gasket: S.S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Bevel Gear, API 600,					D21K			
6. 61	PVVLB68B0105FCAD0ZZ	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, BW, ASTM A105, Trim:STL, Stem:A182 F6a CL.2, Packing:FG, Gasket: S.S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A1W			

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6. 62	PVVLB68B3810FCED0BZ	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, BW, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, API 602, Hydrogen Service					B2E			
6. 63	PVVLB68N0710EJLD0ZZ	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, FNPT, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:PTFE, Gasket:S. S/PTFE SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602,					H3A			
6. 64	PVVLB68R3810AKNC0ZZ	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, RF (ASME B16. 5), ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket:R-PTFE, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 600,					A23D			
6. 65	PVVLB68S0101ACAD0BZ	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, SW, ASTM A105, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602, Hydrogen Service					B1E			
6. 66	PVVLB68S0101ACAD0ZZ	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, SW, ASTM A105, Trim:13Cr, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A1E, A1K, H1A			
6. 67	PVVLB68S0108ACAD0VV	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, SW, ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602, Vacuum					A1AV, A1DV			
6. 68	PVVLB68S0108ACAD0ZZ	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, SW, ASTM A105, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					A1A, A1D, B1A, B1B, D1A			
6. 69	PVVLB68S0710ACLD0VV	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, SW, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602, Vacuum					A2DV			

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6. 70	PVVLB68S0710ACLD0ZZ	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, SW, ASTM A350 LF2 CL1, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602,				A2A, A2D, A2E, B2A, B2B, D2A, H2A				
6. 71	PVVLB68S18081QCD0ZZ	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, SW, ASTM A182 F11 CL2, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:High Temperature Packing for 593℃ Service, Gasket:12Cr5Mo(1Cr5Mo), Bolt/Nut:A193 Gr. B16/A194 Gr. 4, API 602,				B54K				
6. 72	PVVLB68S1808BQCD0ZZ	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, SW, ASTM A182 F11 CL2, Trim:13Cr/STL, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket:12Cr5Mo(1Cr5Mo), Bolt/Nut:A193 Gr. B16/A194 Gr. 4, API 602,				B54H				
6. 73	PVVLB68S3810ACAD0ZZ	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,				A21A				
6. 74	PVVLB68S3810ACND0EV	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602, Extented stem				A21L, B21L				
6. 75	PVVLB68S3810ACND0VV	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602, Vacuum				D21K				
6. 76	PVVLB68S3810ACND0ZZ	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,				A21K, B21K				

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6. 77	PVVLB68S3810FCED0ZZ	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, SW, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:FG, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, API 602,					B21A			
6. 78	PVVLB68S3816AJND0ZZ	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, SW, ASTM A182 F316, Trim:STL, Stem:A182 F316, Packing:GBP, Gasket:S. S/PTFE SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,					A23B			
6. 79	PVVLB68S3910ACND0ZZ	1" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, SW, ASTM A182 F316L, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 602,					D24L			
6. 80	PVVLB68S4438H11DOHT	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, SW, ASTM A182 F347H, Trim:STL, Stem:A182 F347H, Packing:High Temperature Packing for 704℃ Service, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, API 602, HIGH TEMPERATURE SEVICE					B27B			
6. 81	PVVLB68S4438H11DOHV	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL800, SW, ASTM A182 F347H, Trim:STL, Stem:A182 F347H, Packing:High Temperature Packing for 704℃ Service, Gasket:347S. S, Bolt/Nut:B572 UNS N06002/B572 UNS N06002, API 602, FULL VACUUM AT HIGH TEMPERATURE SEVICE					B27A			
6. 82	PVVLB75B5416BCECOZZ	2" - 8"	Globe Valve, BB, OS&Y, CL1500 (PN26. 0Mpa), BW, ASTM A351 CF8M, Trim:STL, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, API 600,					F23A			
6. 83	PVVLB75S3816ACED0ZZ	1/2" - 1-1/2"	Globe Valve, BB, OS&Y, CL1500 (PN26. 0MPa), SW, ASTM A182 F316, Trim:STL, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B8M CLASS 1/A194 Gr. 8M, API 602,					F23A			

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6. 84	PVVLP69B2715FMAK0ZZ	2" - 4"	Globe Valve, PSB, OS&Y, CL900 (PN15. 0MPa), BW, ASTM A217 WC6, Trim:STL, Stem:A182 F304, Packing:FG, Gasket:MFR STD, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, BS 1873,					E54K			
6. 85	PVVLP69B2715FMAK5ZZ	6" - 8"	Globe Valve, PSB, OS&Y, CL900 (PN15. 0MPa), BW, ASTM A217 WC6, Trim:STL, Stem:A182 F304, Packing:FG, Gasket:MFR STD, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, BS 1873,					E54K			
6. 86	PVVLP69J2715FMAK0ZZ	2" - 4"	Globe Valve, PSB, OS&Y, CL900 (PN15. 0MPa), RJ, ASTM A217 WC6, Trim:STL, Stem:A182 F304, Packing:FG, Gasket:MFR STD, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, BS 1873,					E54K			
6. 87	PVVLP69J2715FMAK5ZZ	6" - 12"	Globe Valve, PSB, OS&Y, CL900 (PN15. 0MPa), RJ, ASTM A217 WC6, Trim:STL, Stem:A182 F304, Packing:FG, Gasket:MFR STD, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, BS 1873,					E54K			
6. 88	PVVLP75B1815FMAD0ZZ	1/2" - 1"	Globe Valve, PSB, OS&Y, CL1500 (PN26. 0Mpa), BW, ASTM A182 F11 CL2, Trim:STL, Stem:A182 F304, Packing:FG, Gasket:MFR STD, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 602,					E54K			
6. 89	PVVLP85B3816FMLD0BZ	1/2" - 1-1/2"	Globe Valve, PSB, OS&Y, CL2500 (PN42. 0MPa), BW, ASTM A182 F316, Trim:STL, Stem:A182 F316, Packing:FG, Gasket:MFR STD, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 602, Hydrogen Service					G82E			
6. 90	PVVLP85B5416FMLC0BZ	2"	Globe Valve, PSB, OS&Y, CL2500 (PN42. 0MPa), BW, ASTM A351 CF8M, Trim:STL, Stem:A182 F316, Packing:FG, Gasket:MFR STD, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 600, Hydrogen Service					G82E			
6. 91	PVVLP85B5416FMLC5BZ	3" - 8"	Globe Valve, PSB, OS&Y, CL2500 (PN42. 0MPa), BW, ASTM A351 CF8M, Trim:STL, Stem:A182 F316, Packing:FG, Gasket:MFR STD, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Bevel Gear, API 600, Hydrogen Service					G82E			

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7	PVVN!Needle Valve										
7. 1	PVVNT68N3810ACAN0ZZ	1/2" - 1-1/2"	Needle Valve, TB, OS&Y, CL800, FNPT, ASTM A182 F316, Trim:316, Stem:A182 F316, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, ASME B16. 34,					A24P			
8	PVVP!Plug Valve										
8. 1	PVVPC62R0364A0AB1ZZ	1/2" - 4"	Plug Valve, STD PATN, BB, Cone Plug, nonlubricated, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:316/PTFE, Stem:A182 F316, Packing:GBP, Gasket:316/PFA DIAPHRAGM, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 599,					A1E			
8. 2	PVVPC62R0364A0AB5ZZ	6" - 12"	Plug Valve, STD PATN, BB, Cone Plug, nonlubricated, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:316/PTFE, Stem:A182 F316, Packing:GBP, Gasket:316/PFA DIAPHRAGM, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 599,					A1E			
8. 3	PVVPC62R037DACAB1ZZ	1/2" - 4"	Plug Valve, STD PATN, BB, Cone Plug, nonlubricated, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, PLUG:316, Seat:PTEF, Packing:GBP, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 599,					A1A			
8. 4	PVVPC62R037DBCAB5ZZ	6" - 12"	Plug Valve, STD PATN, BB, Cone Plug, nonlubricated, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, PLUG:316, Seat:PTEF, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Bevel Gear, API 599,					A1A			

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8. 5	PVVPC62R9097EKAB0EV	1" - 4"	Plug Valve, STD PATN, BB, Cone Plug, nonlubricated, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB+PTFE, Trim:304/PTFE LINED/RTFE, Stem:304/PTFE LINED, Packing:PTFE, Gasket:R-PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 599, Extented stem					A01S			
8. 6	PVVPC62R9097EKAB0ZZ	1" - 4"	Plug Valve, STD PATN, BB, Cone Plug, nonlubricated, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB+PTFE, Trim:304/PTFE LINED/RTFE, Stem:304/PTFE LINED, Packing:PTFE, Gasket:R-PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 599,					A01S			
9	PVVU!Butterfly Valve										
9. 1	PVVUB62I0387BCAG3ZZ	36"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 47 A), ASTM A216 WCB, Trim:304/Buna-N, Stem:A182 F304, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					H1A			
9. 2	PVVUB62I0399BCAG3ZZ	30"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 47 A), ASTM A216 WCB, Trim:316/RPTFE, Stem:ASTM A705 17- 4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					A1K			
9. 3	PVVUB62I1199BCLG3ZZ	30" - 48"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 47 A), ASTM A352 LCB, Trim:316/RPTFE, Stem:ASTM A705 17- 4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Worm Gear, API 609,					A2A			

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9. 4	PVVUB62K0368BCAG3VV	30" - 48"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), RF (ASME B16. 47 A), ASTM A216 WCB, Trim:13Cr/RTFE, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609, Vacuum					A1DV			
9. 5	PVVUB62K0368BCAG3ZZ	30" - 48"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), RF (ASME B16. 47 A), ASTM A216 WCB, Trim:13Cr/RTFE, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					A1D			
9. 6	PVVUB62L0364ANAG0ZZ	3" - 4"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Lug (ASME B16. 5), ASTM A216 WCB, Trim:316/PTFE, Stem:A182 F316, Packing:GBP, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 609,					A1E			
9. 7	PVVUB62L0364ANAG3ZZ	6" - 24"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Lug (ASME B16. 5), ASTM A216 WCB, Trim:316/PTFE, Stem:A182 F316, Packing:GBP, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					A1E			
9. 8	PVVUB62L0399BCAG1VV	3" - 6"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Lug (ASME B16. 5), ASTM A216 WCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 609, Vacuum					A1AV			
9. 9	PVVUB62L0399BCAG1ZZ	3" - 6"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Lug (ASME B16. 5), ASTM A216 WCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 609,					A1A, A1K			

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9. 10	PVVUB62L0399BCAG3VV	8" - 24"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Lug (ASME B16. 5), ASTM A216 WCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609, Vacuum					A1AV			
9. 11	PVVUB62L0399BCAG3ZZ	8" - 24"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Lug (ASME B16. 5), ASTM A216 WCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					A1A, A1K			
9. 12	PVVUB62L1199BCLG1ZZ	3" - 6"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Lug (ASME B16. 5), ASTM A352 LCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, LEVER, API 609,					A2A			
9. 13	PVVUB62L1199BCLG3ZZ	8" - 24"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Lug (ASME B16. 5), ASTM A352 LCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Worm Gear, API 609,					A2A			
9. 14	PVVUB62L5499BCAG1ZZ	3" - 6"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Lug (ASME B16. 5), ASTM A351 CF8M, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 609,					A21A			
9. 15	PVVUB62L5499BCAG3ZZ	8" - 24"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Lug (ASME B16. 5), ASTM A351 CF8M, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					A21A			

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9. 16	PVVUB62L5499BCNG1ZZ	3" - 6"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Lug (ASME B16. 5), ASTM A351 CF8M, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, LEVER, API 609,					A21K			
9. 17	PVVUB62L5499BCNG3ZZ	8" - 24"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Lug (ASME B16. 5), ASTM A351 CF8M, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Worm Gear, API 609,					A21K			
9. 18	PVVUB6200399BCAG3ZZ	30"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Lug (ASME B16. 47 A), ASTM A216 WCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					A1K			
9. 19	PVVUB6201199BCLG3ZZ	30" - 48"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Lug (ASME B16. 47 A), ASTM A352 LCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Worm Gear, API 609,					A2A			
9. 20	PVVUB62R0368BCAG0VV	3" - 6"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/RTFE, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 609, Vacuum					A1DV			
9. 21	PVVUB62R0368BCAG0ZZ	3" - 6"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/RTFE, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 609,					A1D			

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9. 22	PVVUB62R0368BCAG3VV	8" - 24"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/RTFE, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609, Vacuum					A1DV			
9. 23	PVVUB62R0368BCAG3ZZ	8" - 24"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A216 WCB, Trim:13Cr/RTFE, Stem:A182 F6a CL. 2, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					A1D			
9. 24	PVVUB62W0364ANAG0ZZ	3" - 4"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Wafer (ASME B16. 5), ASTM A216 WCB, Trim:316/PTFE, Stem:A182 F316, Packing:GBP, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 609,					A1E			
9. 25	PVVUB62W0364ANAG3ZZ	6" - 24"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Wafer (ASME B16. 5), ASTM A216 WCB, Trim:316/PTFE, Stem:A182 F316, Packing:GBP, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					A1E			
9. 26	PVVUB62W0387BCAG0ZZ	3" - 6"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A216 WCB, Trim:304/Buna-N, Stem:A182 F304, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 609,					H1A			
9. 27	PVVUB62W0387BCAG3ZZ	8" - 24"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A216 WCB, Trim:304/Buna-N, Stem:A182 F304, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					H1A			

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9. 28	PVVUB62W0399BCAG1VV	3" - 6"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A216 WCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 609, Vacuum					A1AV			
9. 29	PVVUB62W0399BCAG1ZZ	3" - 6"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A216 WCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 609,					A1A, A1K, H3B			
9. 30	PVVUB62W0399BCAG3VV	8" - 24"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A216 WCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609, Vacuum					A1AV			
9. 31	PVVUB62W0399BCAG3ZZ	8" - 24"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A216 WCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					A1A, A1K, H3B			
9. 32	PVVUB62W1199BCLG1ZZ	3" - 6"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A352 LCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, LEVER, API 609,					A2A, H2A			
9. 33	PVVUB62W1199BCLG3ZZ	8" - 24"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A352 LCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Worm Gear, API 609,					A2A, H2A			

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9. 34	PVVUB62W5499BCAG1ZZ	3" - 6"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A351 CF8M, Trim: 316/RPTFE, Stem: ASTM A705 17-4PH, Packing: GBPR, Gasket: S. S/FG SWG, Bolt/Nut: A193 Gr. B7/A194 Gr. 2H, LEVER, API 609,					A21A			
9. 35	PVVUB62W5499BCAG3ZZ	8" - 24"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A351 CF8M, Trim: 316/RPTFE, Stem: ASTM A705 17-4PH, Packing: GBPR, Gasket: S. S/FG SWG, Bolt/Nut: A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					A21A			
9. 36	PVVUB62W5499BCNG1ZZ	3" - 6"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A351 CF8M, Trim: 316/RPTFE, Stem: ASTM A705 17-4PH, Packing: GBPR, Gasket: S. S/FG SWG, Bolt/Nut: A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, LEVER, API 609,					A21K			
9. 37	PVVUB62W5499BCNG3ZZ	8" - 24"	Butterfly Valve, Bi-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A351 CF8M, Trim: 316/RPTFE, Stem: ASTM A705 17-4PH, Packing: GBPR, Gasket: S. S/FG SWG, Bolt/Nut: A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Worm Gear, API 609,					A21K			
9. 38	PVVUB63L0371BCAG0ZZ	3" - 4"	Butterfly Valve, Bi-offset, CL300 (PN5. 0MPa), Lug (ASME B16. 5), ASTM A216 WCB, Trim: 316/RTFE, Stem: A182 F316, Packing: GBPR, Gasket: S. S/FG SWG, Bolt/Nut: A193 Gr. B7/A194 Gr. 2H, API 609,					B1A			
9. 39	PVVUB63L0371BCAG3ZZ	6" - 24"	Butterfly Valve, Bi-offset, CL300 (PN5. 0MPa), Lug (ASME B16. 5), ASTM A216 WCB, Trim: 316/RTFE, Stem: A182 F316, Packing: GBPR, Gasket: S. S/FG SWG, Bolt/Nut: A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					B1A			

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9. 40	PVVUB63L0399BCAG1BZ	3" - 4"	Butterfly Valve, Bi-offset, CL300 (PN5. 0MPa), Lug (ASME B16. 5), ASTM A216 WCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 609, Hydrogen Service					B1E			
9. 41	PVVUB63L0399BCAG3BZ	6" - 24"	Butterfly Valve, Bi-offset, CL300 (PN5. 0MPa), Lug (ASME B16. 5), ASTM A216 WCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609, Hydrogen Service					B1E			
9. 42	PVVUB63L1199BCLG1ZZ	3" - 4"	Butterfly Valve, Bi-offset, CL300 (PN5. 0MPa), Lug (ASME B16. 5), ASTM A352 LCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, LEVER, API 609,					B2A			
9. 43	PVVUB63L1199BCLG3ZZ	6" - 20"	Butterfly Valve, Bi-offset, CL300 (PN5. 0MPa), Lug (ASME B16. 5), ASTM A352 LCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Worm Gear, API 609,					B2A			
9. 44	PVVUB63L5499BCNG1ZZ	3" - 4"	Butterfly Valve, Bi-offset, CL300 (PN5. 0MPa), Lug (ASME B16. 5), ASTM A351 CF8M, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, LEVER, API 609,					B21K			
9. 45	PVVUB63L5499BCNG3ZZ	6" - 24"	Butterfly Valve, Bi-offset, CL300 (PN5. 0MPa), Lug (ASME B16. 5), ASTM A351 CF8M, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Worm Gear, API 609,					B21K			

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9. 46	PVVUB63W0371BCAG0ZZ	3" - 4"	Butterfly Valve, Bi-offset, CL300 (PN5. 0MPa), Wafer, RF (ASME B16. 5), ASTM A216 WCB, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 609,					B1A			
9. 47	PVVUB63W0371BCAG3ZZ	6" - 24"	Butterfly Valve, Bi-offset, CL300 (PN5. 0MPa), Wafer, RF (ASME B16. 5), ASTM A216 WCB, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					B1A			
9. 48	PVVUB63W0399BCAG1BZ	3" - 4"	Butterfly Valve, Bi-offset, CL300 (PN5. 0MPa), Wafer, RF (ASME B16. 5), ASTM A216 WCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, LEVER, API 609, Hydrogen Service					B1E			
9. 49	PVVUB63W0399BCAG3BZ	6" - 24"	Butterfly Valve, Bi-offset, CL300 (PN5. 0MPa), Wafer, RF (ASME B16. 5), ASTM A216 WCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609, Hydrogen Service					B1E			
9. 50	PVVUB63W1199BCLG1ZZ	3" - 4"	Butterfly Valve, Bi-offset, CL300 (PN5. 0MPa), Wafer, RF (ASME B16. 5), ASTM A352 LCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, LEVER, API 609,					B2A			
9. 51	PVVUB63W1199BCLG3ZZ	6" - 20"	Butterfly Valve, Bi-offset, CL300 (PN5. 0MPa), Wafer, RF (ASME B16. 5), ASTM A352 LCB, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Worm Gear, API 609,					B2A			

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9. 52	PVVUB63W5499BCNG1ZZ	3" - 4"	Butterfly Valve, Bi-offset, CL300 (PN5. 0MPa), Wafer, RF (ASME B16. 5), ASTM A351 CF8M, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, LEVER, API 609,					B21K			
9. 53	PVVUB63W5499BCNG3ZZ	6" - 24"	Butterfly Valve, Bi-offset, CL300 (PN5. 0MPa), Wafer, RF (ASME B16. 5), ASTM A351 CF8M, Trim:316/RPTFE, Stem:ASTM A705 17-4PH, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Worm Gear, API 609,					B21K			
9. 54	PVVUC62L8ZX1EUAG0ZZ	4" - 24"	Butterfly Valve, Center Line, CL150 (PN2. 0MPa), Lug (ASME B16. 5), ASTM A126 GR. B, Trim:LINED WITH PFA SEAT, DUCTILE IRON DISC COATED PFA, Packing:PTFE, Gasket:PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 609,					H61B			
9. 55	PVVUC62L8ZX1EUAG3ZZ	16"	Butterfly Valve, Center Line, CL150 (PN2. 0MPa), Lug (ASME B16. 5), ASTM A126 GR. B, Trim:LINED WITH PFA SEAT, DUCTILE IRON DISC COATED PFA, Packing:PTFE, Gasket:PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					H61C			
9. 56	PVVUC62L9097EKAG0ZZ	2" - 4"	Butterfly Valve, Center Line, CL150 (PN2. 0MPa), Lug (ASME B16. 5), ASTM A216 WCB+PTFE, Trim:304/PTFE LINED/RTFE, Stem:304/PTFE LINED, Packing:PTFE, Gasket:R-PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 609,					A01S			
9. 57	PVVUC62L9097EKAG3ZZ	6" - 24"	Butterfly Valve, Center Line, CL150 (PN2. 0MPa), Lug (ASME B16. 5), ASTM A216 WCB+PTFE, Trim:304/PTFE LINED/RTFE, Stem:304/PTFE LINED, Packing:PTFE, Gasket:R-PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					A01S			

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						Contr. 图号 Contr. DRW. NO.		108184-0000-MA-SPC-0002		日 期 DATE	2012. 12. 14
						KPI 图号 KPI DRW. NO.		108184-0000-MA-SPC-0002			
序号 NO.	阀 号 VALVE NUMBER	公 称 直 径 NPS	描 述 DESCRIPTION					管道等级 SPEC.	修改号 REV.	备 注 REMARK	
9. 58	PVVUC62W9097EKAG0ZZ	2" - 4"	Butterfly Valve, Center Line, CL150 (PN2. OMPa), Wafer, RF (ASME B16. 5), ASTM A216 WCB+PTFE, Trim:304/PTFE LINED/RTFE, Stem:304/PTFE LINED, Packing:PTFE, Gasket:R-PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 609,					A01S			
9. 59	PVVUC62W9097EKAG3ZZ	6" - 8"	Butterfly Valve, Center Line, CL150 (PN2. OMPa), Wafer, RF (ASME B16. 5), ASTM A216 WCB+PTFE, Trim:304/PTFE LINED/RTFE, Stem:304/PTFE LINED, Packing:PTFE, Gasket:R-PTFE, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					A01S			
9. 60	PVVUT62L1171BCLG0VV	3" - 6"	Butterfly Valve, Tri-offset, CL150 (PN2. OMPa), Lug (ASME B16. 5), ASTM A352 LCB, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 609, Vacuum					A2DV			
9. 61	PVVUT62L1171BCLG0ZZ	3" - 6"	Butterfly Valve, Tri-offset, CL150 (PN2. OMPa), Lug (ASME B16. 5), ASTM A352 LCB, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, API 609,					A2D, A2E, B2B			
9. 62	PVVUT62L1171BCLG3VV	8" - 24"	Butterfly Valve, Tri-offset, CL150 (PN2. OMPa), Lug (ASME B16. 5), ASTM A352 LCB, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Worm Gear, API 609, Vacuum					A2DV			
9. 63	PVVUT62L1171BCLG3ZZ	6" - 24"	Butterfly Valve, Tri-offset, CL150 (PN2. OMPa), Lug (ASME B16. 5), ASTM A352 LCB, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. L7/A194 Gr. 4 S3, Worm Gear, API 609,					A2D, A2E, B2B			

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						KPI 图号 KPI DRW. NO.		108184-0000-MA-SPC-0002			
序号 NO.	阀 号 VALVE NUMBER	公 称 直 径 NPS	描 述 DESCRIPTION					管道等级 SPEC.	修改号 REV.	备 注 REMARK	
9. 64	PVVUT62R5464BCNG0EV	3" - 6"	Butterfly Valve, Tri-offset, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim: 316/PTFE, Stem: A182 F316, Packing: GBPR, Gasket: S. S/FG SWG, Bolt/Nut: A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 609, Extented stem					A21L			
9. 65	PVVUT62R5464BCNG3EV	8" - 24"	Butterfly Valve, Tri-offset, CL150 (PN2. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim: 316/PTFE, Stem: A182 F316, Packing: GBPR, Gasket: S. S/FG SWG, Bolt/Nut: A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Worm Gear, API 609, Extented stem					A21L			
9. 66	PVVUT62W1171BCLG0VV	3" - 6"	Butterfly Valve, Tri-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A352 LCB, Trim: 316/RTFE, Stem: A182 F316, Packing: GBPR, Gasket: S. S/FG SWG, Bolt/Nut: A320 Gr. L7/A194 Gr. 4 S3, API 609, Vacuum					A2DV			
9. 67	PVVUT62W1171BCLG0ZZ	3" - 6"	Butterfly Valve, Tri-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A352 LCB, Trim: 316/RTFE, Stem: A182 F316, Packing: GBPR, Gasket: S. S/FG SWG, Bolt/Nut: A320 Gr. L7/A194 Gr. 4 S3, API 609,					A2D, A2E, B2B			
9. 68	PVVUT62W1171BCLG3VV	8" - 24"	Butterfly Valve, Tri-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A352 LCB, Trim: 316/RTFE, Stem: A182 F316, Packing: GBPR, Gasket: S. S/FG SWG, Bolt/Nut: A320 Gr. L7/A194 Gr. 4 S3, Worm Gear, API 609, Vacuum					A2DV			
9. 69	PVVUT62W1171BCLG3ZZ	6" - 24"	Butterfly Valve, Tri-offset, CL150 (PN2. 0MPa), Wafer, RF (ASME B16. 5), ASTM A352 LCB, Trim: 316/RTFE, Stem: A182 F316, Packing: GBPR, Gasket: S. S/FG SWG, Bolt/Nut: A320 Gr. L7/A194 Gr. 4 S3, Worm Gear, API 609,					A2D, A2E, B2B			

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						校 对 CHKD		设计阶段 PHASE	OBE	版 次 REV.	D01
						Contr. 图号 Contr. DRW. NO.		108184-0000-MA-SPC-0002		日 期 DATE	2012. 12. 14
						KPI 图号 KPI DRW. NO.		108184-0000-MA-SPC-0002			
序号 NO.	阀 号 VALVE NUMBER	公 称 直 径 NPS	描 述 DESCRIPTION					管道等级 SPEC.	修改号 REV.	备 注 REMARK	
9. 70	PVVUT63L0371BCAG0ZZ	3" - 4"	Butterfly Valve, Tri-offset, CL300 (PN5. 0MPa), Lug (ASME B16. 5), ASTM A216 WCB, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 609,					B1B			
9. 71	PVVUT63L0371BCAG3ZZ	6" - 24"	Butterfly Valve, Tri-offset, CL300 (PN5. 0MPa), Lug (ASME B16. 5), ASTM A216 WCB, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					B1B			
9. 72	PVVUT63R5464BCNG0EV	3" - 4"	Butterfly Valve, Tri-offset, CL300 (PN5. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316/PTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, API 609, Extented stem					B21L			
9. 73	PVVUT63R5464BCNG3EV	6" - 24"	Butterfly Valve, Tri-offset, CL300 (PN5. 0MPa), RF (ASME B16. 5), ASTM A351 CF8M, Trim:316/PTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A320 Gr. B8 CLASS 2/A194 Gr. 8 S1, Worm Gear, API 609, Extented stem					B21L			
9. 74	PVVUT63W0371BCAG0ZZ	3" - 4"	Butterfly Valve, Tri-offset, CL300 (PN5. 0MPa), Wafer, RF (ASME B16. 5), ASTM A216 WCB, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, API 609,					B1B			
9. 75	PVVUT63W0371BCAG3ZZ	6" - 24"	Butterfly Valve, Tri-offset, CL300 (PN5. 0MPa), Wafer, RF (ASME B16. 5), ASTM A216 WCB, Trim:316/RTFE, Stem:A182 F316, Packing:GBPR, Gasket: S. S/FG SWG, Bolt/Nut:A193 Gr. B7/A194 Gr. 2H, Worm Gear, API 609,					B1B			